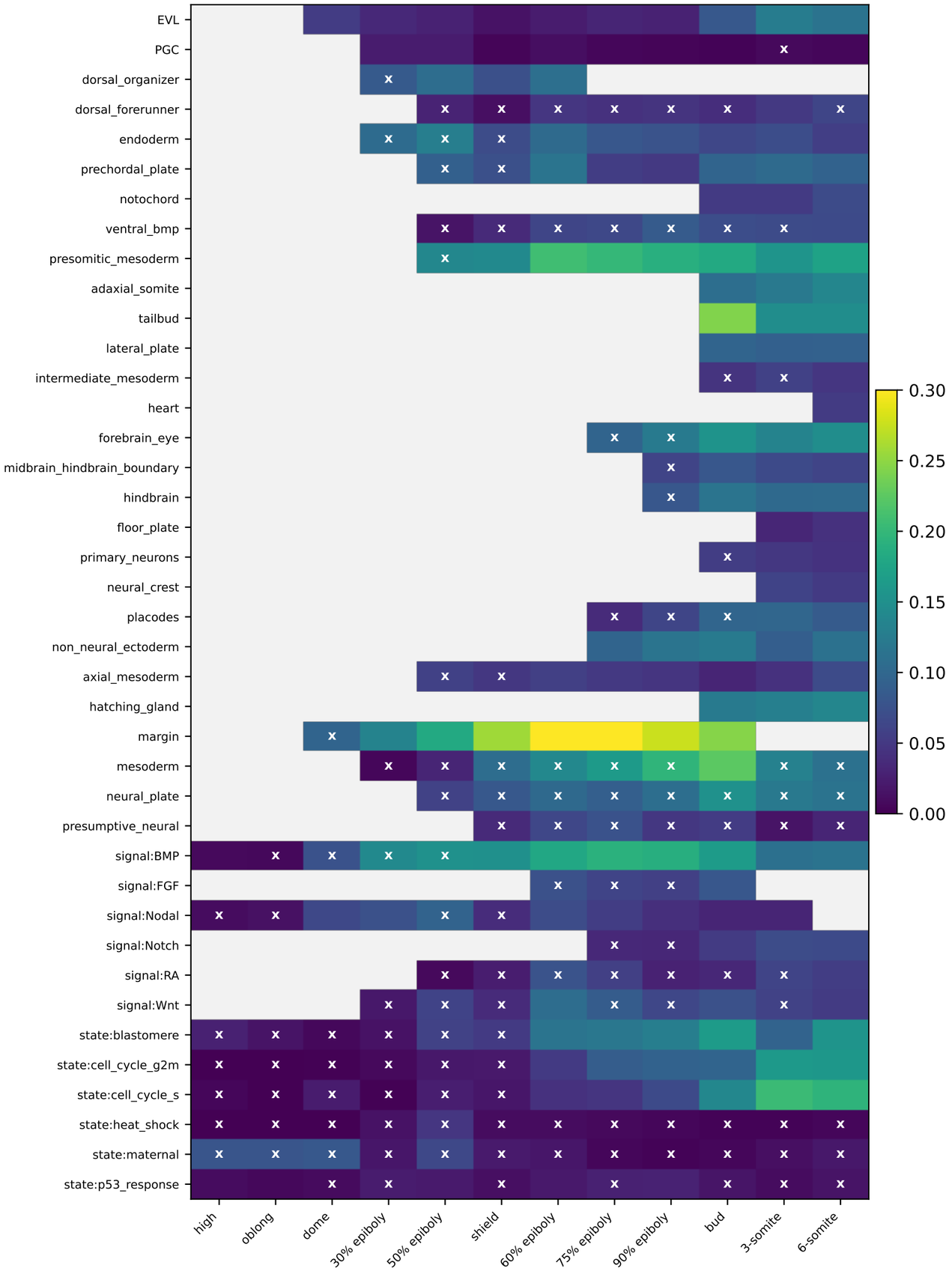
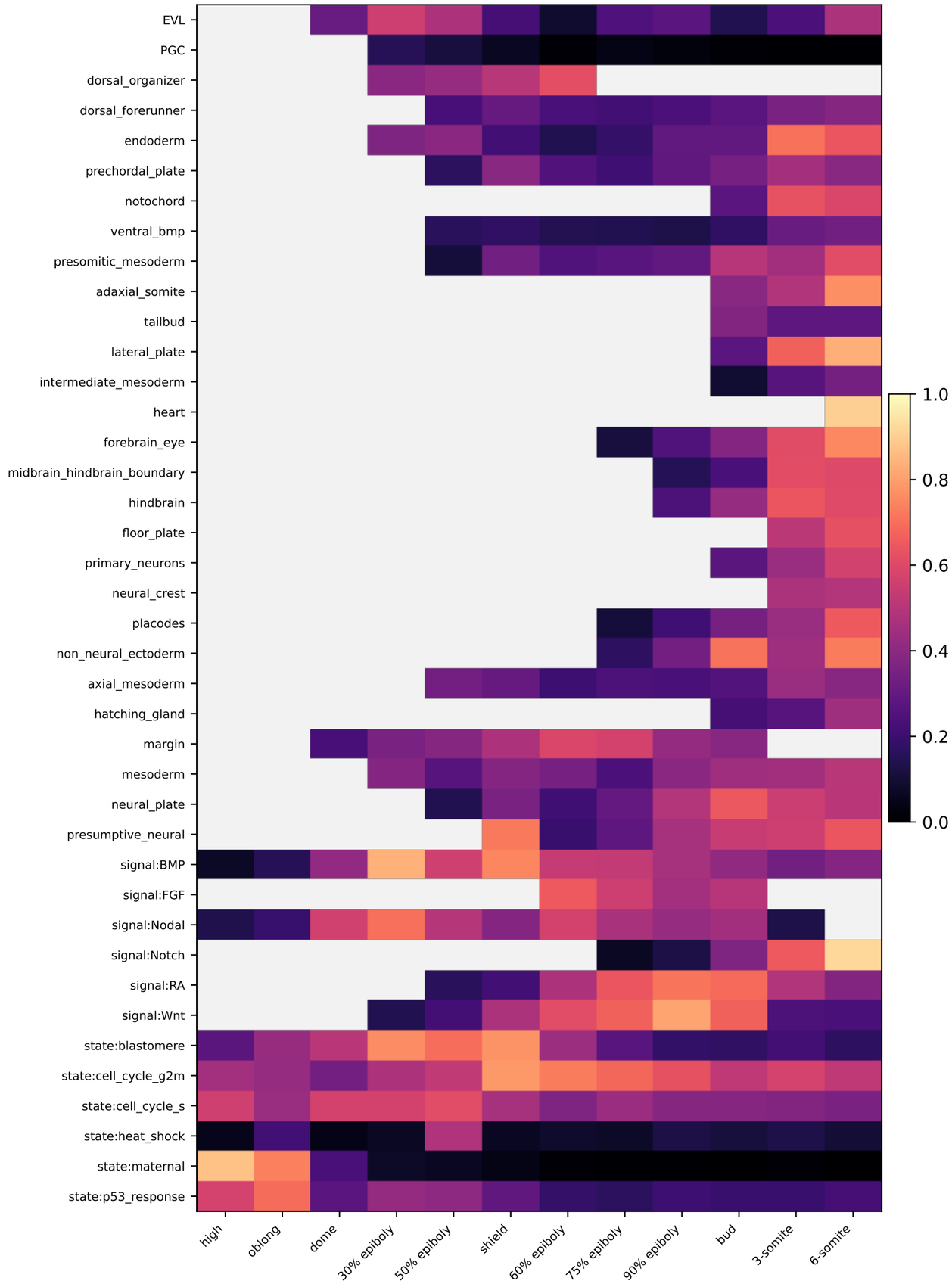
share of cells with the program active (median library); x: not supported

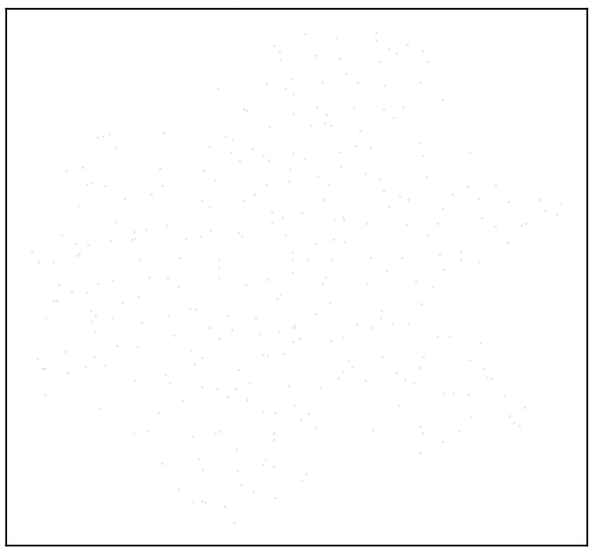


library level (pseudobulk relative to the maximum over libraries)

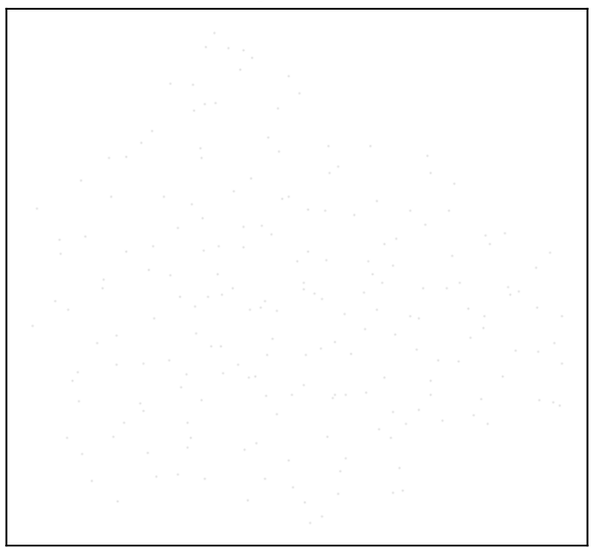


EVL

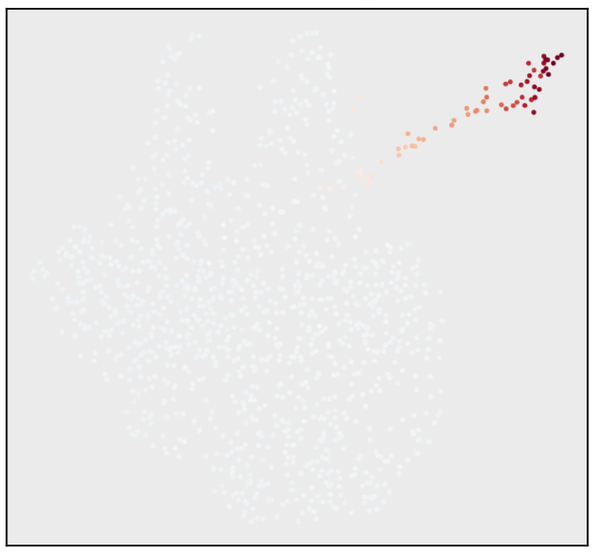
high: not defined



oblong: not defined

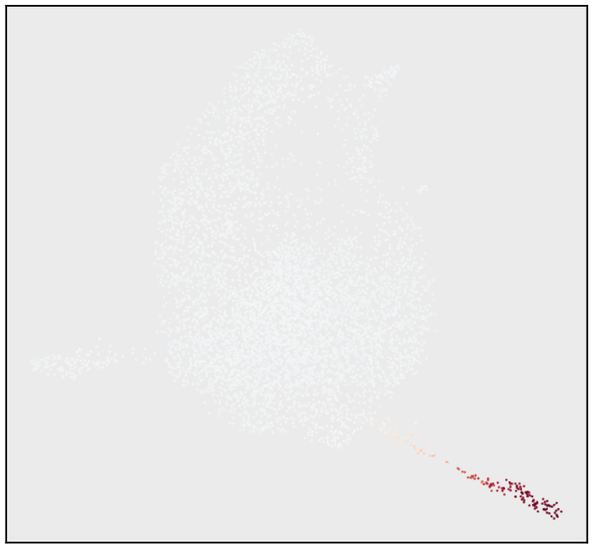


dome: 28 genes, supported
active in 5% of cells, library level 0.31

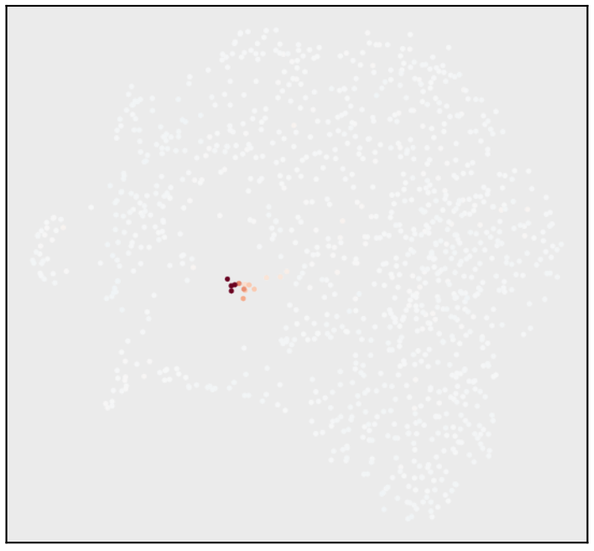


30% epiboly: 30 genes, supported
active in 4% of cells, library level 0.55

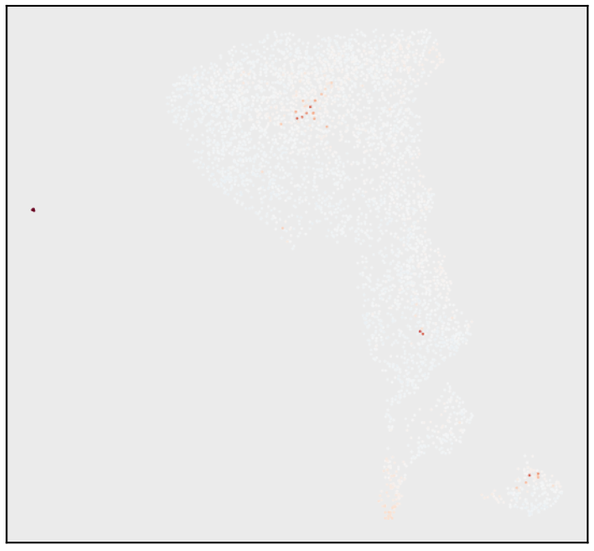
50% epiboly: 30 genes, supported
active in 3% of cells, library level 0.48



shield: 30 genes, supported
active in 1% of cells, library level 0.23



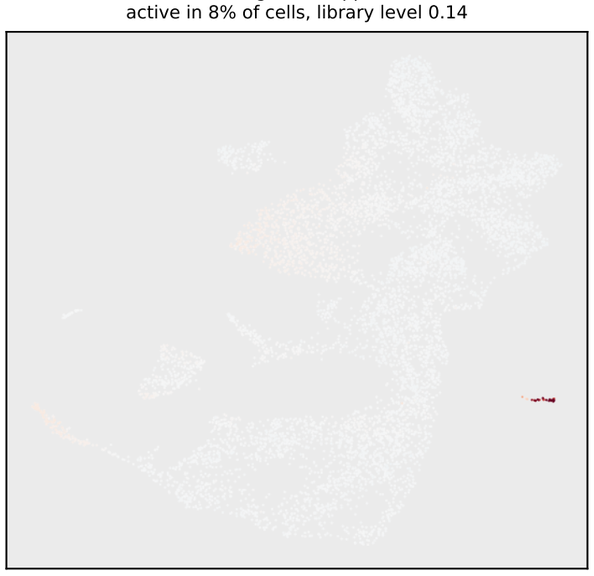
60% epiboly: 30 genes, supported
active in 2% of cells, library level 0.09



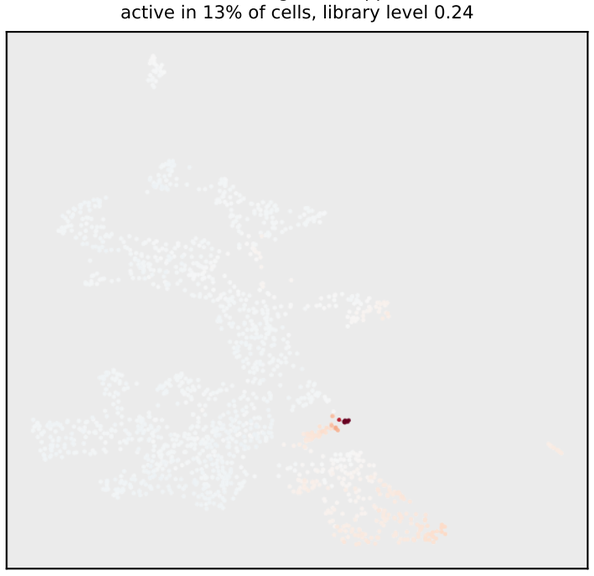
75% epiboly: 30 genes, supported
active in 3% of cells, library level 0.25

90% epiboly: 30 genes, supported
active in 3% of cells, library level 0.27

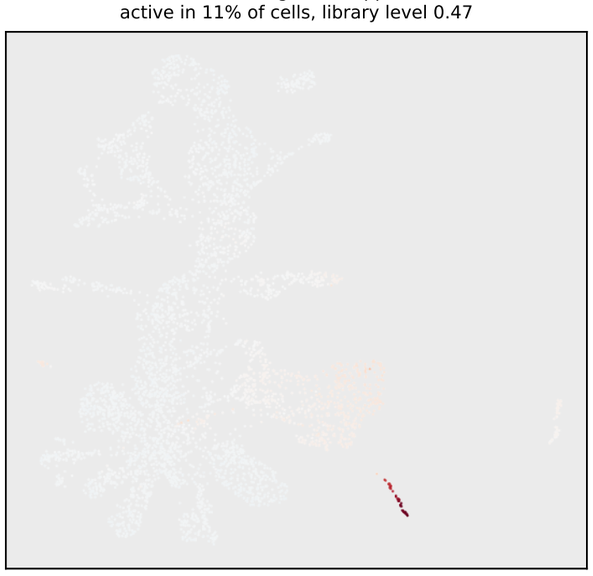
bud: 30 genes, supported
active in 8% of cells, library level 0.14



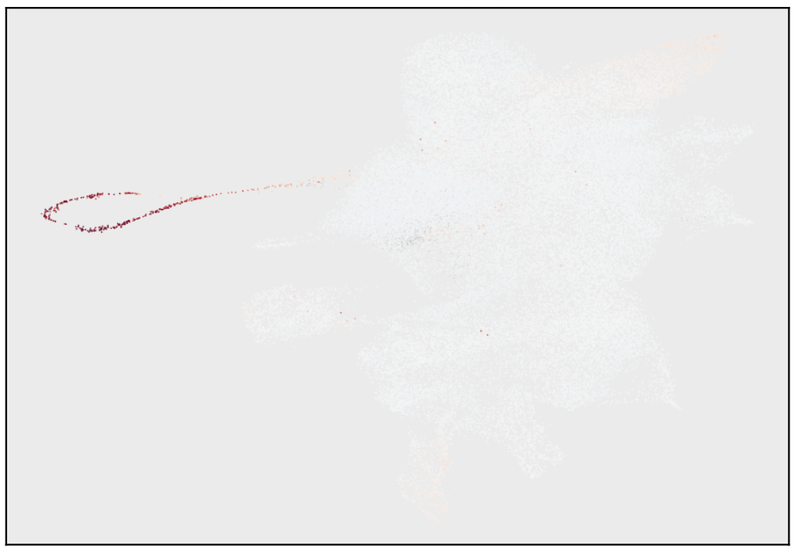
3-somite: 29 genes, supported
active in 13% of cells, library level 0.24



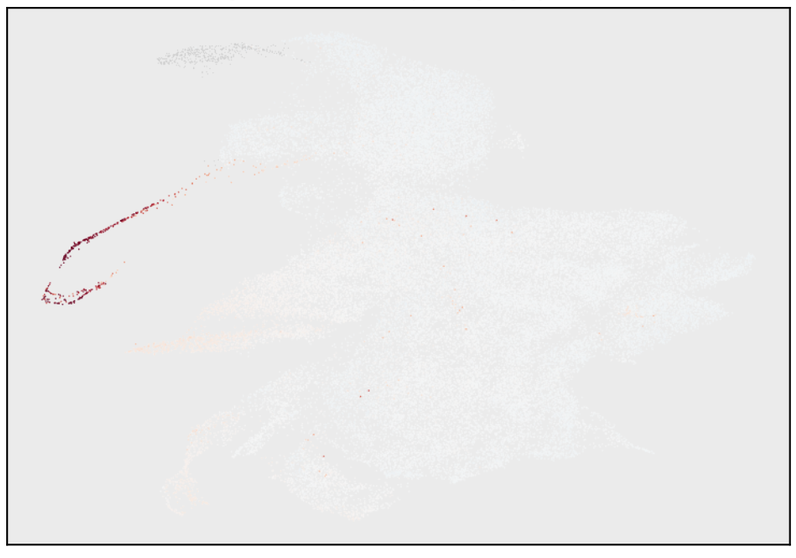
6-somite: 30 genes, supported
active in 11% of cells, library level 0.47



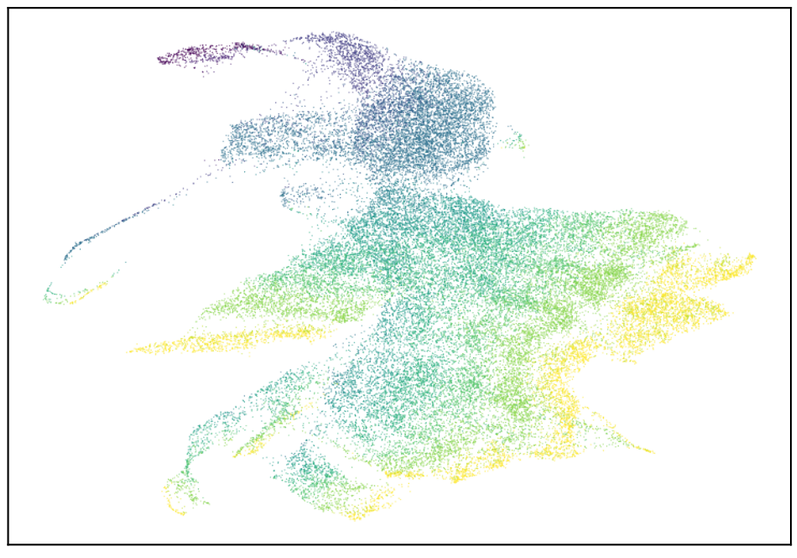
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



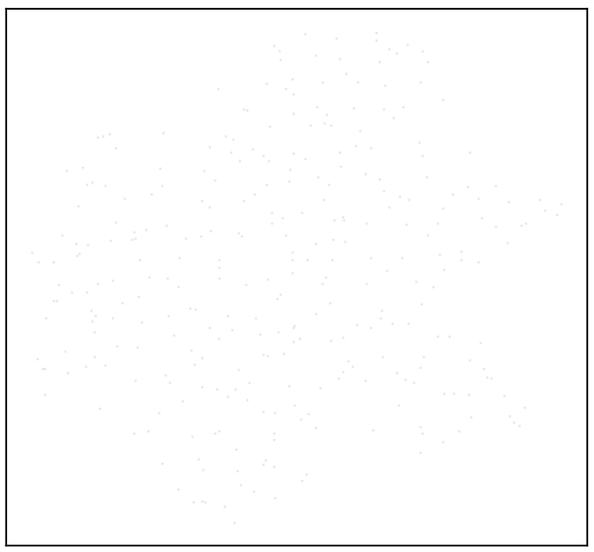
atlas, unaligned: stages



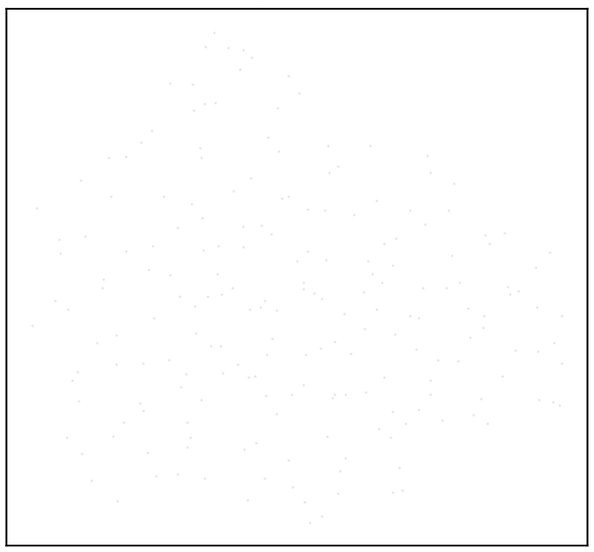
Genes (number of stages): krt4 (10), krt18 (10), krt8 (10), krt5 (10), cldne (10), capn9 (10), cldnb (10), cyt1 (10), evpla (10), eppk1 (10), ftr83 (10), krt97 (10), lye (10), pfn1 (10), rab25a (10), sult6b1 (10), zgc:101000 (10), zgc:110333 (10), znf185 (10), cebpb (10), epcam (10), ppl (10), gbgt1l4 (10), mgll (10), gmds (10), zgc:162964 (10), szl (10), krt17 (9), uox (9), elovl7b (9)

PGC

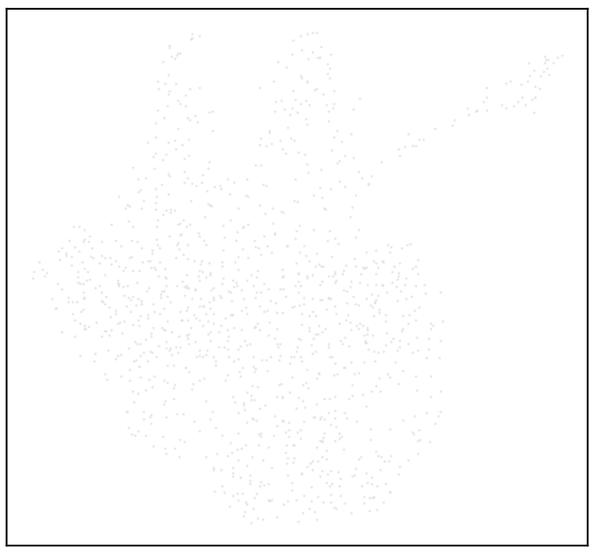
high: not defined



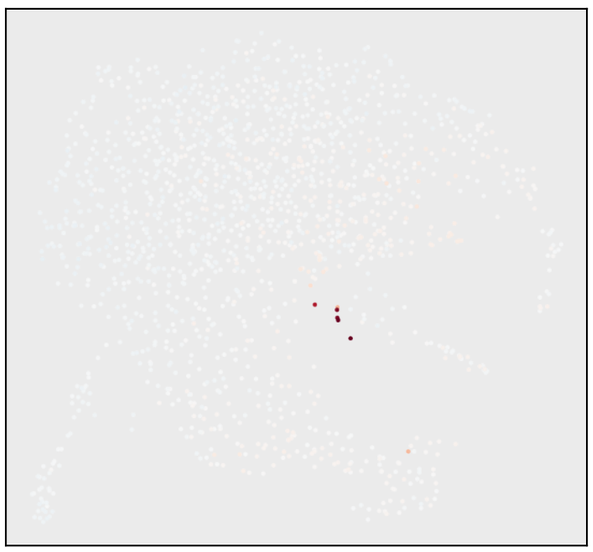
oblong: not defined



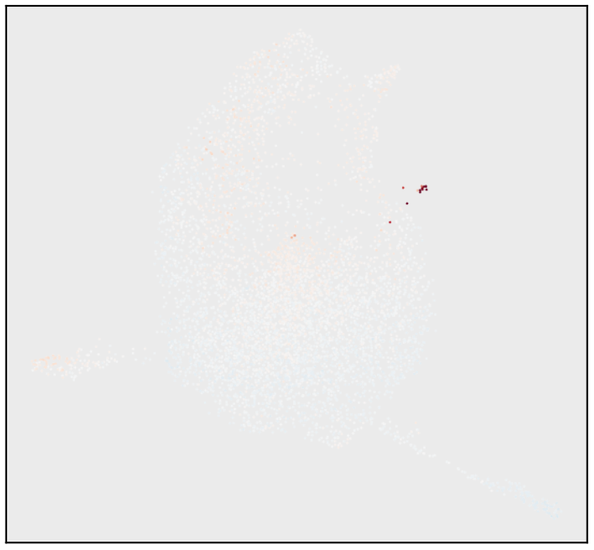
dome: not defined



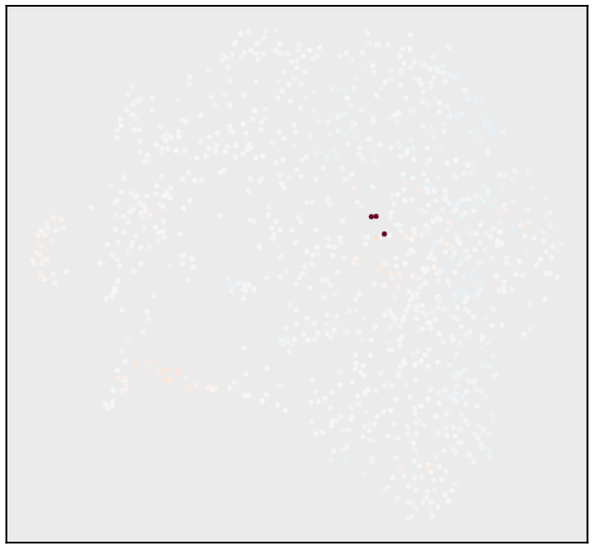
30% epiboly: 11 genes, supported
active in 2% of cells, library level 0.15



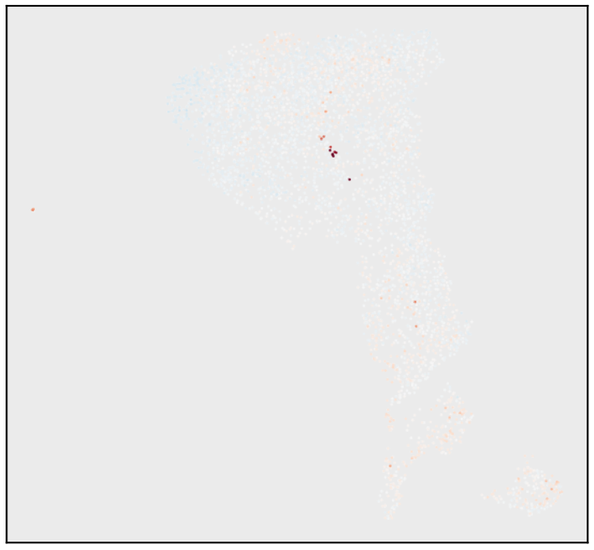
50% epiboly: 12 genes, supported
active in 2% of cells, library level 0.12



shield: 10 genes, supported
active in 0% of cells, library level 0.06

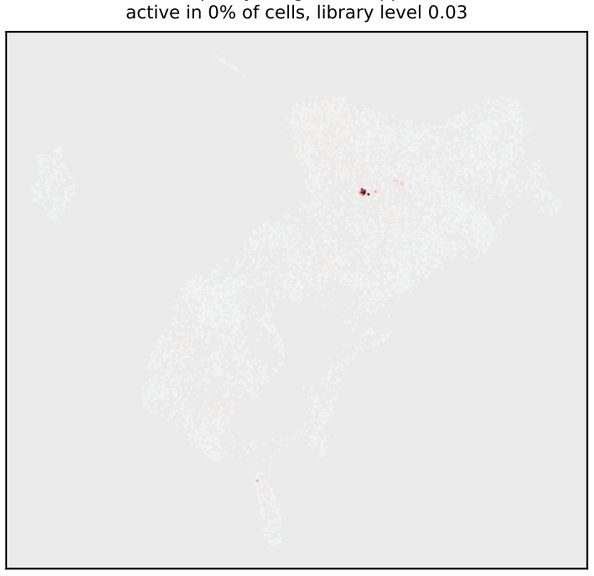


60% epiboly: 11 genes, supported
active in 1% of cells, library level 0.02

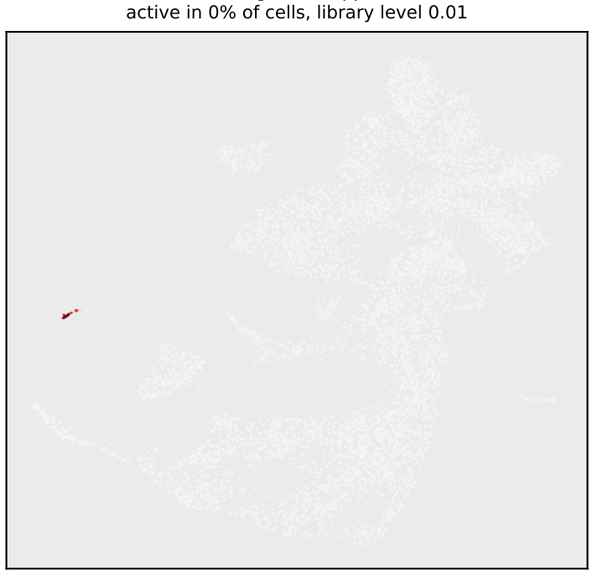


75% epiboly: 12 genes, supported
active in 1% of cells, library level 0.04

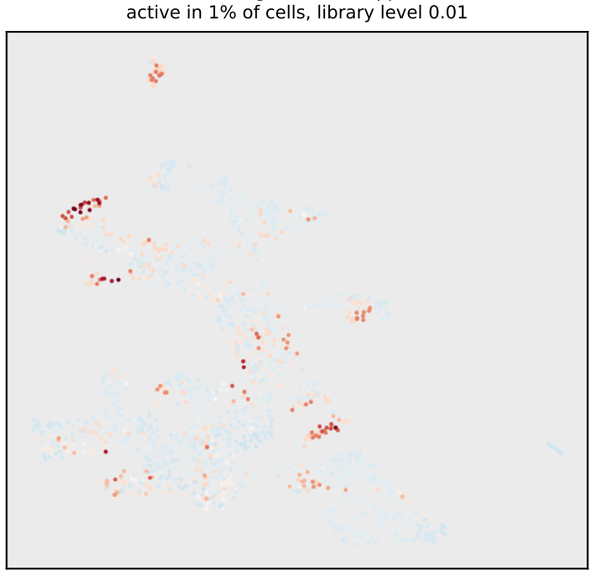
90% epiboly: 11 genes, supported
active in 0% of cells, library level 0.03



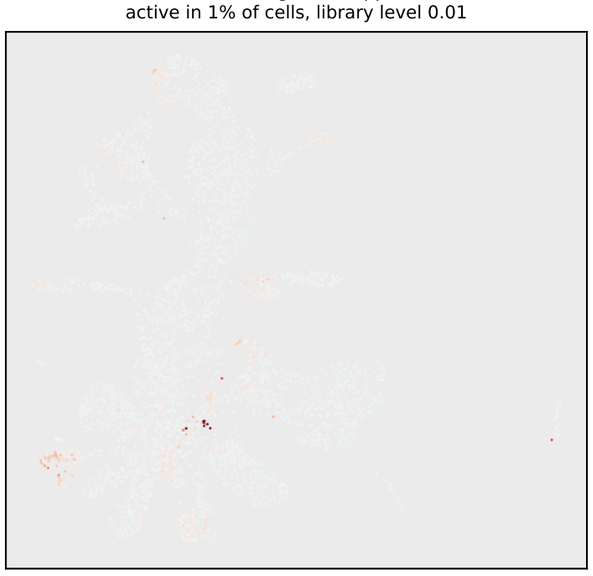
bud: 11 genes, supported
active in 0% of cells, library level 0.01



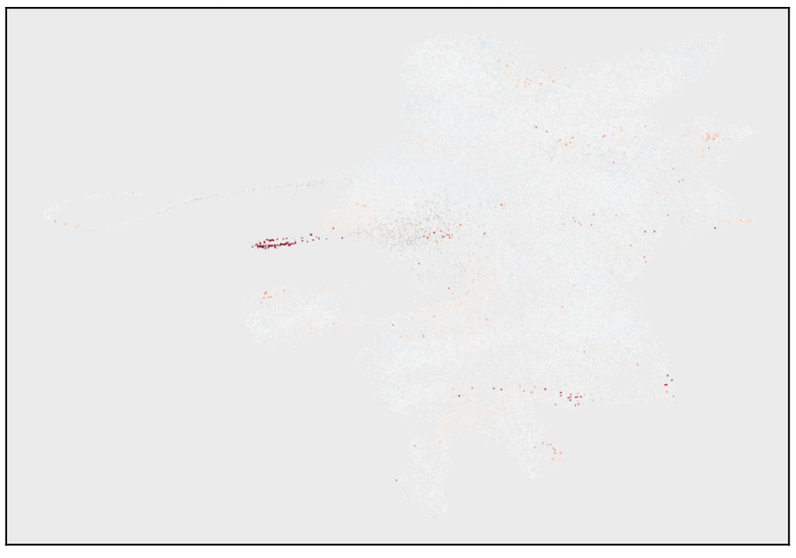
3-somite: 6 genes, not supported
active in 1% of cells, library level 0.01



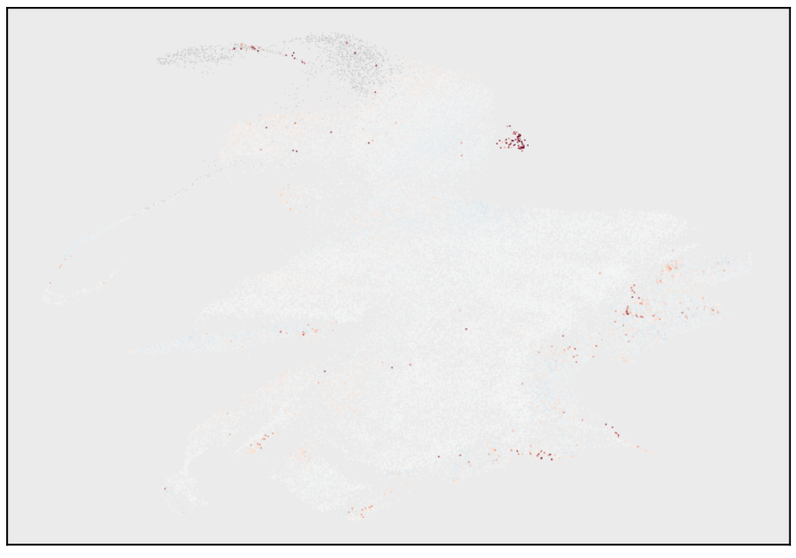
6-somite: 11 genes, supported
active in 1% of cells, library level 0.01



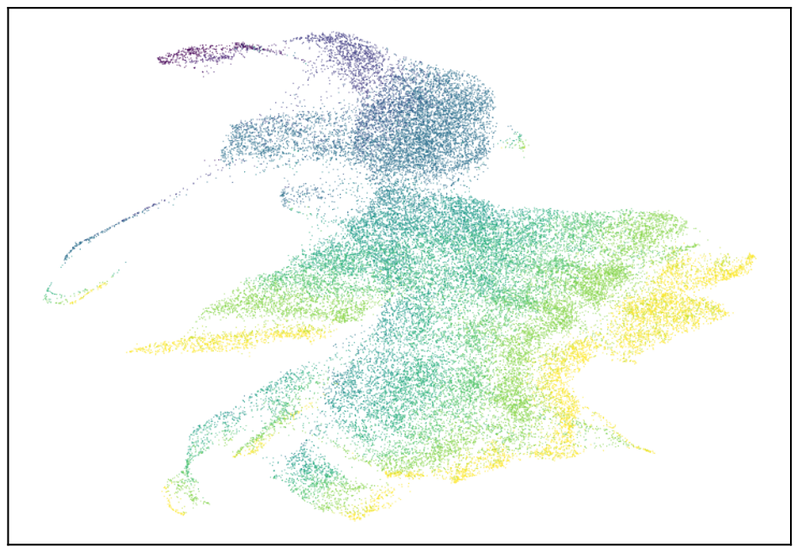
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



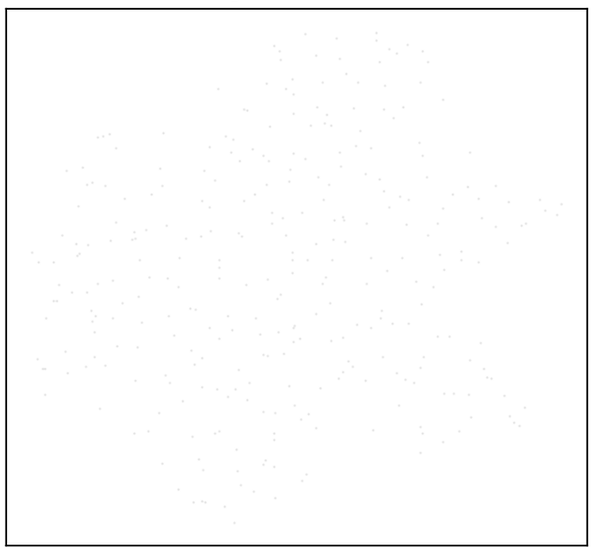
atlas, unaligned: stages



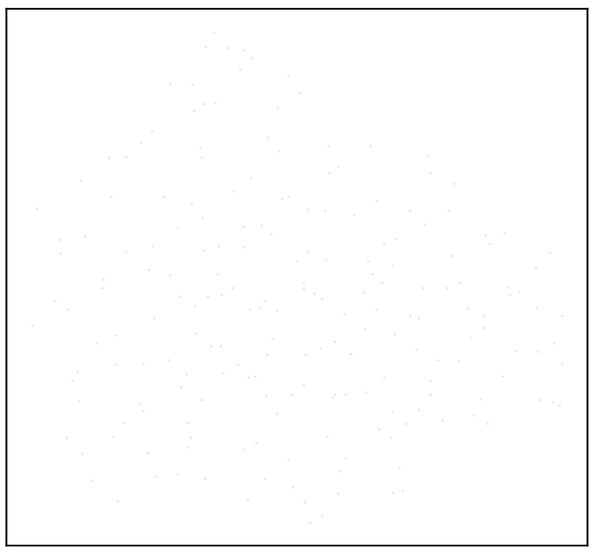
Genes (number of stages): ddx4 (9), gra (9), h1m (9), slc16a3 (9), org (9), nanos3 (8), dnd1 (8), ca15b (8), tdrd7a (8), nanog (8), nanos1 (6), rgs14a (4)

dorsal_organizer

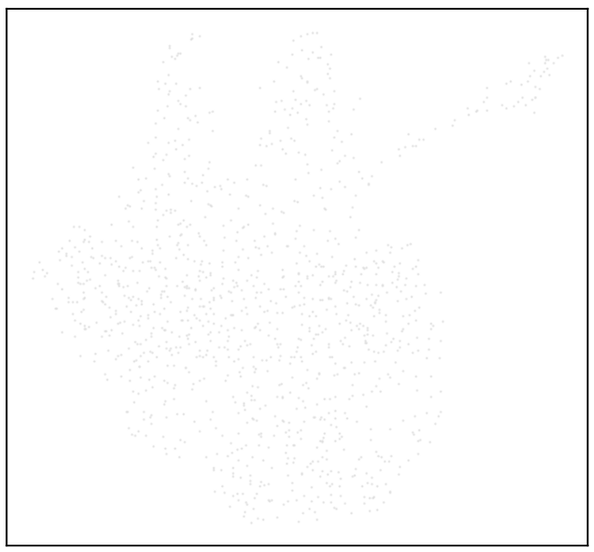
high: not defined



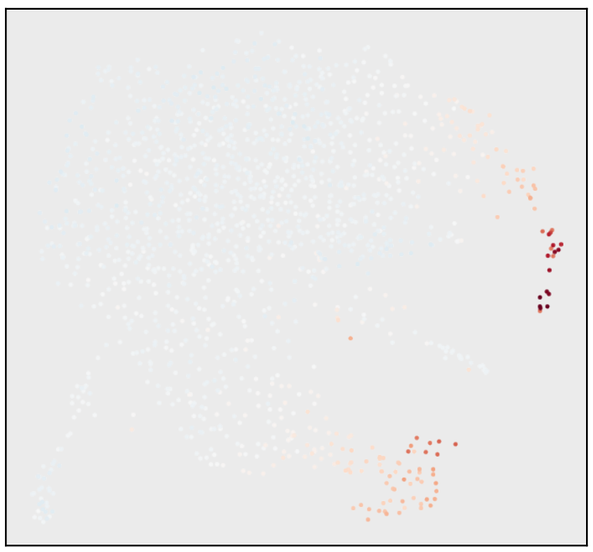
oblong: not defined



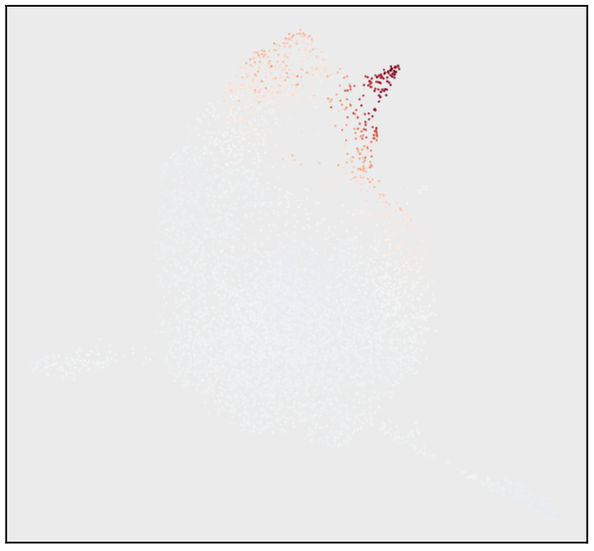
dome: not defined



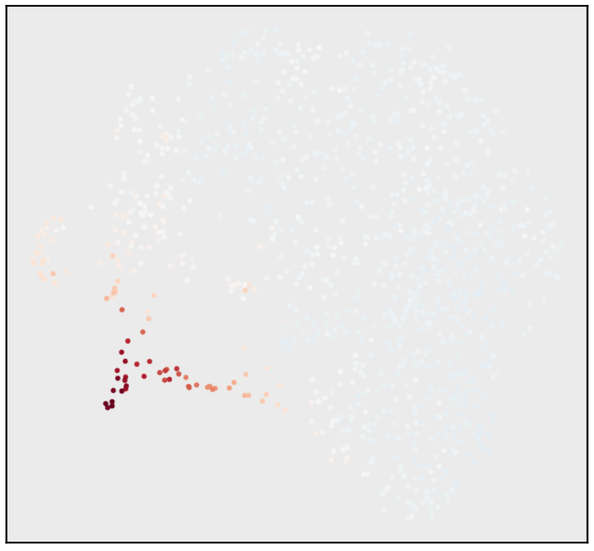
30% epiboly: 25 genes, not supported
active in 9% of cells, library level 0.40



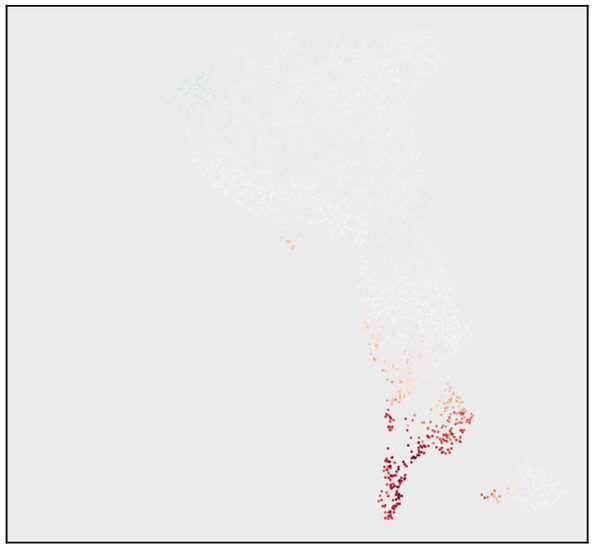
50% epiboly: 30 genes, supported
active in 11% of cells, library level 0.42



shield: 28 genes, supported
active in 7% of cells, library level 0.50



60% epiboly: 30 genes, supported
active in 11% of cells, library level 0.62

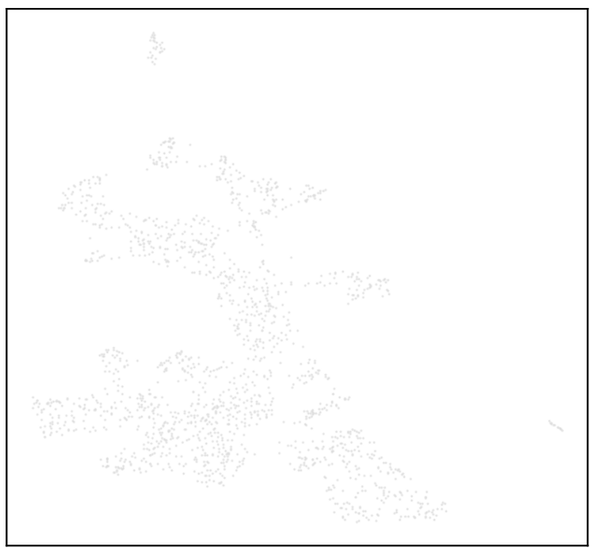


75% epiboly: not defined

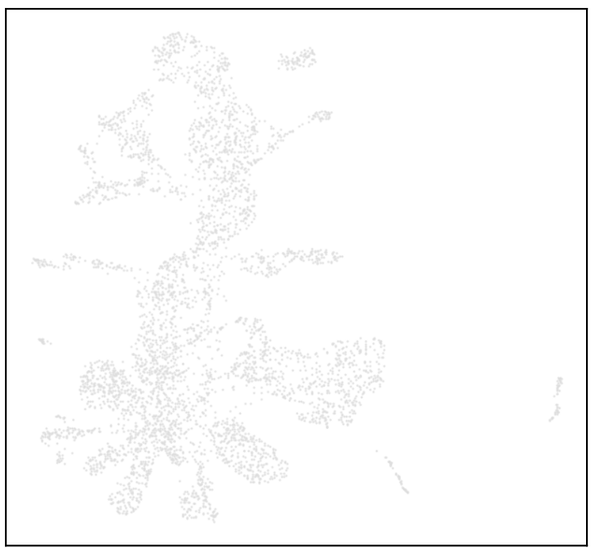
90% epiboly: not defined

bud: not defined

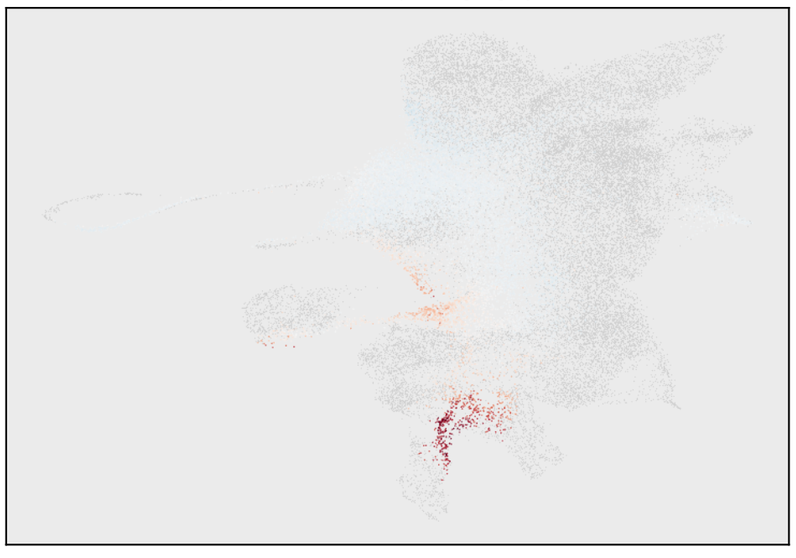
3-somite: not defined



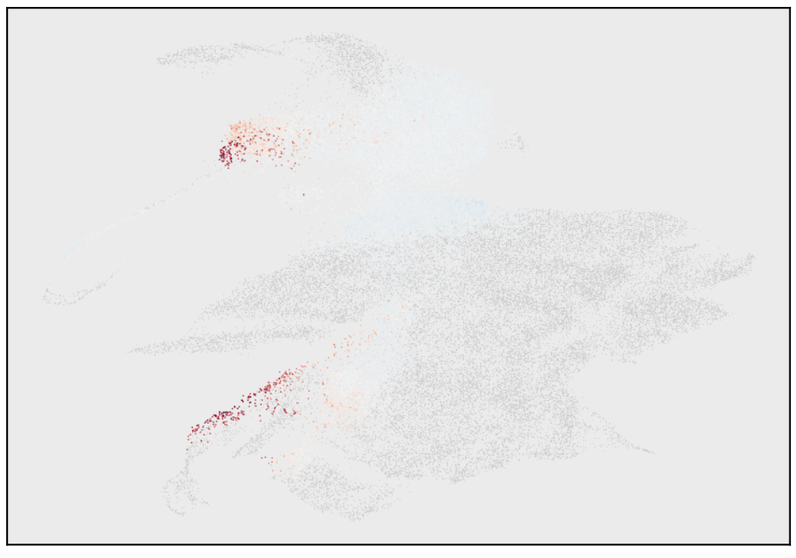
6-somite: not defined



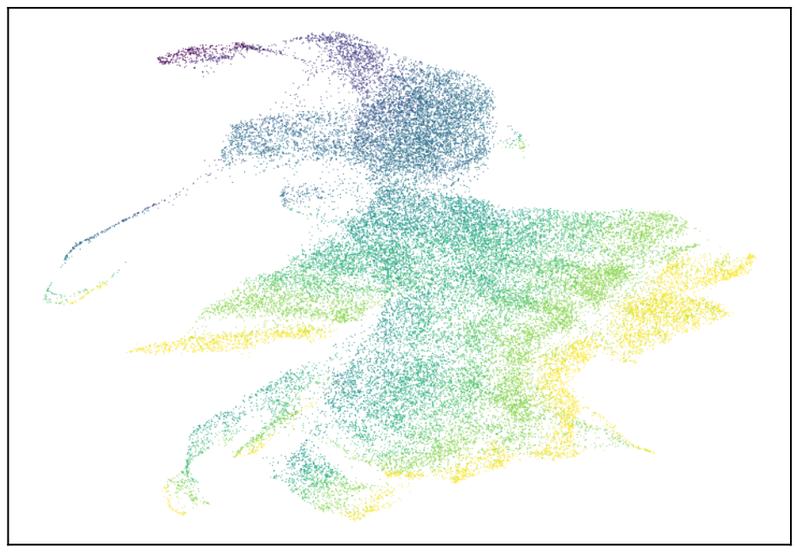
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



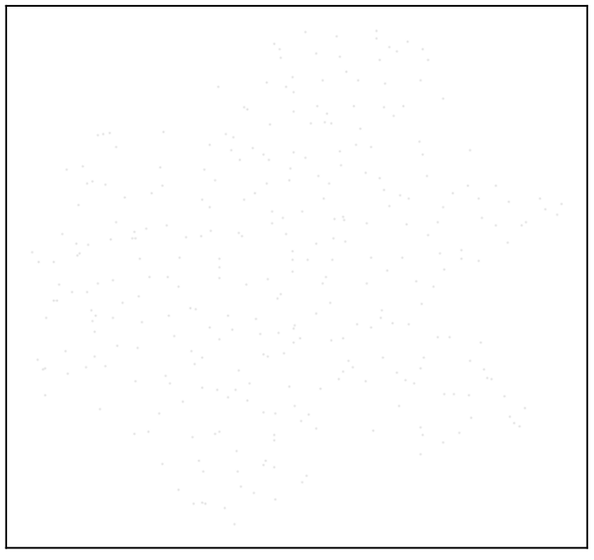
atlas, unaligned: stages



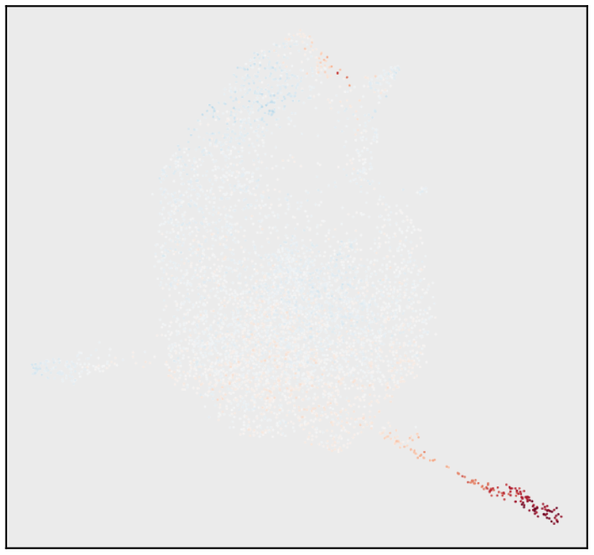
Genes (number of stages): gsc (4), chd (4), noto (4), dkk1b (4), admp (4), frzb (4), nog1 (4), insb (4), ripply1 (4), xbp1 (4), id3 (4), pitx2 (4), efnb2a (4), foxd3 (4), lhx1a (4), gadd45ba (4), kirrel3l (4), fgf3 (4), id1 (4), bambia (4), otx1a (4), foxd5 (4), nnr (4), dact2 (4), rasl11b (4), foxj1a (3), tbx1 (3), shha (3), tbr1b (2), twist2 (2)

dorsal_forerunner

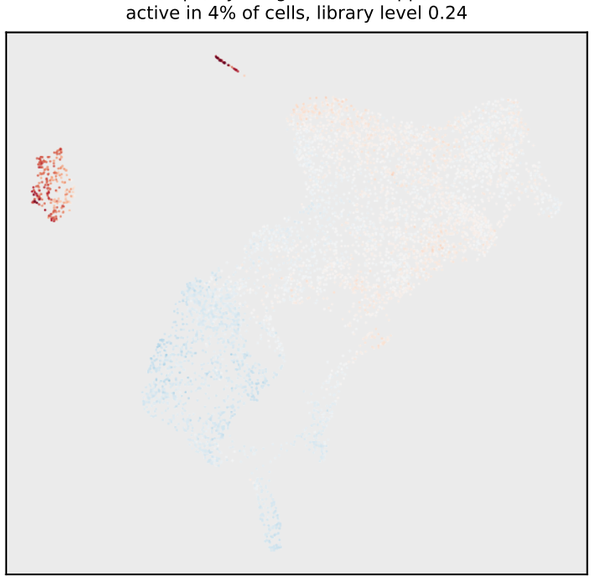
high: not defined



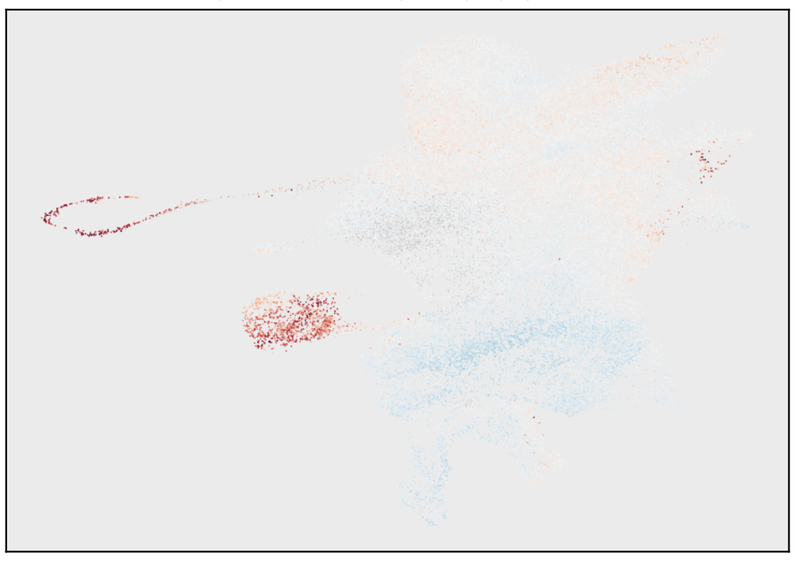
50% epiboly: 26 genes, not supported
active in 3% of cells, library level 0.23



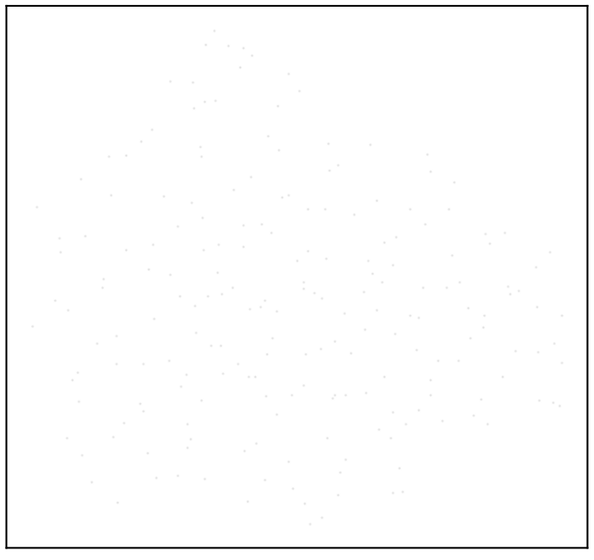
90% epiboly: 27 genes, not supported
active in 4% of cells, library level 0.24



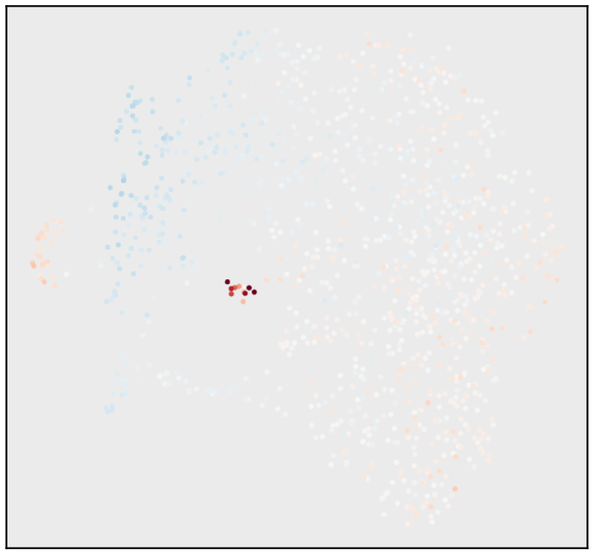
atlas, aligned (scores scaled per stage; grey: not defined)



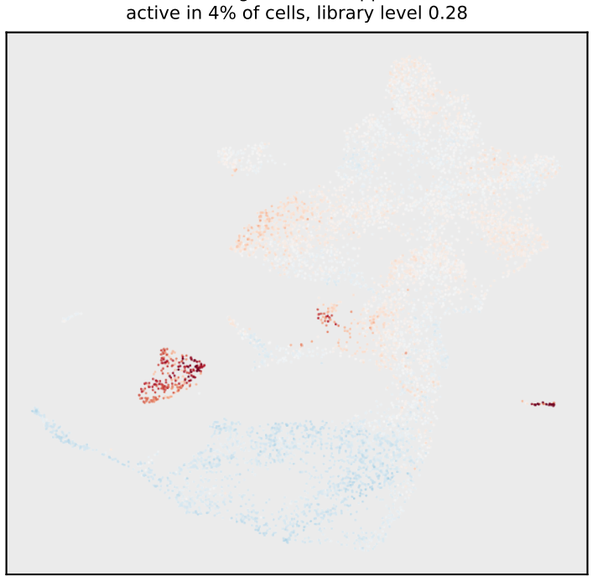
oblong: not defined



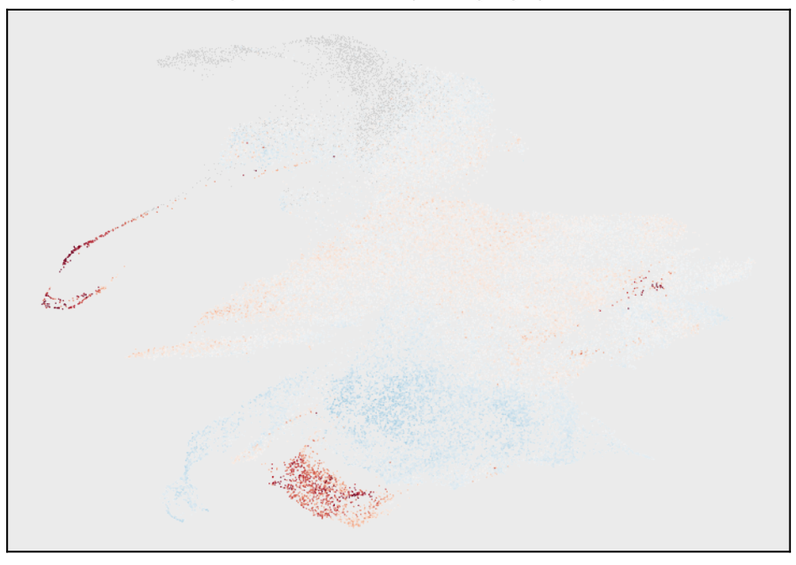
shield: 21 genes, not supported
active in 1% of cells, library level 0.31



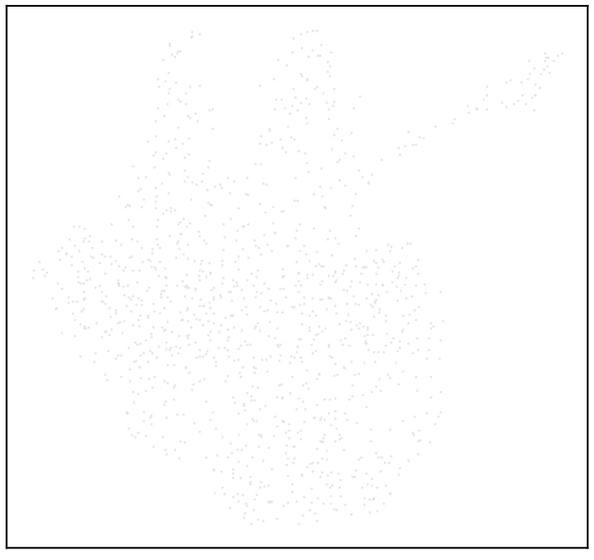
bud: 30 genes, not supported
active in 4% of cells, library level 0.28



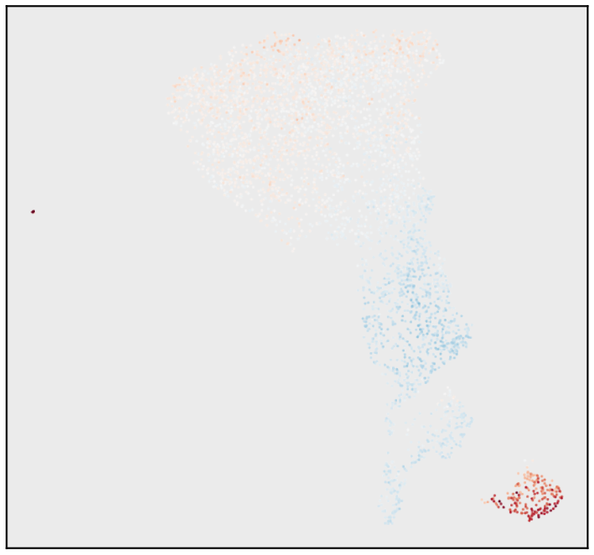
atlas, unaligned (scores scaled per stage; grey: not defined)



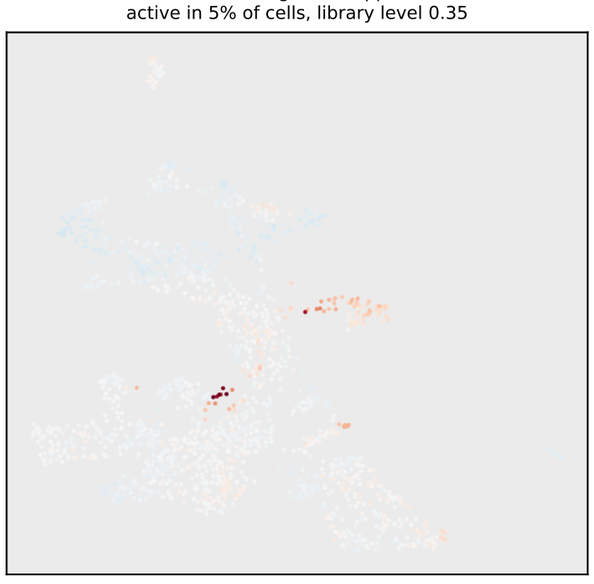
dome: not defined



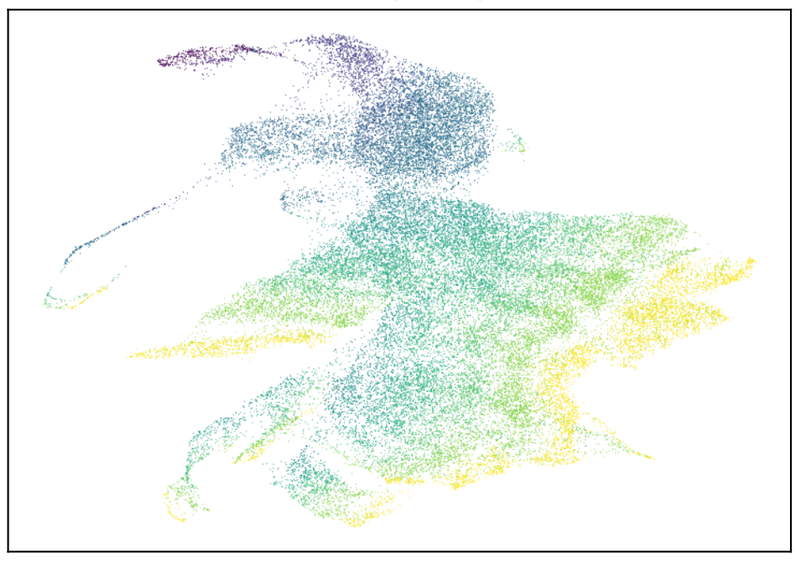
60% epiboly: 26 genes, not supported
active in 5% of cells, library level 0.23



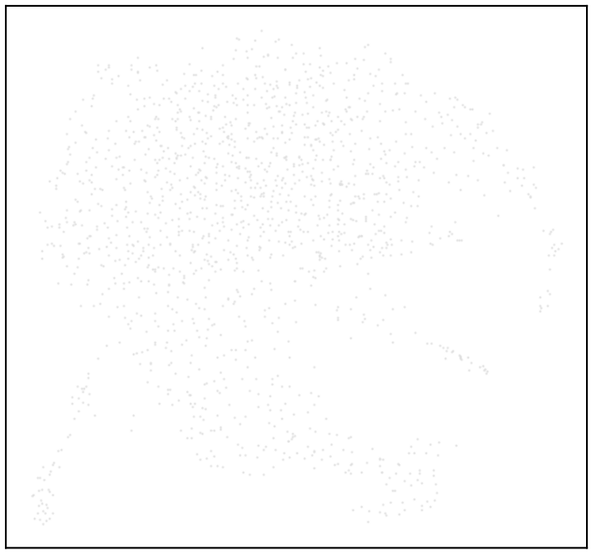
3-somite: 27 genes, supported
active in 5% of cells, library level 0.35



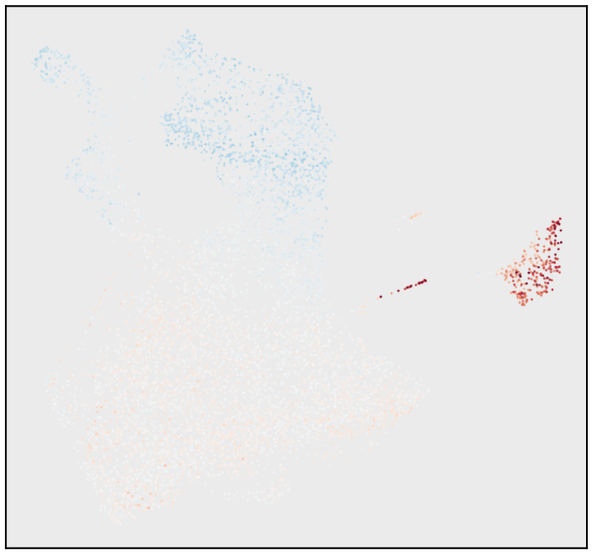
atlas, unaligned: stages



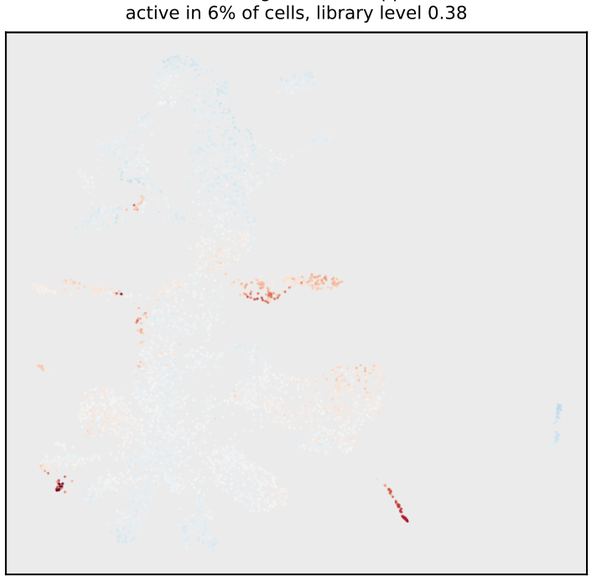
30% epiboly: not defined



75% epiboly: 28 genes, not supported
active in 4% of cells, library level 0.22



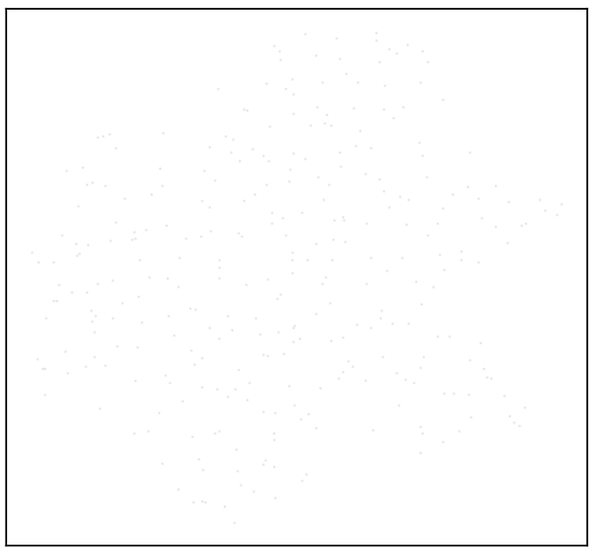
6-somite: 29 genes, not supported
active in 6% of cells, library level 0.38



Genes (number of stages): sult6b1 (8), sox17 (8), foxj1a (8), si:ch73-234b20.5 (8), vgll4l (8), si:ch211-152c2.3 (8), tjp3 (8), rassf7b (8), ca9 (8), jupa (8), cldn5a (8), cfap53 (8), dnmt3bb.1 (8), rock2b (8), tmem106ba (8), zgc:55461 (8), arg2 (8), pfkfb4b (8), ahi1 (8), relt (8), cfap45 (7), ribc2 (7), nnr (7), tagln3b (6), fgf4 (5), ccdc151 (5), cav1 (5), capsla (5), dnaaf1 (4), abcc6a (3)

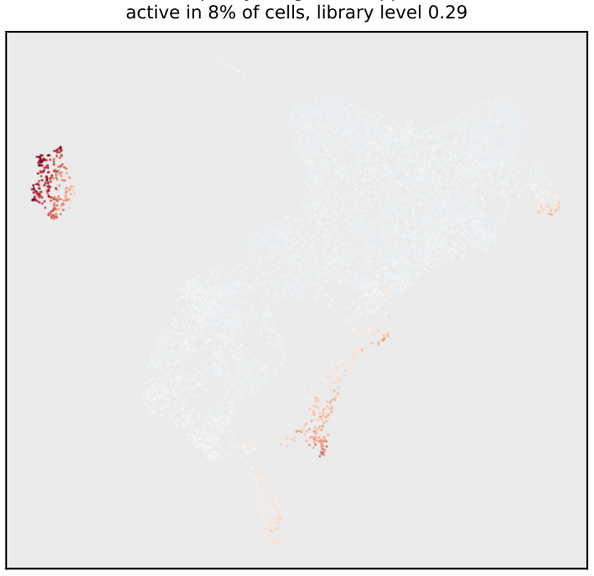
endoderm

high: not defined

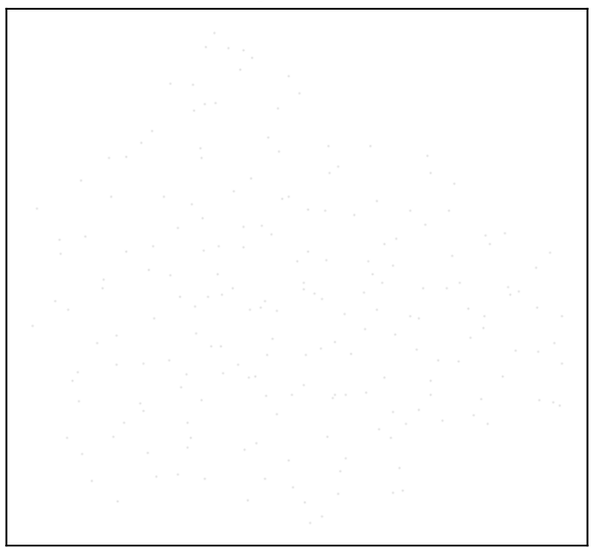


50% epiboly: 9 genes, not supported
active in 13% of cells, library level 0.40

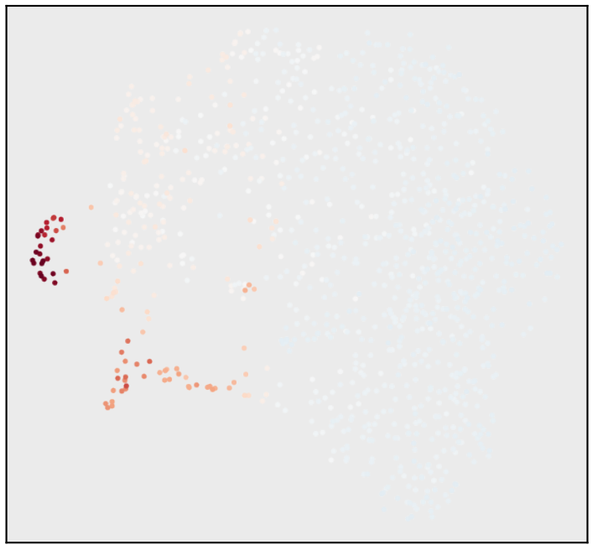
90% epiboly: 15 genes, supported
active in 8% of cells, library level 0.29



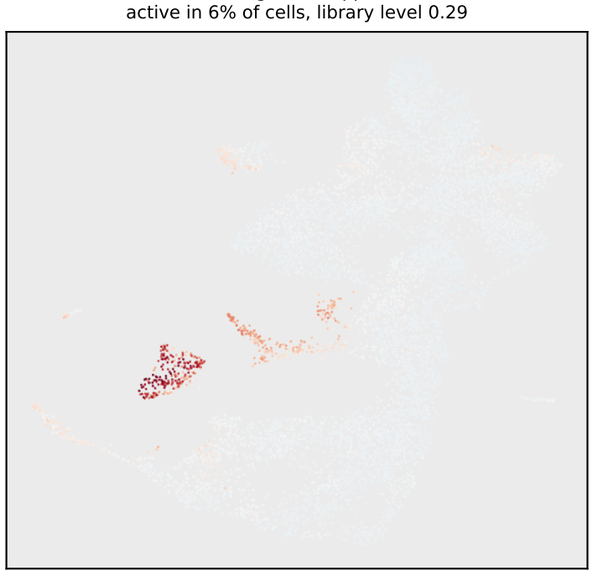
oblong: not defined



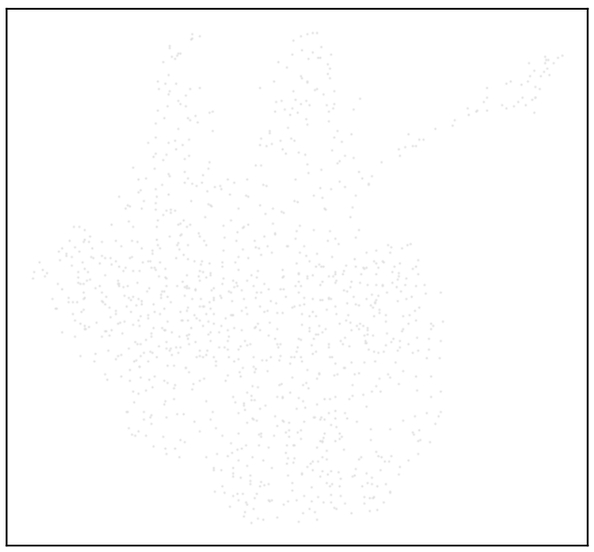
shield: 7 genes, not supported
active in 7% of cells, library level 0.22



bud: 15 genes, supported
active in 6% of cells, library level 0.29

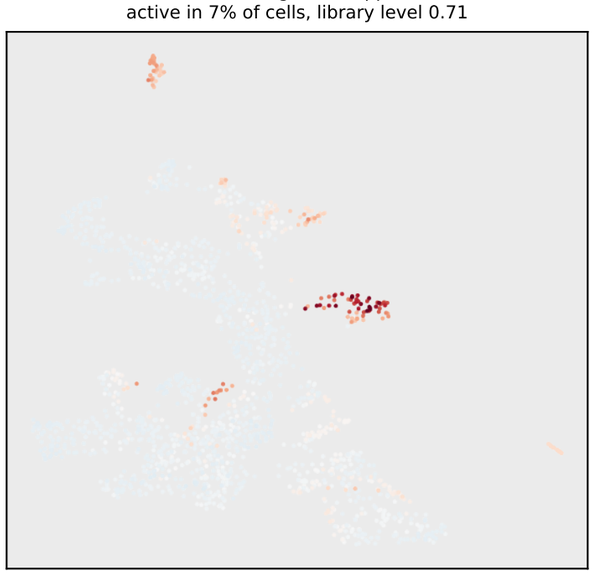


dome: not defined

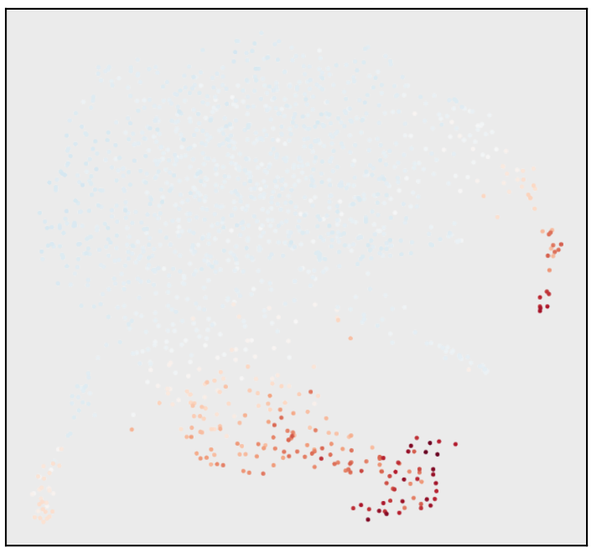


60% epiboly: 12 genes, supported
active in 10% of cells, library level 0.14

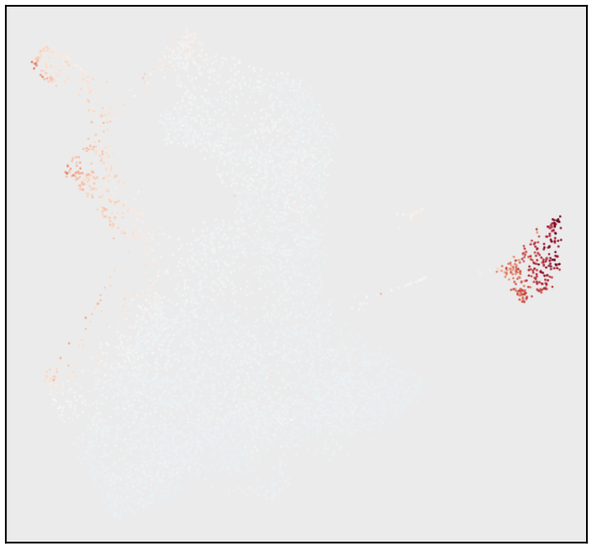
3-somite: 10 genes, supported
active in 7% of cells, library level 0.71



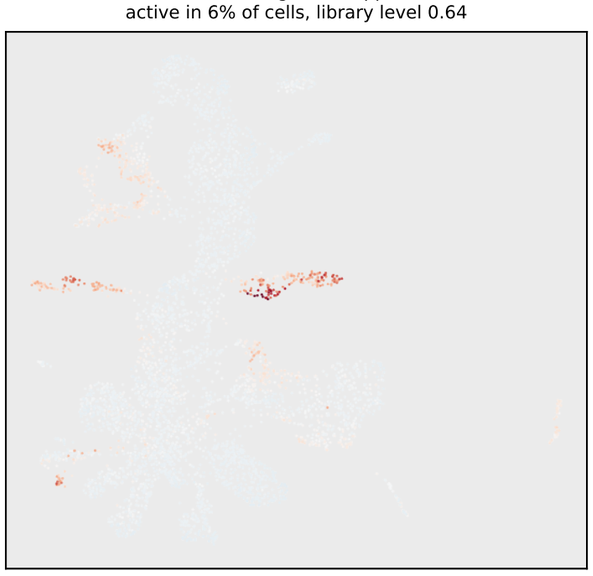
30% epiboly: 8 genes, not supported
active in 10% of cells, library level 0.36



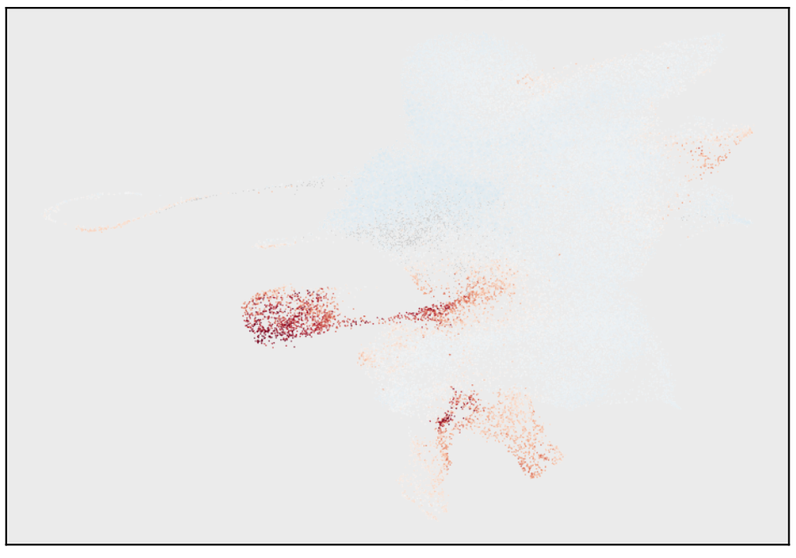
75% epiboly: 14 genes, supported
active in 8% of cells, library level 0.18



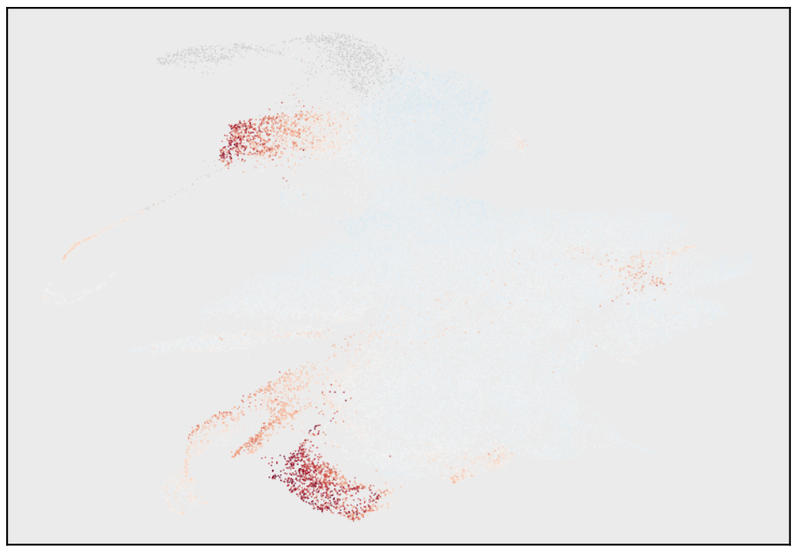
6-somite: 12 genes, supported
active in 6% of cells, library level 0.64



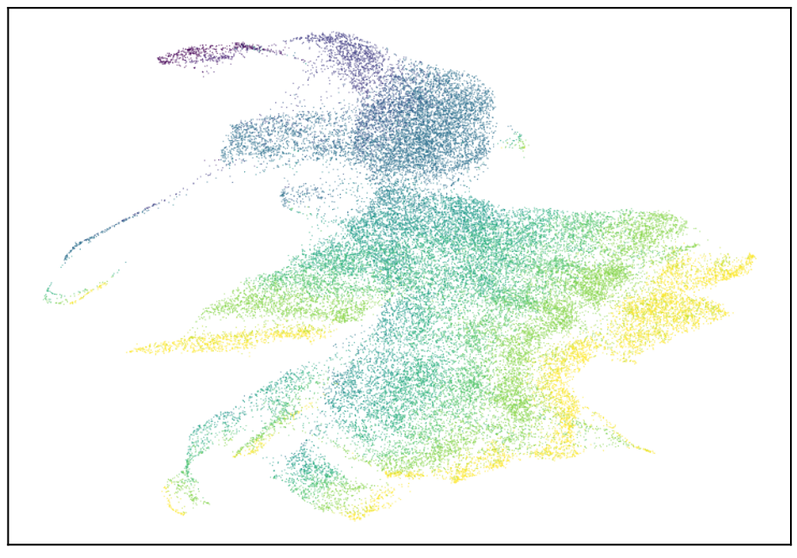
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



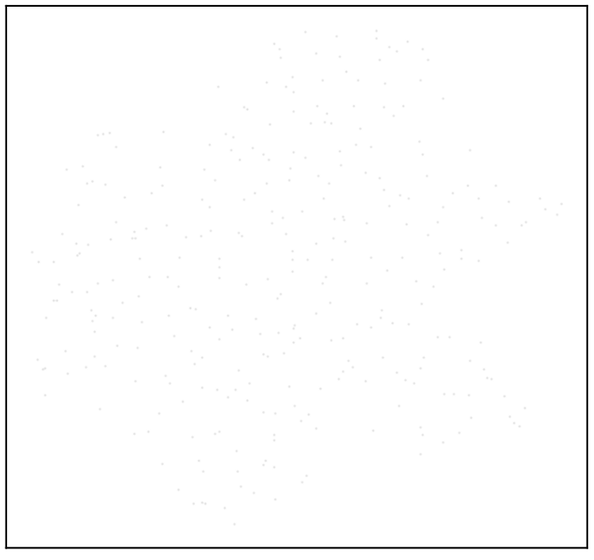
atlas, unaligned: stages



Genes (number of stages): foxa2 (9), foxa3 (9), sox17 (8), ptgs2a (8), nme4 (8), sox32 (7), mixl1 (7), sebox (7), nipsnap1 (7), foxa1 (6), nkx6.2 (6), fgfrl1b (6), mnx1 (5), mnx2b (5), bmp6 (4)

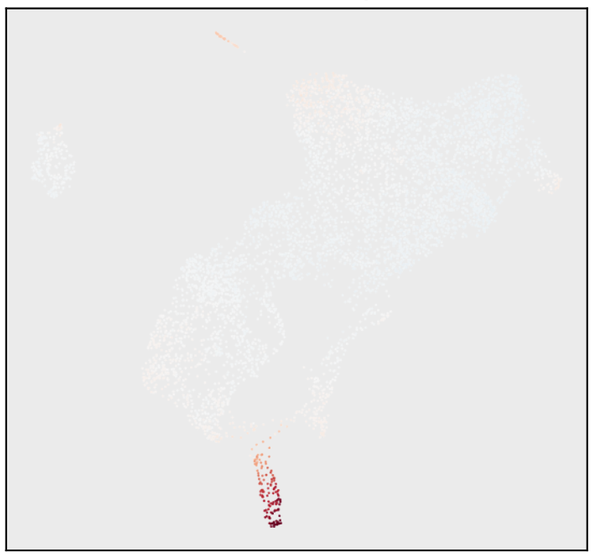
prechordal_plate

high: not defined

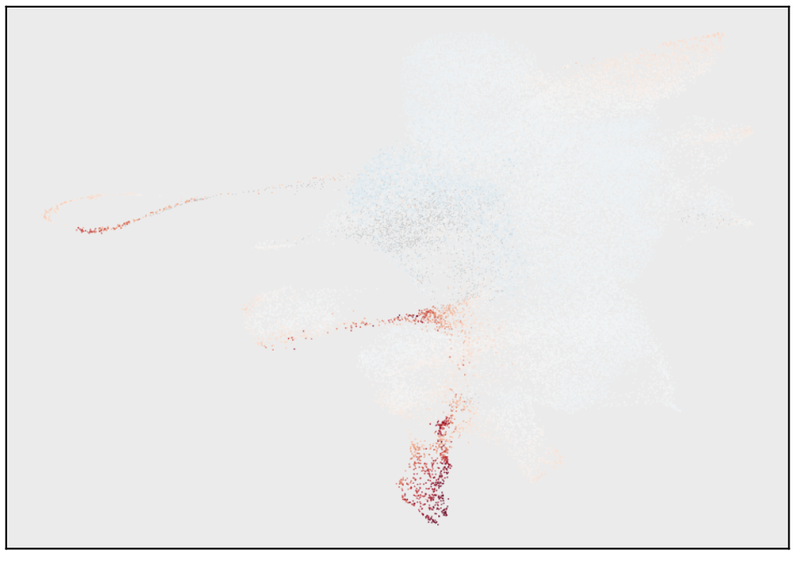


50% epiboly: 26 genes, not supported
active in 9% of cells, library level 0.17

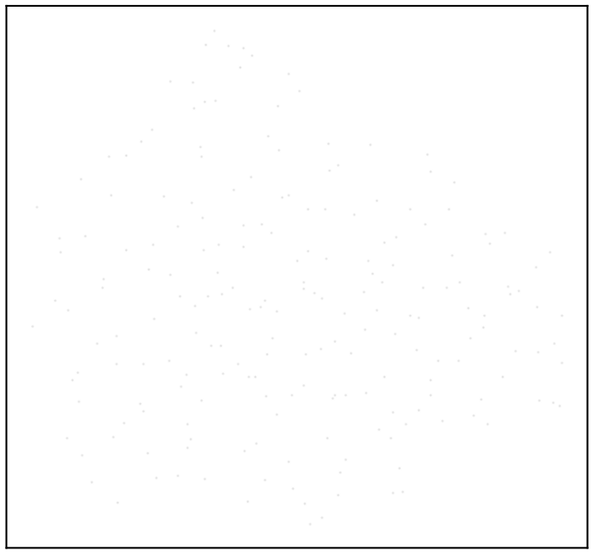
90% epiboly: 30 genes, supported
active in 5% of cells, library level 0.29



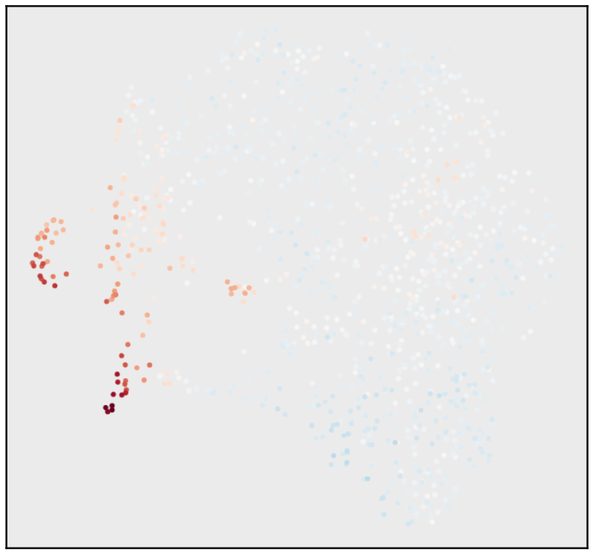
atlas, aligned (scores scaled per stage; grey: not defined)



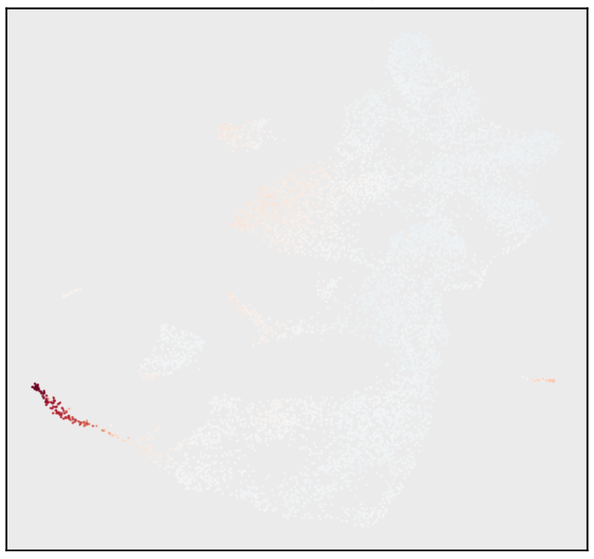
oblong: not defined



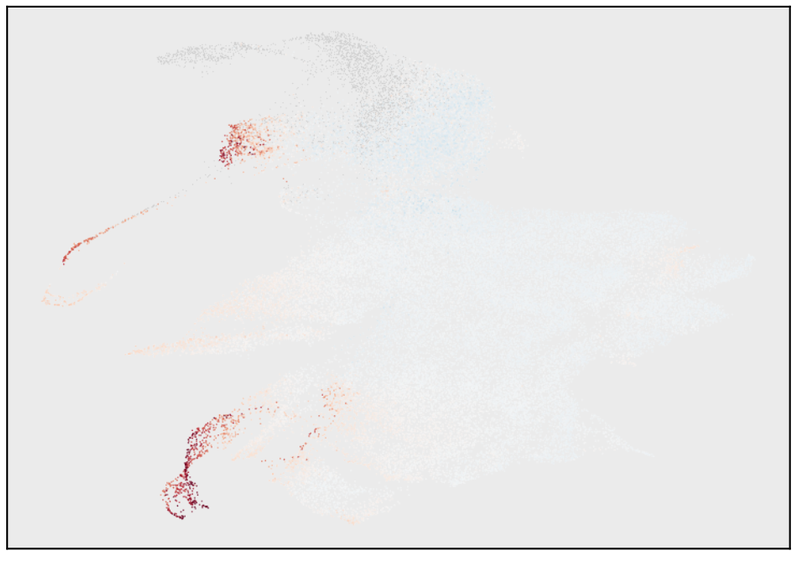
shield: 19 genes, not supported
active in 7% of cells, library level 0.39



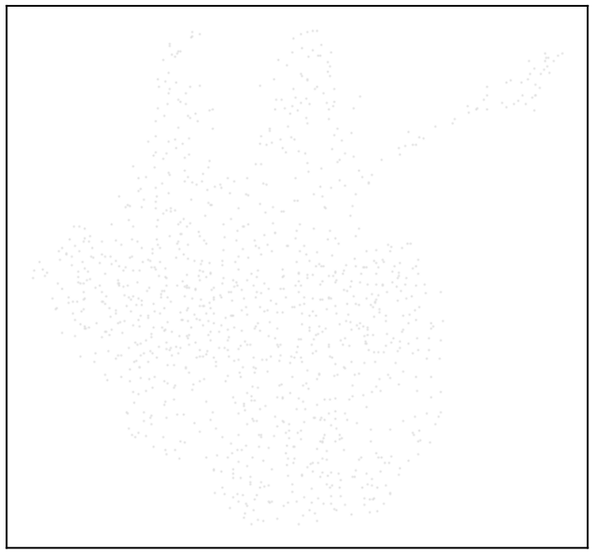
bud: 30 genes, supported
active in 10% of cells, library level 0.35



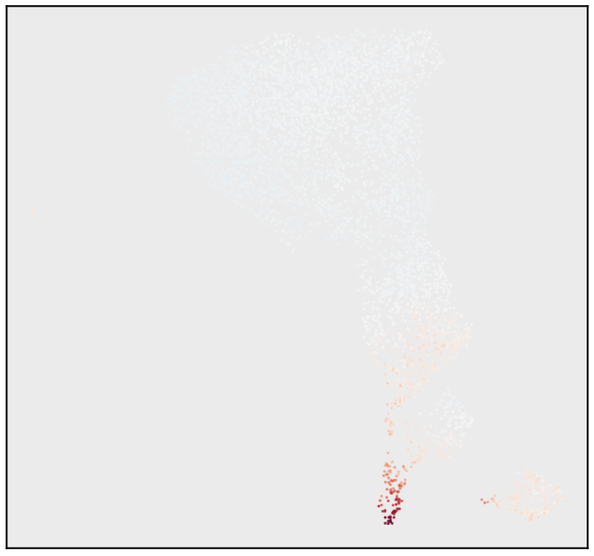
atlas, unaligned (scores scaled per stage; grey: not defined)



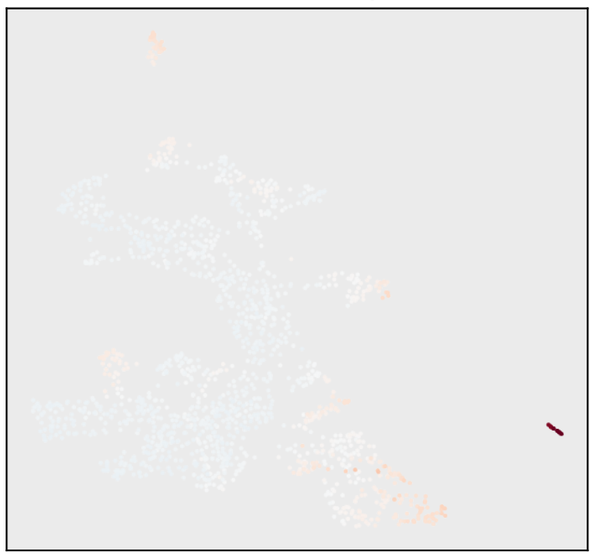
dome: not defined



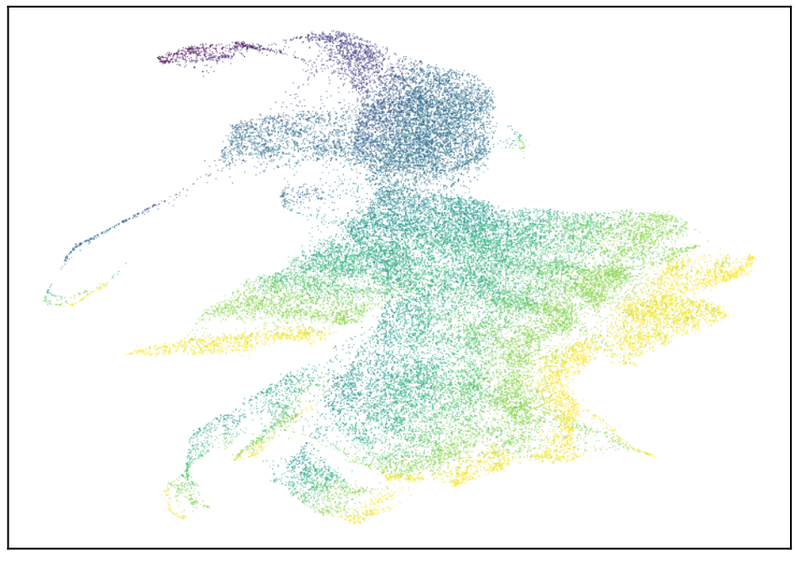
60% epiboly: 27 genes, supported
active in 12% of cells, library level 0.26



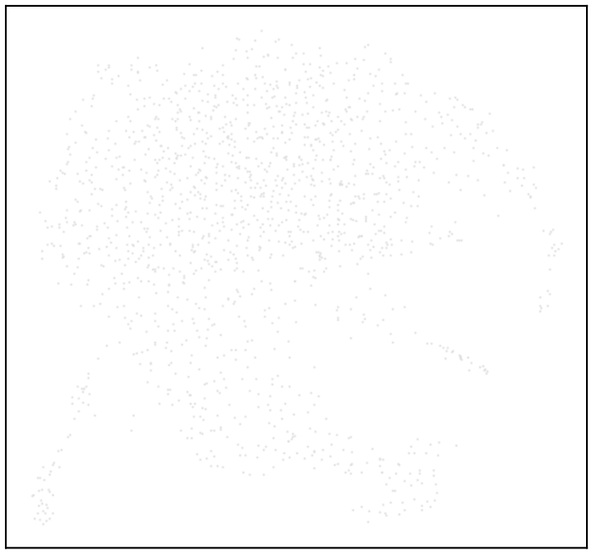
3-somite: 27 genes, supported
active in 10% of cells, library level 0.46



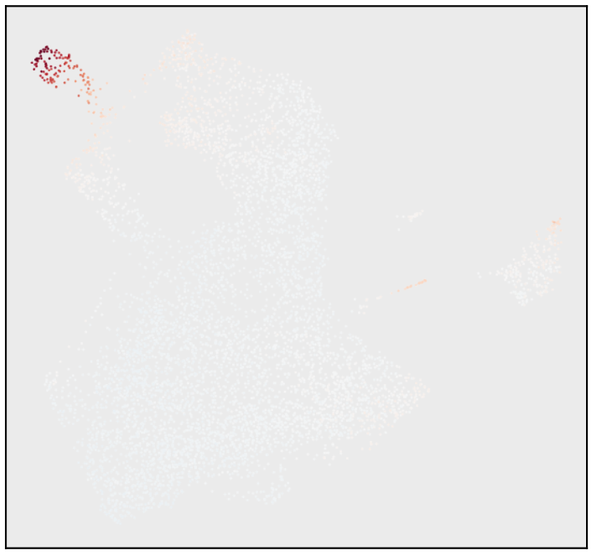
atlas, unaligned: stages



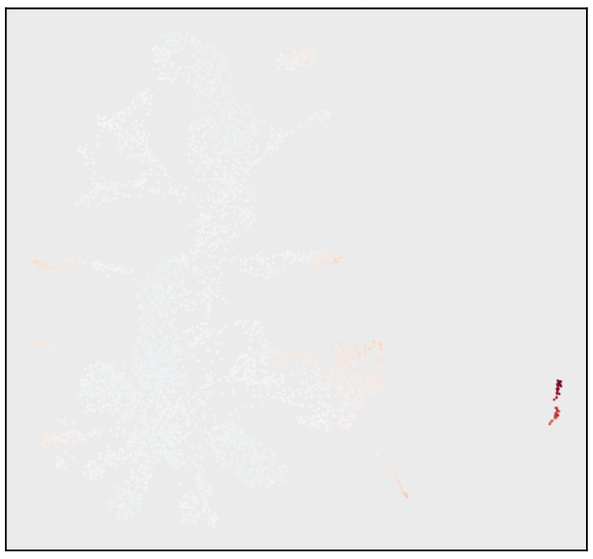
30% epiboly: not defined



75% epiboly: 30 genes, supported
active in 5% of cells, library level 0.21



6-somite: 30 genes, supported
active in 9% of cells, library level 0.39

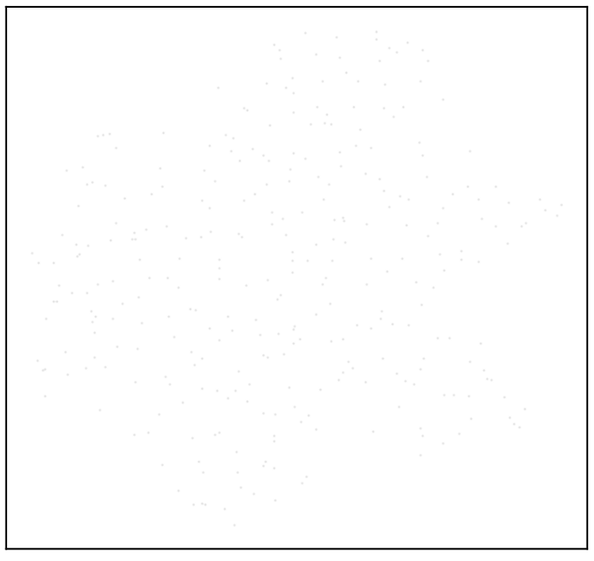


high
oblong
dome
30% epiboly
50% epiboly
shield
60% epiboly
75% epiboly
90% epiboly
bud
3-somite
6-somite

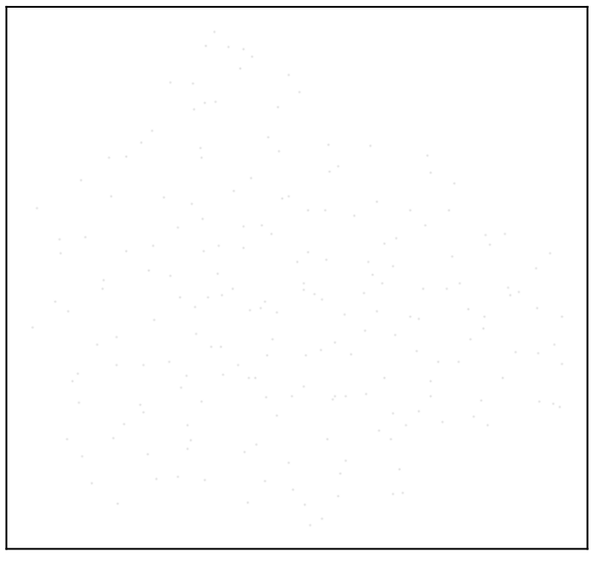
Genes (number of stages): id3 (8), tagln2 (8), pitx2 (8), elovl6 (8), calr3b (8), mllt1b (8), gadd45ga (8), cd63 (8), b3gnt5b (8), hyou1 (8), cmtm6 (8), ezrb (8), sdc4 (8), si:dkey-27i16.2 (8), pdlim1 (8), ckbb (8), hdlbpa (8), icn (7), rdh8a (7), ovol1 (7), epha7 (7), pcdh18a (7), tmem120a (7), lmo1 (7), ctslb (6), col8a1a (6), nkx2.2b (6), si:ch211-163l21.8 (6), inhbaa (5), nmt1b (5)

notochord

high: not defined



oblong: not defined

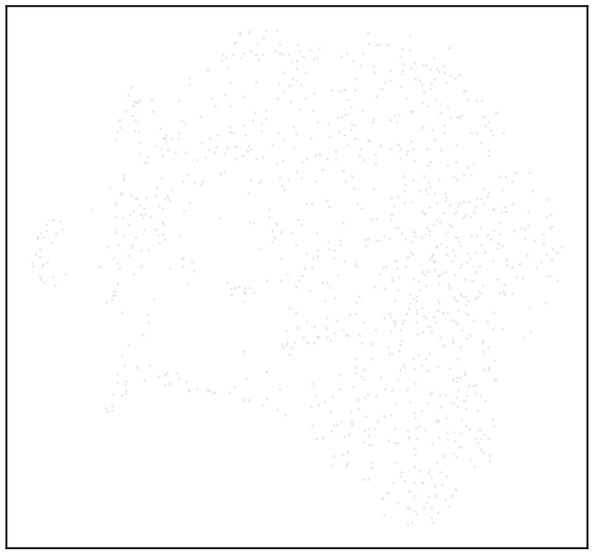


dome: not defined

30% epiboly: not defined

50% epiboly: not defined

shield: not defined

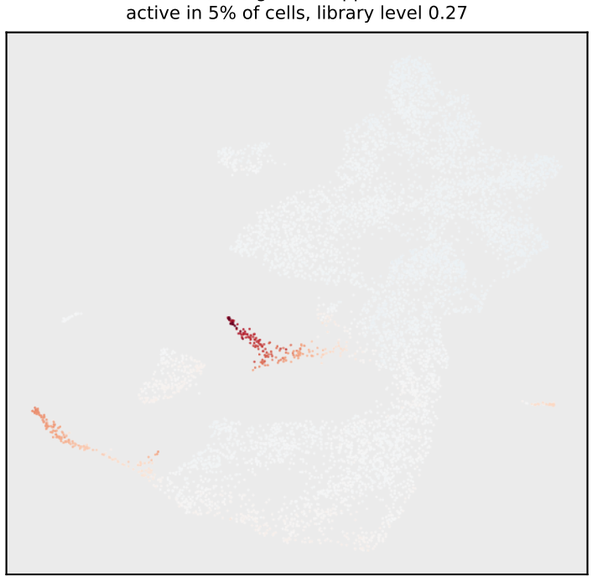


60% epiboly: not defined

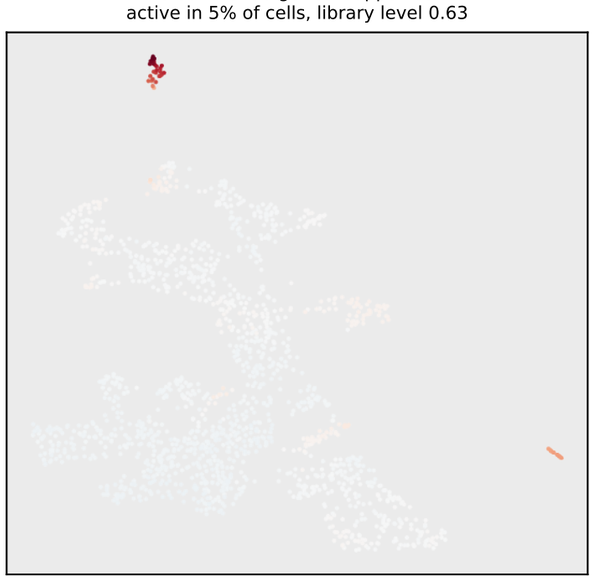
75% epiboly: not defined

90% epiboly: not defined

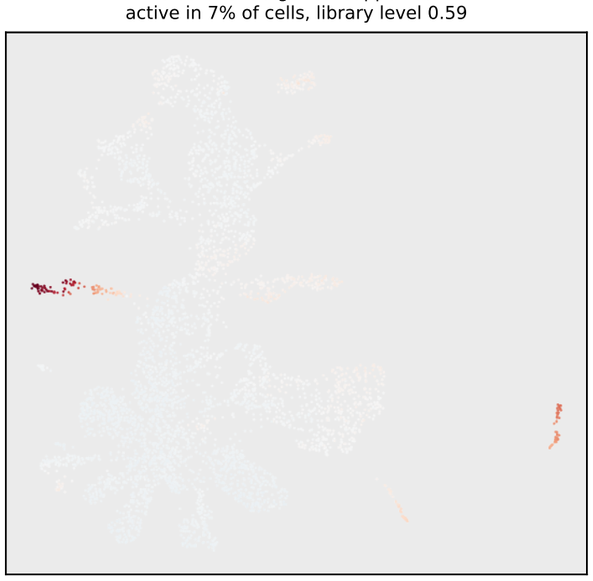
bud: 30 genes, supported
active in 5% of cells, library level 0.27



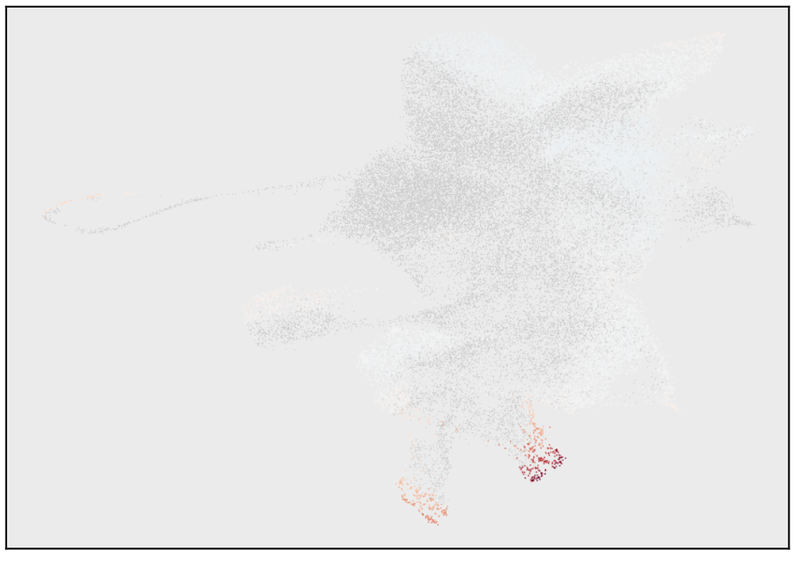
3-somite: 30 genes, supported
active in 5% of cells, library level 0.63



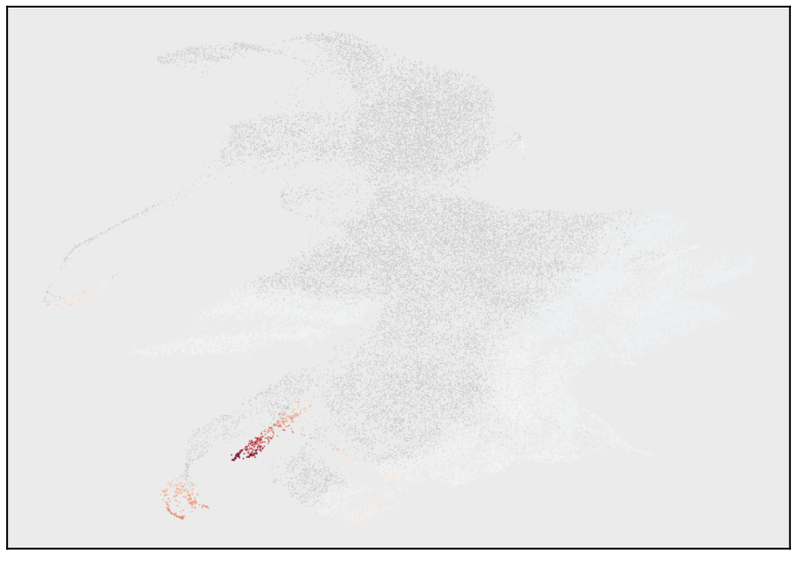
6-somite: 30 genes, supported
active in 7% of cells, library level 0.59



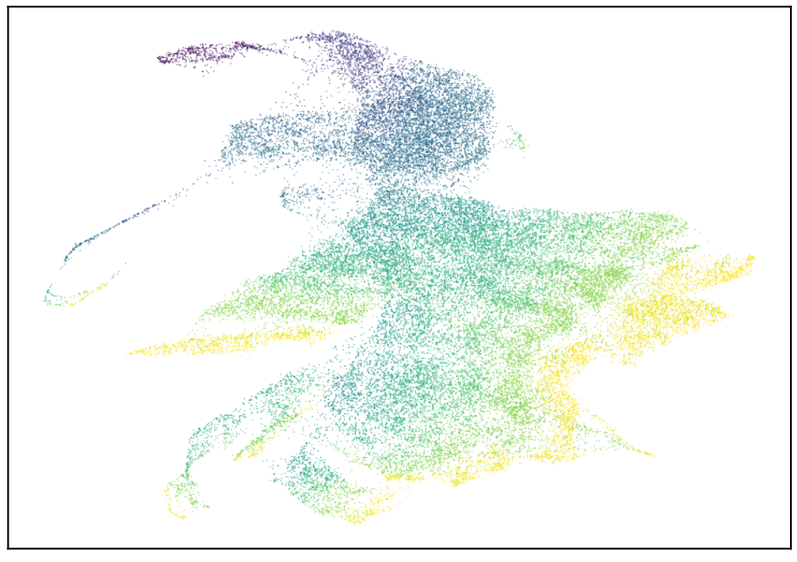
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



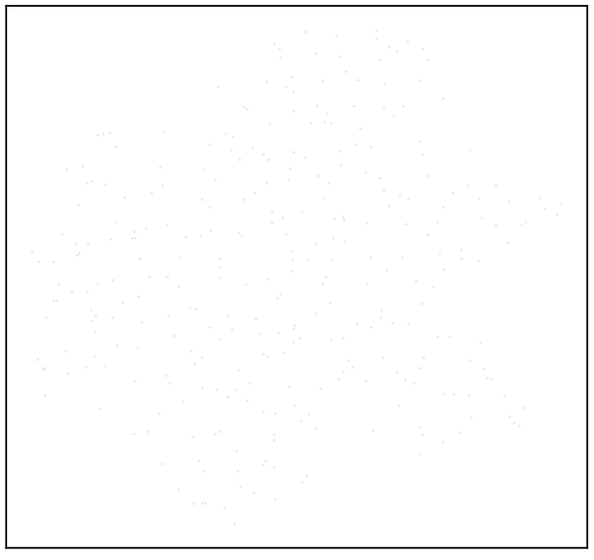
atlas, unaligned: stages



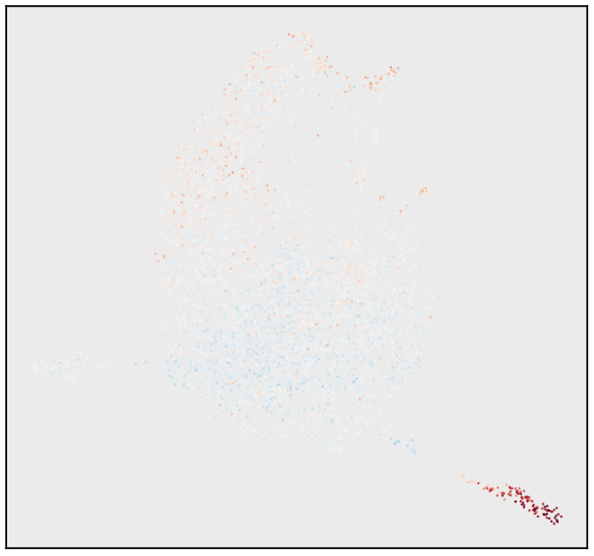
Genes (number of stages): ta (3), hspa5 (3), xbp1 (3), col8a1a (3), hfe2 (3), nog2 (3), pak1 (3), si:dkey-261h17.1 (3), anxa2a (3), calr3b (3), emilin3a (3), ntd5 (3), pmp22b (3), rcn3 (3), rrbp1b (3), stk10 (3), admp (3), aff3 (3), asph (3), lect1 (3), loxl5b (3), pdgfbb (3), sec61g (3), zgc:162964 (3), dnajc3a (3), sec61b (3), col2a1a (3), mnx1 (3), calr (3), cast (3)

ventral_bmp

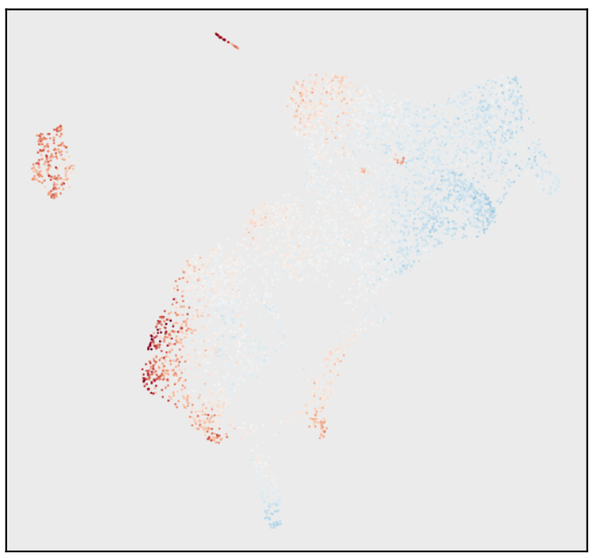
high: not defined



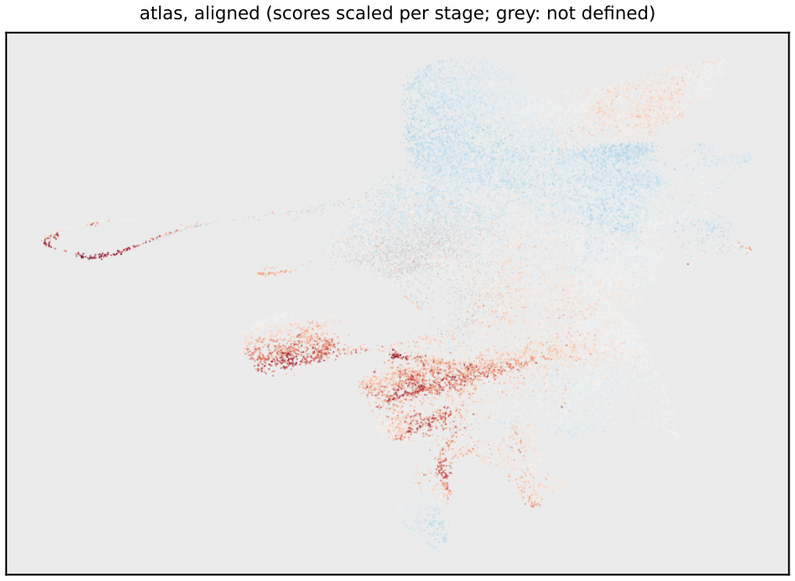
50% epiboly: 14 genes, not supported
active in 2% of cells, library level 0.16



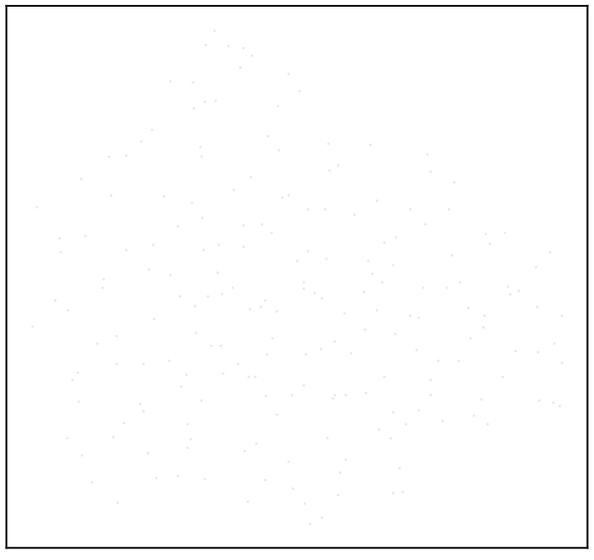
90% epiboly: 14 genes, not supported
active in 9% of cells, library level 0.13



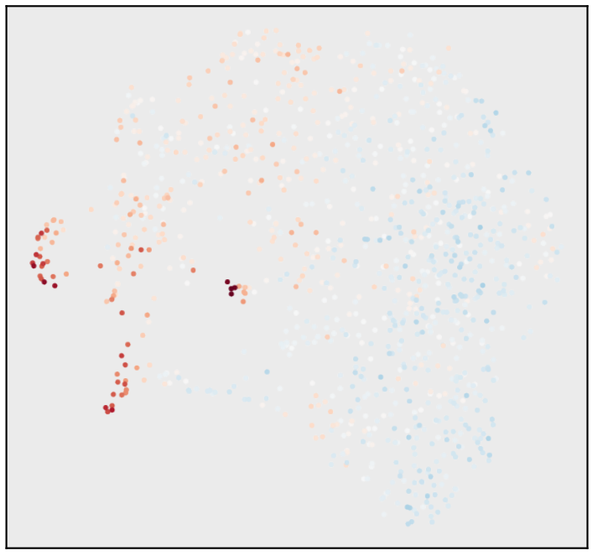
atlas, aligned (scores scaled per stage; grey: not defined)



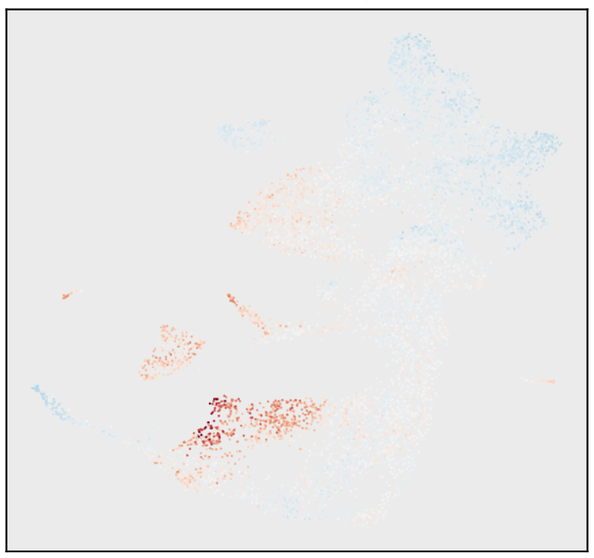
oblong: not defined



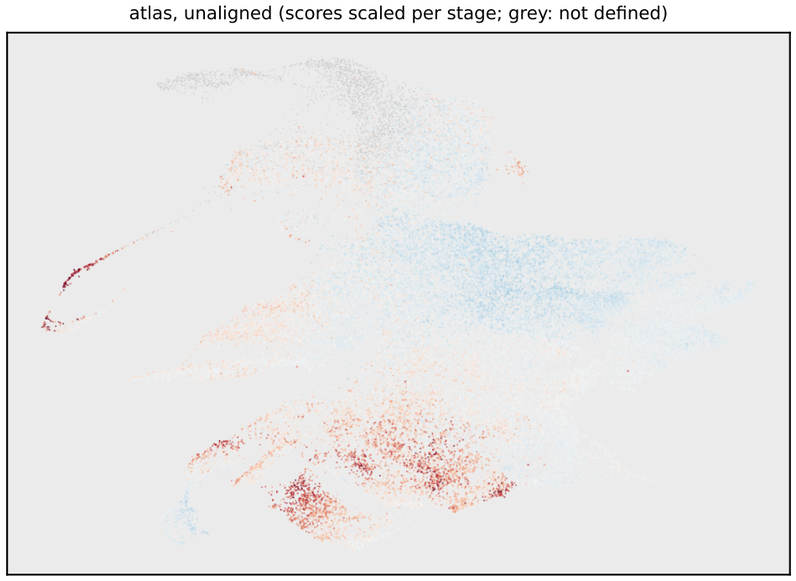
shield: 12 genes, not supported
active in 4% of cells, library level 0.17



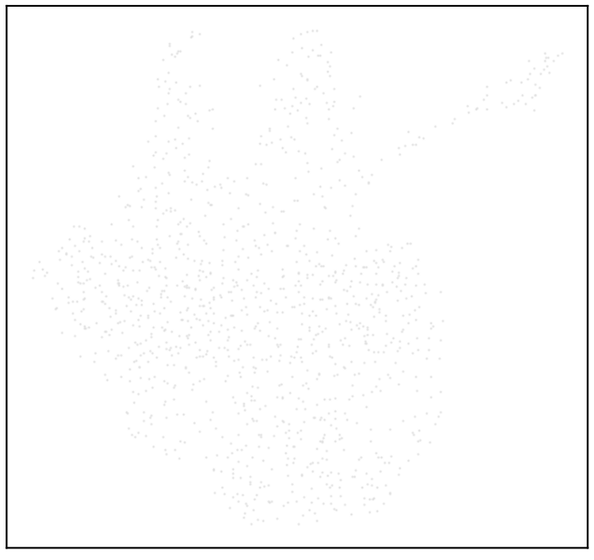
bud: 15 genes, not supported
active in 7% of cells, library level 0.17



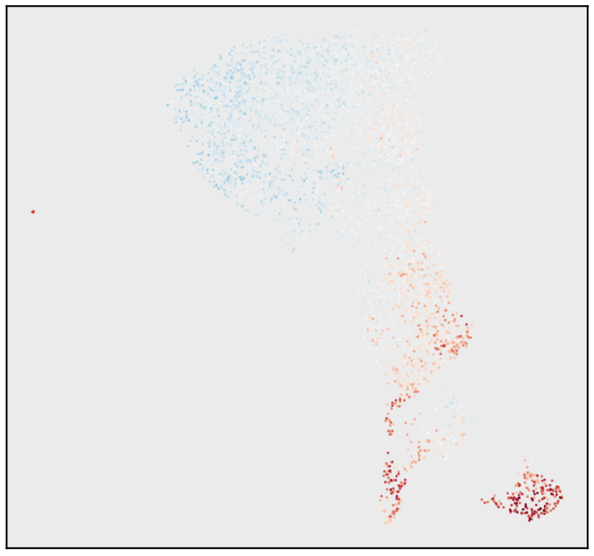
atlas, unaligned (scores scaled per stage; grey: not defined)



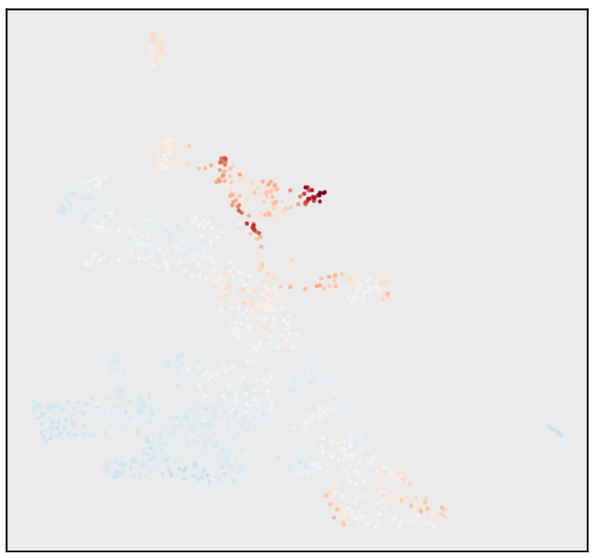
dome: not defined



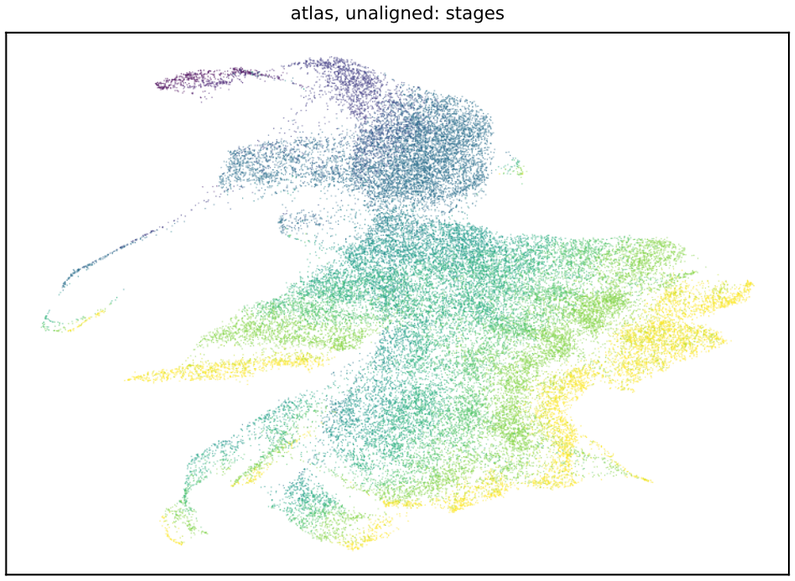
60% epiboly: 13 genes, not supported
active in 6% of cells, library level 0.14



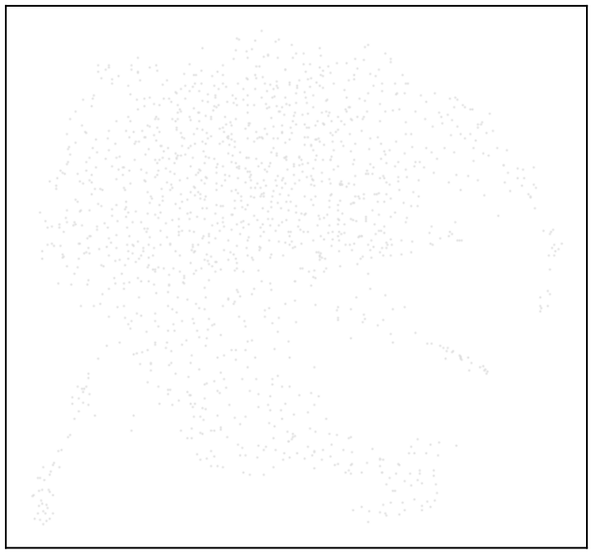
3-somite: 16 genes, not supported
active in 7% of cells, library level 0.32



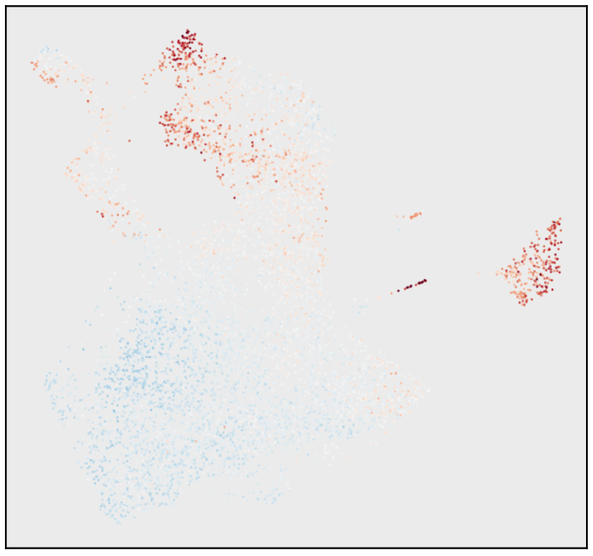
atlas, unaligned: stages



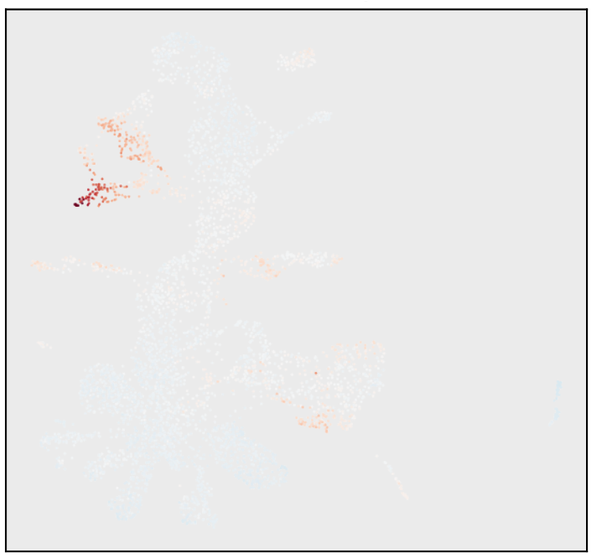
30% epiboly: not defined



75% epiboly: 14 genes, not supported
active in 6% of cells, library level 0.14



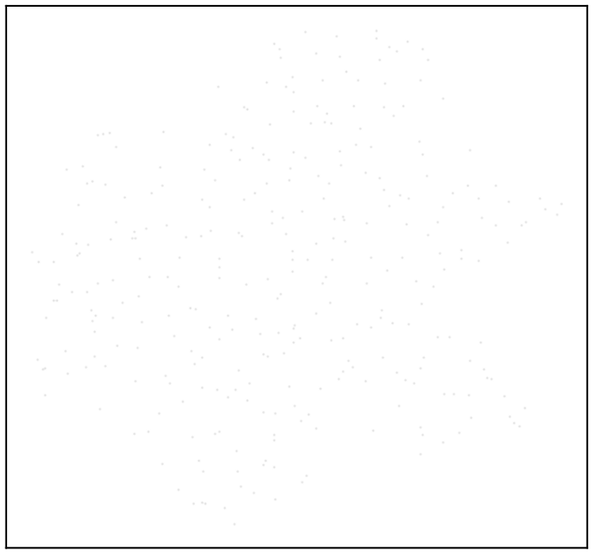
6-somite: 17 genes, supported
active in 7% of cells, library level 0.34



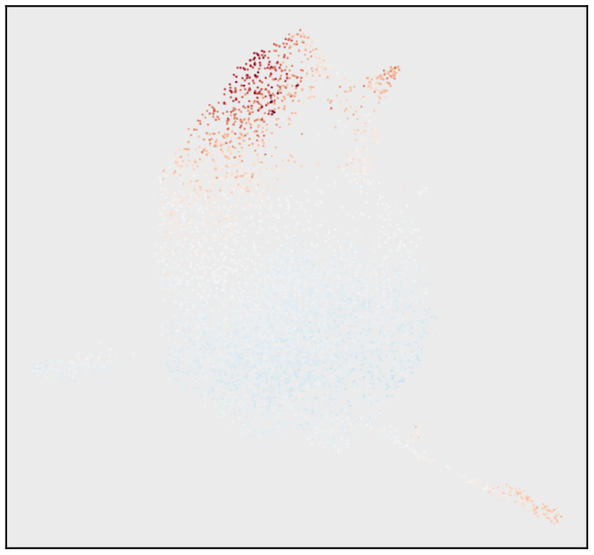
Genes (number of stages): dhrs3b (8), klf6a (8), si:ch211-106e7.2 (8), stmn1a (8), si:ch211-197h24.9 (8), morc3b (8), hk1 (8), ppp1r14ab (8), cldng (8), hdr (8), rbpms2a (8), si:dkey-52l18.4 (7), micall2a (6), dusp5 (5), bmp6 (4), jam2b (3), gfi1aa (2)

presomitic_mesoderm

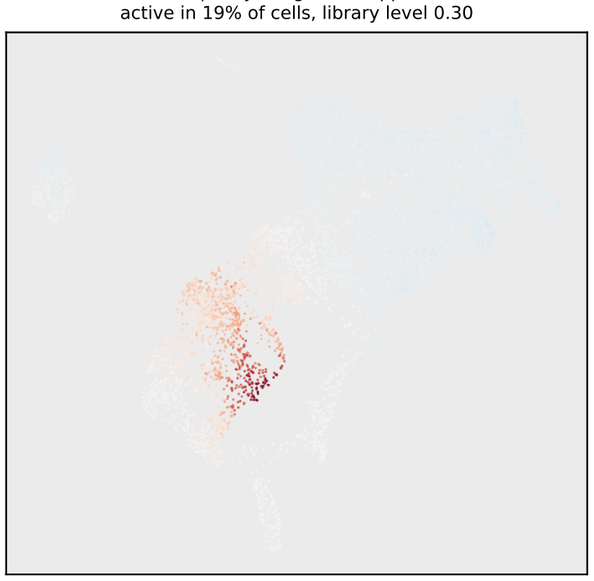
high: not defined



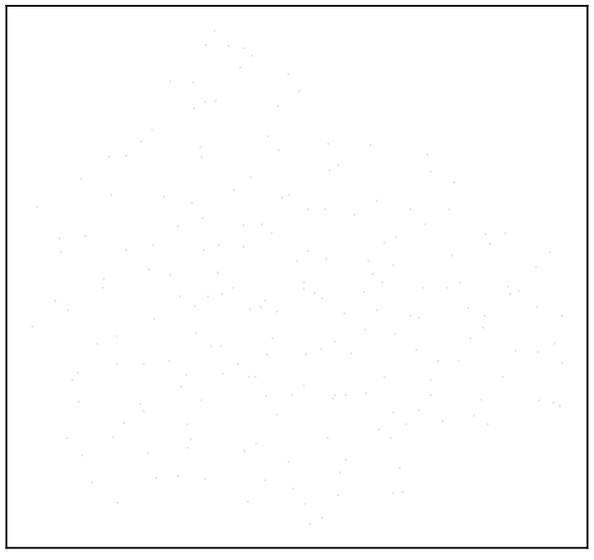
50% epiboly: 26 genes, not supported
active in 14% of cells, library level 0.11



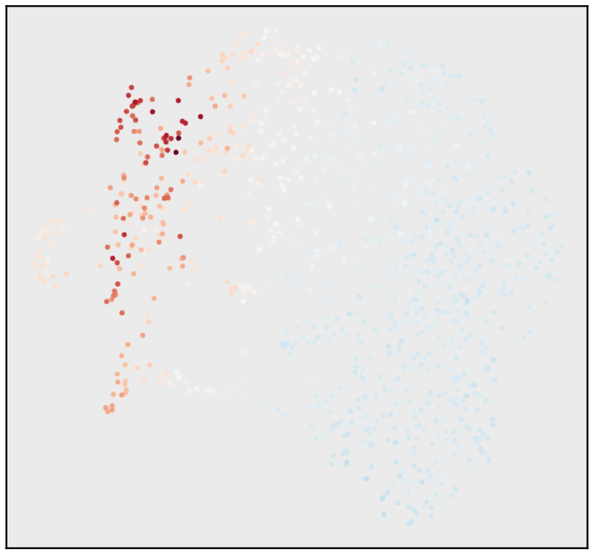
90% epiboly: 30 genes, supported
active in 19% of cells, library level 0.30



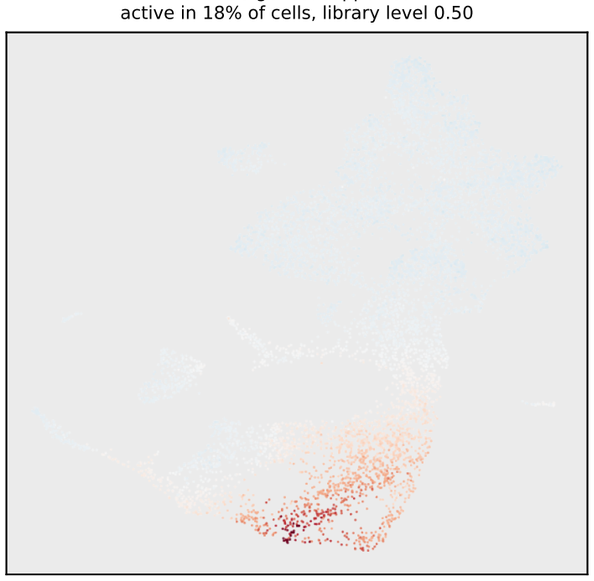
oblong: not defined



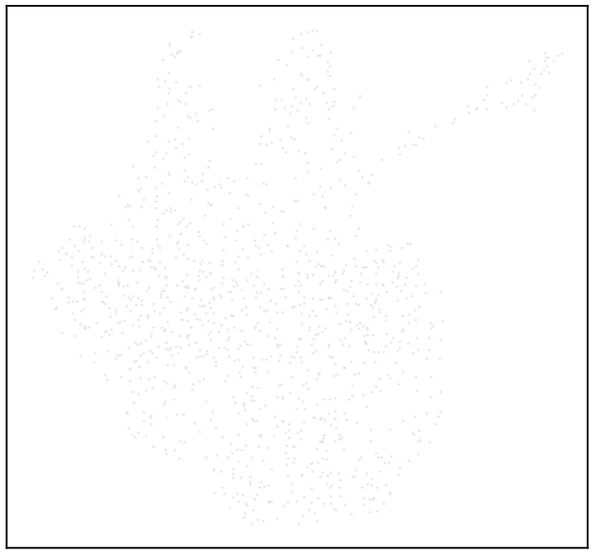
shield: 23 genes, supported
active in 14% of cells, library level 0.33



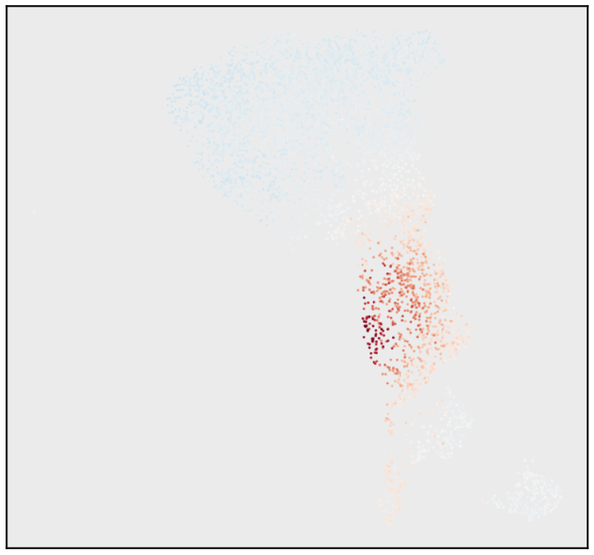
bud: 30 genes, supported
active in 18% of cells, library level 0.50



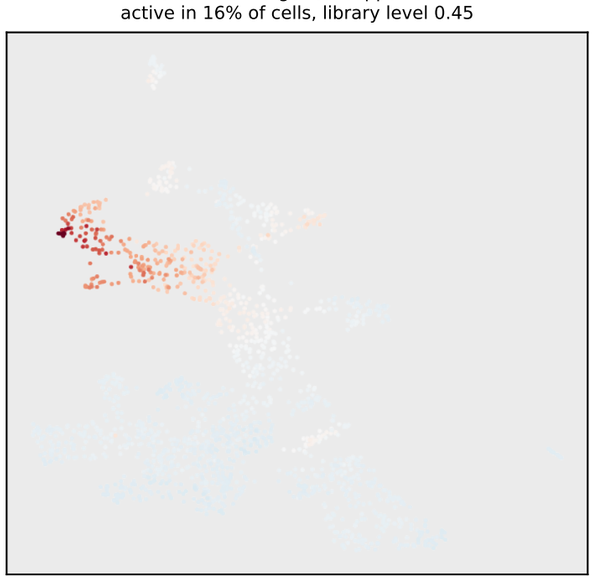
dome: not defined



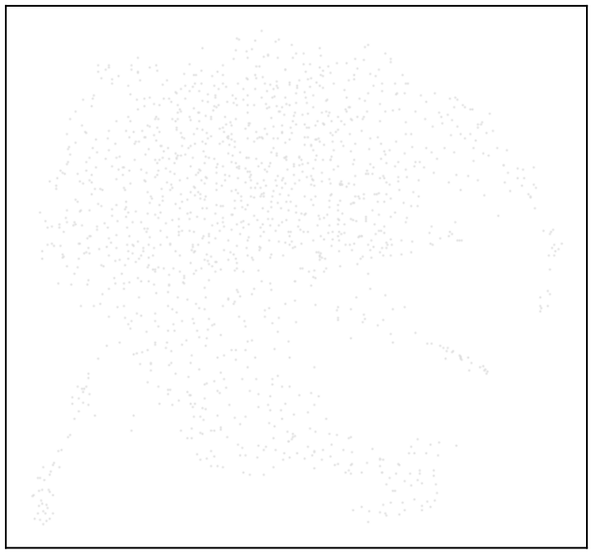
60% epiboly: 29 genes, supported
active in 21% of cells, library level 0.25



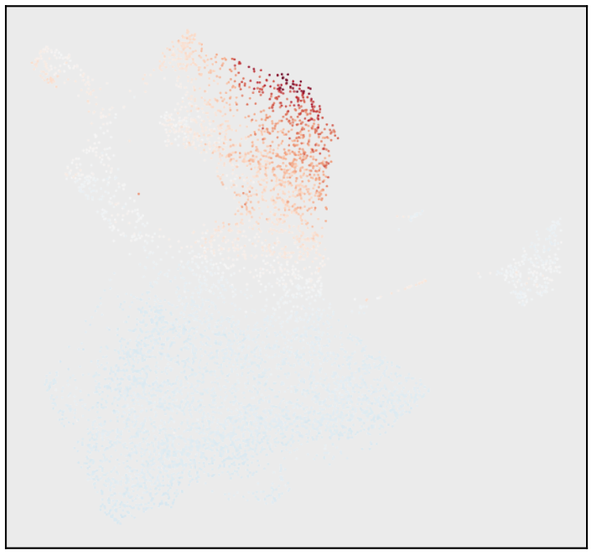
3-somite: 29 genes, supported
active in 16% of cells, library level 0.45



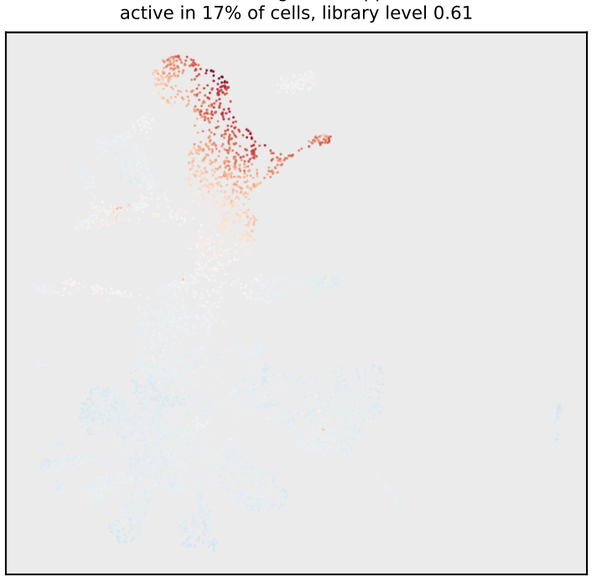
30% epiboly: not defined



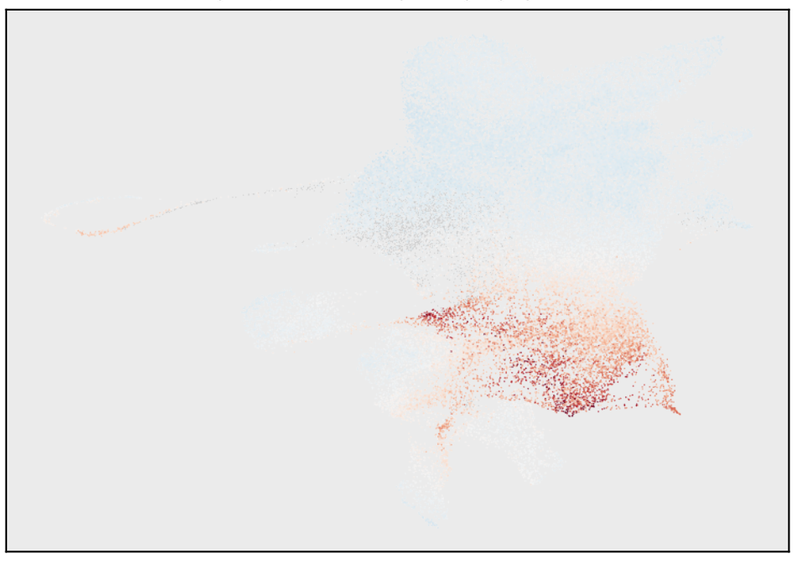
75% epiboly: 30 genes, supported
active in 20% of cells, library level 0.27



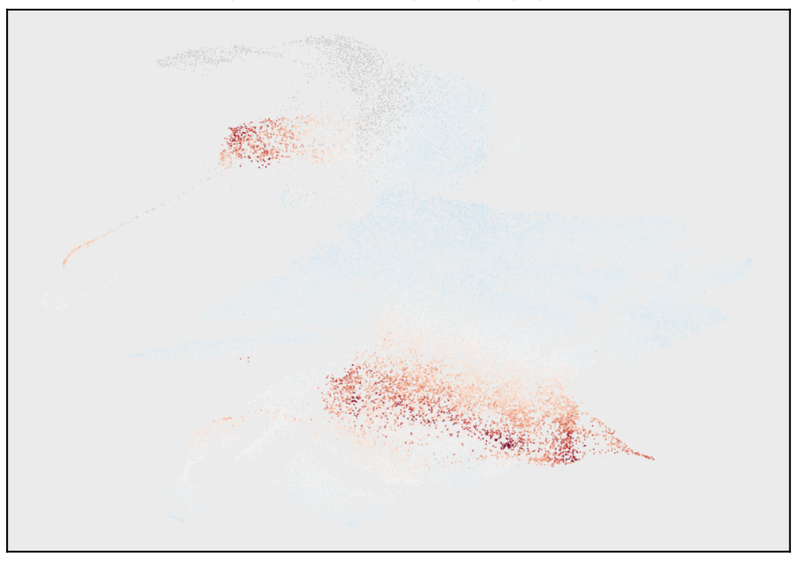
6-somite: 30 genes, supported
active in 17% of cells, library level 0.61



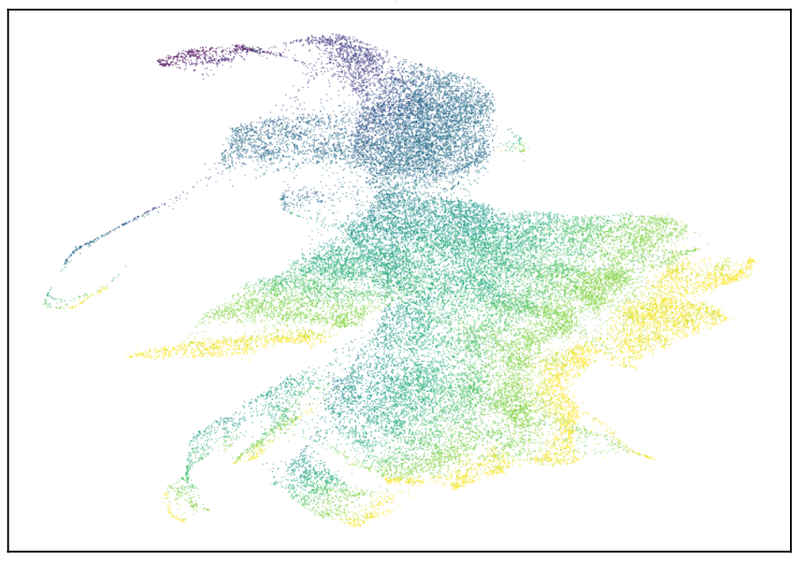
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



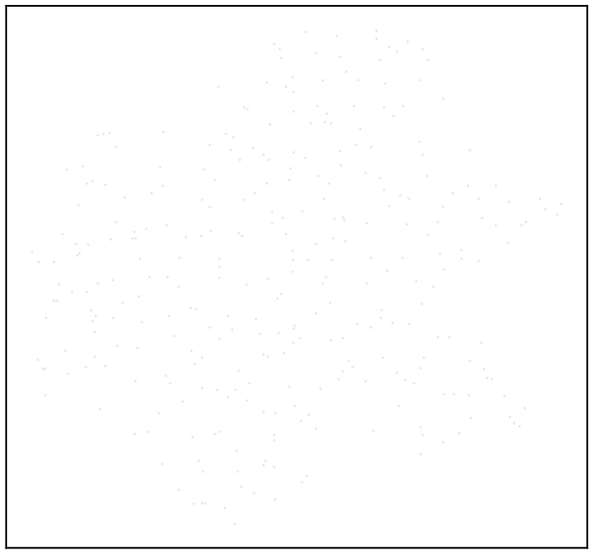
atlas, unaligned: stages



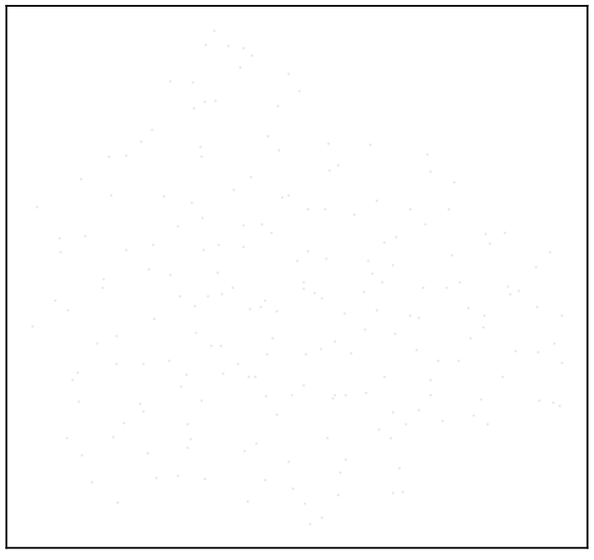
Genes (number of stages): myf5 (8), msgn1 (8), ripply1 (8), mespab (8), hes6 (8), rdh10a (8), pcdh8 (8), mespaa (8), her7 (8), ripply2 (8), her1 (8), mdka (8), mespba (8), efemp2b (8), dlc (8), foxc1a (8), arl4aa (8), blf (8), unc45b (8), fgf17 (8), gipc2 (8), lhfp (8), tbx6 (7), tcf15 (7), mespbb (7), slc2a12 (7), meox1 (6), lbx2 (6), hoxb7a (6), hoxb6b (5)

adaxial_somite

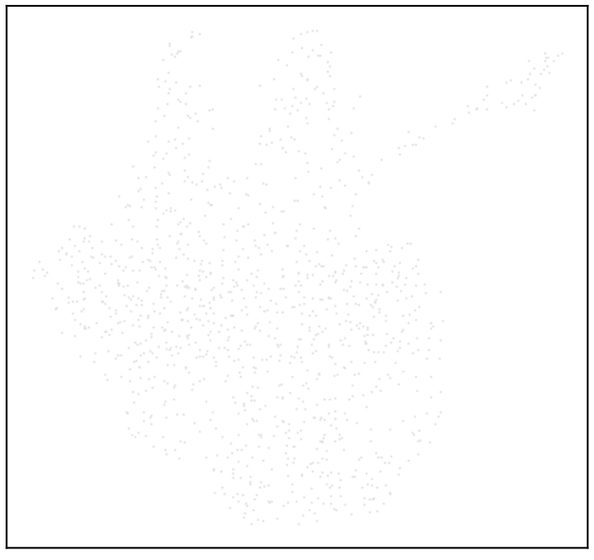
high: not defined



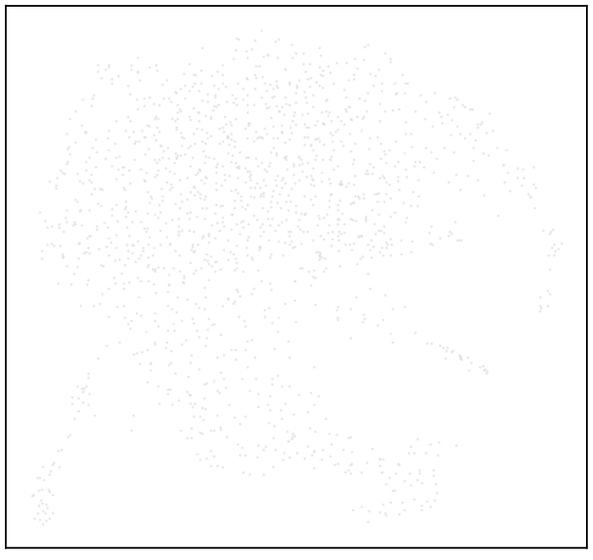
oblong: not defined



dome: not defined

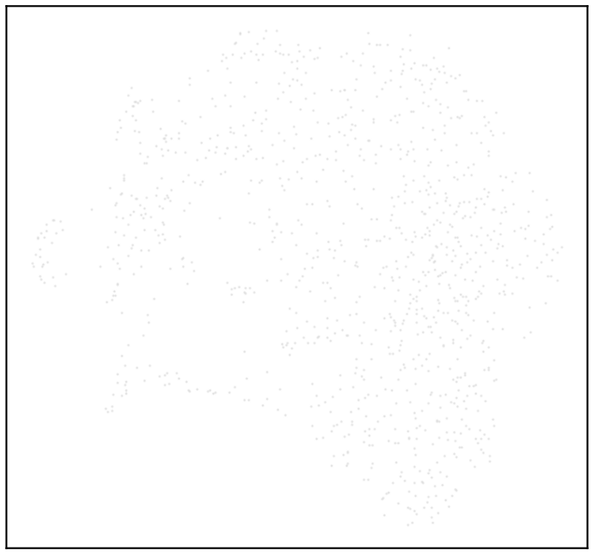


30% epiboly: not defined



50% epiboly: not defined

shield: not defined

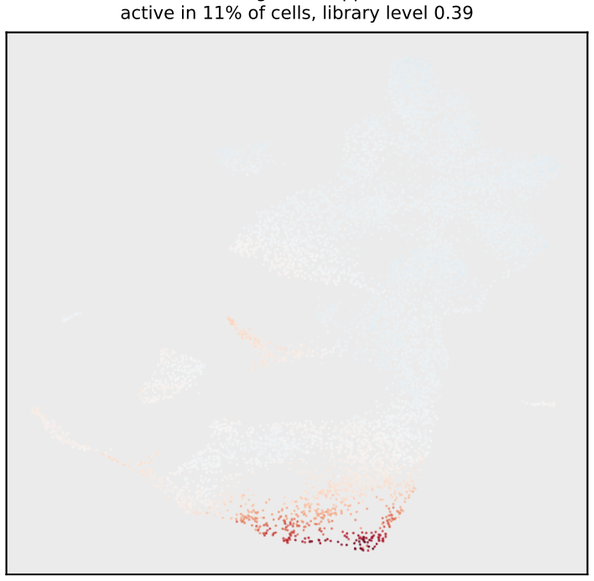


60% epiboly: not defined

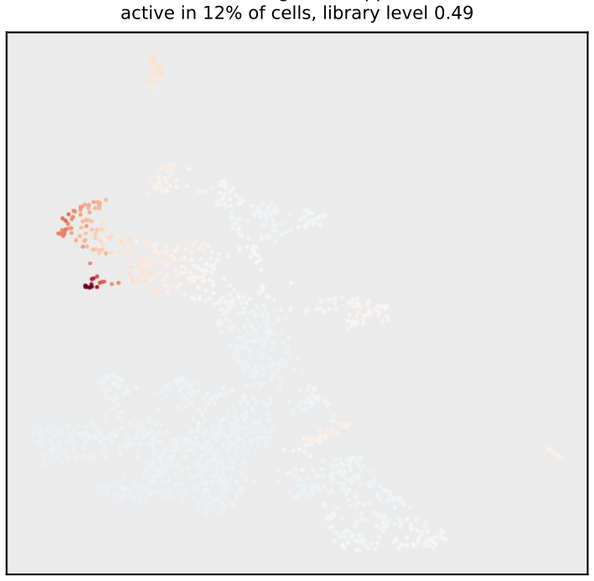
75% epiboly: not defined

90% epiboly: not defined

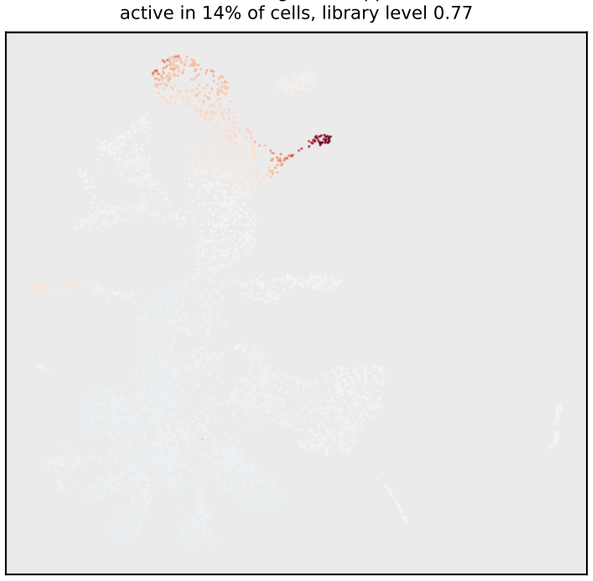
bud: 28 genes, supported
active in 11% of cells, library level 0.39



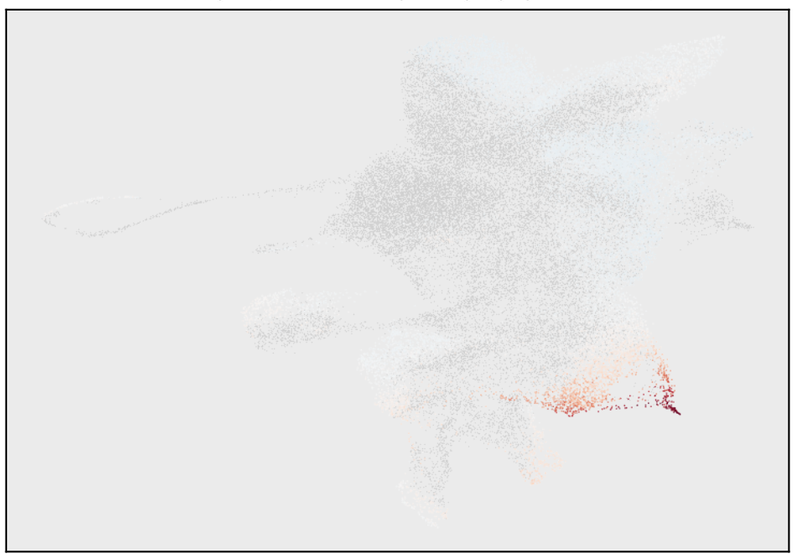
3-somite: 30 genes, supported
active in 12% of cells, library level 0.49



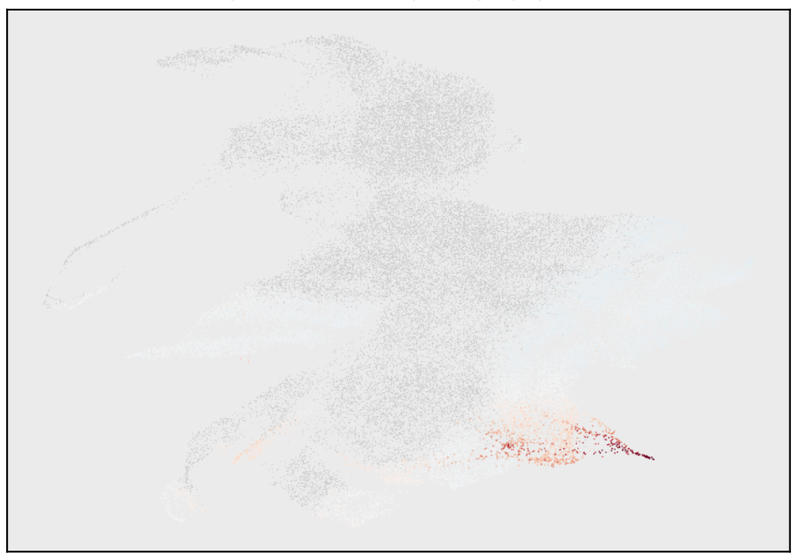
6-somite: 30 genes, supported
active in 14% of cells, library level 0.77



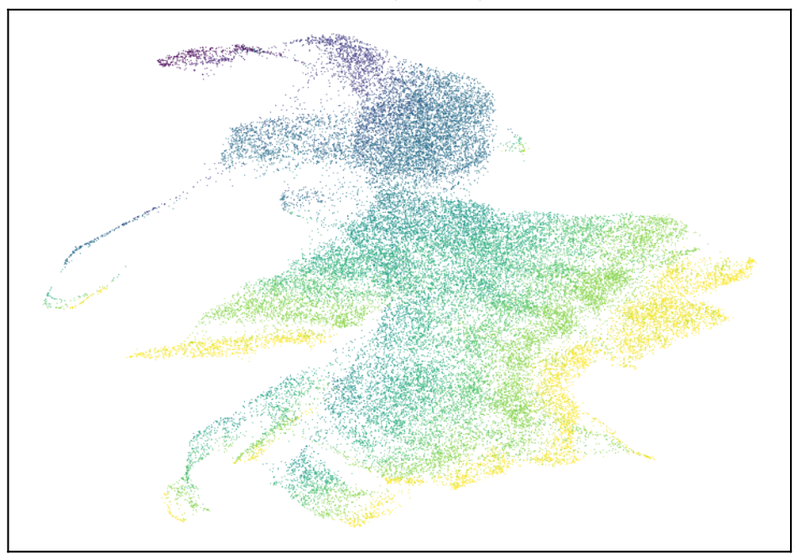
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



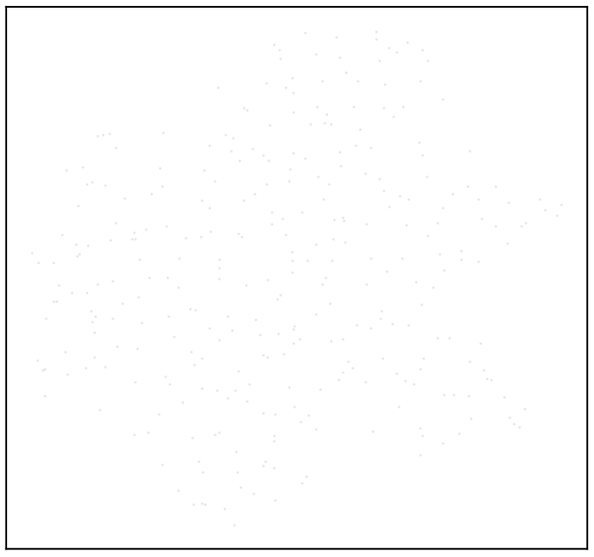
atlas, unaligned: stages



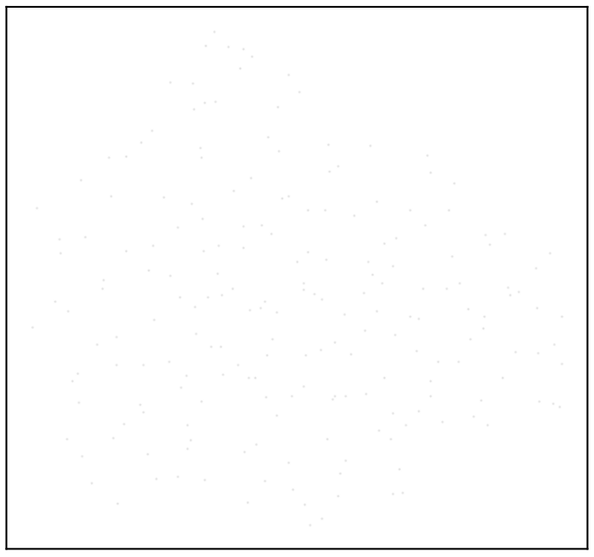
Genes (number of stages): myod1 (3), myf5 (3), hsp90aa1.1 (3), efemp2a (3), ripply1 (3), mef2d (3), rbm24a (3), tcf15 (3), acta1a (3), actc1a (3), alpi.1 (3), meox1 (3), myl10 (3), ntd5 (3), rdh10a (3), trim109 (3), ved (3), ttna (3), fstl1a (3), unc45b (3), asph (3), prdm1a (3), mespaa (3), ttnb (3), arl4aa (3), cpn1 (3), zgc:110340 (3), uncx4.1 (3), myog (2), smyhc1 (2)

tailbud

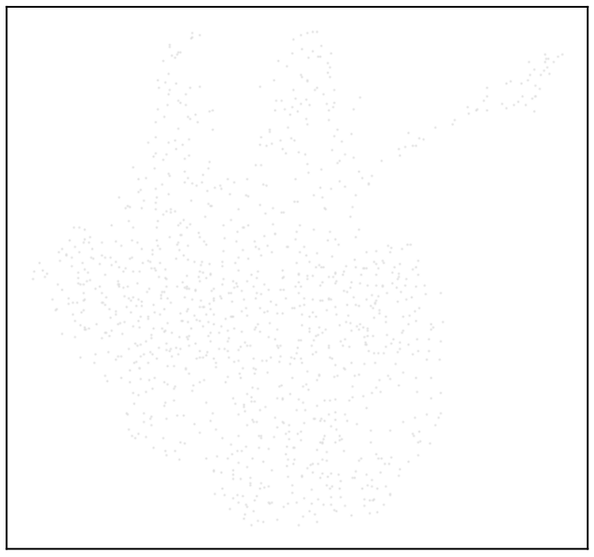
high: not defined



oblong: not defined



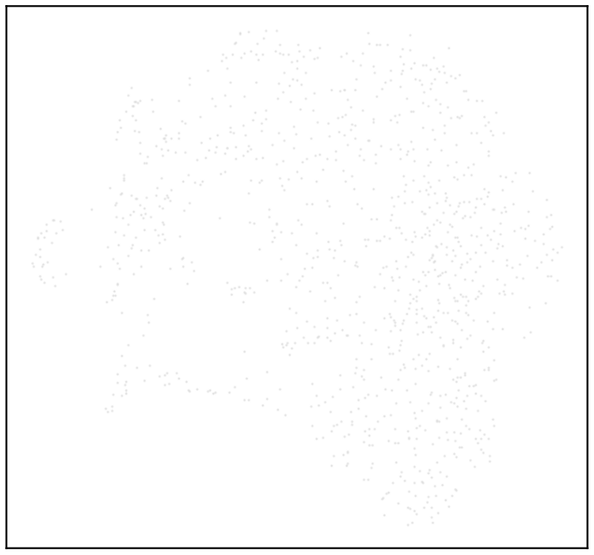
dome: not defined



30% epiboly: not defined

50% epiboly: not defined

shield: not defined

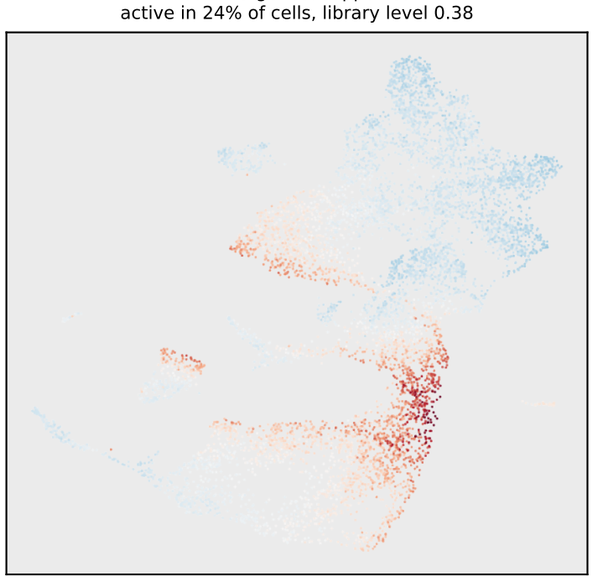


60% epiboly: not defined

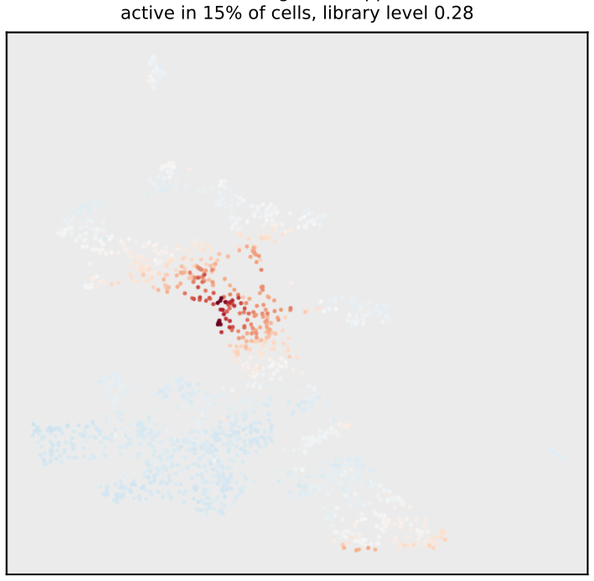
75% epiboly: not defined

90% epiboly: not defined

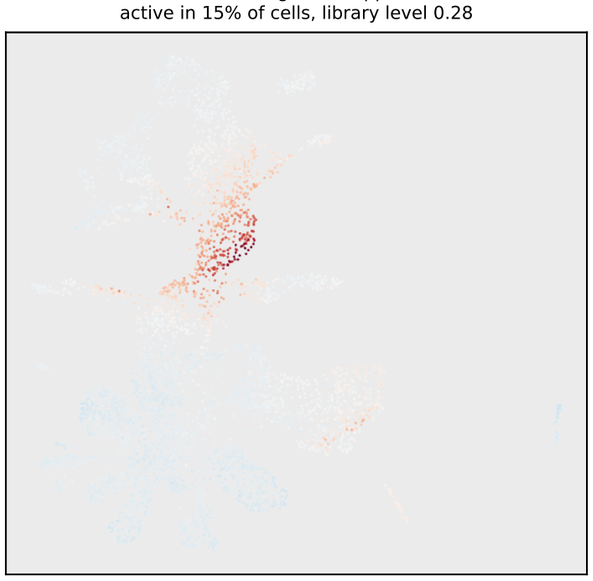
bud: 29 genes, supported
active in 24% of cells, library level 0.38



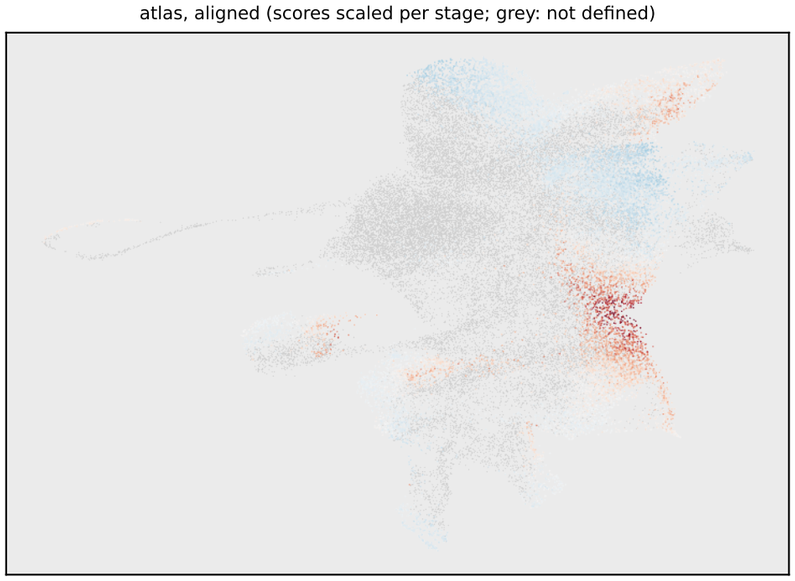
3-somite: 30 genes, supported
active in 15% of cells, library level 0.28



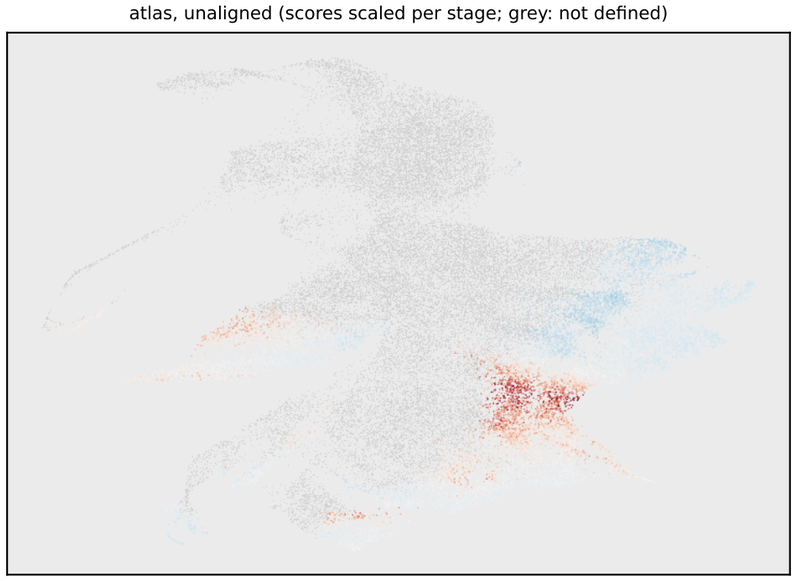
6-somite: 30 genes, supported
active in 15% of cells, library level 0.28



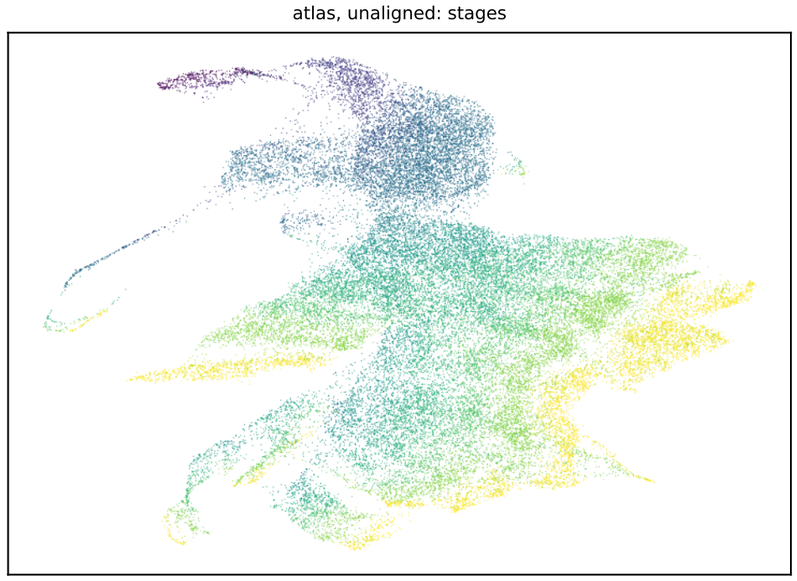
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



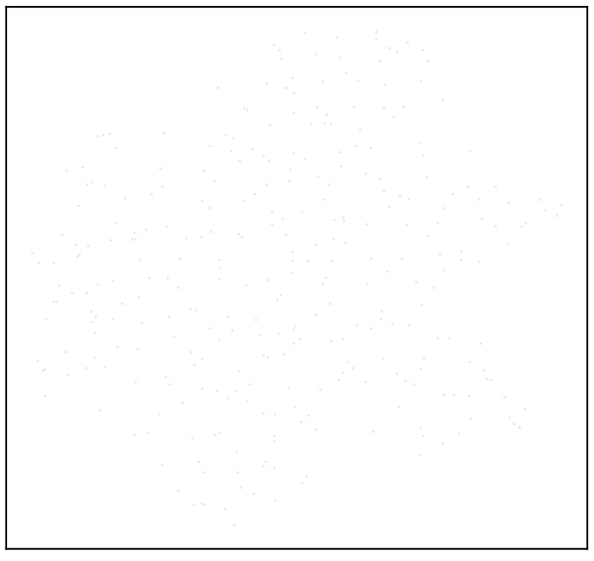
atlas, unaligned: stages



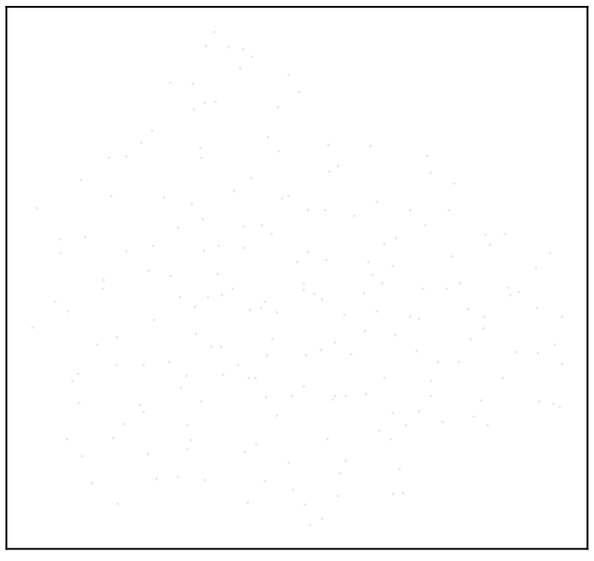
Genes (number of stages): hes6 (3), vox (3), sult6b1 (3), szl (3), eve1 (3), foxd5 (3), tbx6l (3), adam8a (3), wnt8a (3), bmp2b (3), bmp7a (3), p4ha2 (3), stmn1a (3), tagln3b (3), bmp4 (3), msx2b (3), bmp2a (3), fzd10 (3), rasl11b (3), ccnb1 (3), hoxd11a (3), lpar1 (3), vent (3), rarga (3), mllt3 (3), aif1l (3), cmtm6 (3), nfkbiaa (3), tjp2b (3), fgf10a (2)

lateral_plate

high: not defined



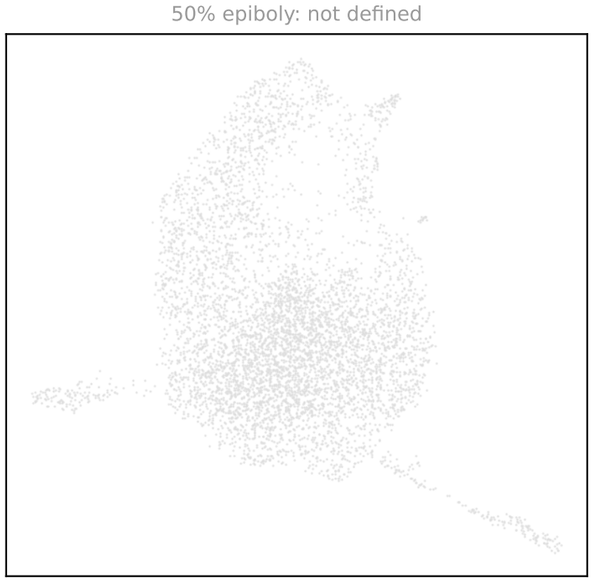
oblong: not defined



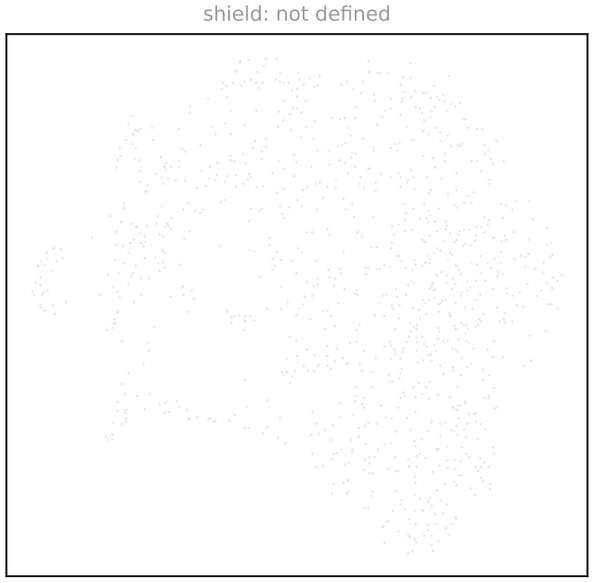
dome: not defined

30% epiboly: not defined

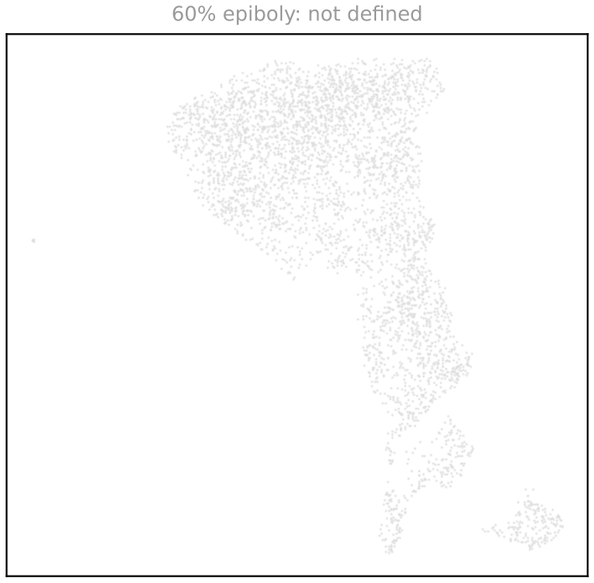
50% epiboly: not defined



shield: not defined



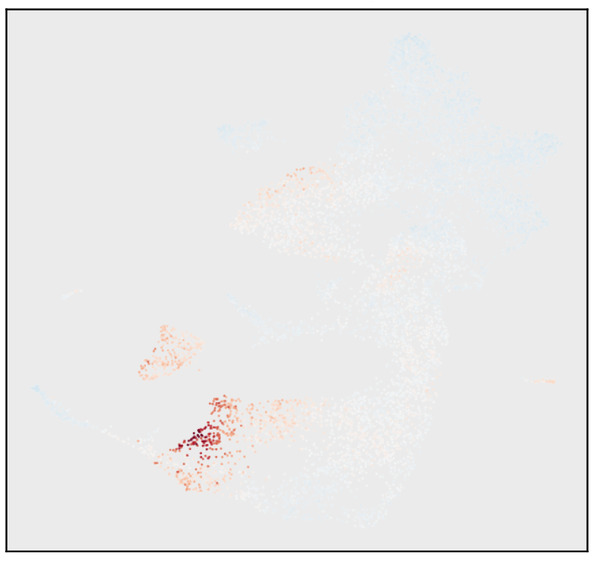
60% epiboly: not defined



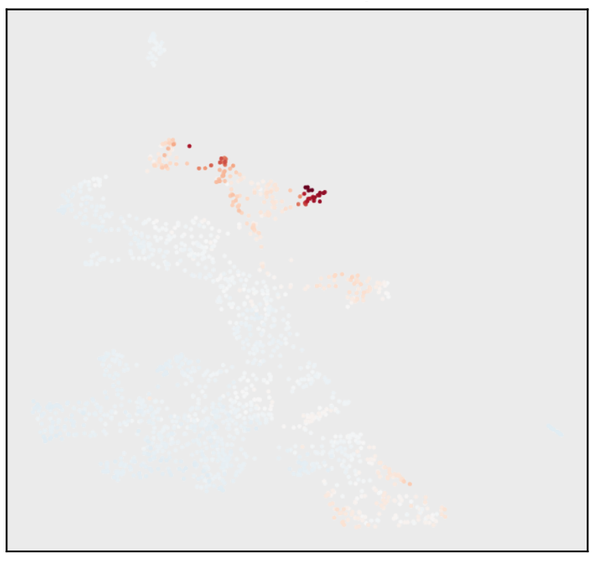
75% epiboly: not defined

90% epiboly: not defined

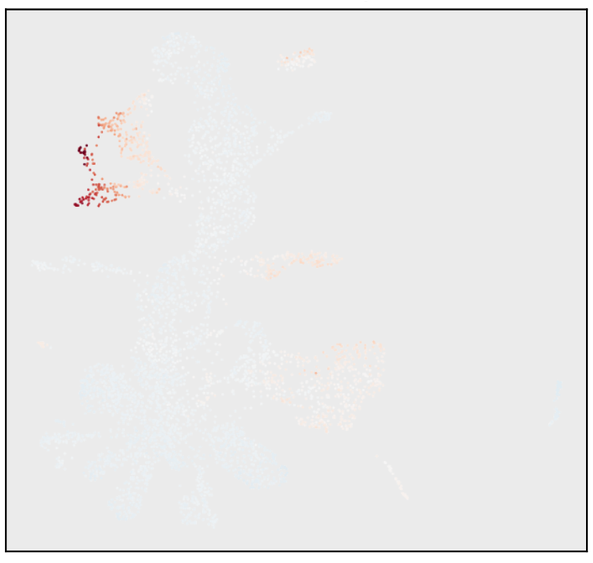
bud: 24 genes, supported
active in 10% of cells, library level 0.27



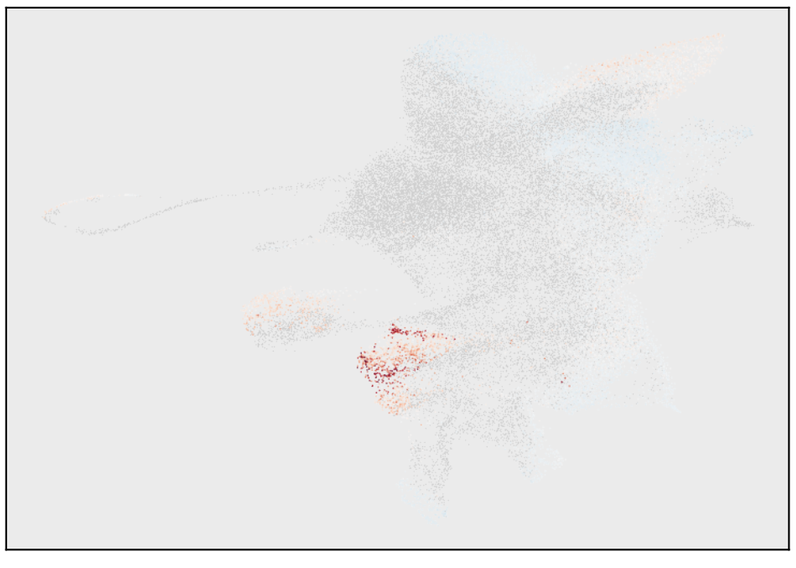
3-somite: 24 genes, supported
active in 9% of cells, library level 0.67



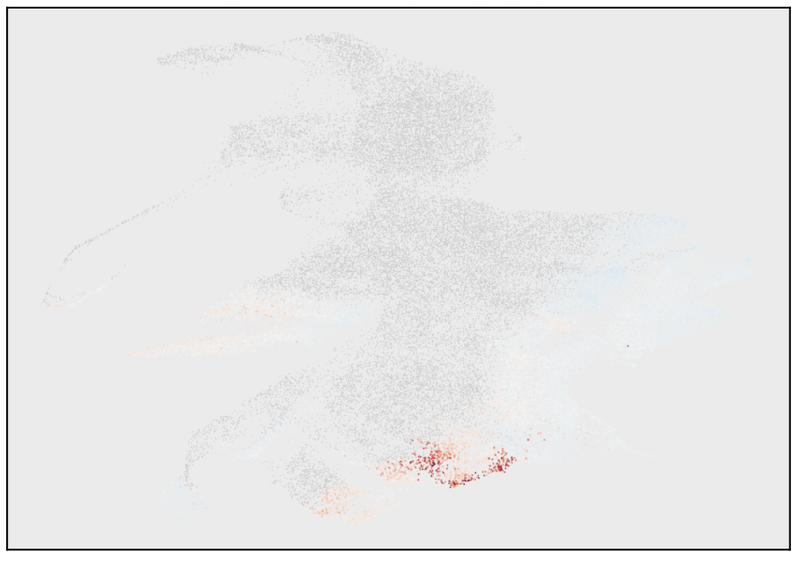
6-somite: 30 genes, supported
active in 9% of cells, library level 0.83



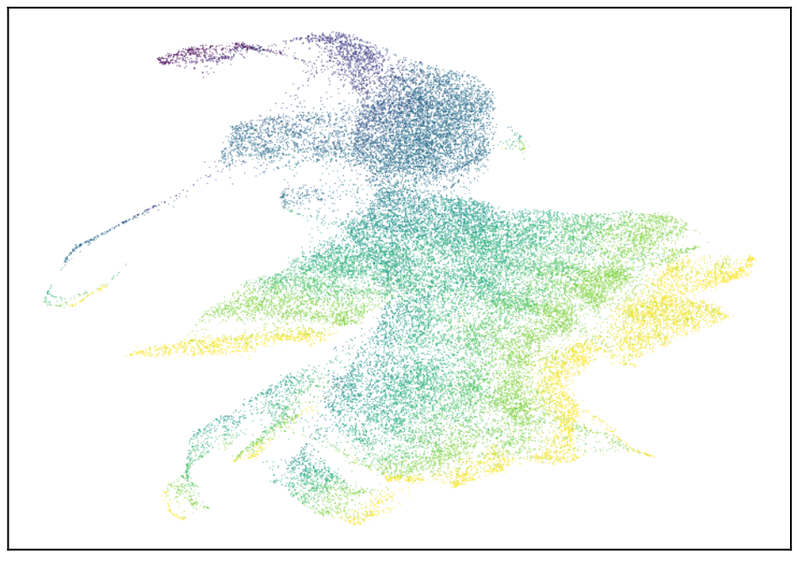
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



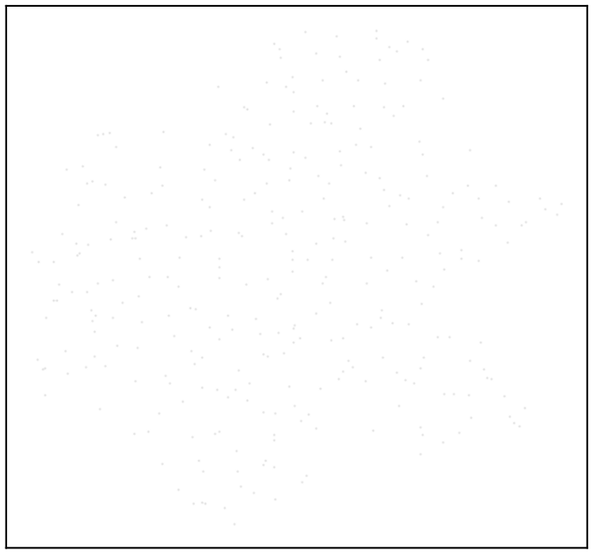
atlas, unaligned: stages



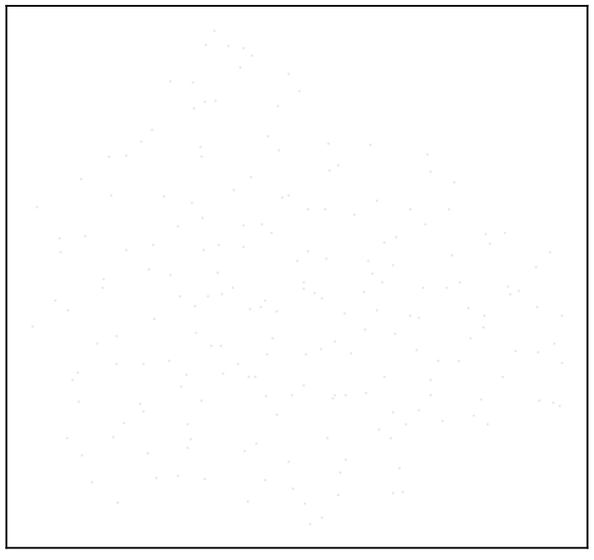
Genes (number of stages): hand2 (3), tmem88a (3), nkx2.7 (3), gata5 (3), gata6 (3), gata2a (3), lmo2 (3), etv2 (3), sox7 (3), tbx1 (3), dhrs3a (3), tpm4a (3), dusp5 (3), myb (3), blf (3), ptgs1 (3), hoxb5b (3), atf3 (3), fli1a (3), hoxa13b (3), hoxb5a (3), mnx2b (3), tal1 (2), gfi1aa (2), cebpa (2), hbbe1.1 (2), gata1a (1), nkx2.5 (1), spi1b (1), isl2b (1)

intermediate_mesoderm

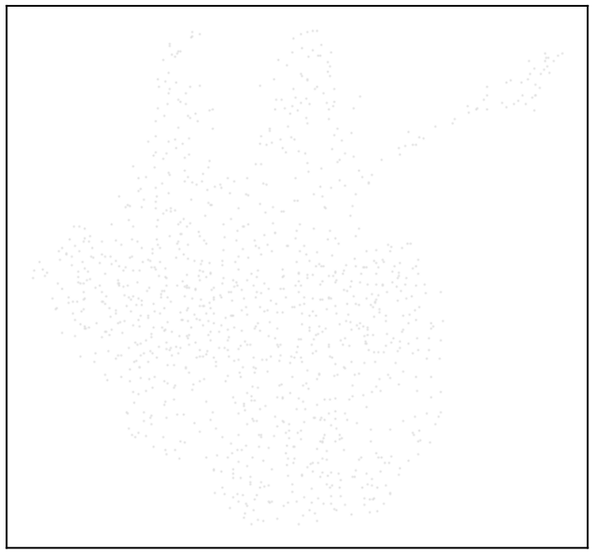
high: not defined



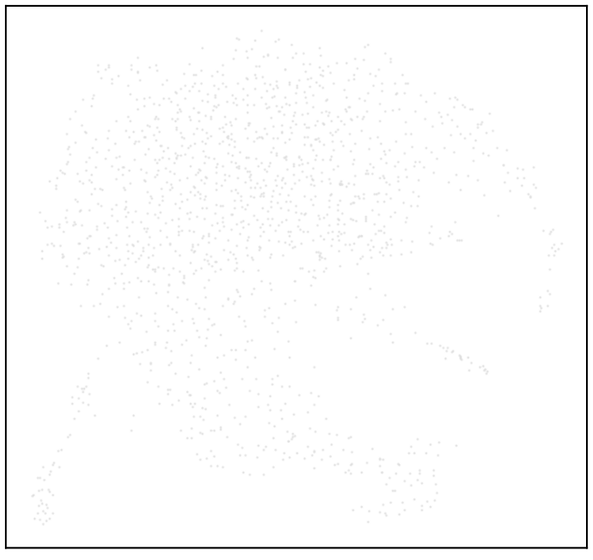
oblong: not defined



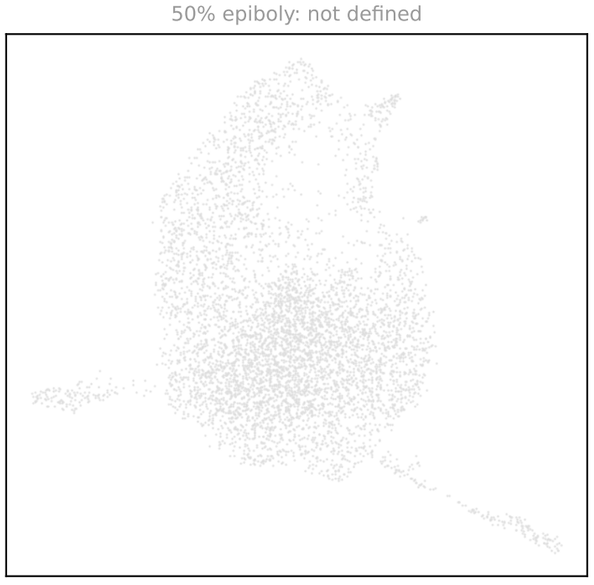
dome: not defined



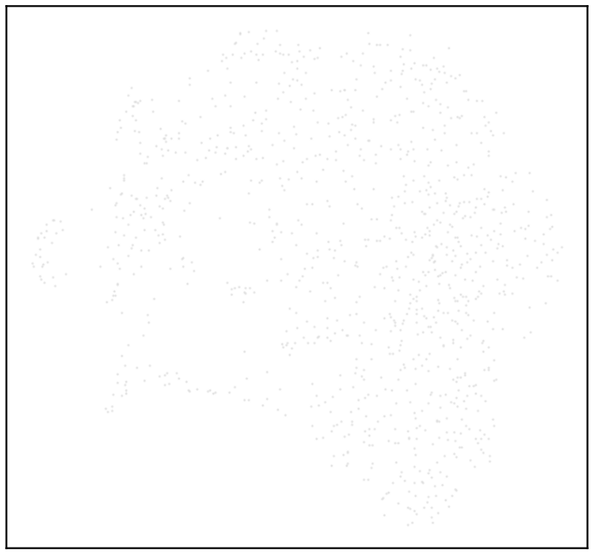
30% epiboly: not defined



50% epiboly: not defined



shield: not defined

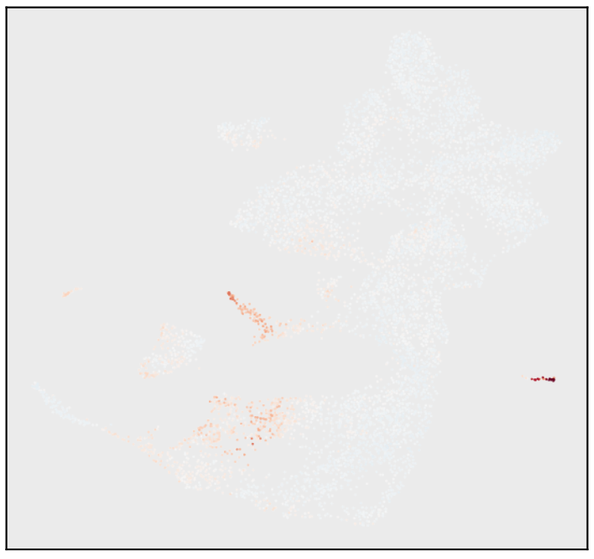


60% epiboly: not defined

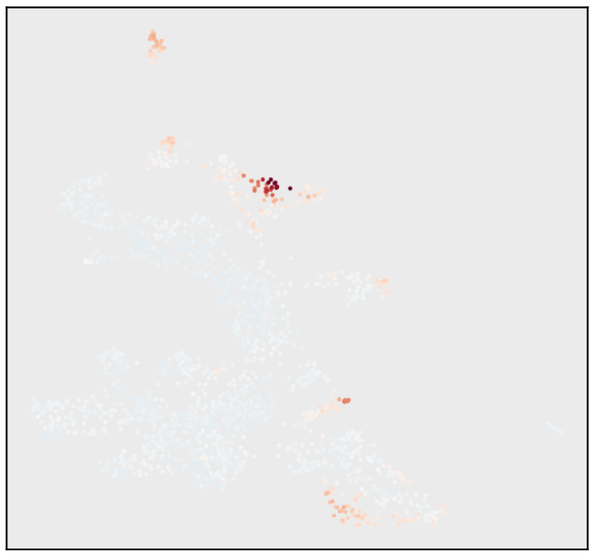
75% epiboly: not defined

90% epiboly: not defined

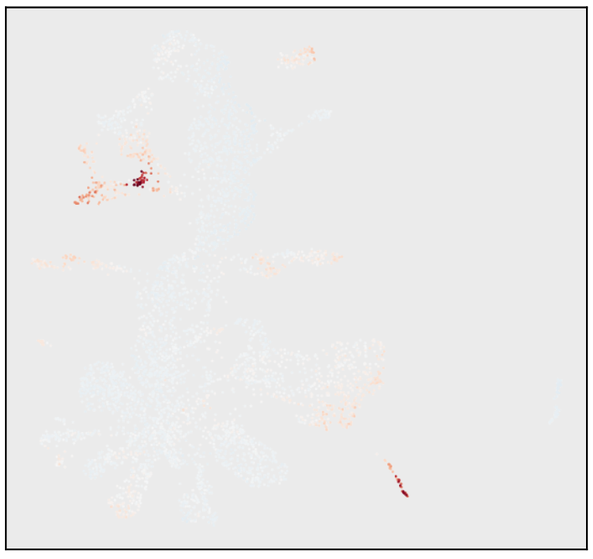
bud: 12 genes, not supported
active in 4% of cells, library level 0.09



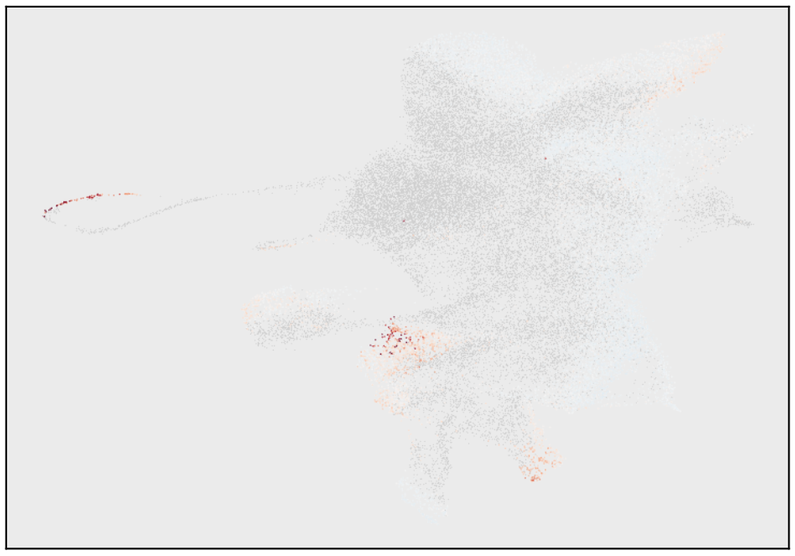
3-somite: 9 genes, not supported
active in 6% of cells, library level 0.27



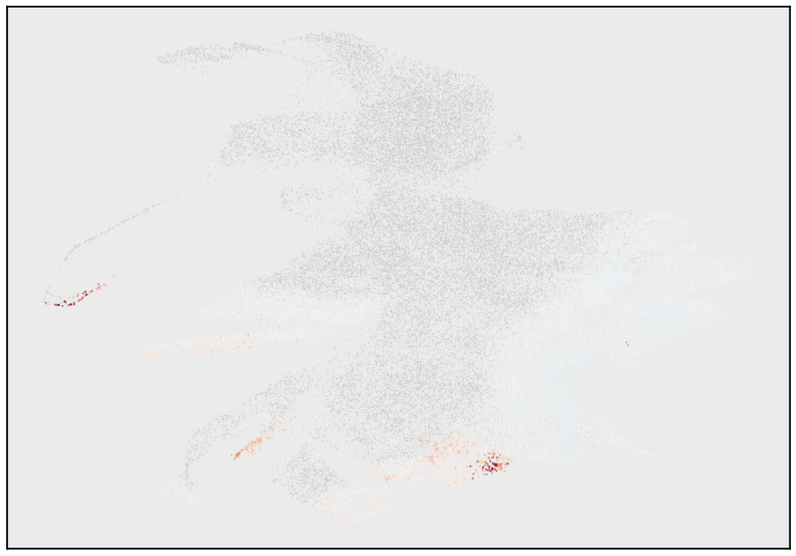
6-somite: 12 genes, supported
active in 5% of cells, library level 0.34



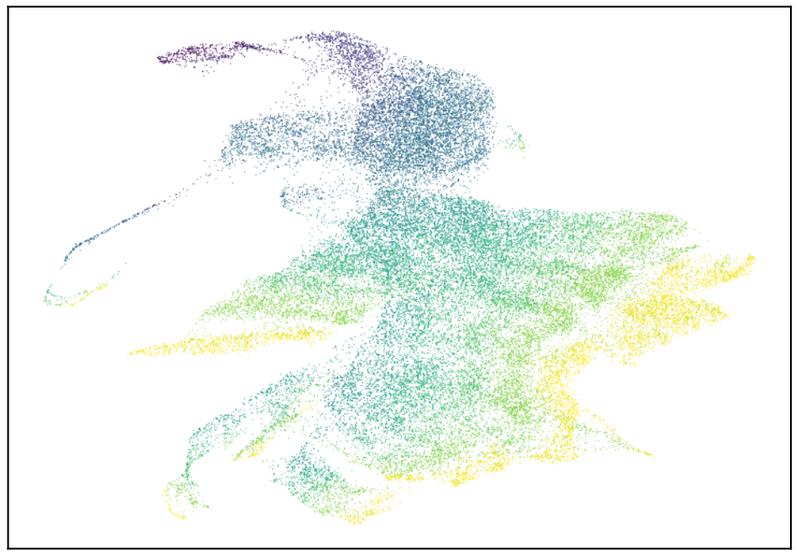
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



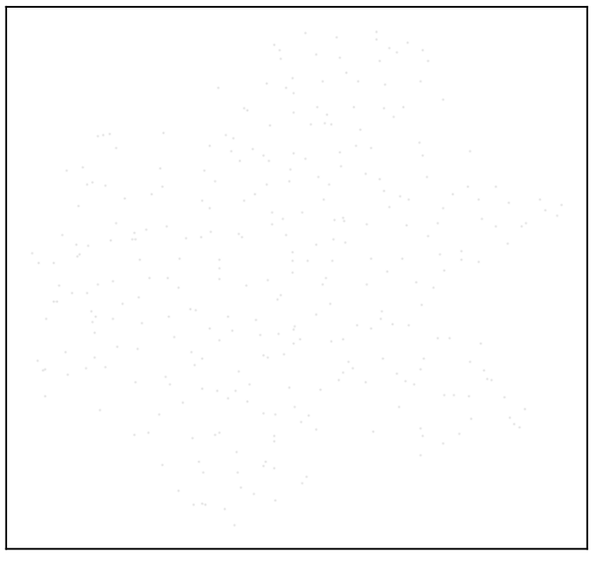
atlas, unaligned: stages



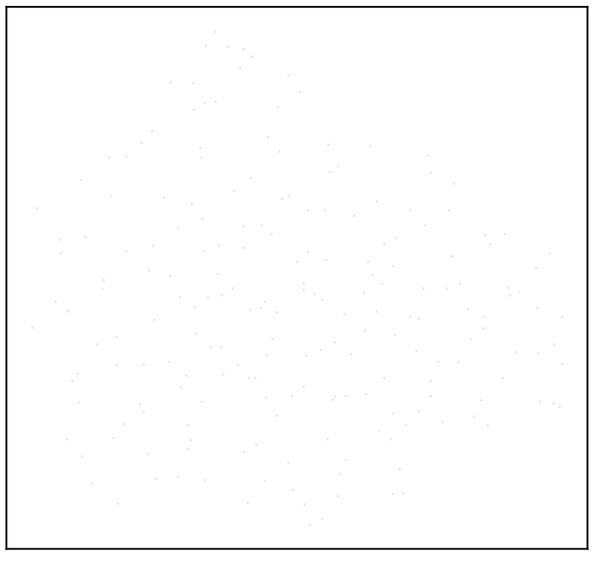
Genes (number of stages): ftr83 (3), osr1 (3), lhx1a (3), hnf1bb (3), acy3.2 (3), ptgs1 (3), emilin1a (3), calm1a (3), gadd45ab (3), cebpa (2), ndrg1a (2), ponzr1 (2)

heart

high: not defined



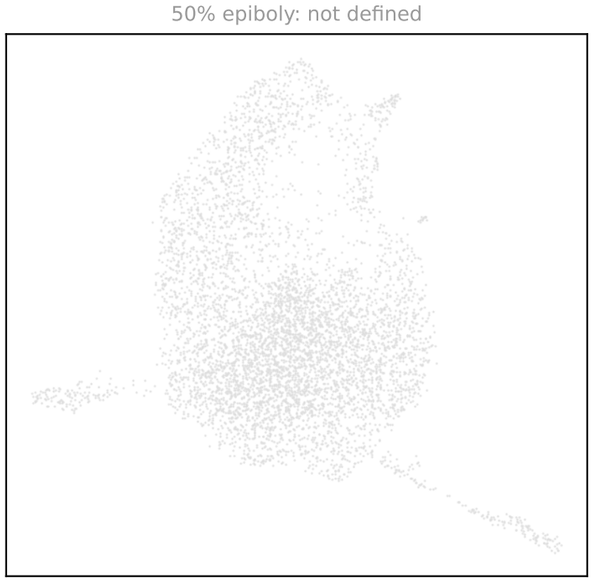
oblong: not defined



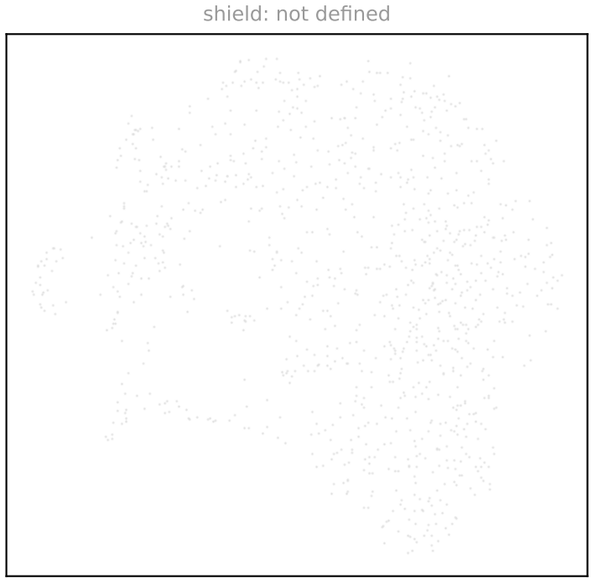
dome: not defined

30% epiboly: not defined

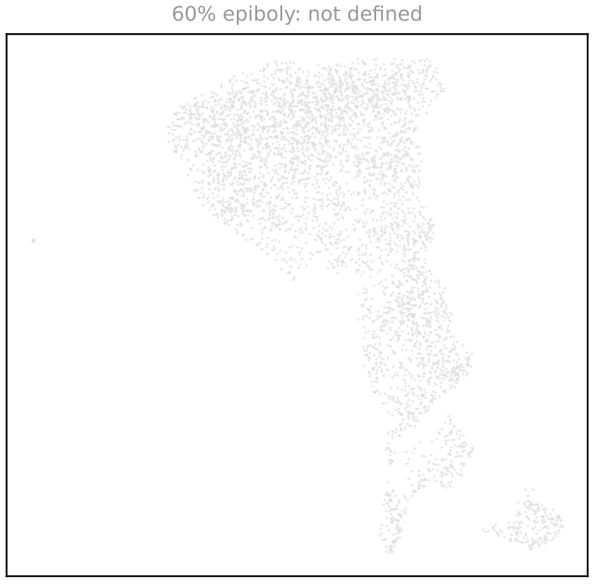
50% epiboly: not defined



shield: not defined



60% epiboly: not defined

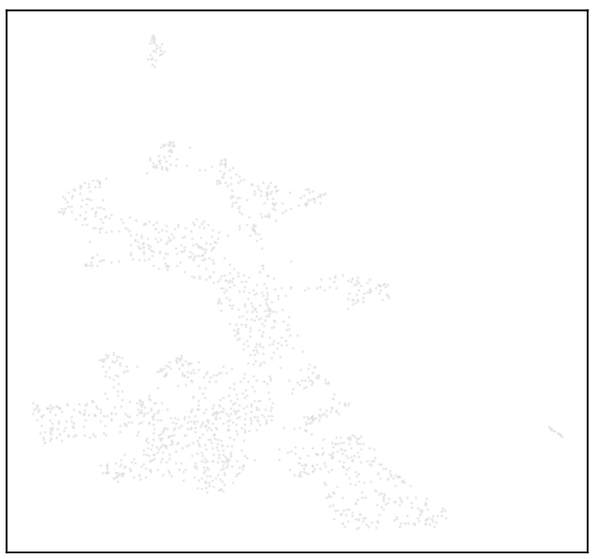


75% epiboly: not defined

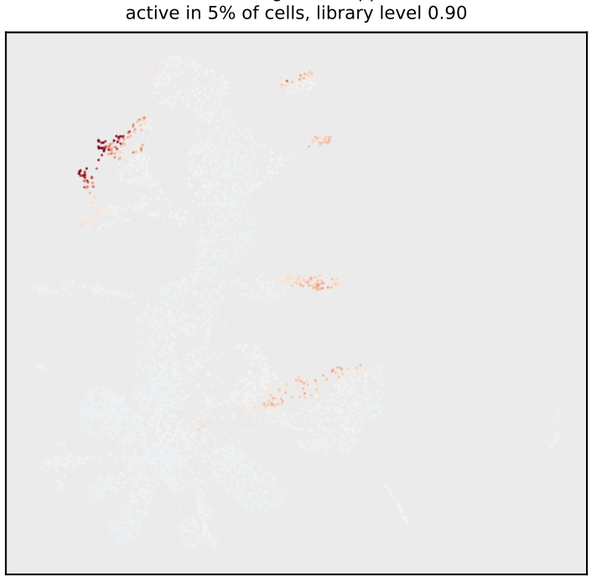
90% epiboly: not defined

bud: not defined

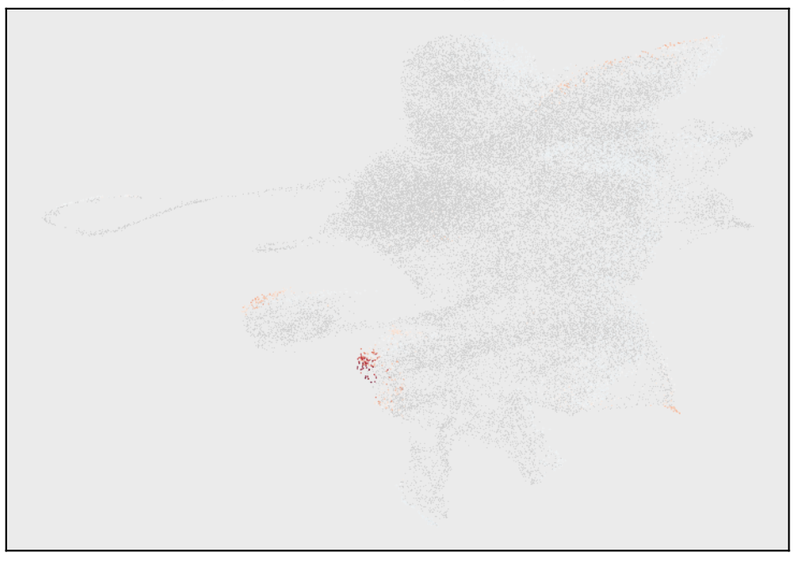
3-somite: not defined



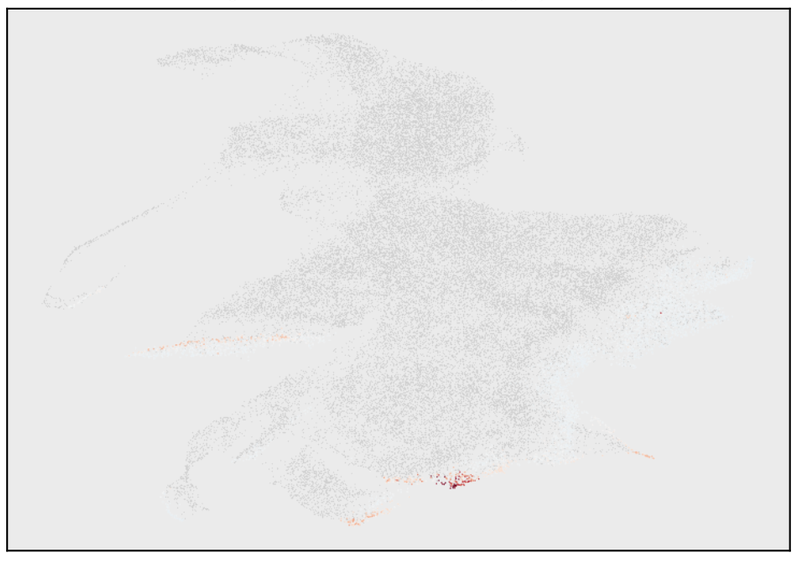
6-somite: 5 genes, supported
active in 5% of cells, library level 0.90



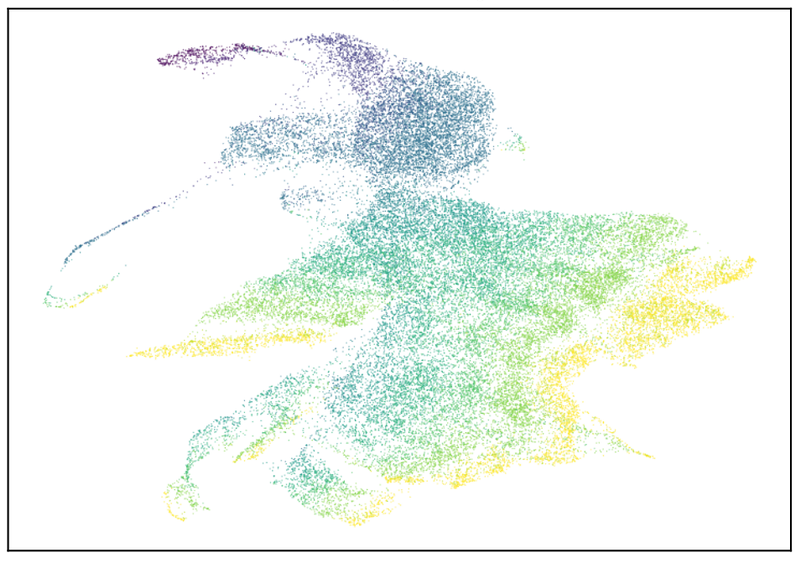
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



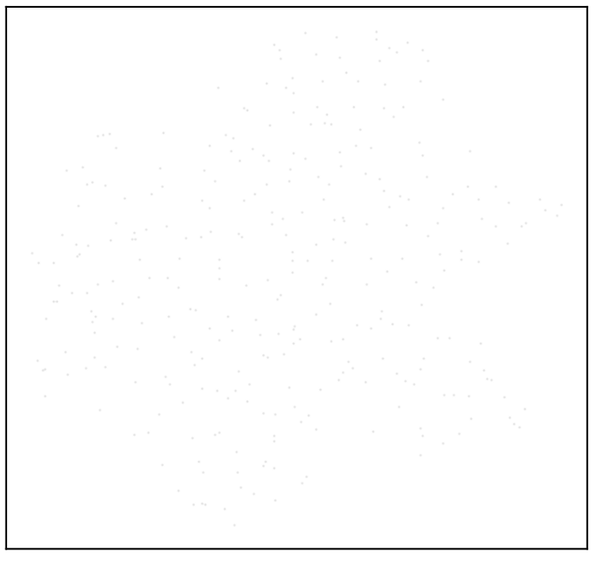
atlas, unaligned: stages



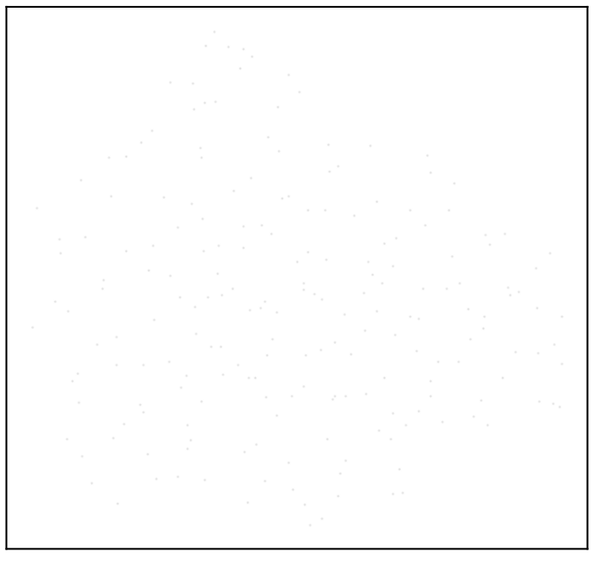
Genes (number of stages): nkx2.5 (1), tmem88a (1), nkx2.7 (1), rbfox1l (1), tbx20 (1)

forebrain_eye

high: not defined



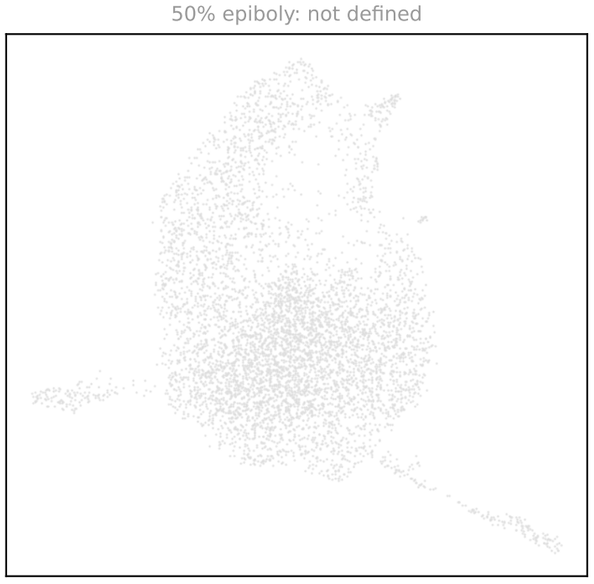
oblong: not defined



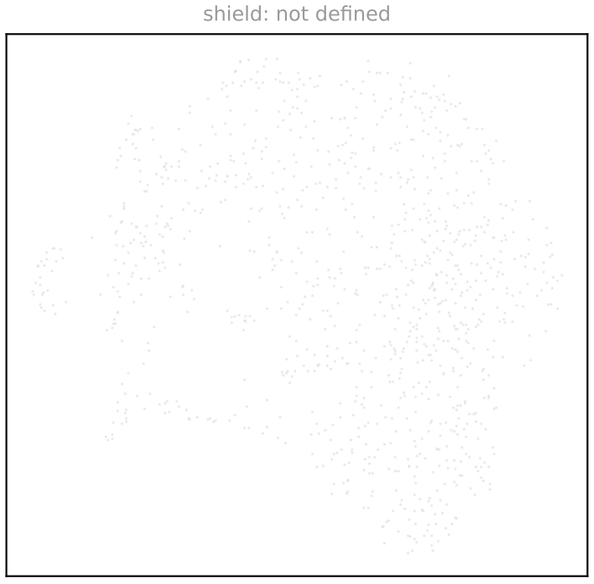
dome: not defined

30% epiboly: not defined

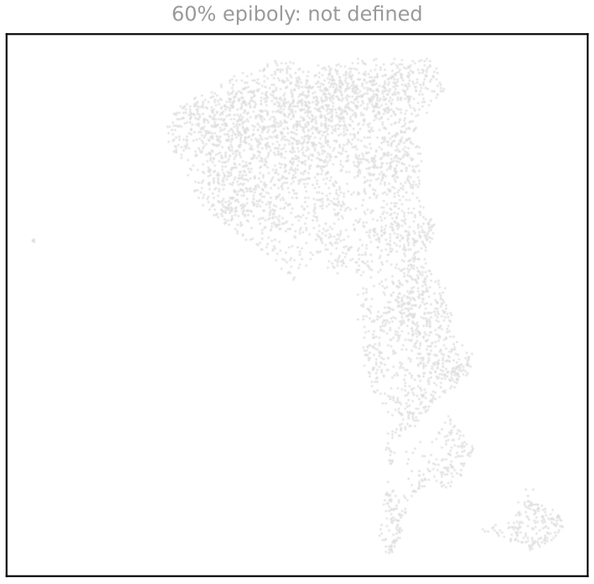
50% epiboly: not defined



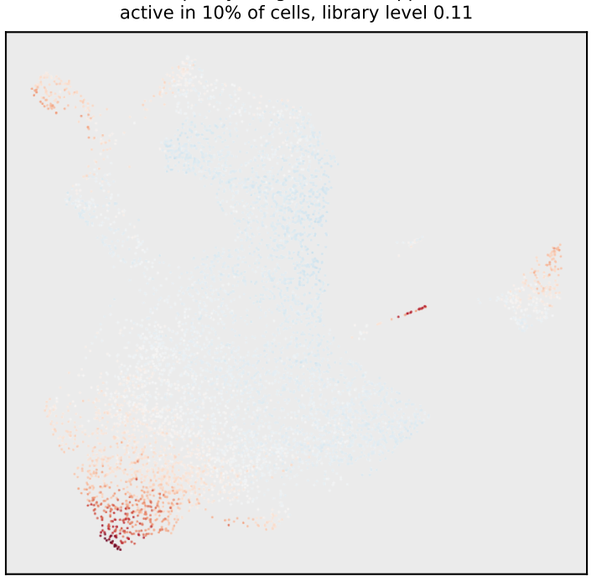
shield: not defined



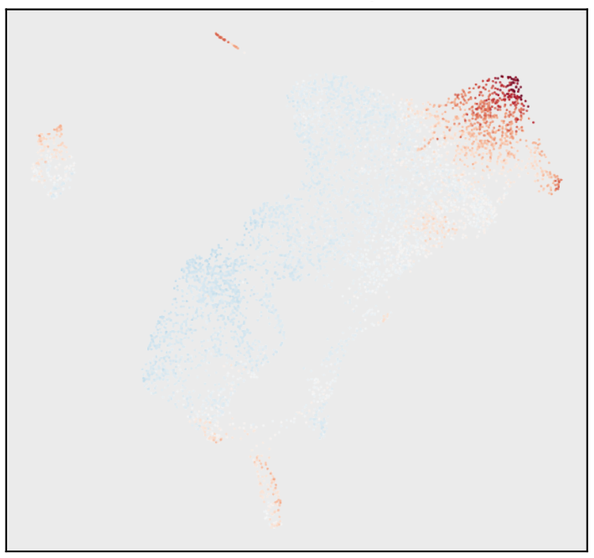
60% epiboly: not defined



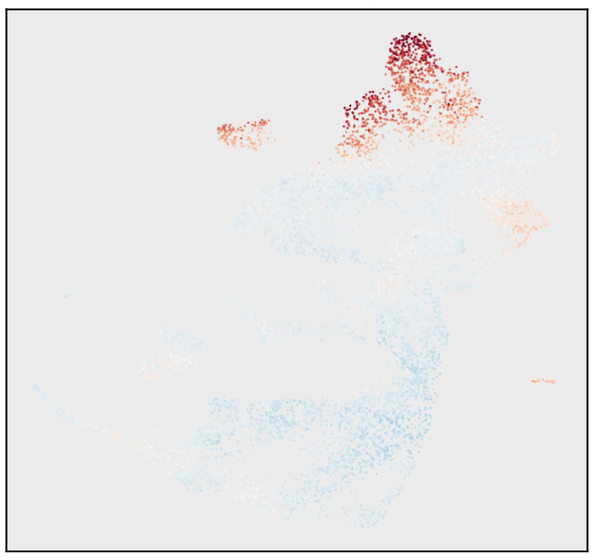
75% epiboly: 22 genes, not supported
active in 10% of cells, library level 0.11



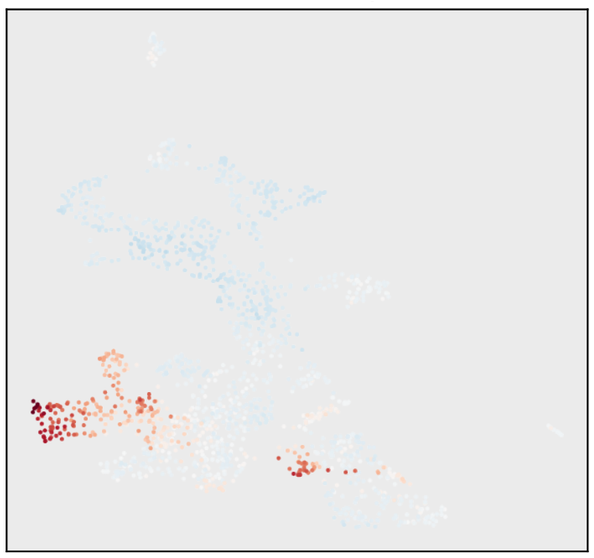
90% epiboly: 25 genes, not supported
active in 12% of cells, library level 0.25



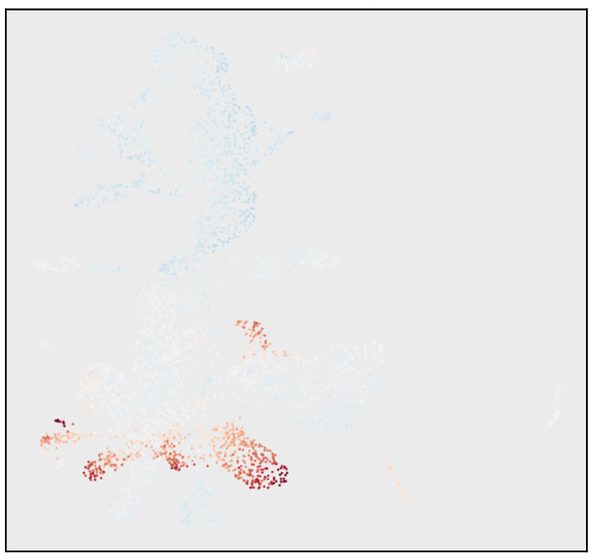
bud: 29 genes, supported
active in 15% of cells, library level 0.38



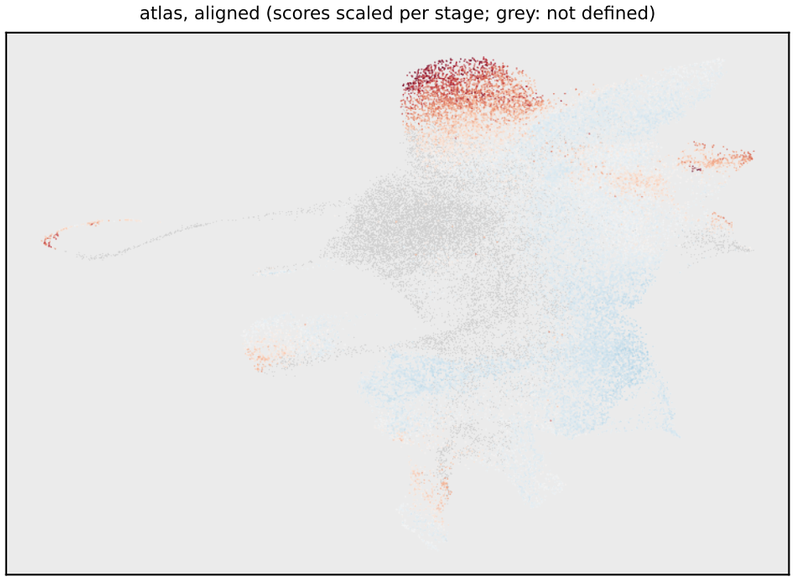
3-somite: 30 genes, supported
active in 13% of cells, library level 0.61



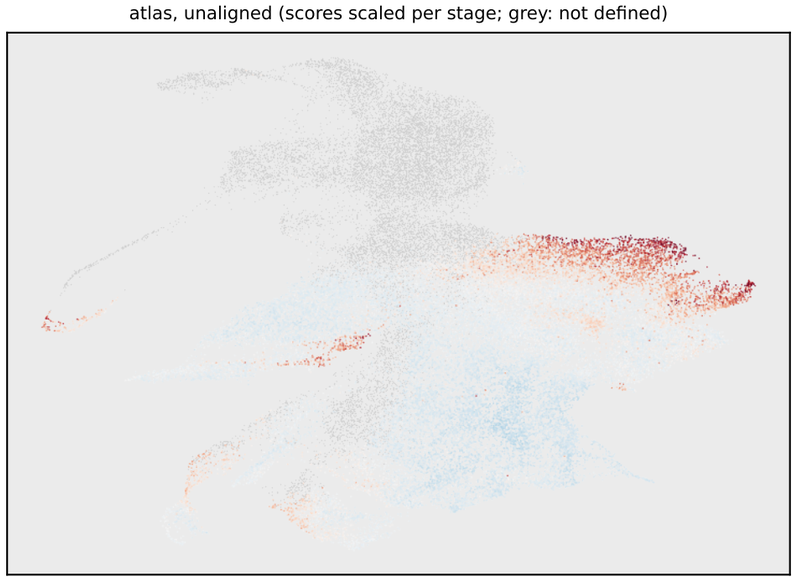
6-somite: 30 genes, supported
active in 15% of cells, library level 0.75



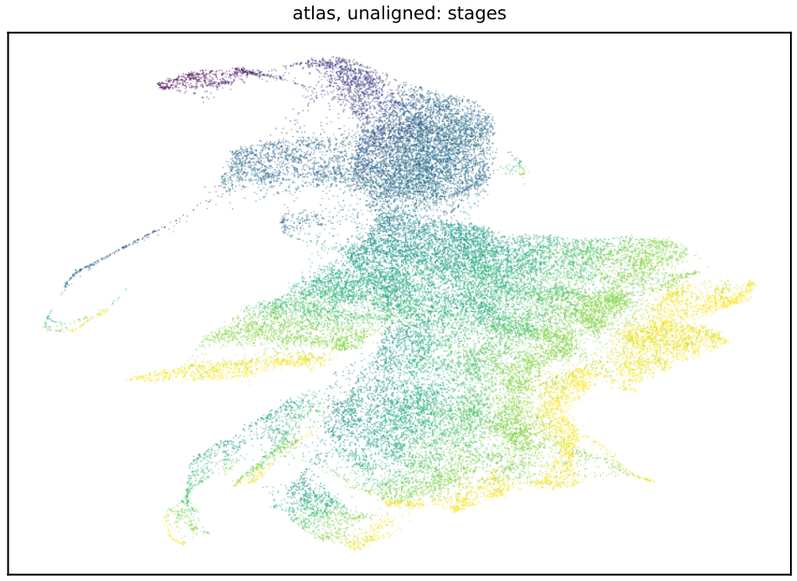
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



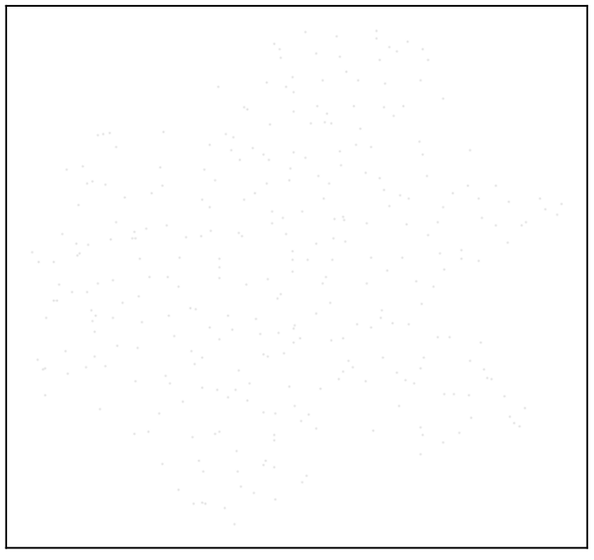
atlas, unaligned: stages



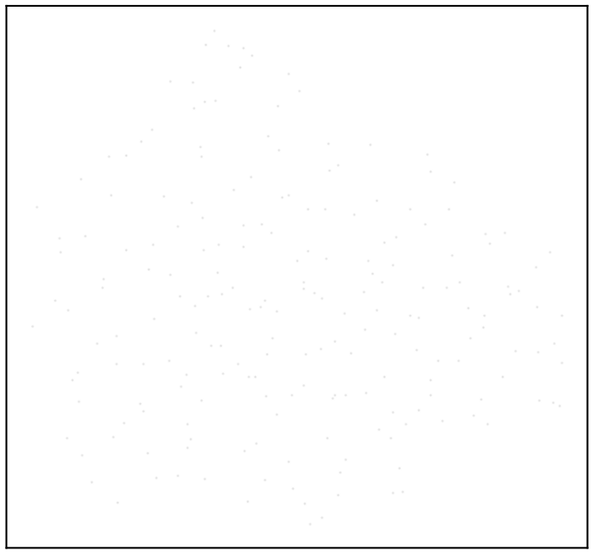
Genes (number of stages): rx3 (5), si:ch211-153b23.5 (5), six3b (5), hesx1 (5), pax6b (5), stm (5), nkx2.4b (5), olig3 (5), dmbx1a (5), sox4a (5), pitx3 (5), pax6a (5), swap70b (5), barhl2 (5), stc2a (5), alcamb (5), ascl1a (5), dbx1b (5), sox13 (5), efnb2a (5), nat8l (5), rspo3 (5), emx3 (4), pdgfbb (4), arx (4), otpb (3), foxg1a (3), mab21l2 (3), insm1a (3), rx2 (2)

midbrain_hindbrain_boundary

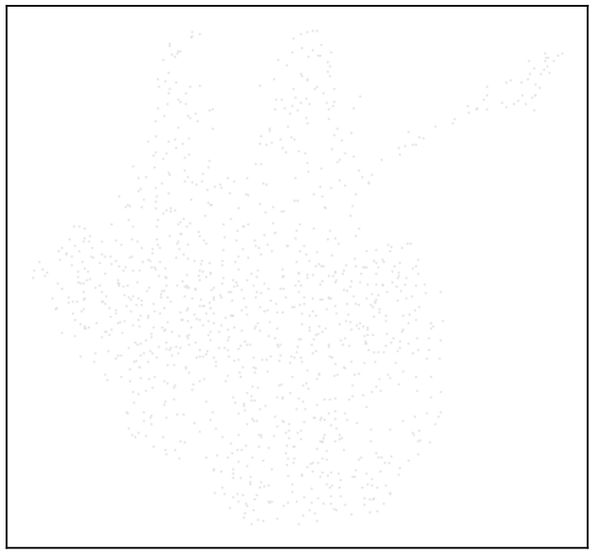
high: not defined



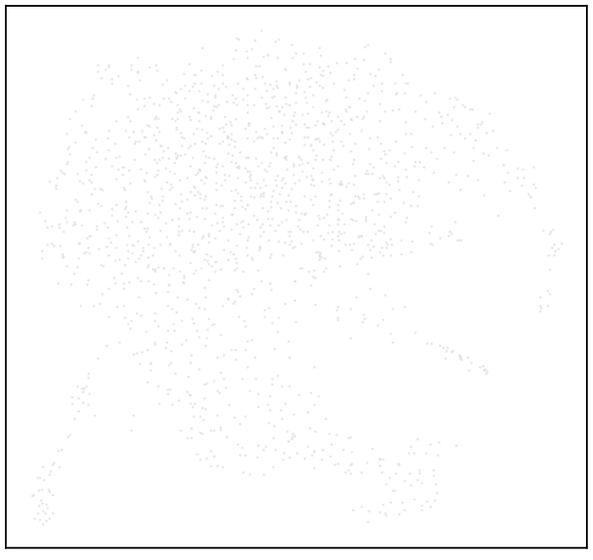
oblong: not defined



dome: not defined

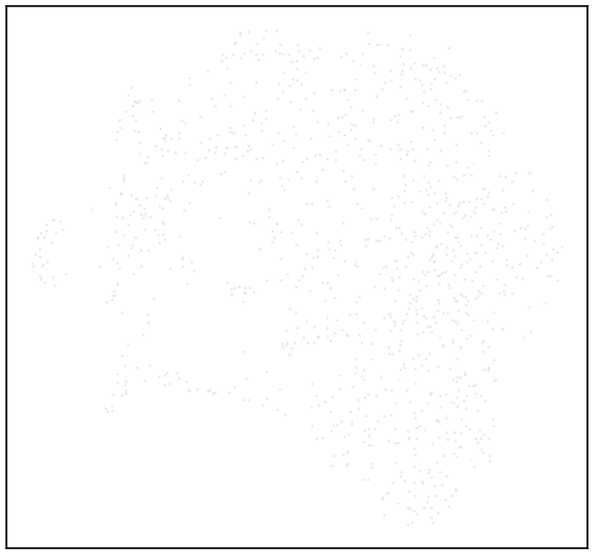


30% epiboly: not defined



50% epiboly: not defined

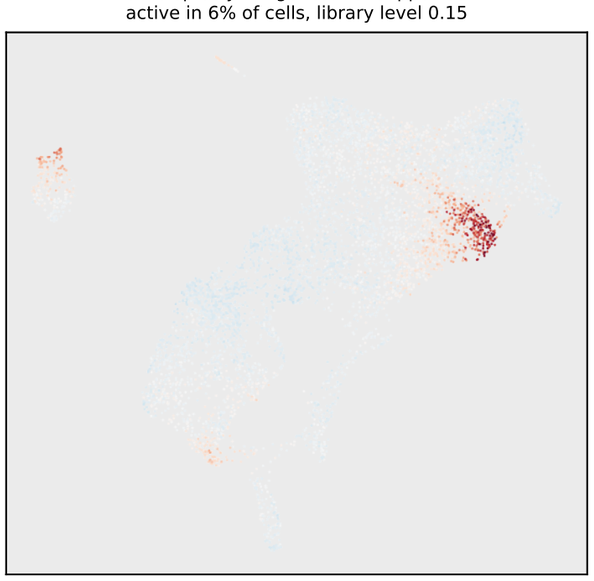
shield: not defined



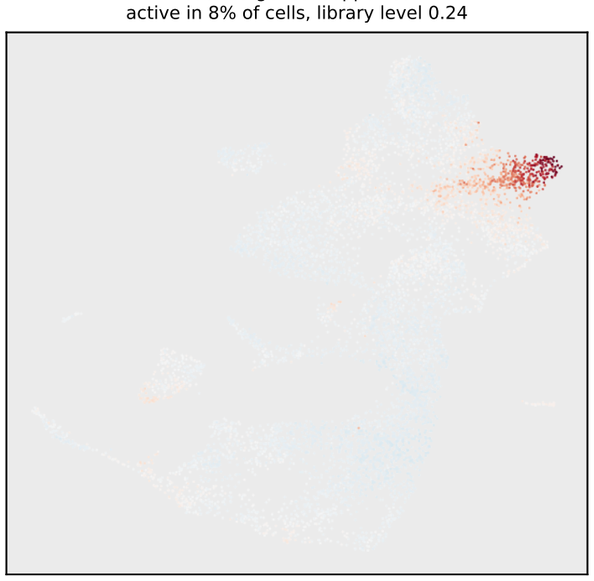
60% epiboly: not defined

75% epiboly: not defined

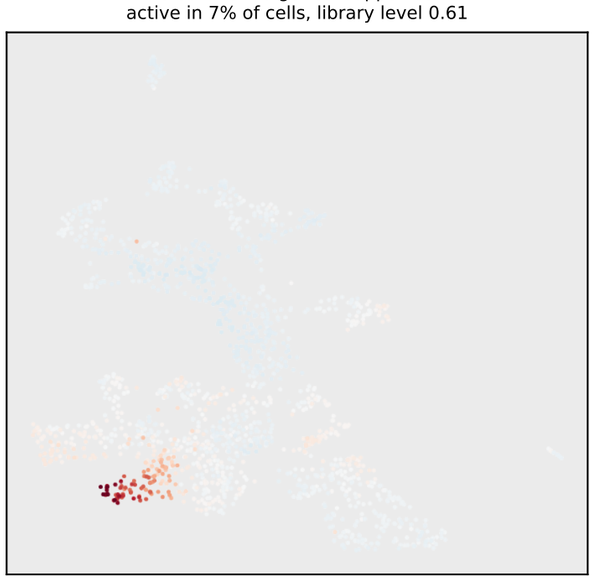
90% epiboly: 21 genes, not supported
active in 6% of cells, library level 0.15



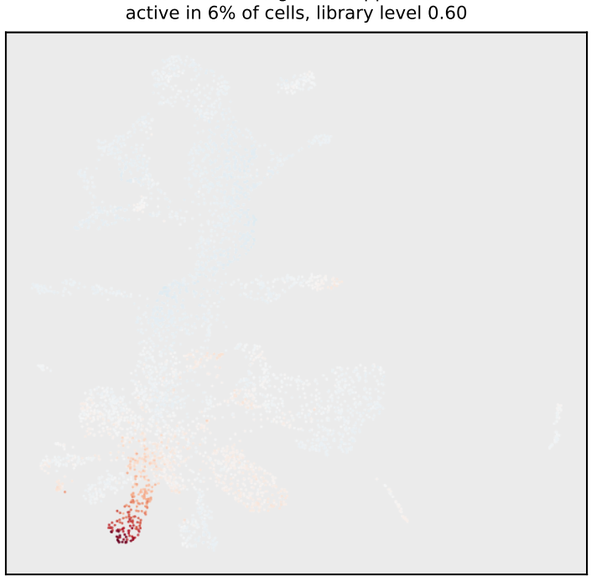
bud: 25 genes, supported
active in 8% of cells, library level 0.24



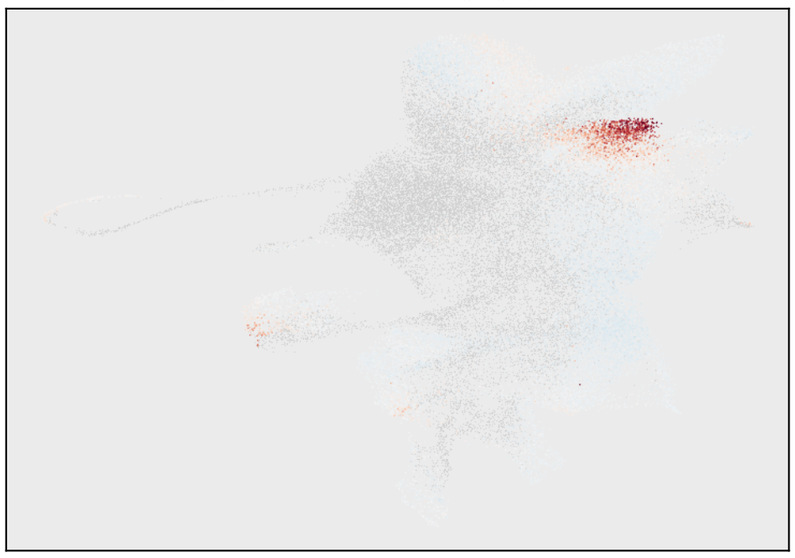
3-somite: 26 genes, supported
active in 7% of cells, library level 0.61



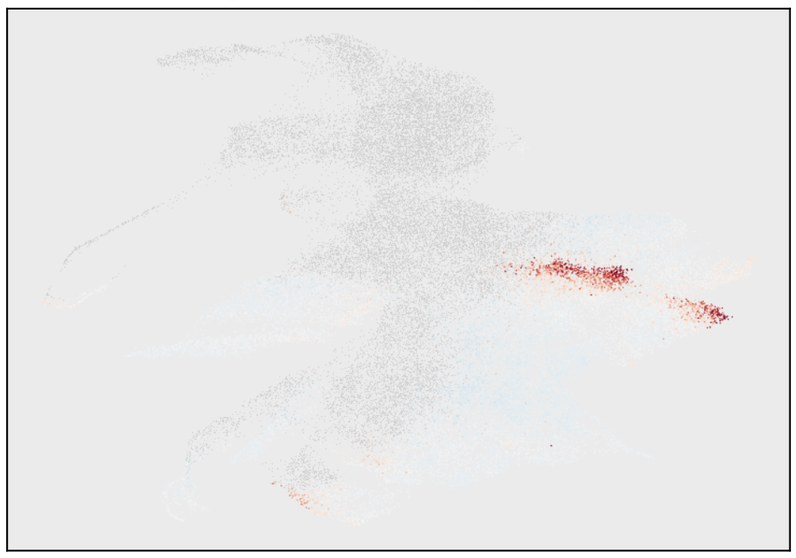
6-somite: 26 genes, supported
active in 6% of cells, library level 0.60



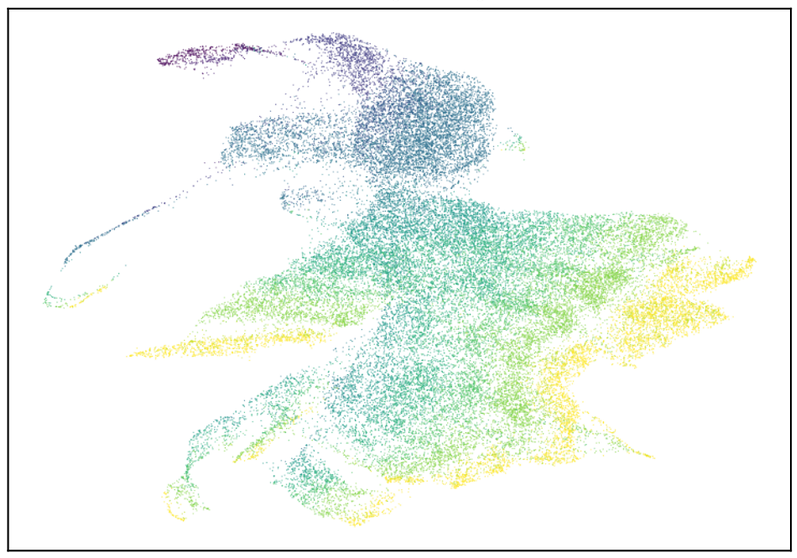
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



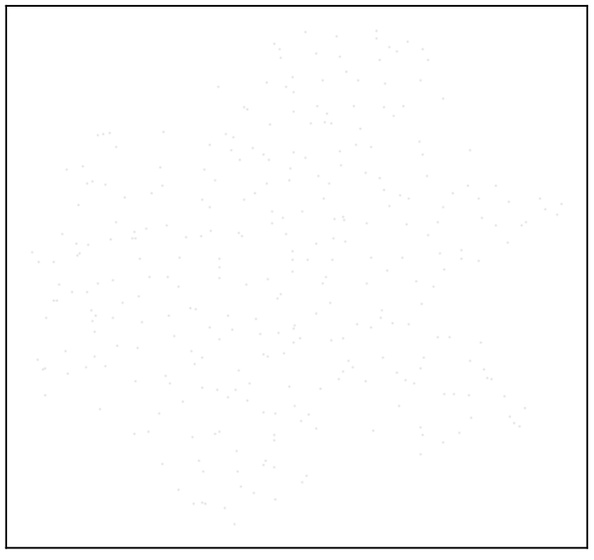
atlas, unaligned: stages



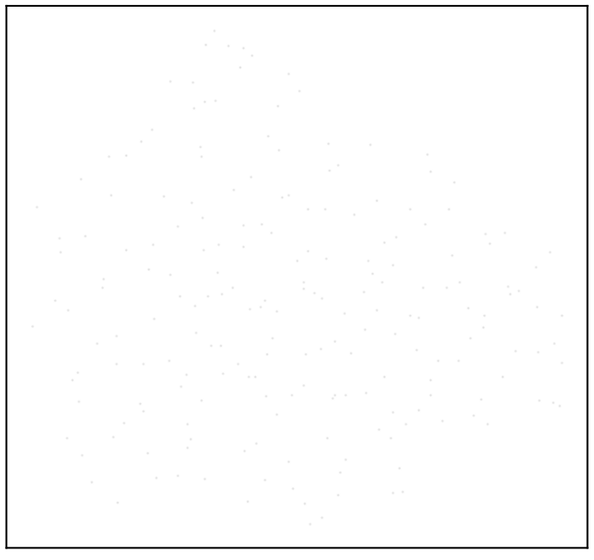
Genes (number of stages): her5 (4), eng2a (4), sox21a (4), pax2a (4), dmbx1a (4), her11 (4), cnpy1 (4), phox2a (4), prdm12b (4), wnt8b (4), emilin1a (4), lmx1bb (4), tubb5 (4), stmn4 (4), uacab (4), bmpr1ba (4), calm1a (4), ier5l (4), pou3f2b (4), wnk1b (4), zfhx4 (4), eng2b (3), mab21l2 (3), wnt7ba (3), gfap (3), bhlhe22 (2)

hindbrain

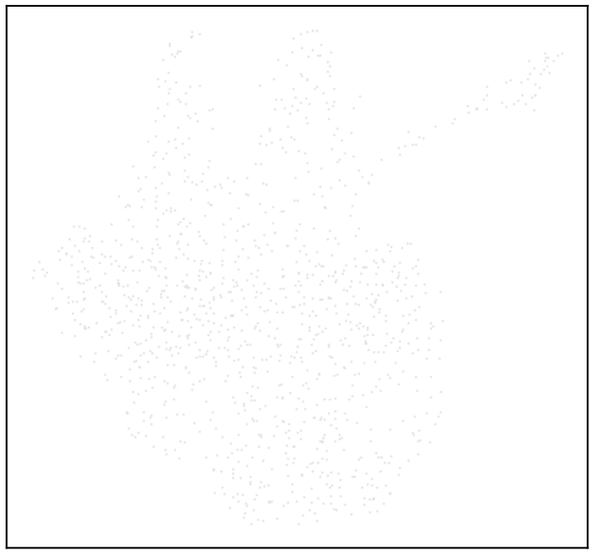
high: not defined



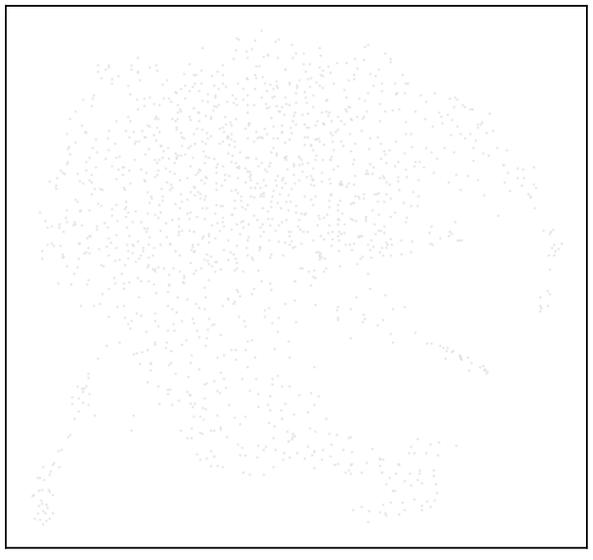
oblong: not defined



dome: not defined

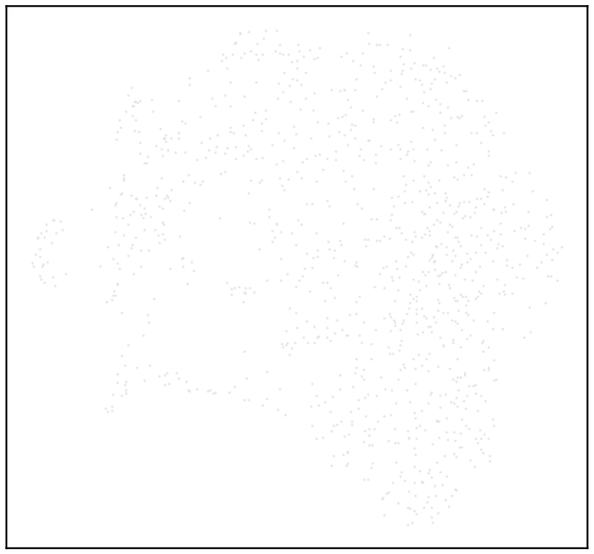


30% epiboly: not defined



50% epiboly: not defined

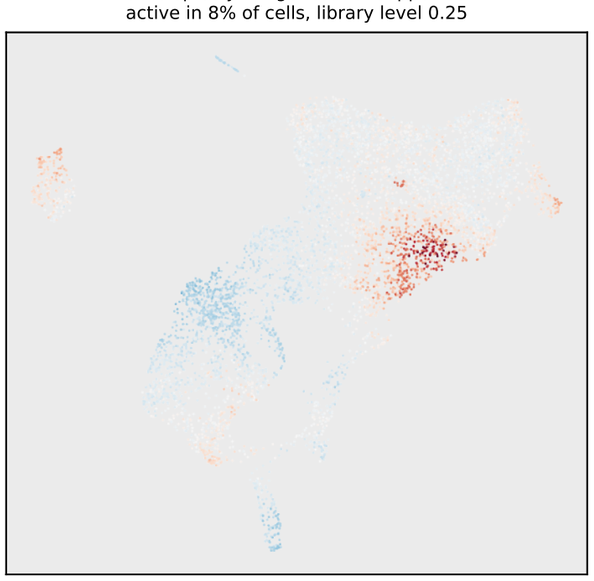
shield: not defined



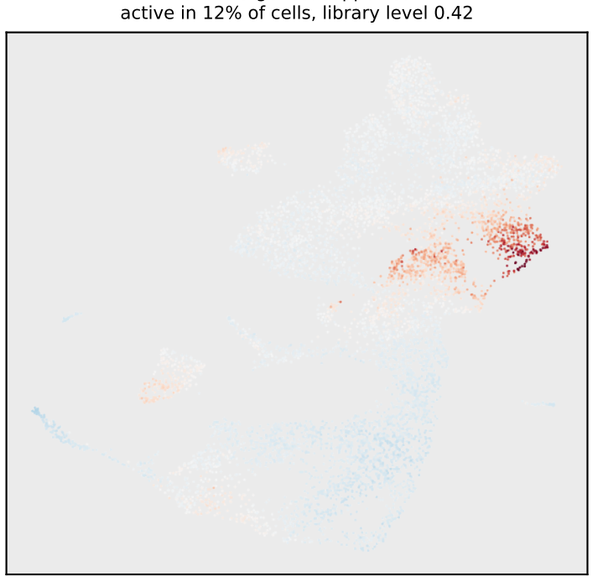
60% epiboly: not defined

75% epiboly: not defined

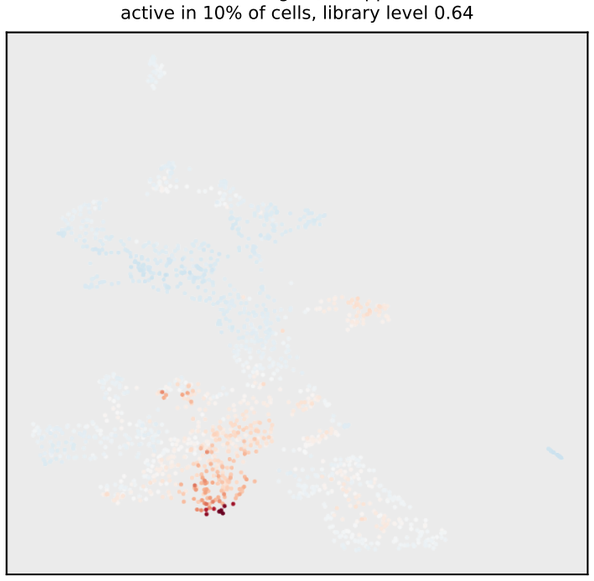
90% epiboly: 27 genes, not supported
active in 8% of cells, library level 0.25



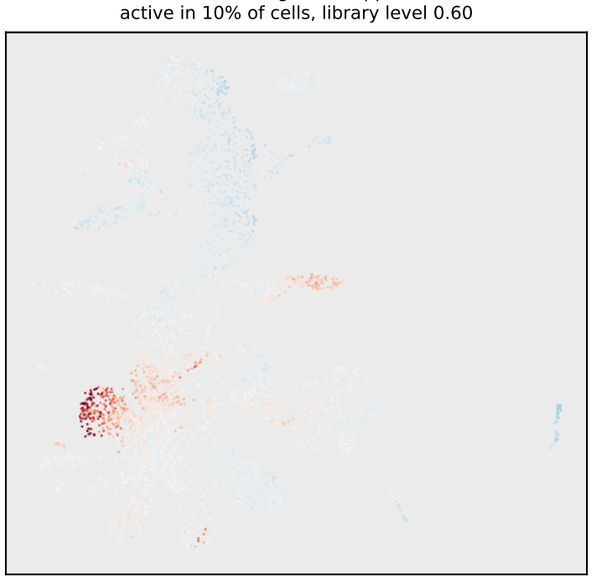
bud: 29 genes, supported
active in 12% of cells, library level 0.42



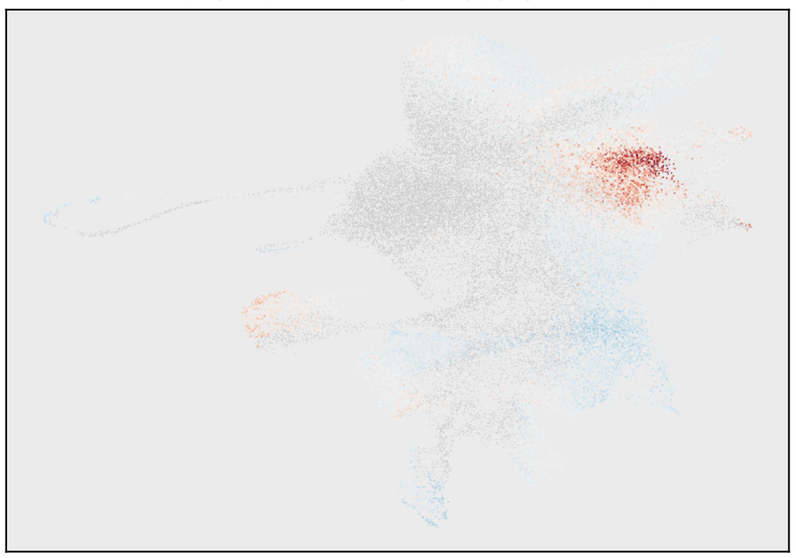
3-somite: 29 genes, supported
active in 10% of cells, library level 0.64



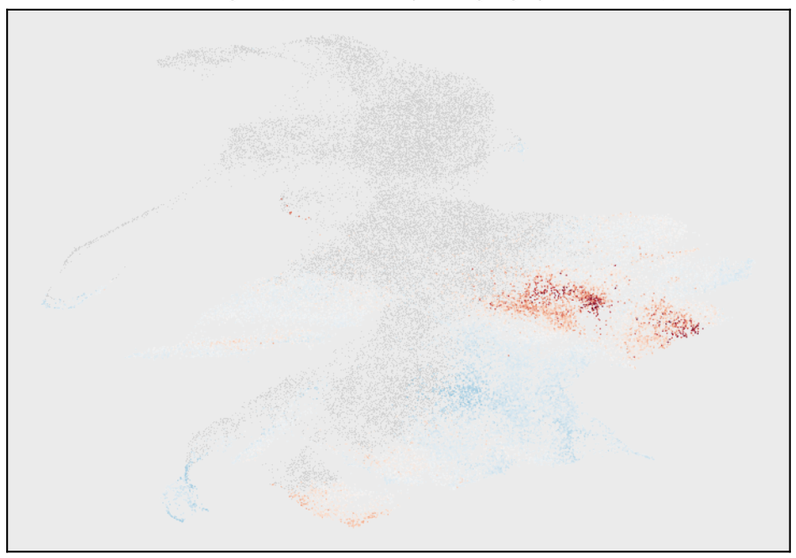
6-somite: 30 genes, supported
active in 10% of cells, library level 0.60



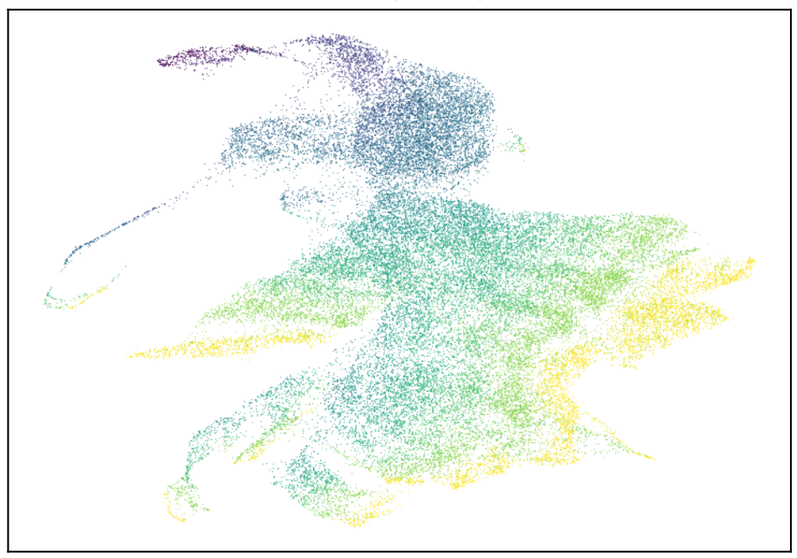
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



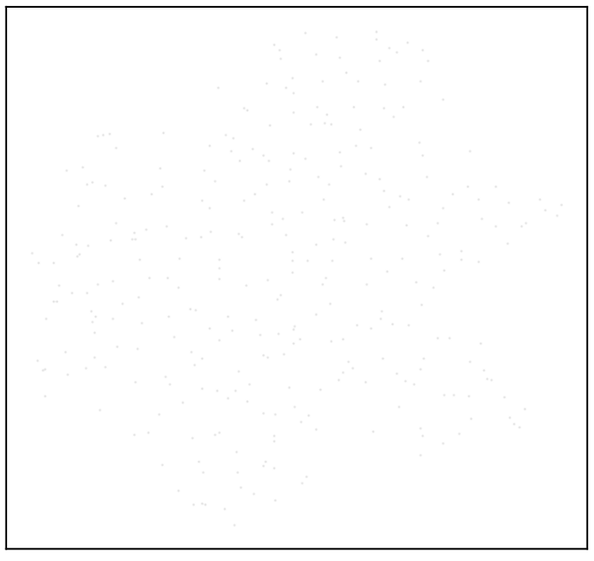
atlas, unaligned: stages



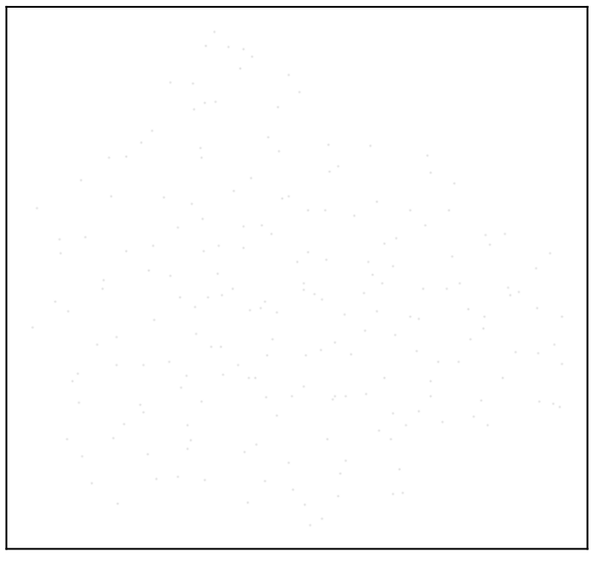
Genes (number of stages): cyp26c1 (4), olig3 (4), fgf3 (4), si:ch211-152c2.3 (4), mafba (4), cyp26b1 (4), sox21a (4), phlda1 (4), dusp2 (4), gsx1 (4), phox2a (4), meis3 (4), hoxb1a (4), gbx1 (4), irx5a (4), dbx1b (4), sox11b (4), hoxd4a (4), hoxb3a (4), sox13 (4), hnf1ba (4), efnb2a (4), hoxa3a (4), btg1 (4), pcdh19 (4), epha7 (4), gpd1b (4), egr2b (3), otpb (3), atoh1a (1)

floor_plate

high: not defined



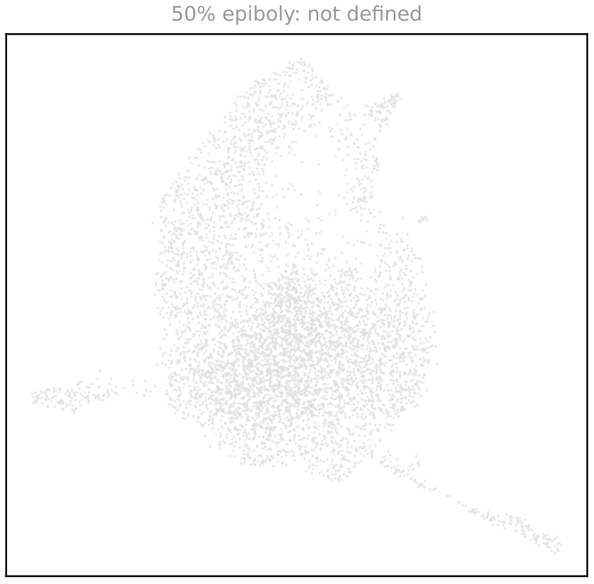
oblong: not defined



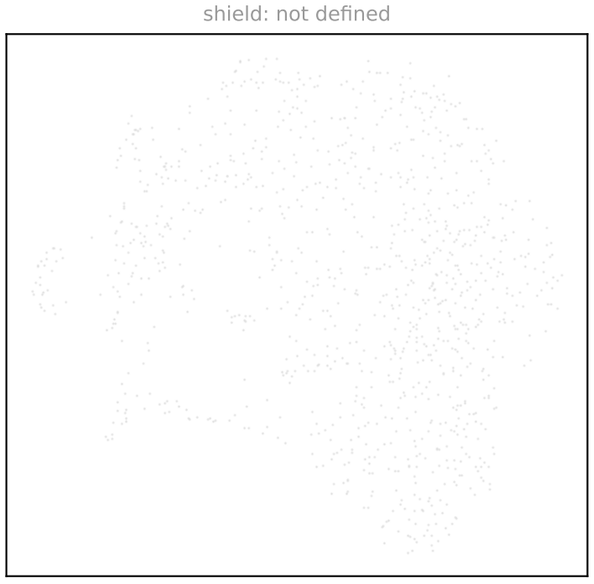
dome: not defined

30% epiboly: not defined

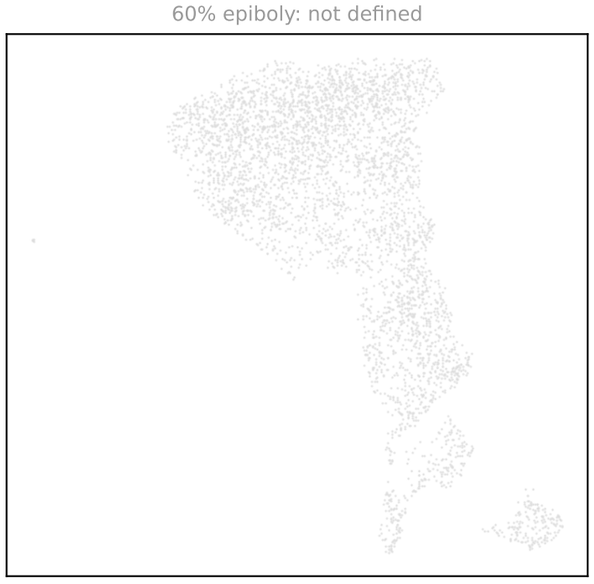
50% epiboly: not defined



shield: not defined



60% epiboly: not defined

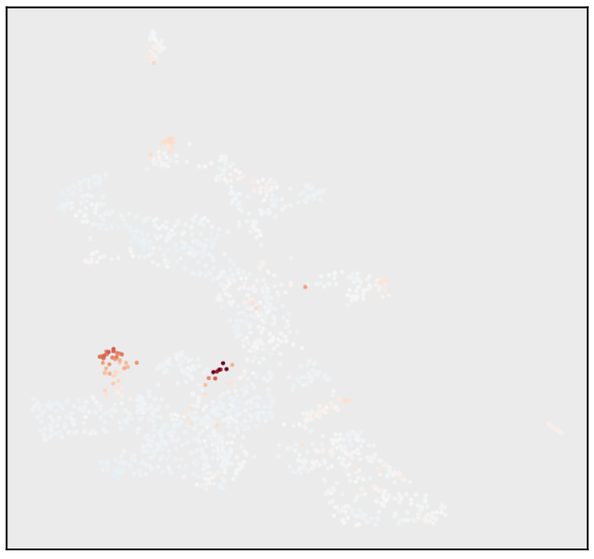


75% epiboly: not defined

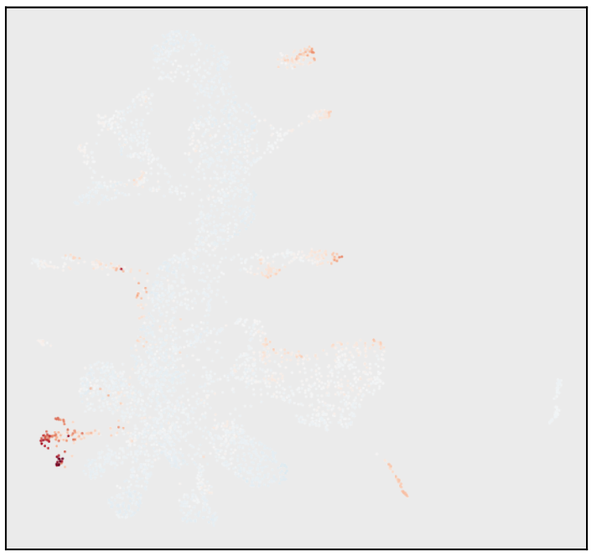
90% epiboly: not defined

bud: not defined

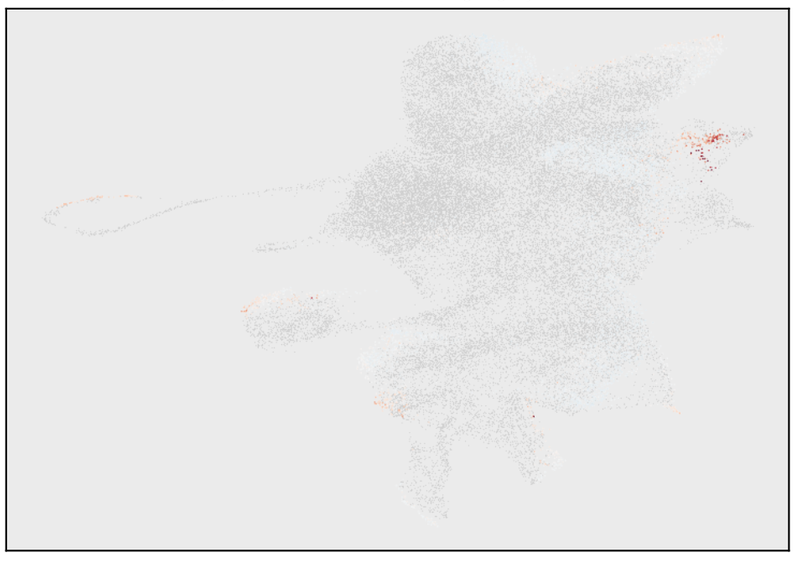
3-somite: 8 genes, supported
active in 3% of cells, library level 0.51



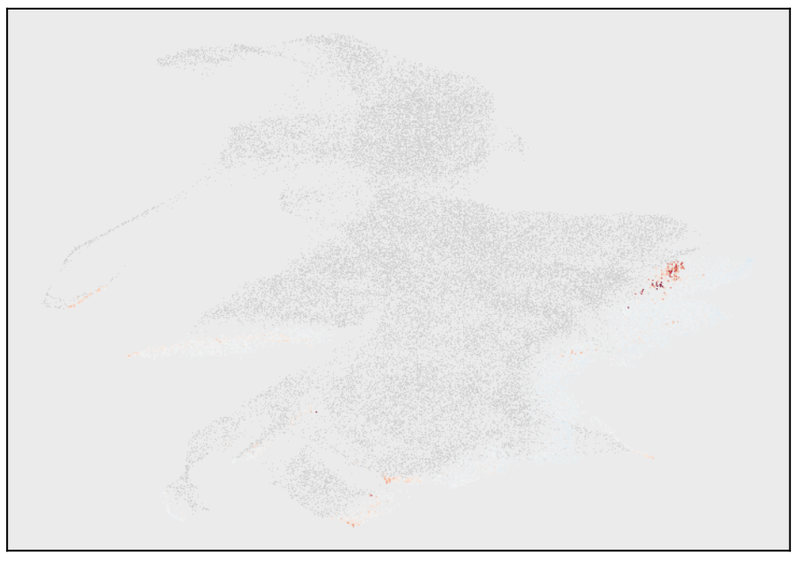
6-somite: 9 genes, supported
active in 4% of cells, library level 0.62



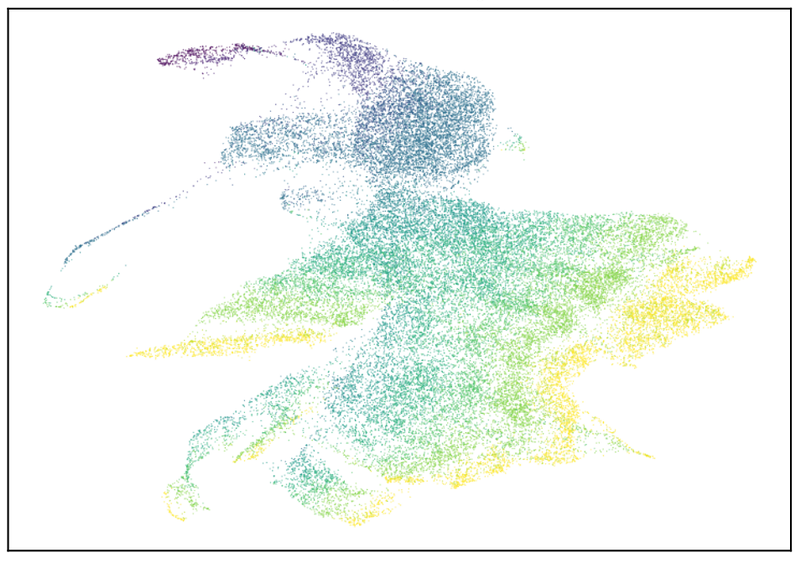
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



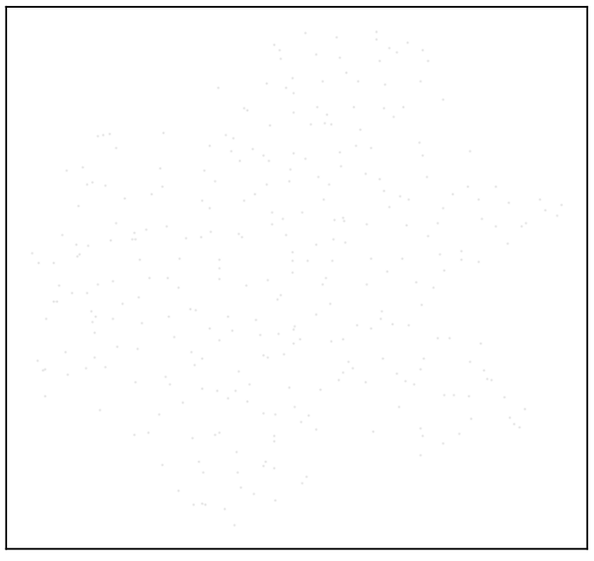
atlas, unaligned: stages



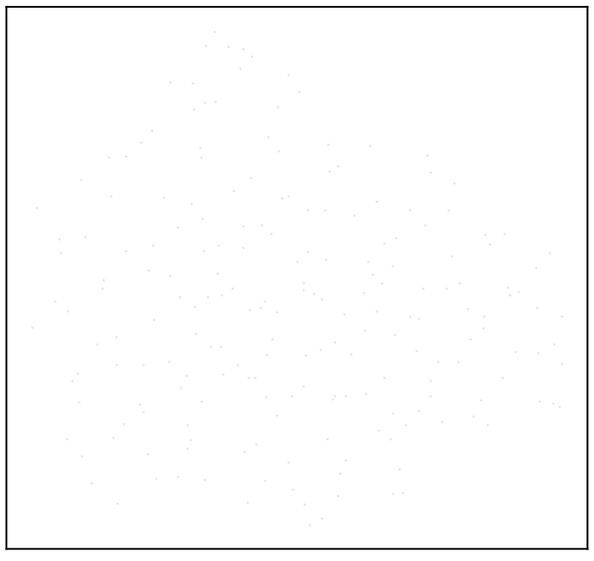
Genes (number of stages): nkx2.2b (2), shhb (2), ctgfa (2), igf2b (2), lft2 (2), pih1d3 (2), ribc2 (2), sepw1 (2), nme5 (1)

primary_neurons

high: not defined



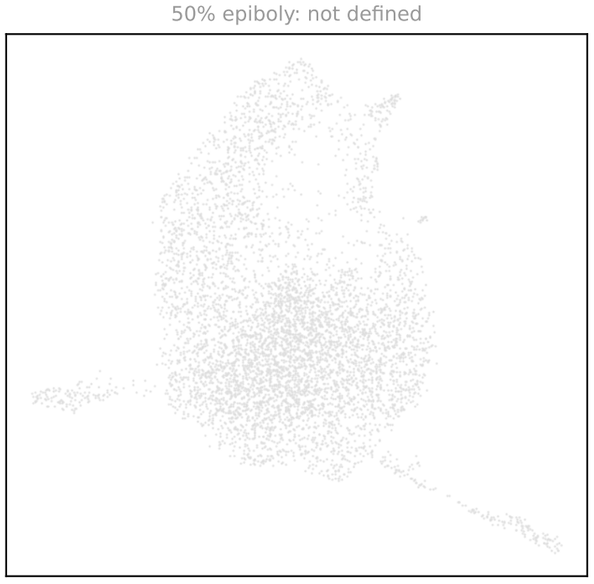
oblong: not defined



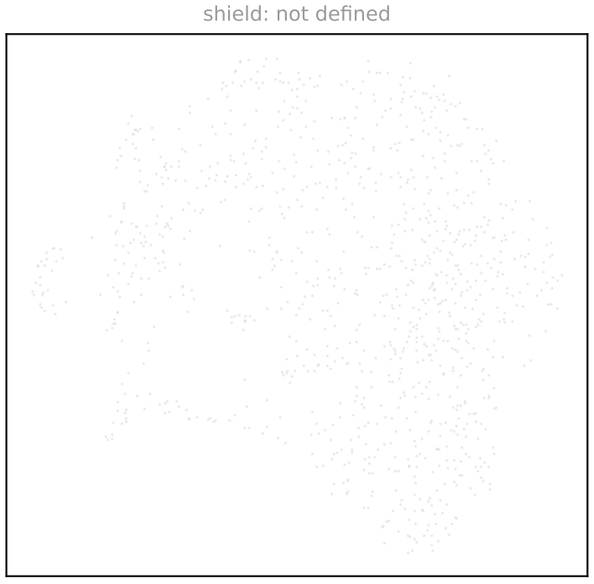
dome: not defined

30% epiboly: not defined

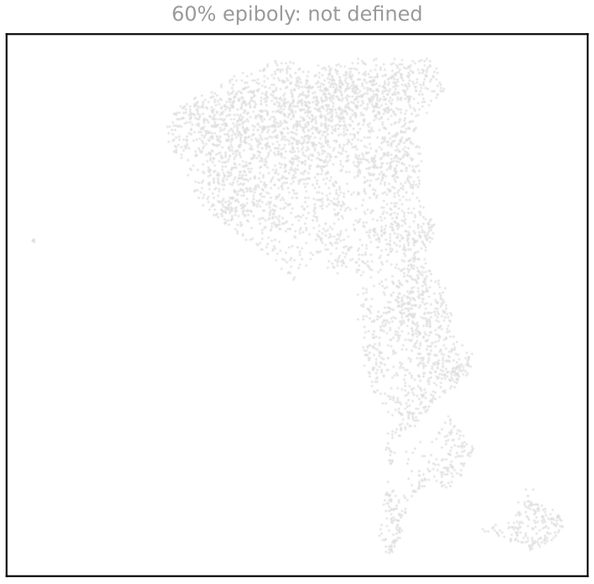
50% epiboly: not defined



shield: not defined



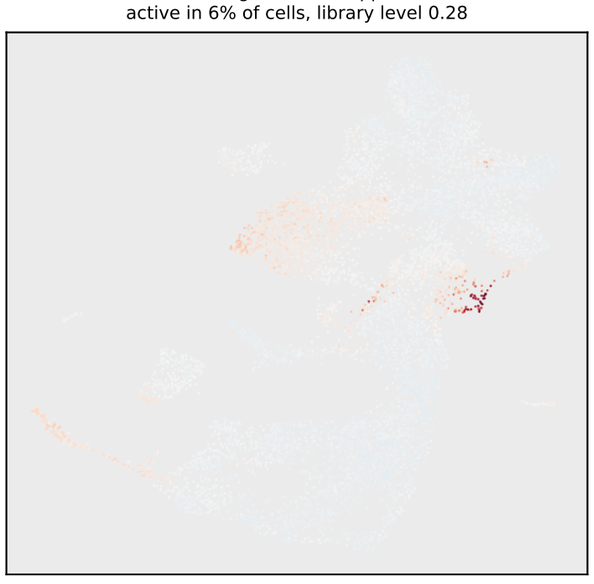
60% epiboly: not defined



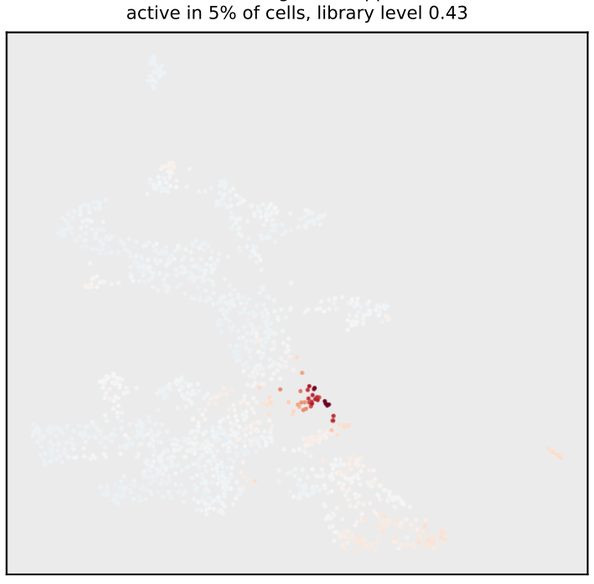
75% epiboly: not defined

90% epiboly: not defined

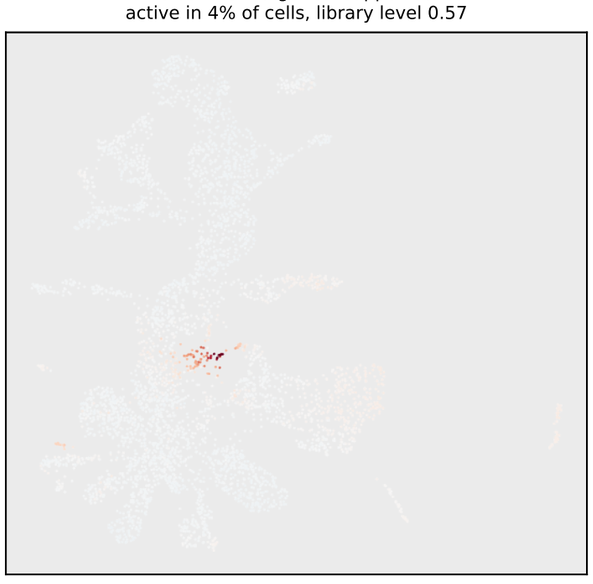
bud: 19 genes, not supported
active in 6% of cells, library level 0.28



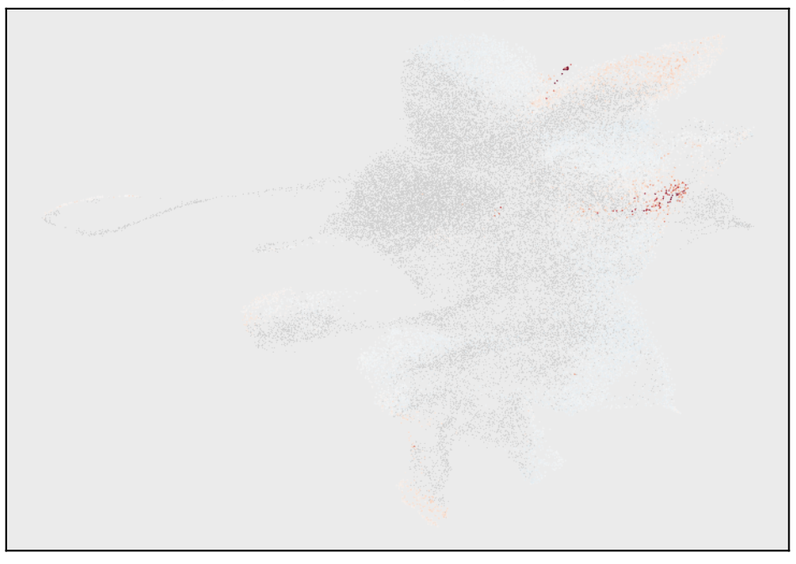
3-somite: 18 genes, supported
active in 5% of cells, library level 0.43



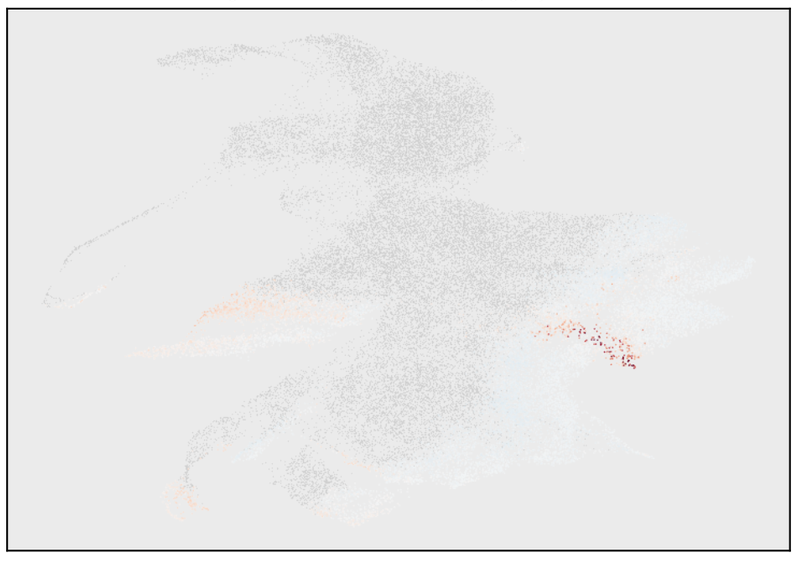
6-somite: 19 genes, supported
active in 4% of cells, library level 0.57



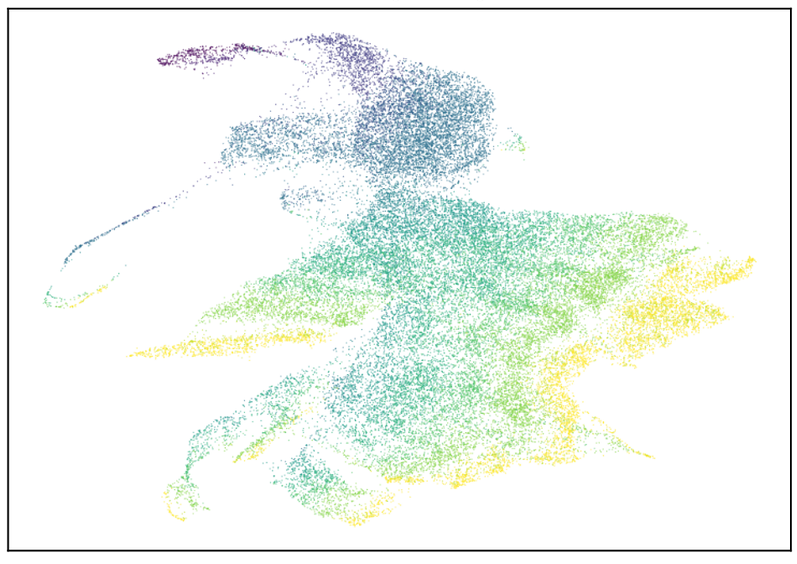
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



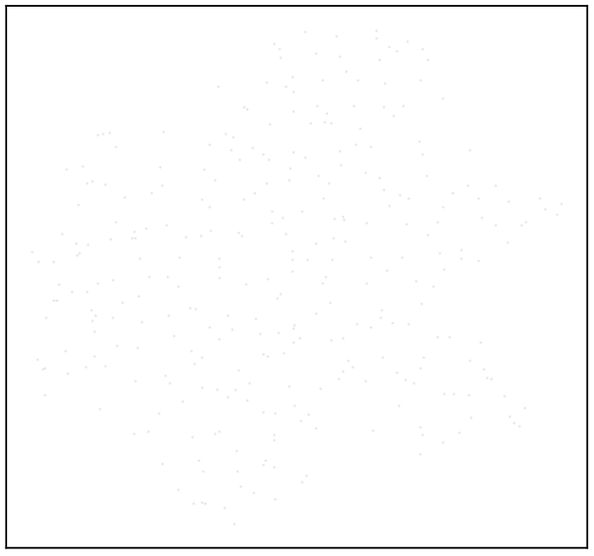
atlas, unaligned: stages



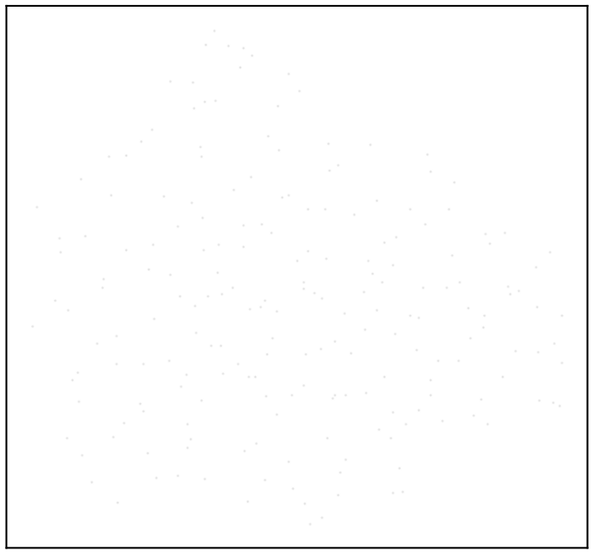
Genes (number of stages): dlb (3), neurod4 (3), gata3 (3), pnx (3), isl1 (3), elavl3 (3), pnocb (3), vim (3), tlx2 (3), gadd45gb.1 (3), insm1a (3), scrt2 (3), nfasca (3), tlx3b (3), chodl (3), btbd6b (3), cbfb (3), hacd3 (3), prdm14 (2)

neural_crest

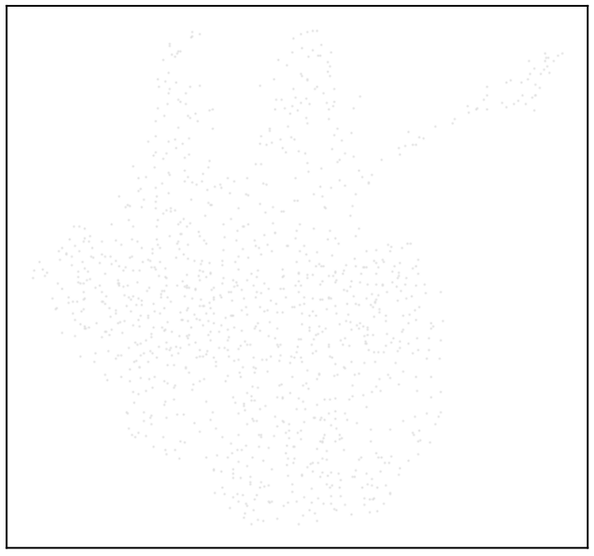
high: not defined



oblong: not defined



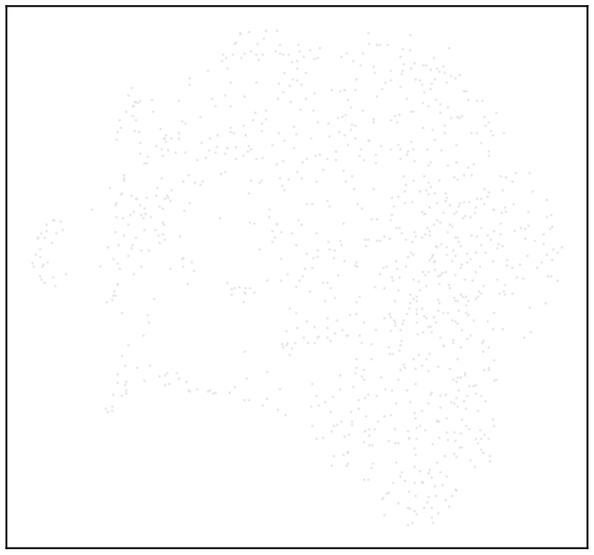
dome: not defined



30% epiboly: not defined

50% epiboly: not defined

shield: not defined



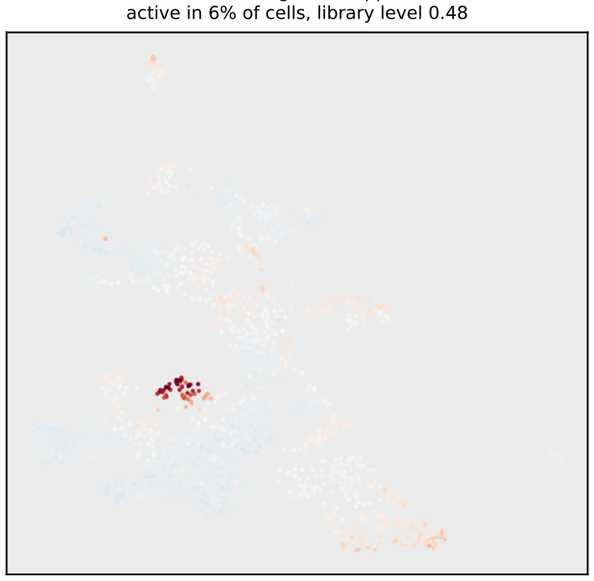
60% epiboly: not defined

75% epiboly: not defined

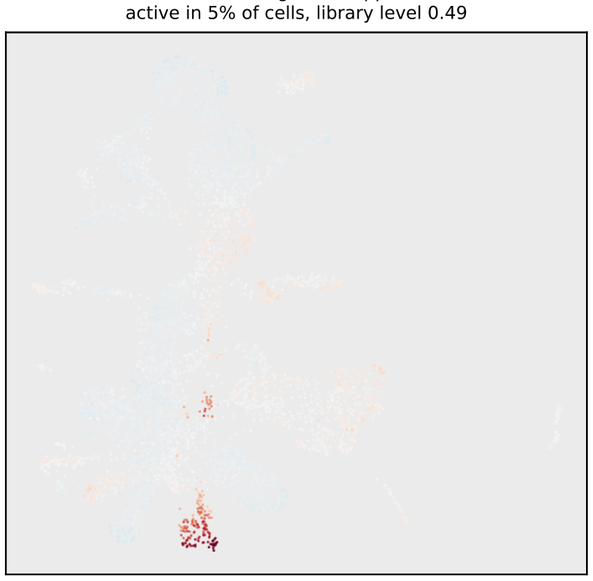
90% epiboly: not defined

bud: not defined

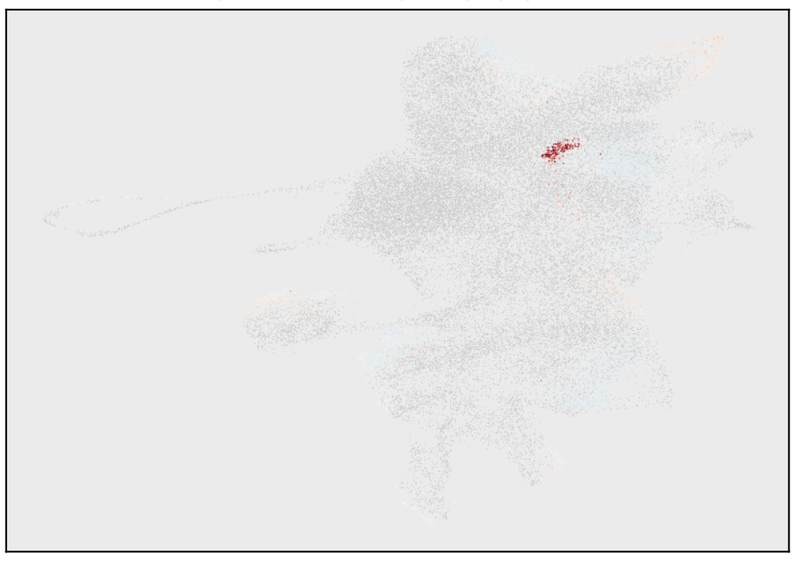
3-somite: 16 genes, supported
active in 6% of cells, library level 0.48



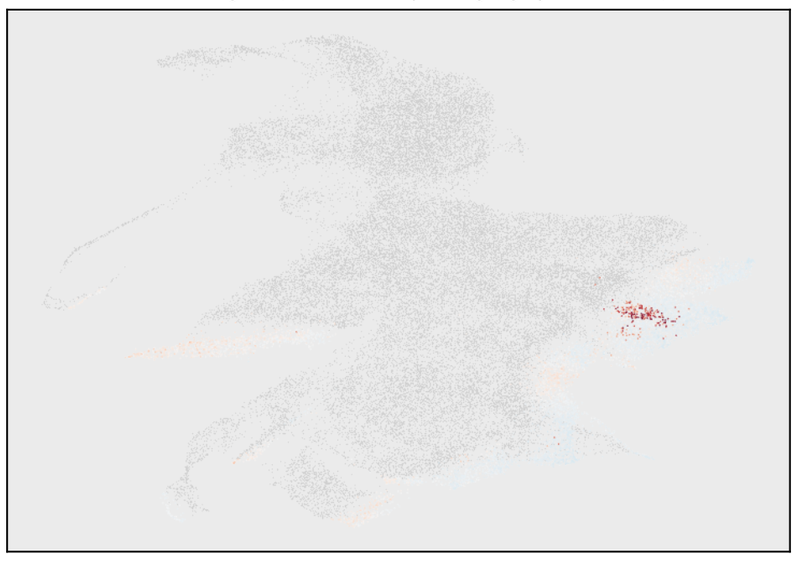
6-somite: 17 genes, supported
active in 5% of cells, library level 0.49



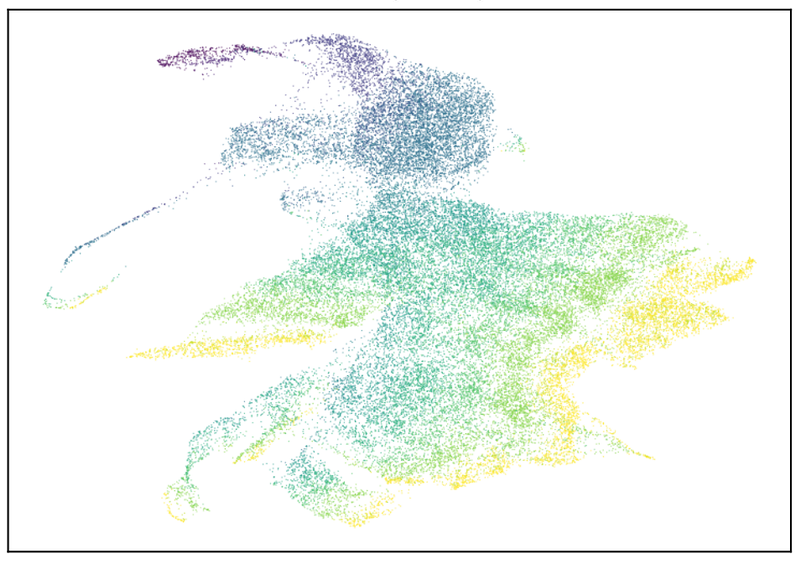
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



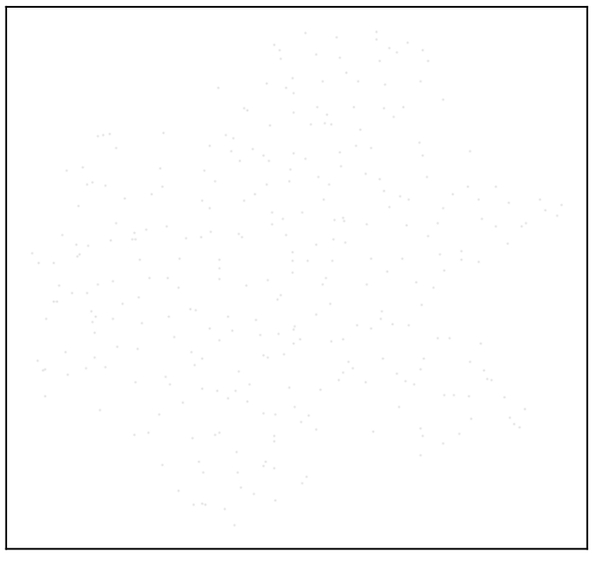
atlas, unaligned: stages



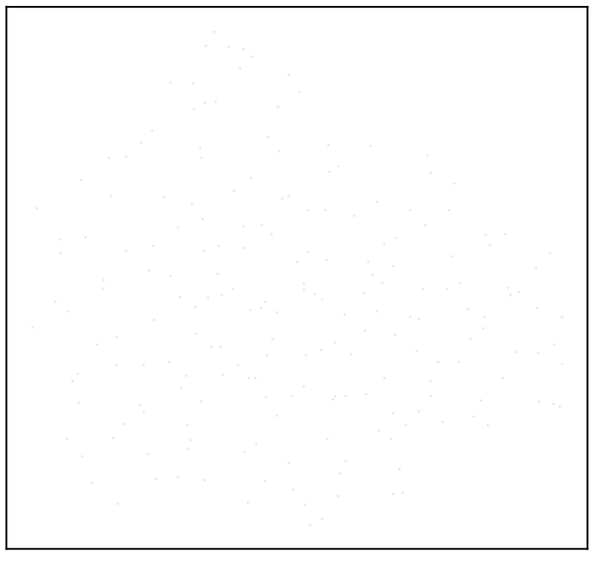
Genes (number of stages): foxd3 (2), sox10 (2), cdh6 (2), ved (2), alx1 (2), twist1b (2), fam212aa (2), fam46ba (2), lhx5 (2), sox9b (2), tfap2a (2), kctd15a (2), nes (2), ovol1 (2), sox1b (2), crip2 (2), ednrab (1)

placodes

high: not defined



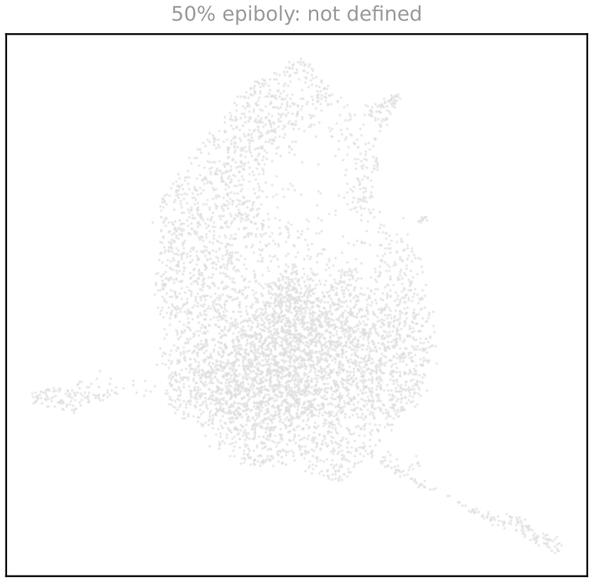
oblong: not defined



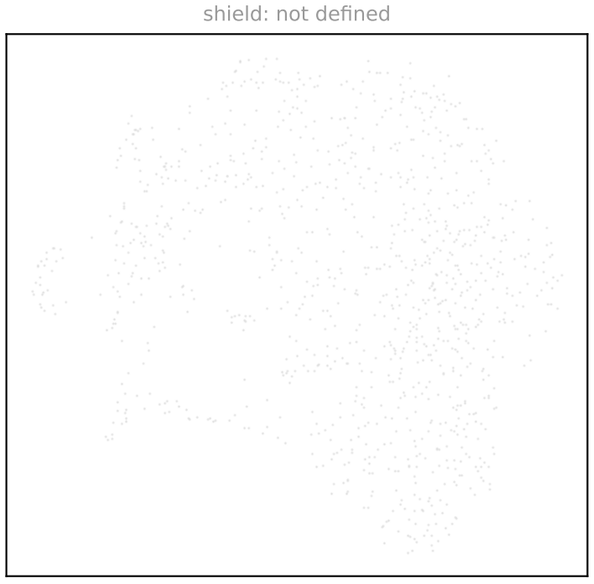
dome: not defined

30% epiboly: not defined

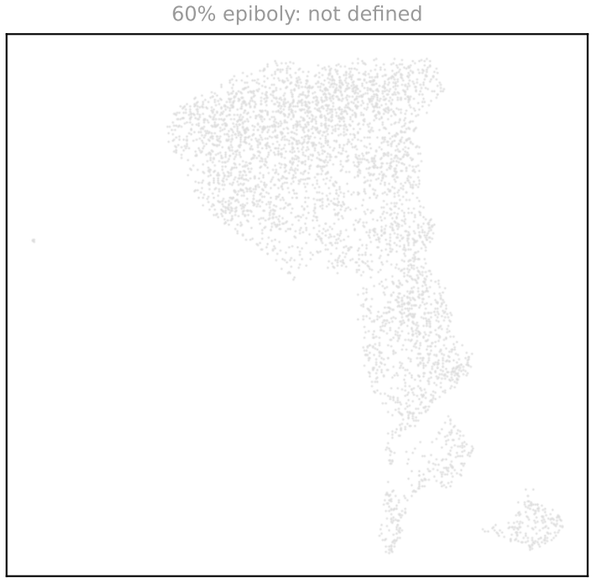
50% epiboly: not defined



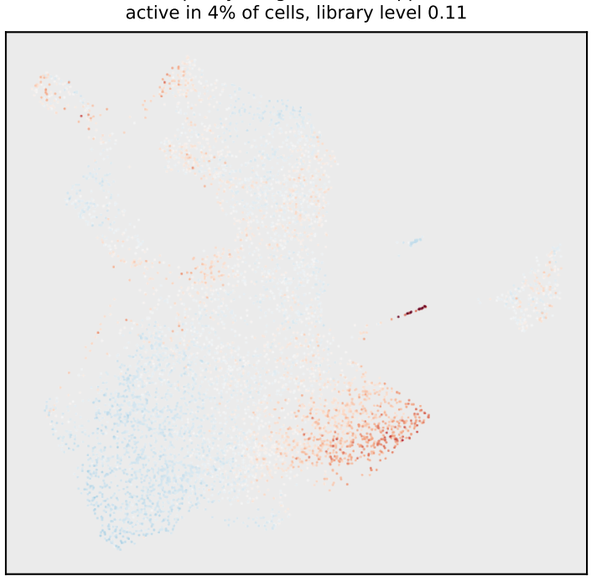
shield: not defined



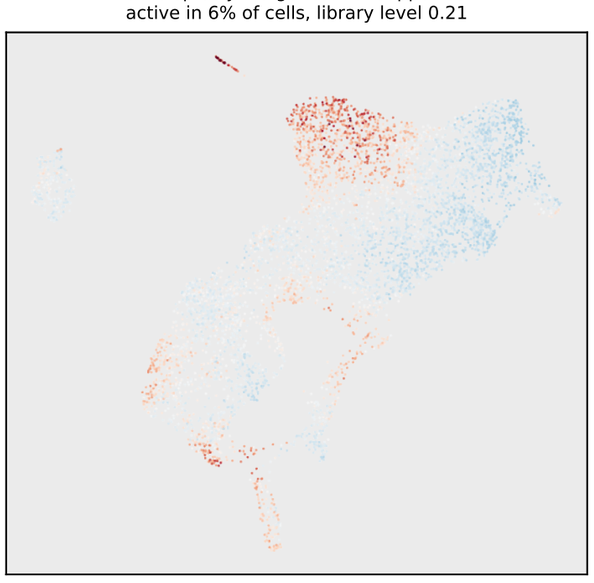
60% epiboly: not defined



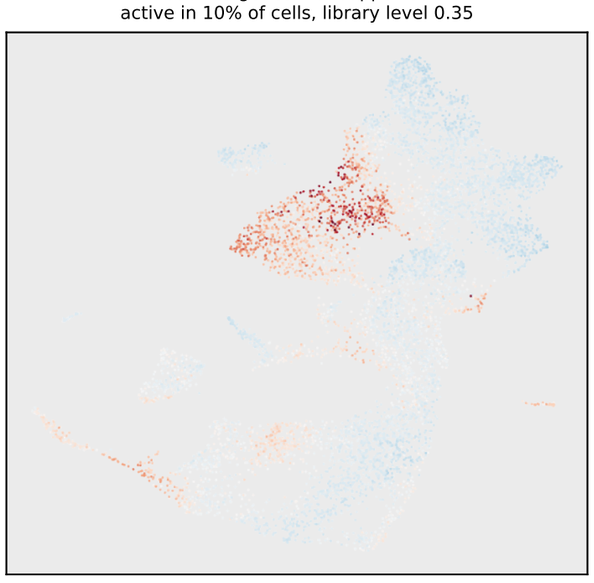
75% epiboly: 19 genes, not supported
active in 4% of cells, library level 0.11



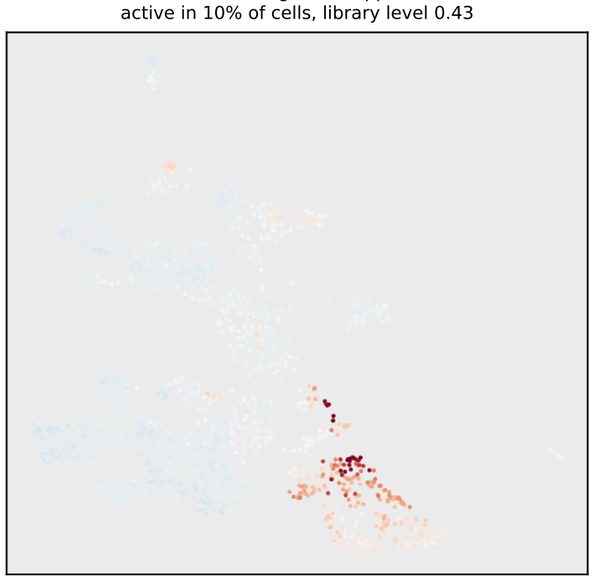
90% epiboly: 20 genes, not supported
active in 6% of cells, library level 0.21



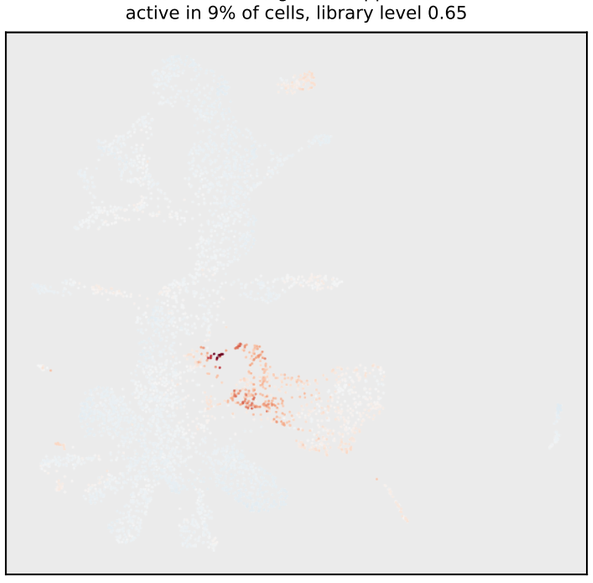
bud: 23 genes, not supported
active in 10% of cells, library level 0.35



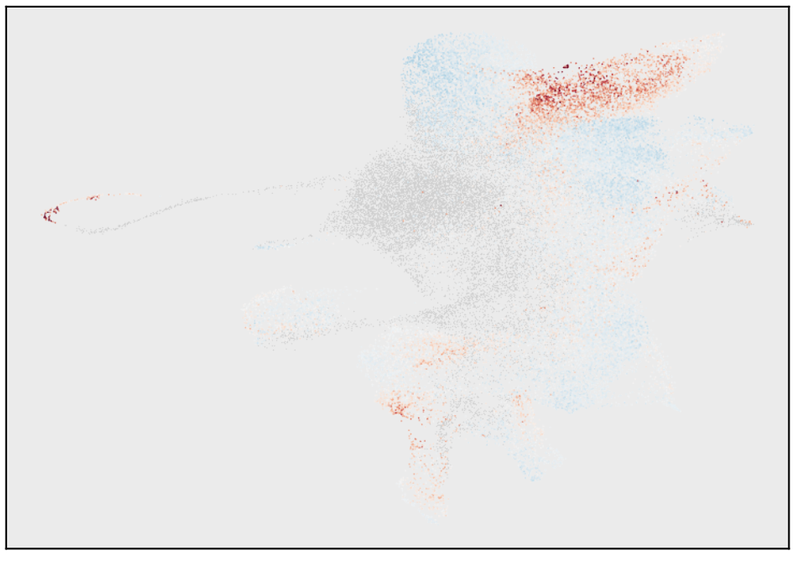
3-somite: 25 genes, supported
active in 10% of cells, library level 0.43



6-somite: 28 genes, supported
active in 9% of cells, library level 0.65

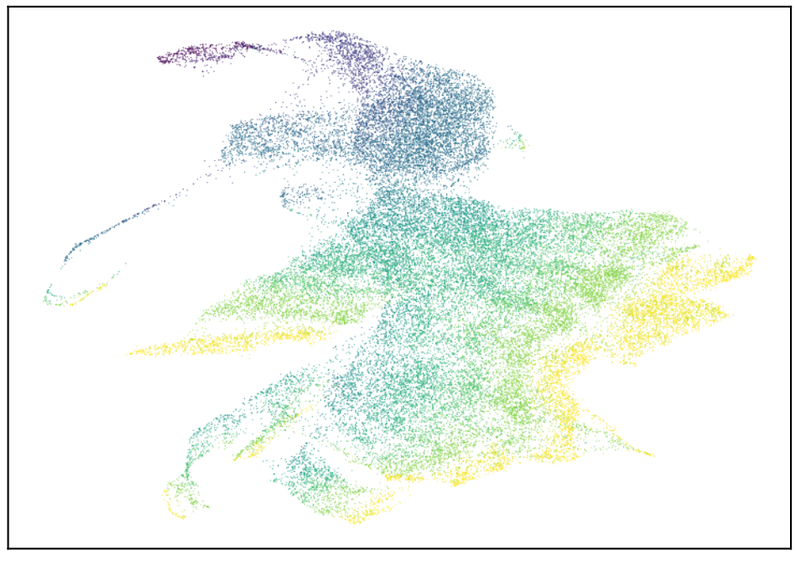


atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)

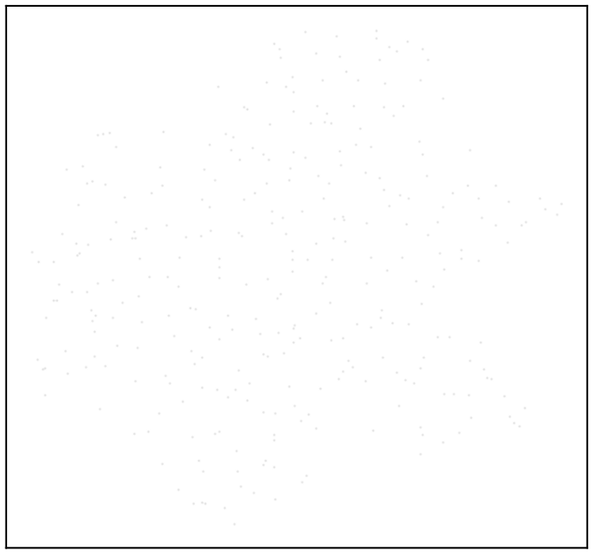
atlas, unaligned: stages



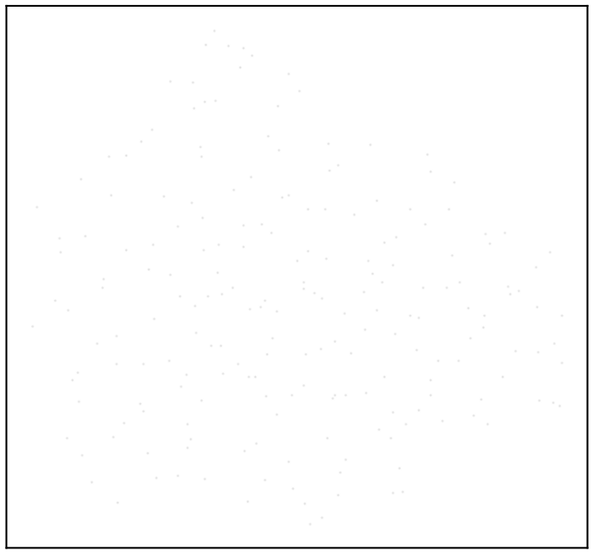
Genes (number of stages): foxi1 (5), pnocb (5), pax8 (5), pitx3 (5), tfap2c (5), cldna (5), her13 (5), ptgs2a (5), fgf24 (5), sdpra (5), sp5a (5), foxj1b (5), dlx4b (5), eya1 (5), insm1b (5), chodl (5), pih1d3 (5), six4b (5), irf6 (5), drgx (3), insm1a (3), scrt2 (3), tlx3b (3), sfrp2 (3), atoh1b (2), atoh1a (1), isl2b (1), hmx3a (1)

non_neural_ecto

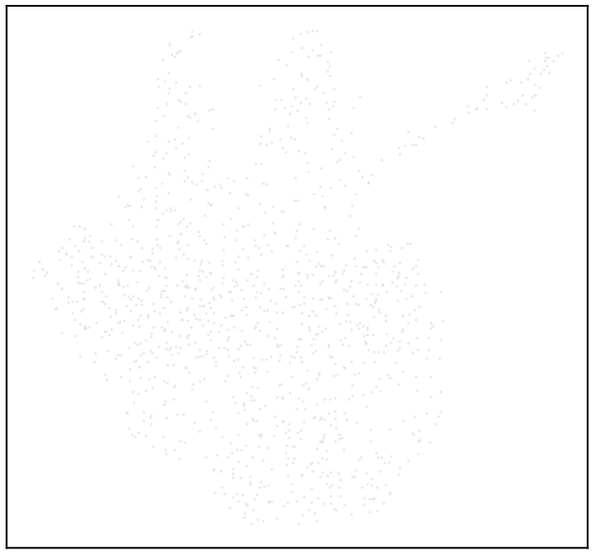
high: not defined



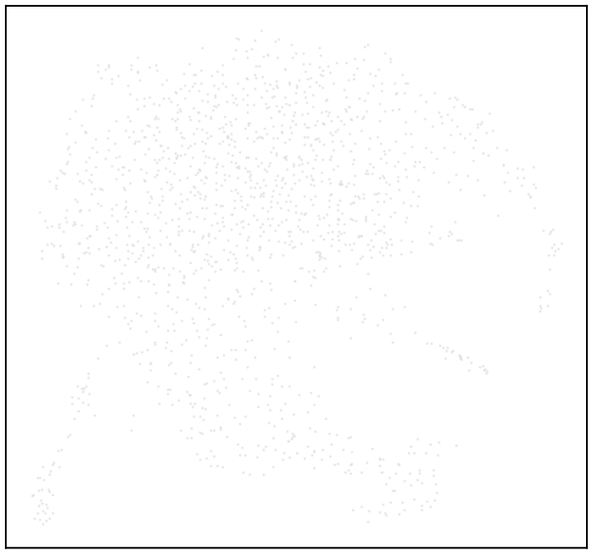
oblong: not defined



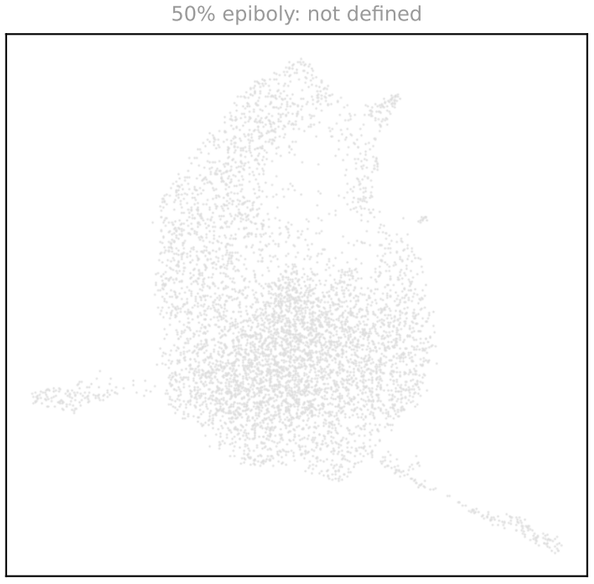
dome: not defined



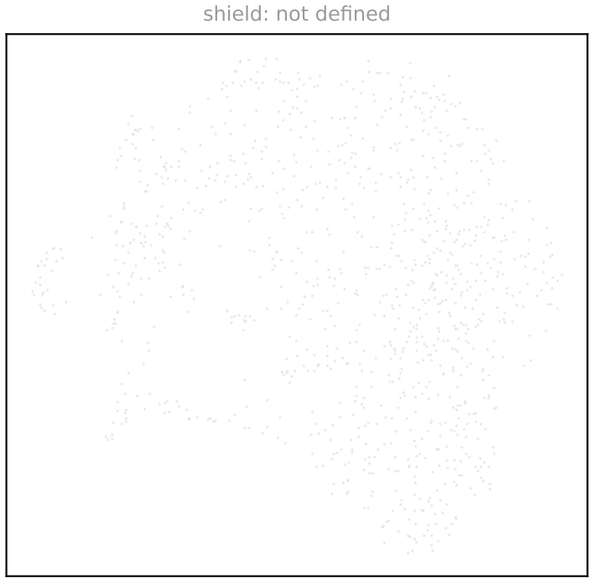
30% epiboly: not defined



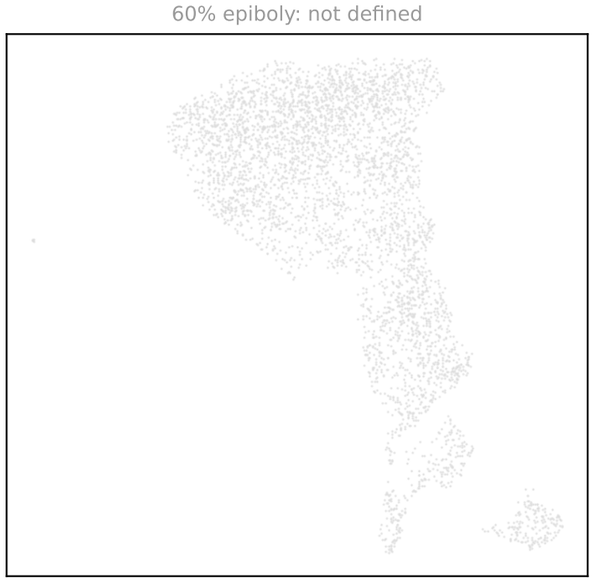
50% epiboly: not defined



shield: not defined

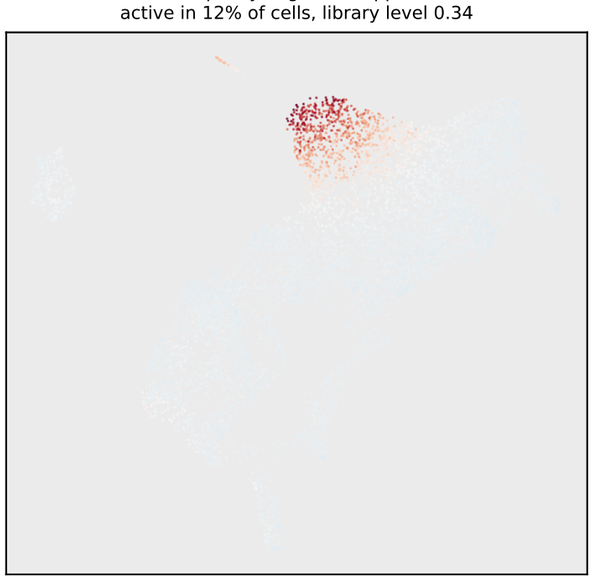


60% epiboly: not defined

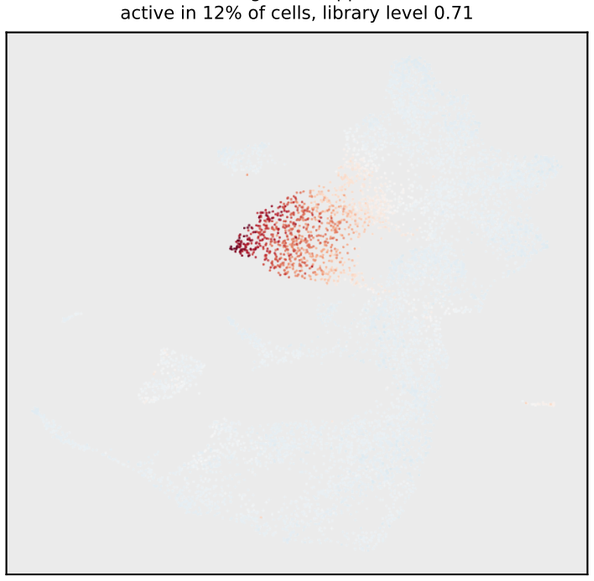


75% epiboly: 7 genes, supported
active in 10% of cells, library level 0.17

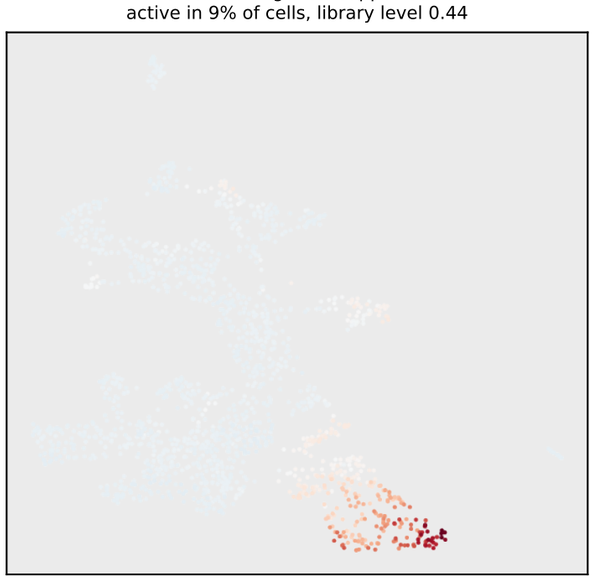
90% epiboly: 7 genes, supported
active in 12% of cells, library level 0.34



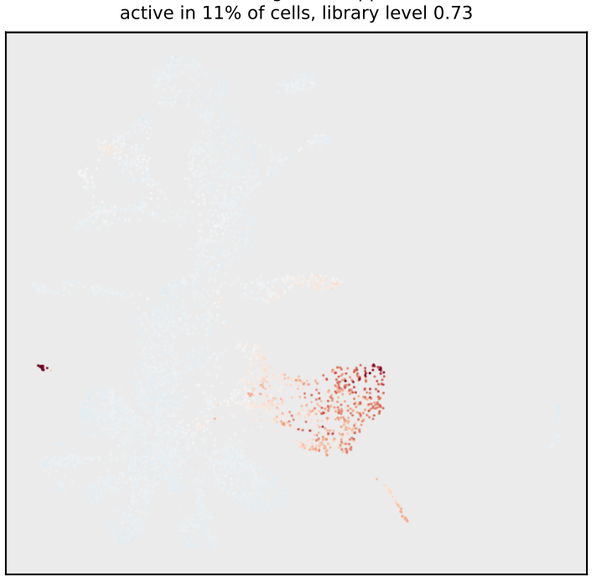
bud: 6 genes, supported
active in 12% of cells, library level 0.71



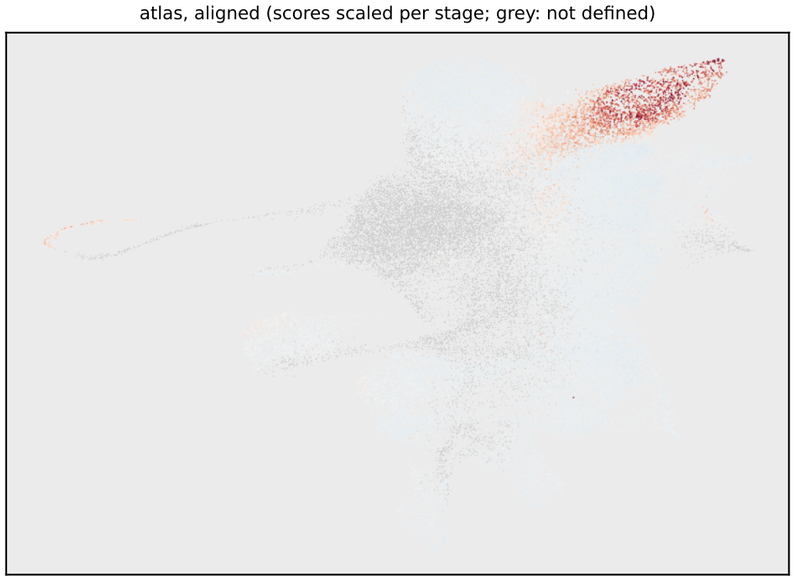
3-somite: 7 genes, supported
active in 9% of cells, library level 0.44



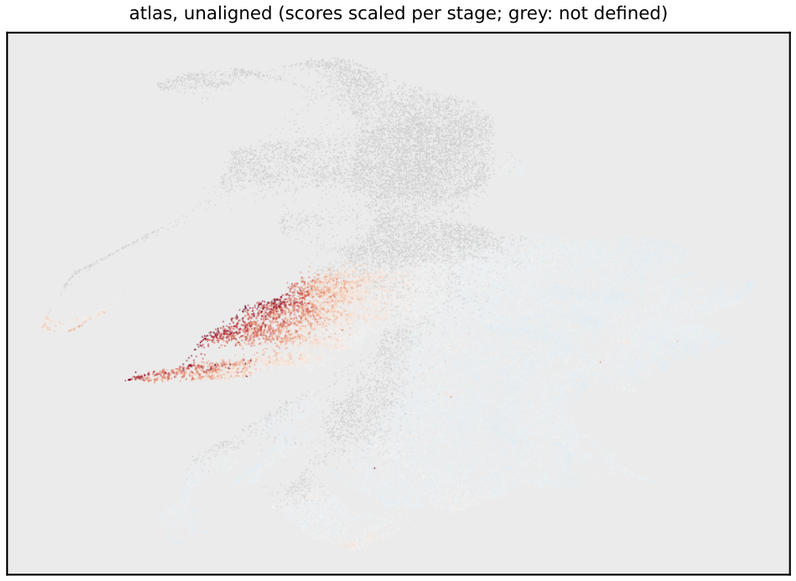
6-somite: 8 genes, supported
active in 11% of cells, library level 0.73



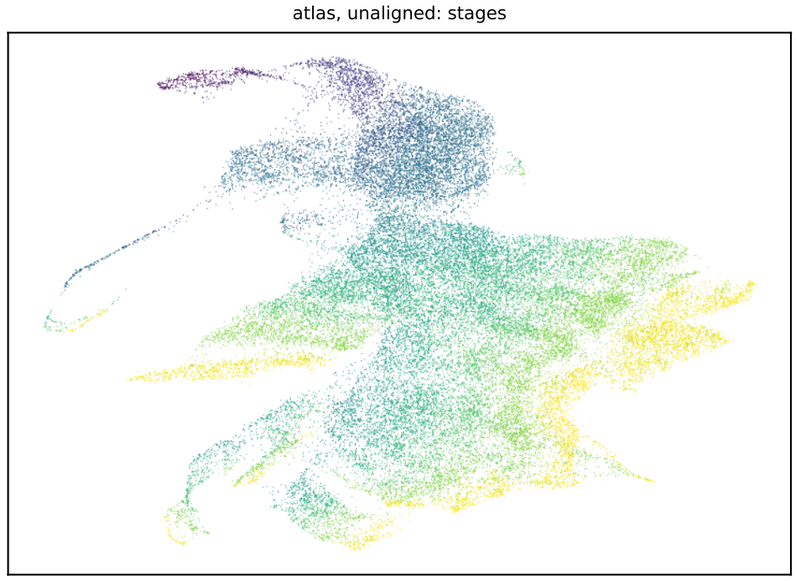
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



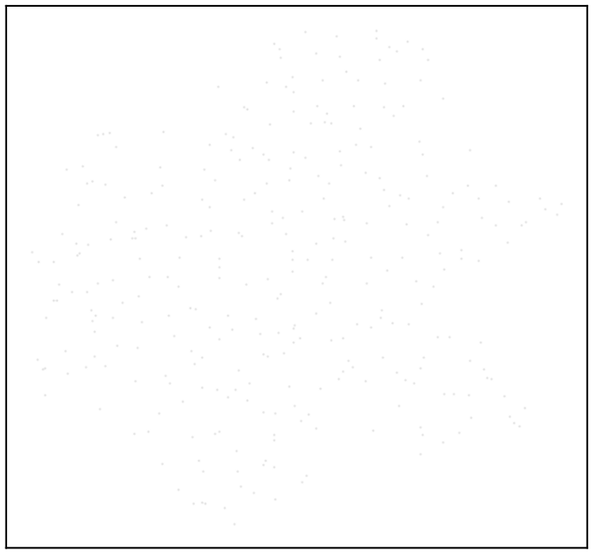
atlas, unaligned: stages



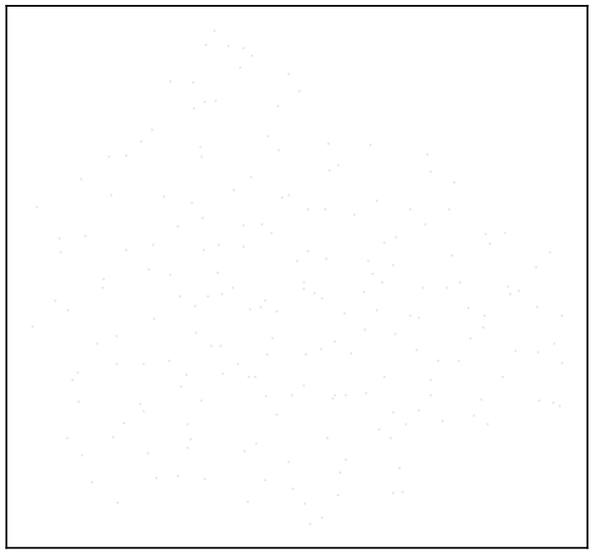
Genes (number of stages): gata3 (5), tp63 (5), foxi3a (5), tfap2c (5), fermt1 (5), bambib (5), grhl1 (4), foxi3b (1)

axial_mesoderm

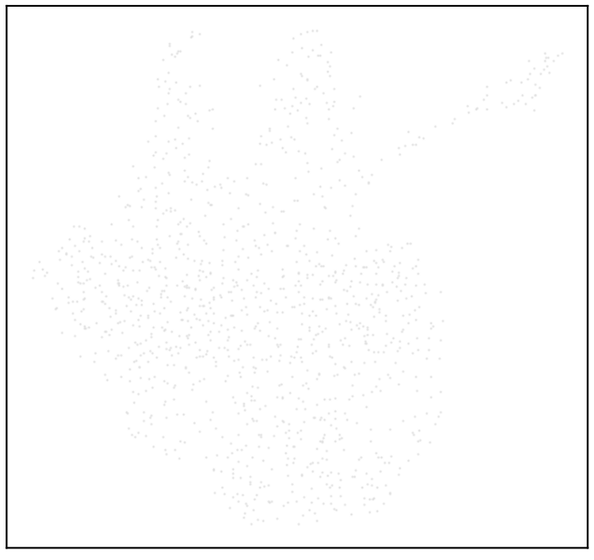
high: not defined



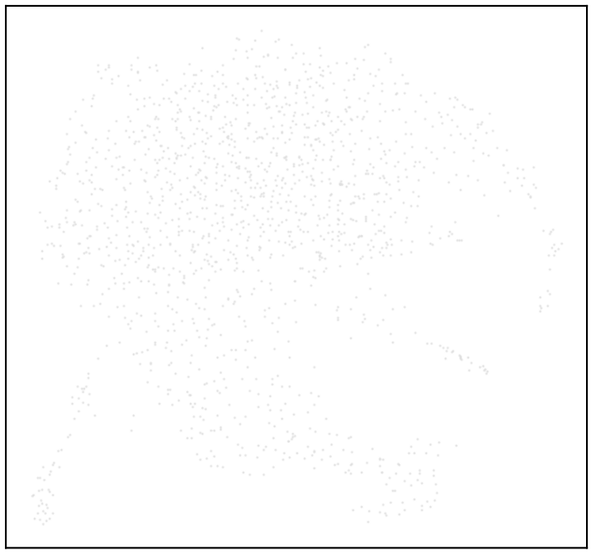
oblong: not defined



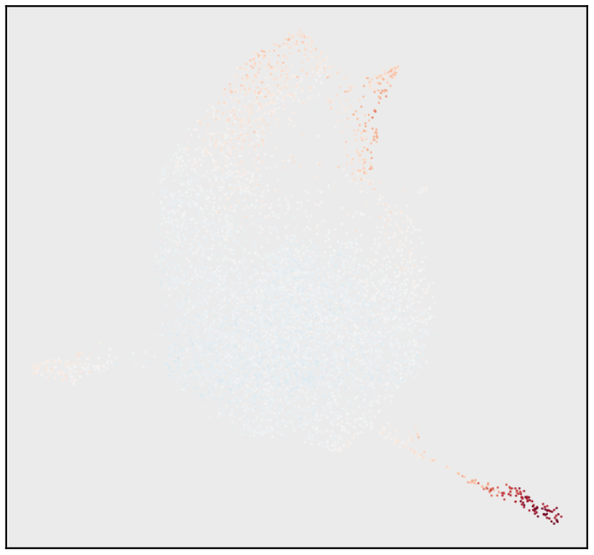
dome: not defined



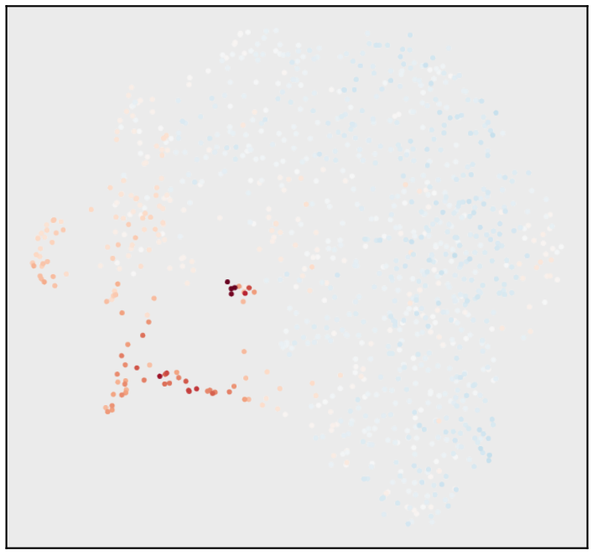
30% epiboly: not defined



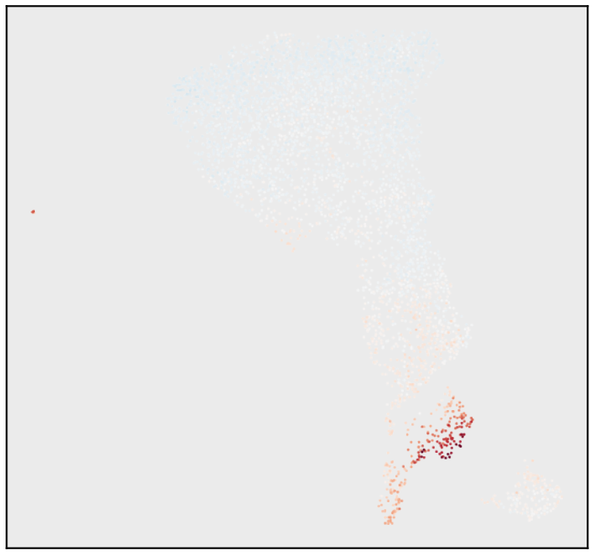
50% epiboly: 20 genes, not supported
active in 6% of cells, library level 0.34



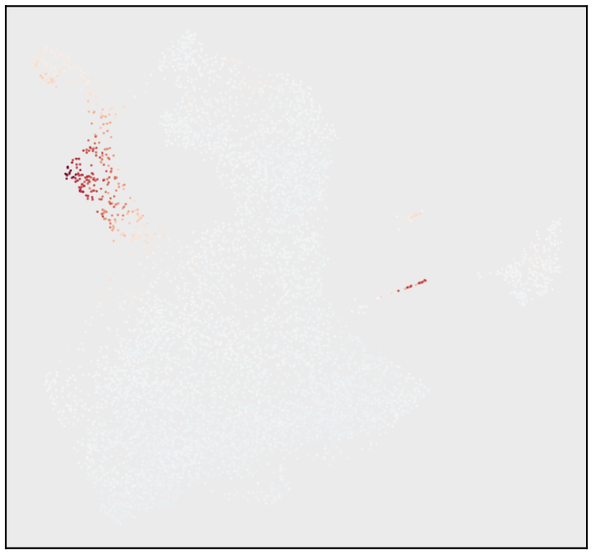
shield: 17 genes, not supported
active in 5% of cells, library level 0.31



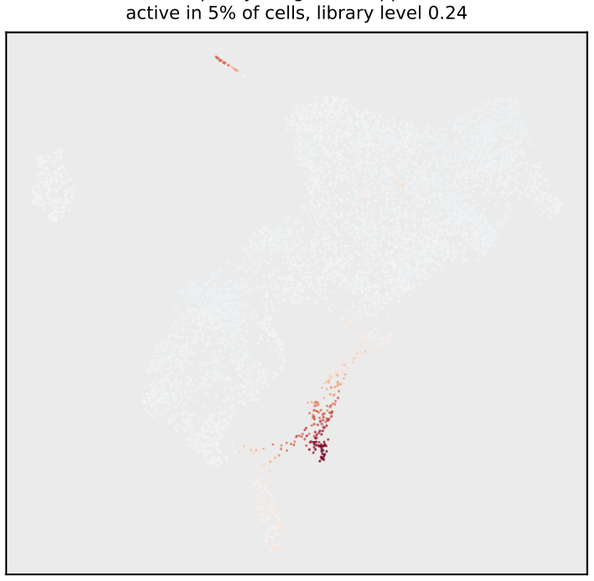
60% epiboly: 23 genes, supported
active in 6% of cells, library level 0.21



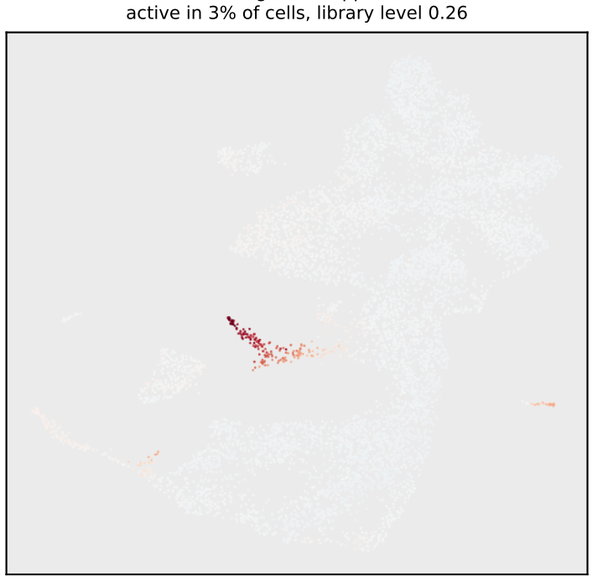
75% epiboly: 24 genes, supported
active in 5% of cells, library level 0.25



90% epiboly: 24 genes, supported
active in 5% of cells, library level 0.24

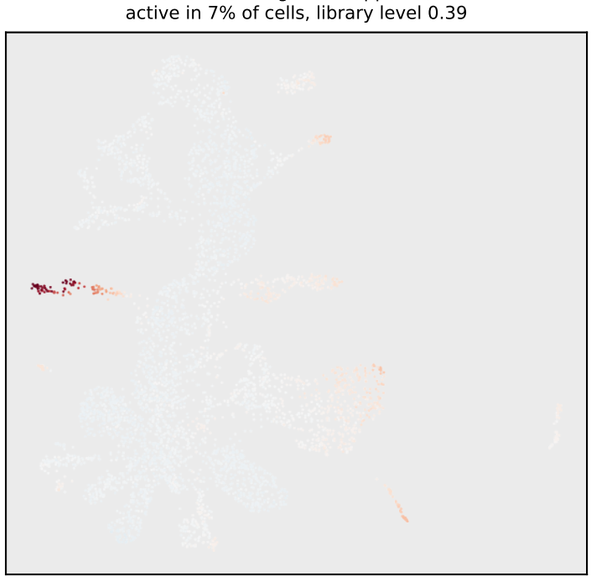


bud: 24 genes, supported
active in 3% of cells, library level 0.26

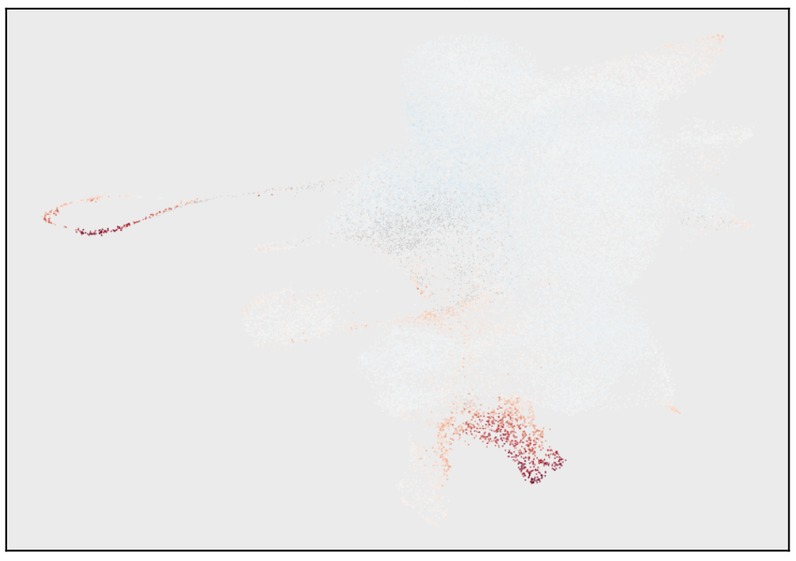


3-somite: 24 genes, supported
active in 4% of cells, library level 0.43

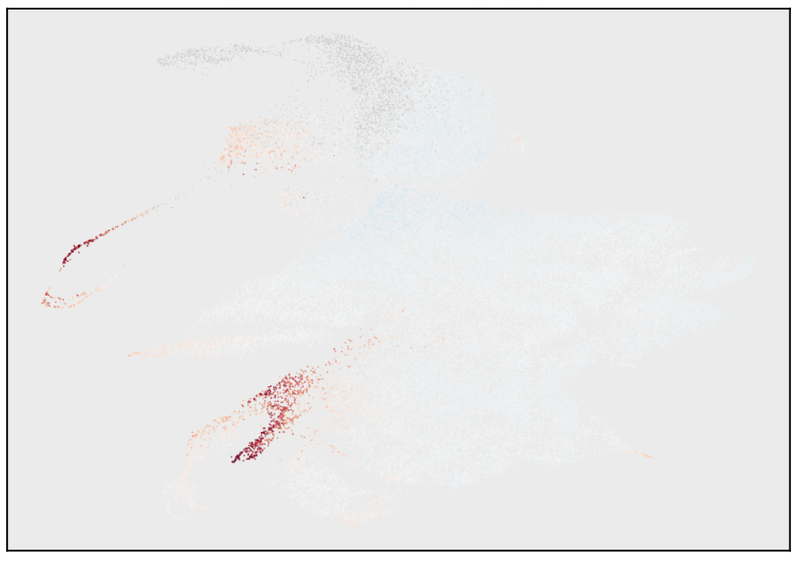
6-somite: 24 genes, supported
active in 7% of cells, library level 0.39



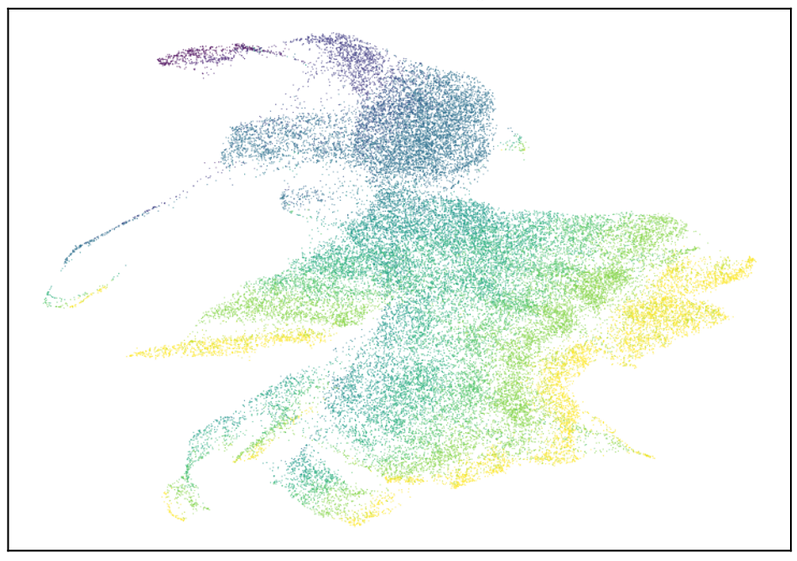
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



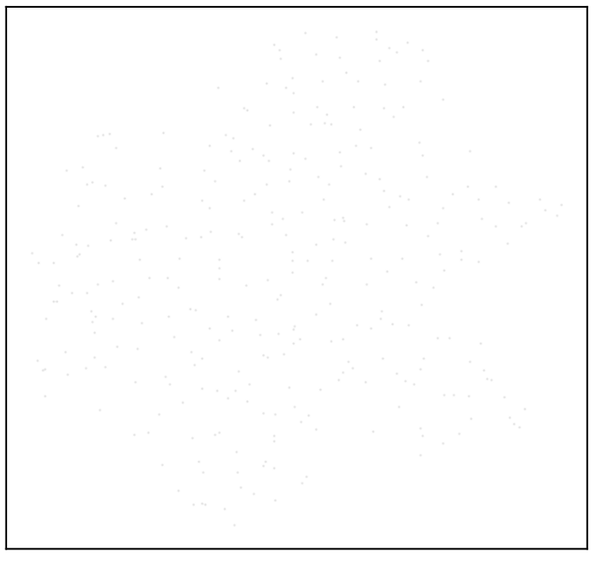
atlas, unaligned: stages



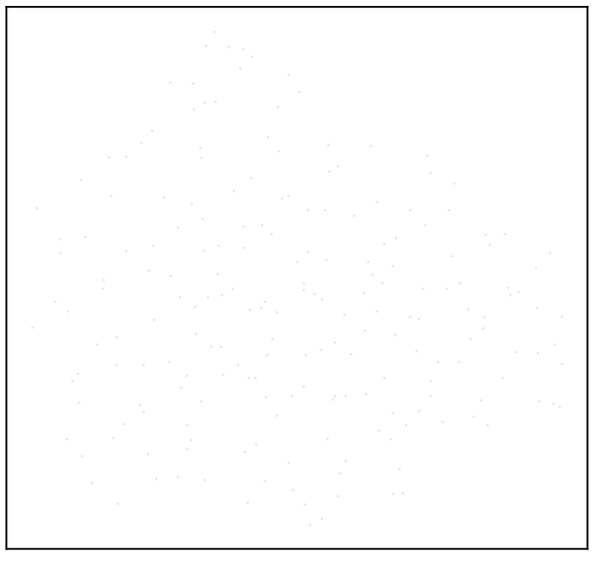
Genes (number of stages): cnn2 (8), ntd5 (8), fam212aa (8), ckap4 (8), perp (8), rcn3 (8), pak1 (8), trib3 (8), arg2 (8), egln3 (8), fam3c (8), ip6k2b (8), spty2d1 (8), si:dkey-112a7.4 (8), acvr1ba (8), dnajb11 (8), twist2 (7), fbln2 (7), creb3l1 (7), slit2 (6), itga6a (6), bcam (6), tph1b (6), hfe2 (5), apobec2a (2)

hatching_gland

high: not defined



oblong: not defined

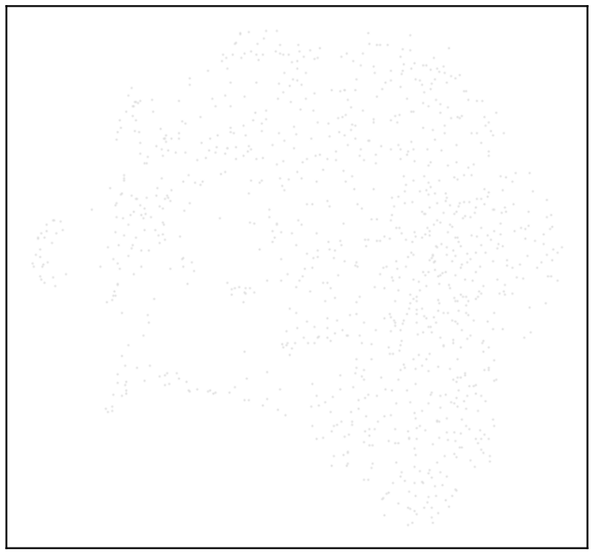


dome: not defined

30% epiboly: not defined

50% epiboly: not defined

shield: not defined

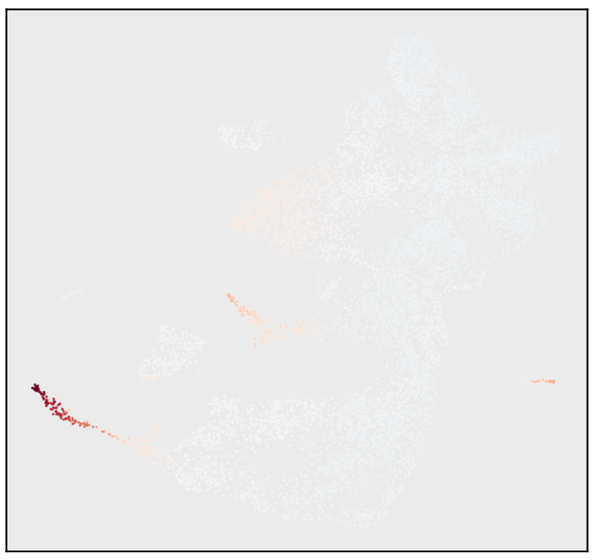


60% epiboly: not defined

75% epiboly: not defined

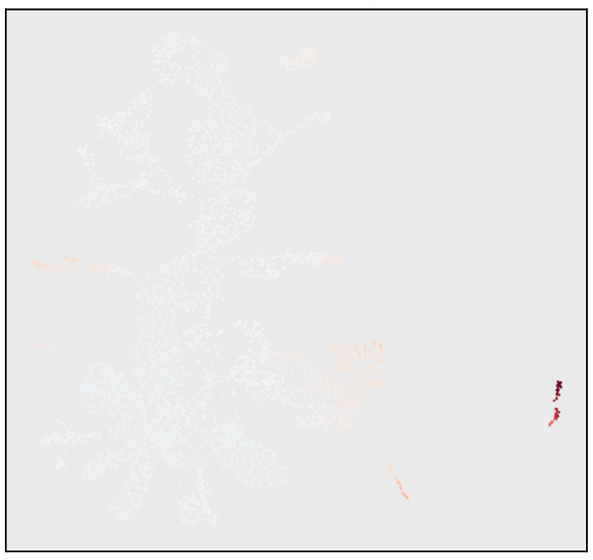
90% epiboly: not defined

bud: 30 genes, supported
active in 12% of cells, library level 0.22



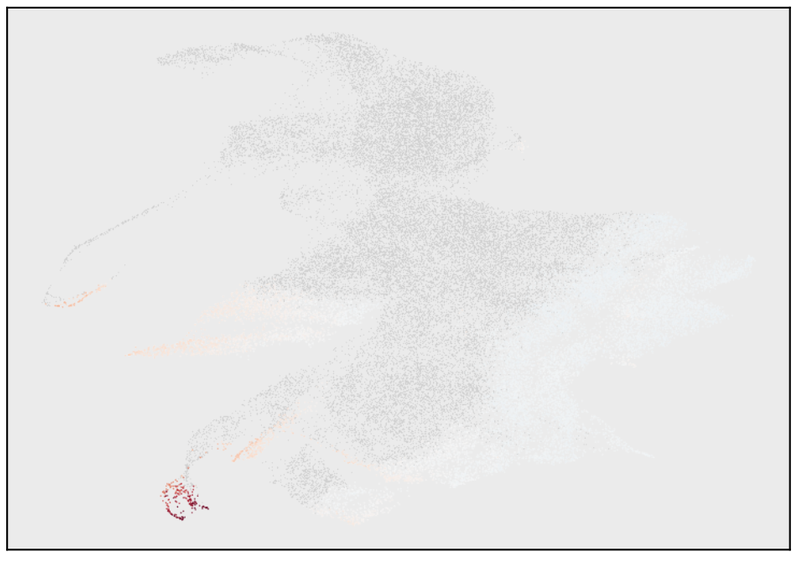
3-somite: 30 genes, supported
active in 13% of cells, library level 0.27

6-somite: 30 genes, supported
active in 14% of cells, library level 0.44

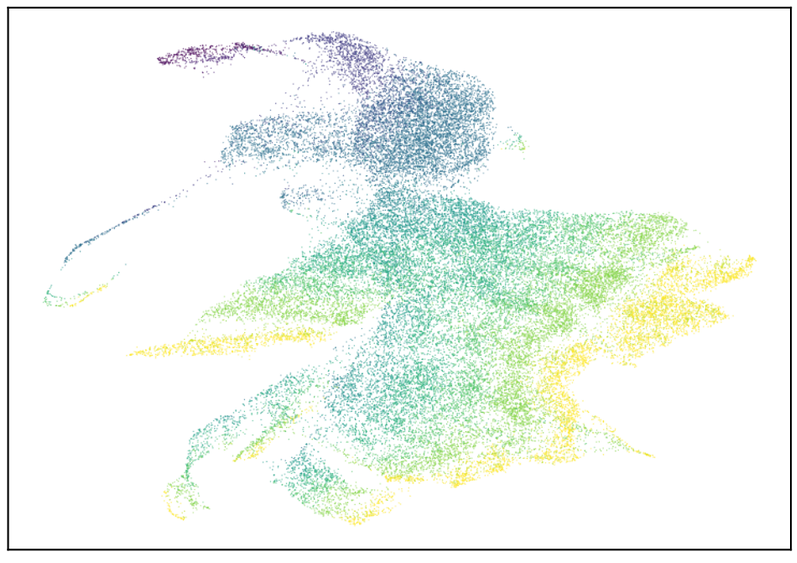


atlas, aligned (scores scaled per stage; grey: not defined)

atlas, unaligned (scores scaled per stage; grey: not defined)



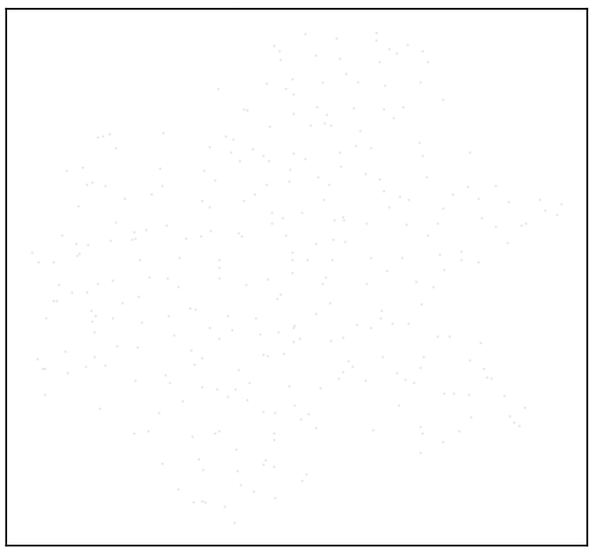
atlas, unaligned: stages



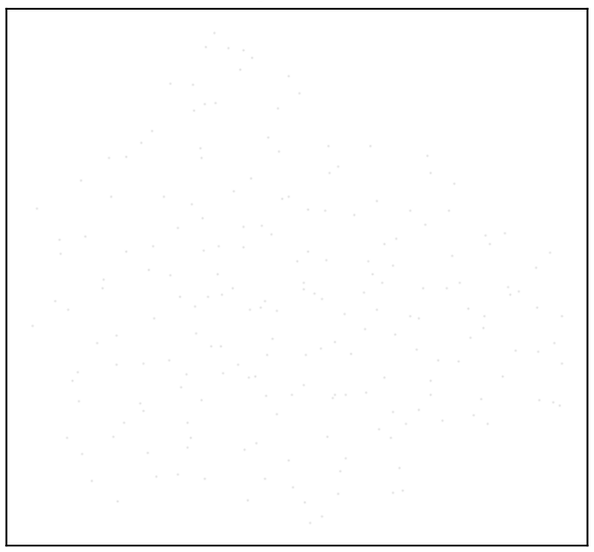
Genes (number of stages): hspa5 (3), pitx2 (3), xbp1 (3), cd63 (3), frzb (3), he1a (3), id3 (3), tagln2 (3), calr3b (3), fbxo2 (3), gadd45ga (3), pfn1 (3), rrbp1b (3), si:ch211-153b23.5 (3), stk10 (3), hsp90b1 (3), bhlha15 (3), sec61g (3), zgc:162964 (3), bambia (3), capns1a (3), dnajc3a (3), inhbaa (3), sec61b (3), cldnh (3), calr (3), hyou1 (3), pdia3 (3), ssr4 (3), cnpy1 (3)

margin

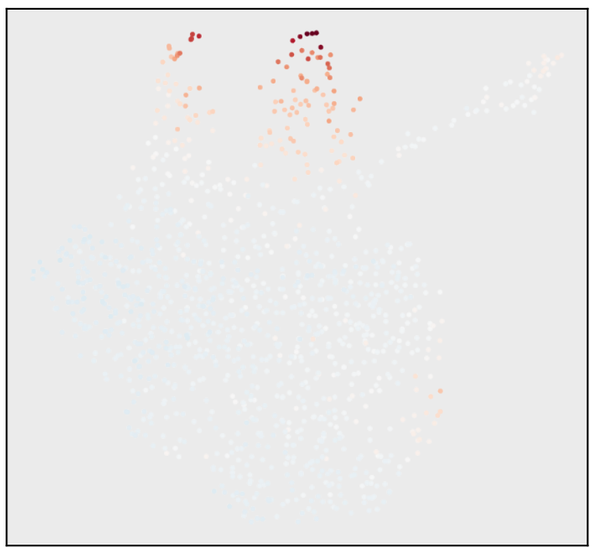
high: not defined



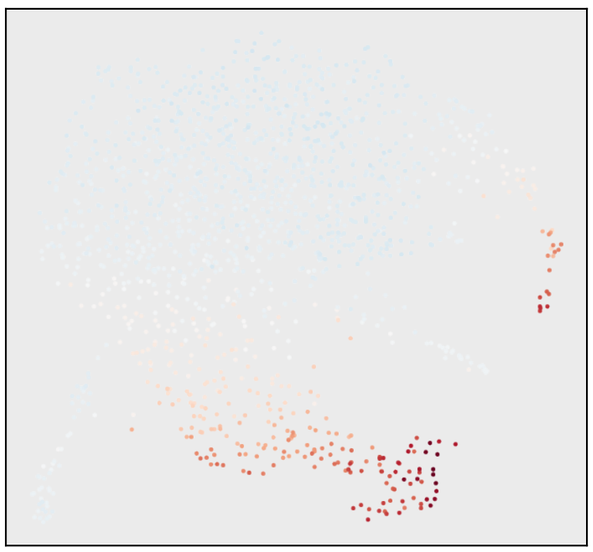
oblong: not defined



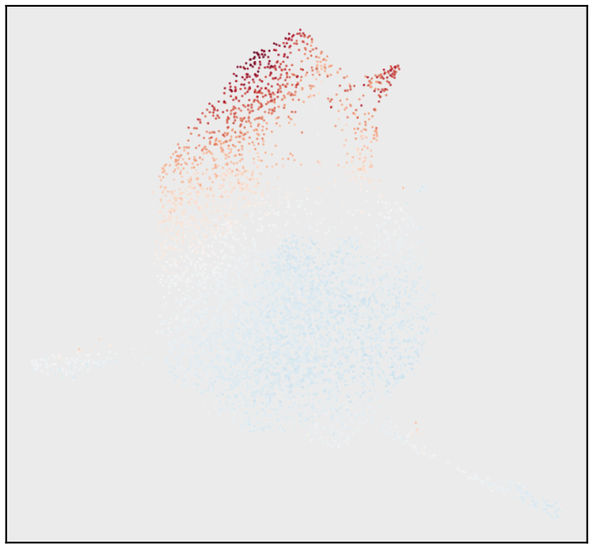
dome: 26 genes, not supported
active in 10% of cells, library level 0.23



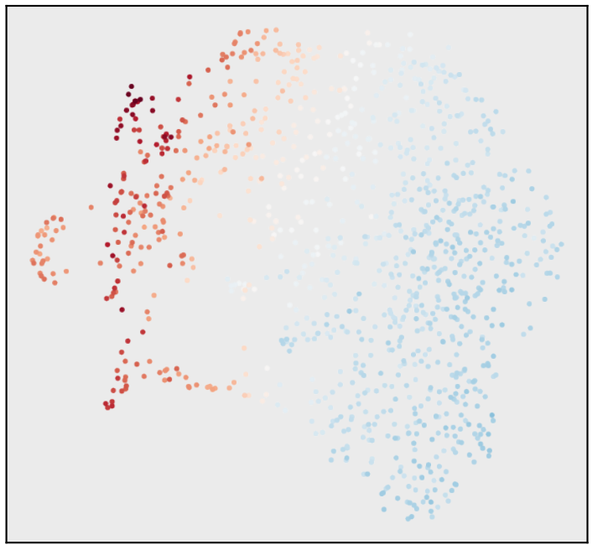
30% epiboly: 29 genes, supported
active in 13% of cells, library level 0.36



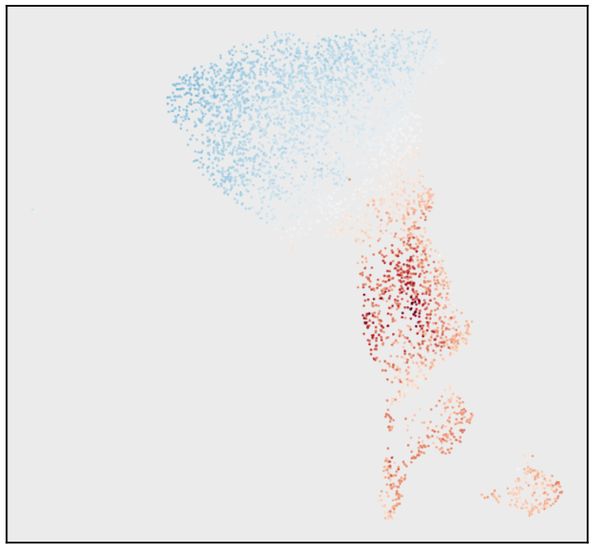
50% epiboly: 30 genes, supported
active in 18% of cells, library level 0.38



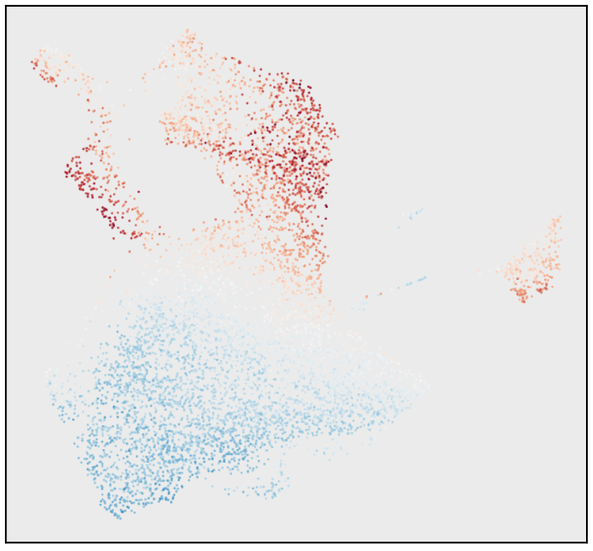
shield: 30 genes, supported
active in 26% of cells, library level 0.48



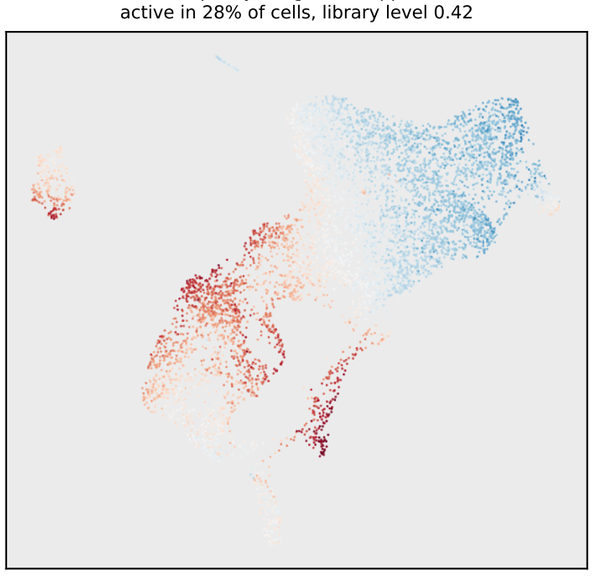
60% epiboly: 30 genes, supported
active in 33% of cells, library level 0.59



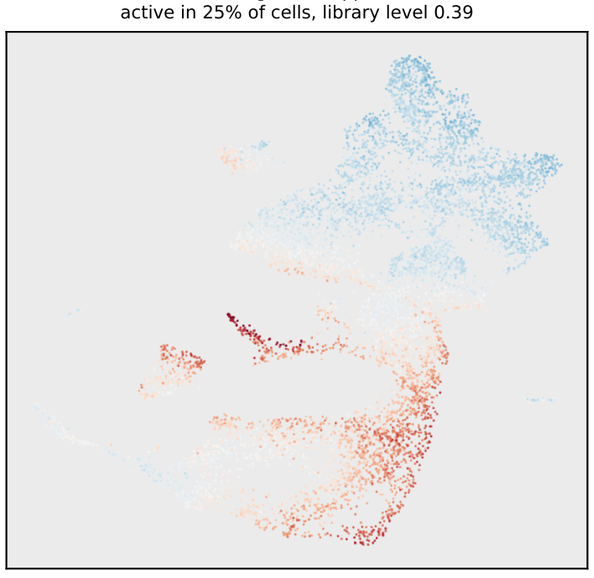
75% epiboly: 30 genes, supported
active in 33% of cells, library level 0.57



90% epiboly: 30 genes, supported
active in 28% of cells, library level 0.42

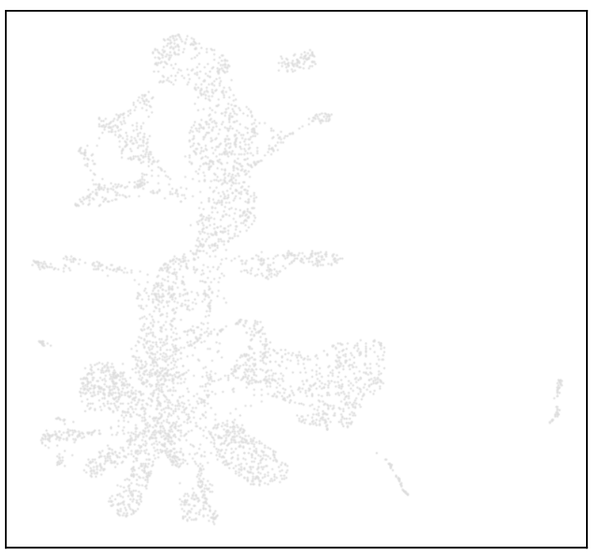


bud: 30 genes, supported
active in 25% of cells, library level 0.39

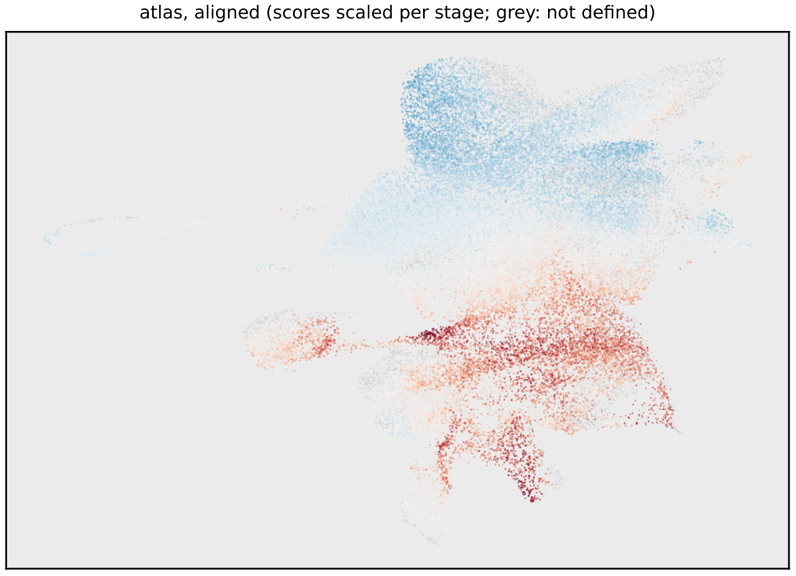


3-somite: not defined

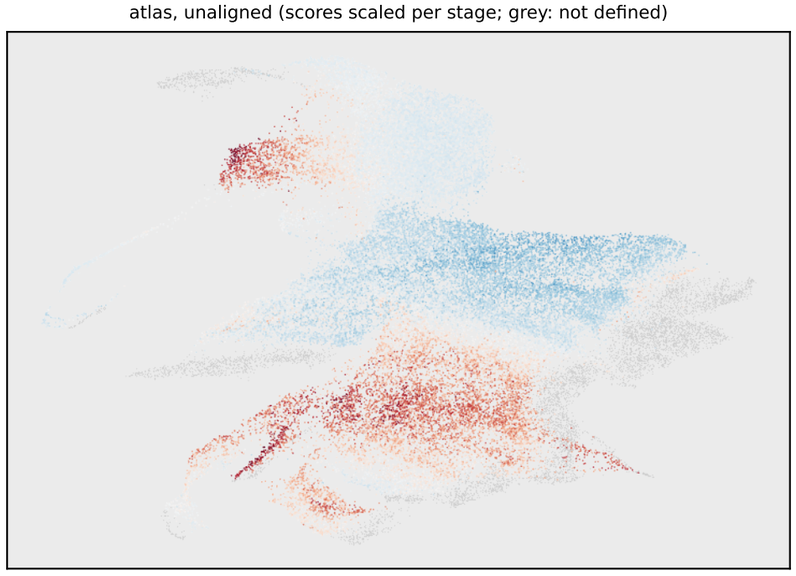
6-somite: not defined



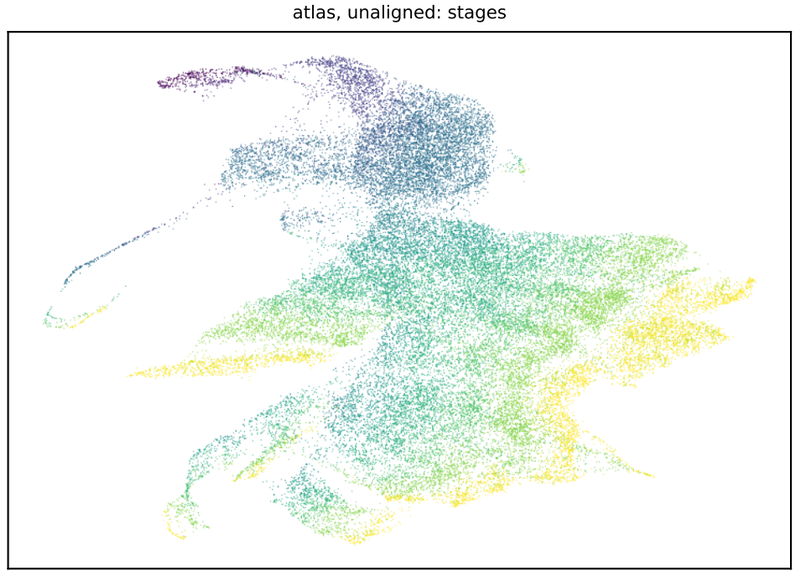
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



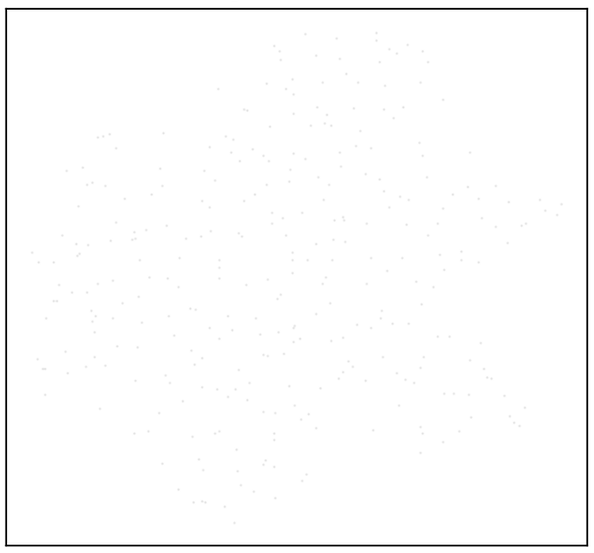
atlas, unaligned: stages



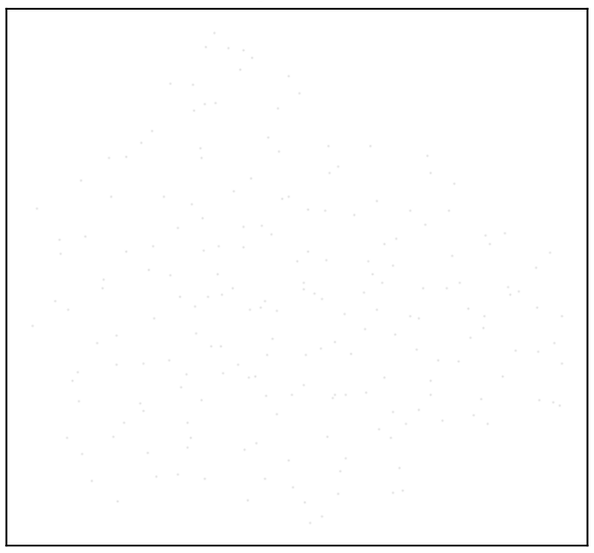
Genes (number of stages): eve1 (8), lft2 (8), mespab (8), ved (8), foxa (8), lft1 (8), cdx4 (8), xbp1 (8), wnt8a (8), fscn1a (8), gata6 (8), hes6 (8), rdh10a (8), arl4ab (8), vox (8), aplnrb (8), efnb2a (8), cdkn1ca (8), gadd45bb (8), p4ha2 (8), dhrs3b (8), flrt3 (8), rnd1l (8), mixl1 (8), sp5l (8), cyp27c1 (8), mespaa (7), msgn1 (7), serpinh1b (7), cpn1 (6)

mesoderm

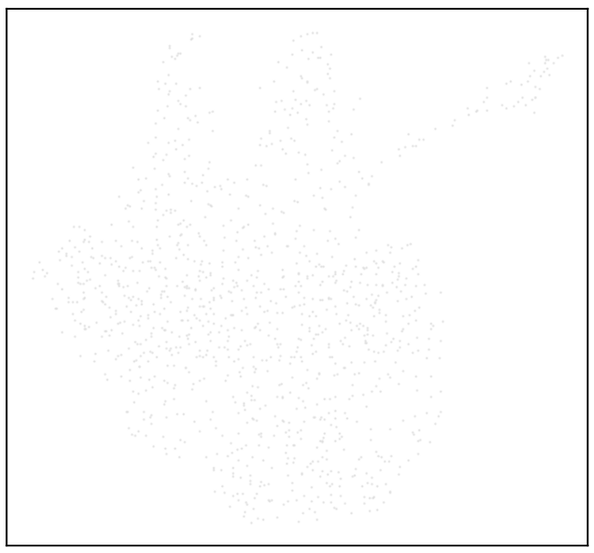
high: not defined



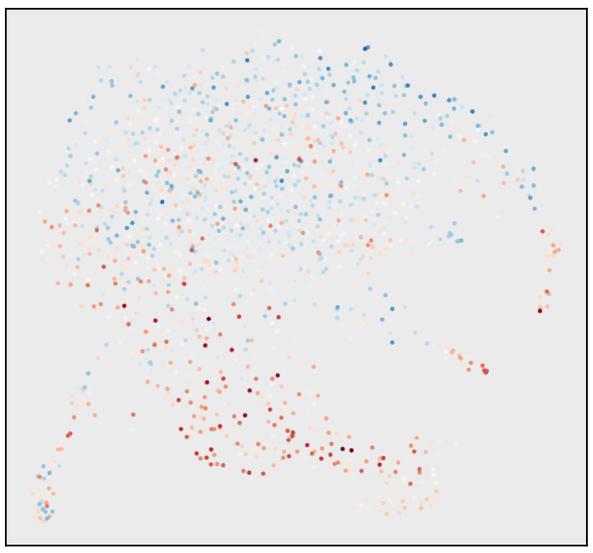
oblong: not defined



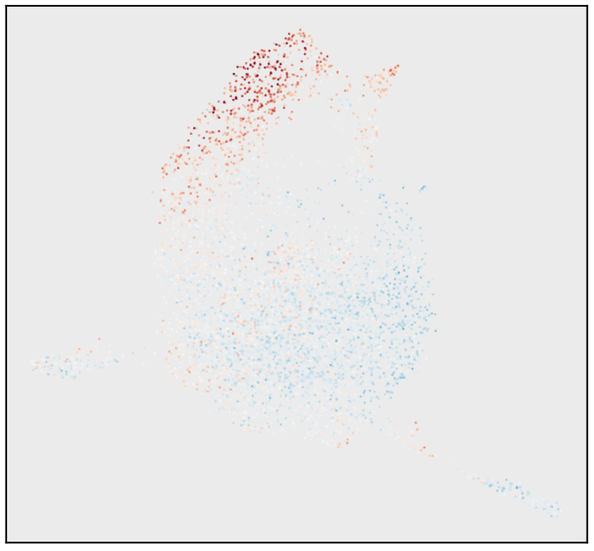
dome: not defined



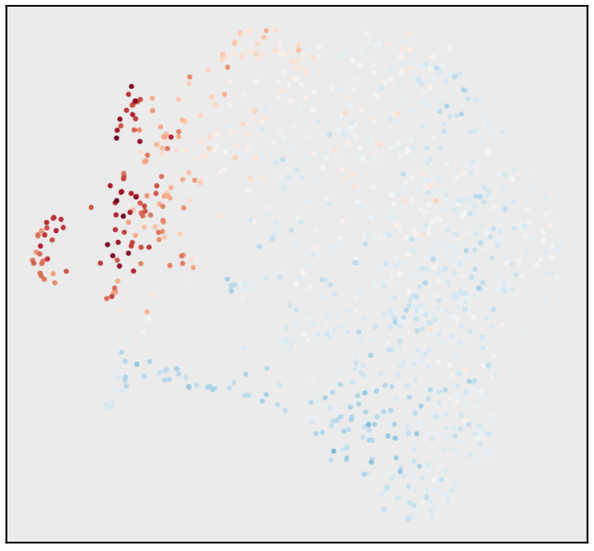
30% epiboly: 20 genes, not supported
active in 0% of cells, library level 0.38



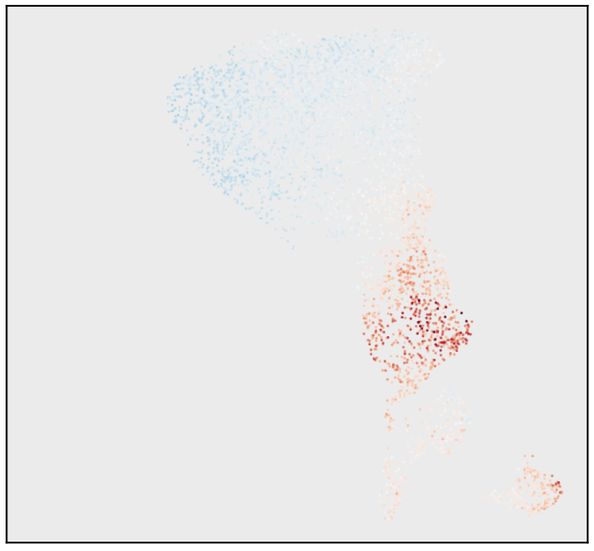
50% epiboly: 22 genes, not supported
active in 3% of cells, library level 0.27



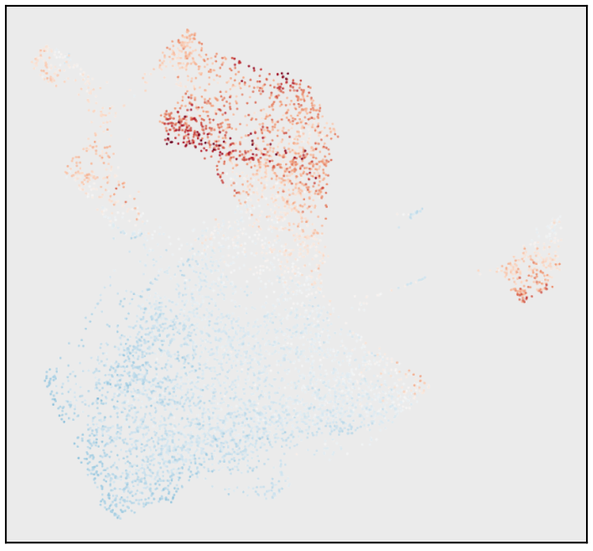
shield: 19 genes, not supported
active in 11% of cells, library level 0.38



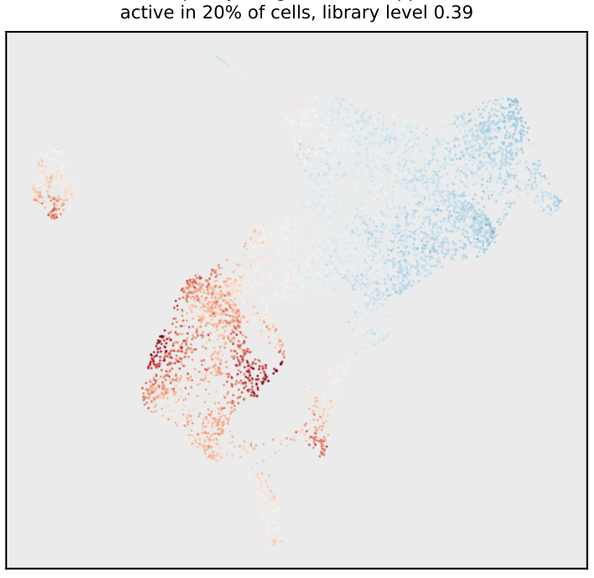
60% epiboly: 24 genes, not supported
active in 14% of cells, library level 0.35



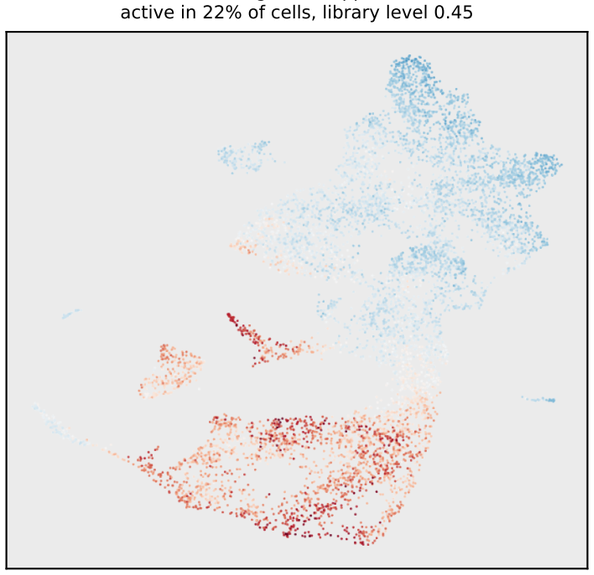
75% epiboly: 27 genes, not supported
active in 16% of cells, library level 0.24



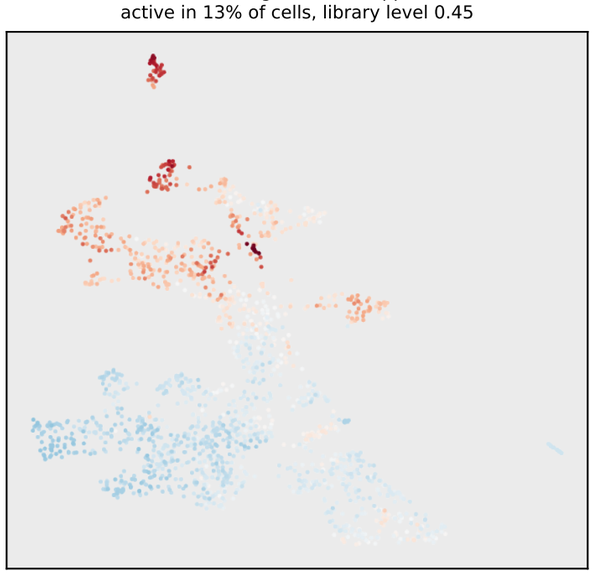
90% epiboly: 28 genes, not supported
active in 20% of cells, library level 0.39



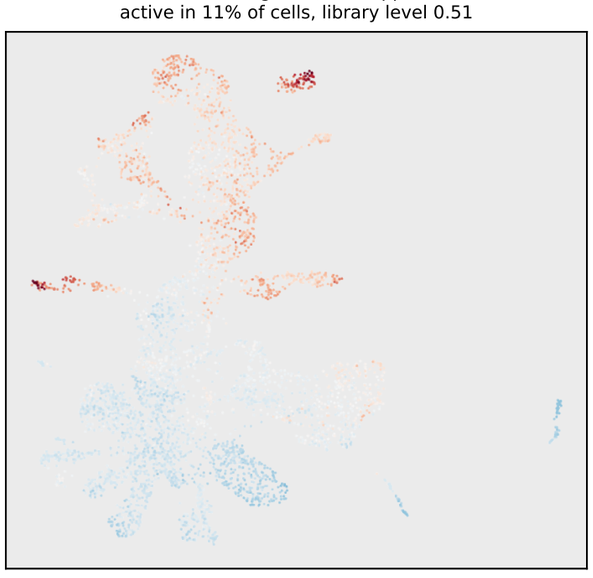
bud: 29 genes, supported
active in 22% of cells, library level 0.45



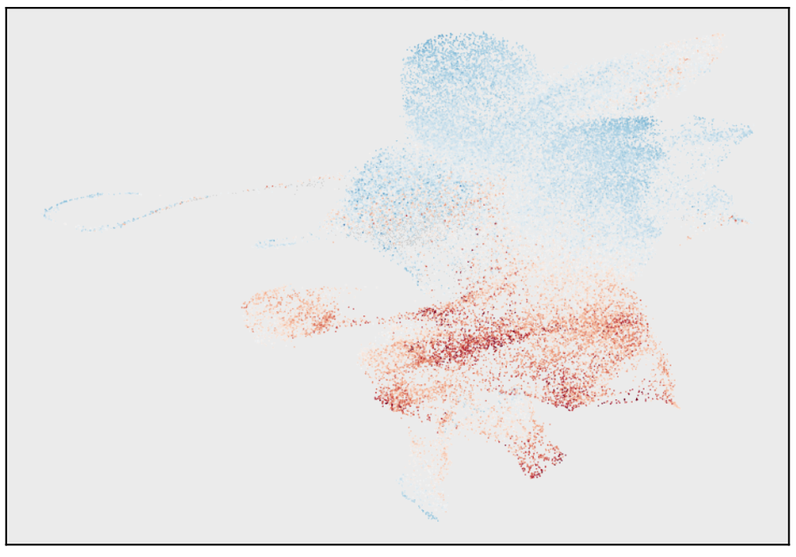
3-somite: 29 genes, not supported
active in 13% of cells, library level 0.45



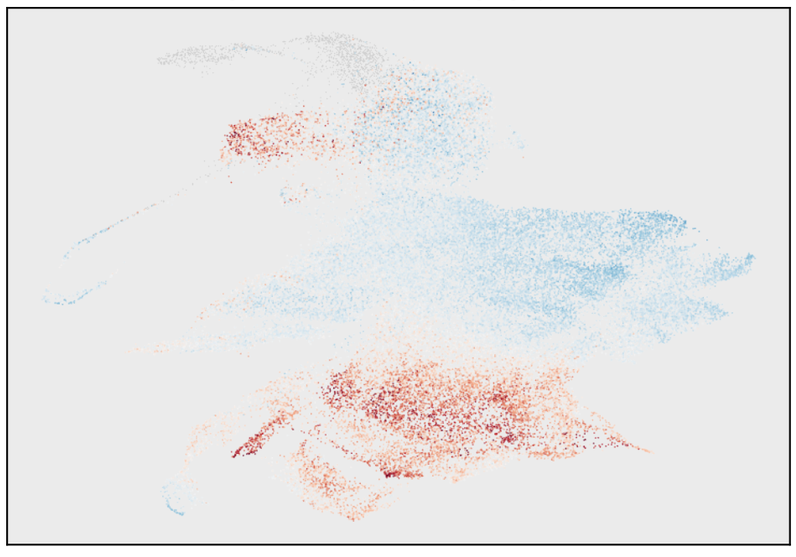
6-somite: 30 genes, not supported
active in 11% of cells, library level 0.51



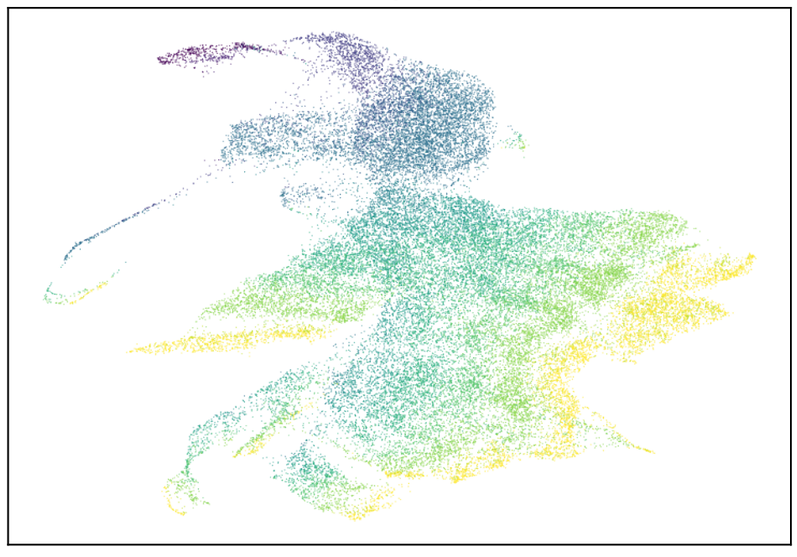
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



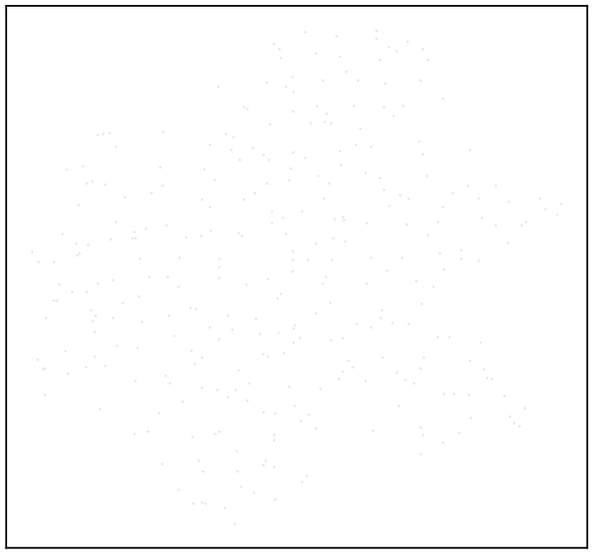
atlas, unaligned: stages



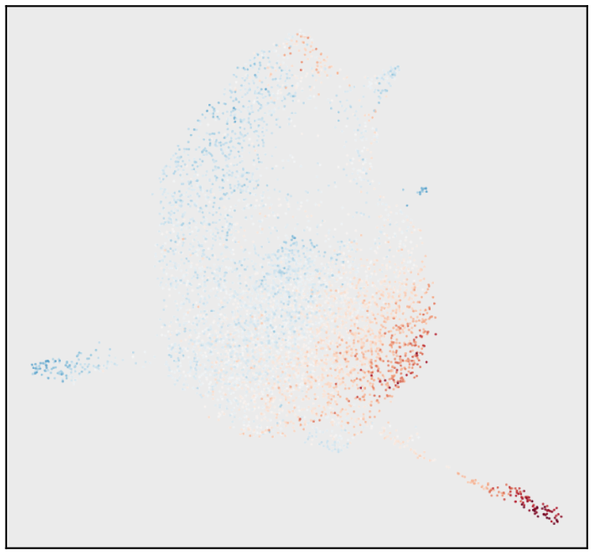
Genes (number of stages): szl (9), cxcl12b (9), si:dkey-261j4.5 (9), blf (9), rcn3 (9), lpar1 (9), adam8a (9), cmtm6 (9), si:dkey-261j4.4 (9), ip6k2b (9), map1b (9), si:dkey-27i16.2 (9), si:ch211-197h24.9 (9), hmgb3a (9), nrarpa (9), pdlim1 (9), dhrs9 (9), sept15 (9), tmem136a (9), hey1 (8), tbx6 (7), epha7 (7), angptl7 (6), lect1 (6), hbae3 (5), hoxb5b (5), hoxb5a (5), foxl2 (3), sst1.1 (3), lrrc17 (2)

neural_plate

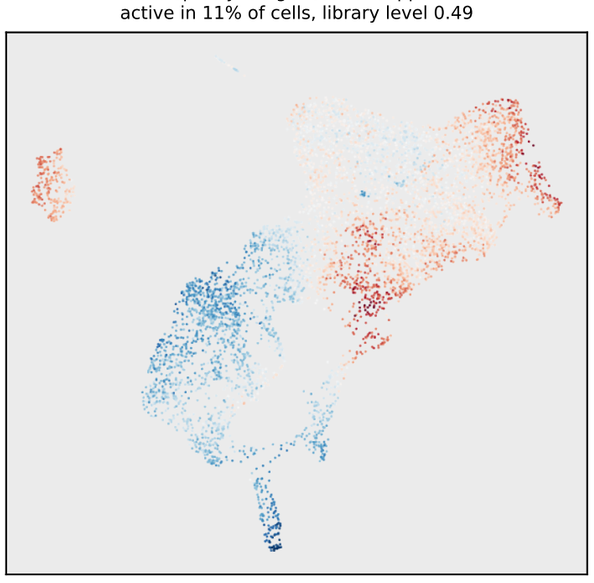
high: not defined



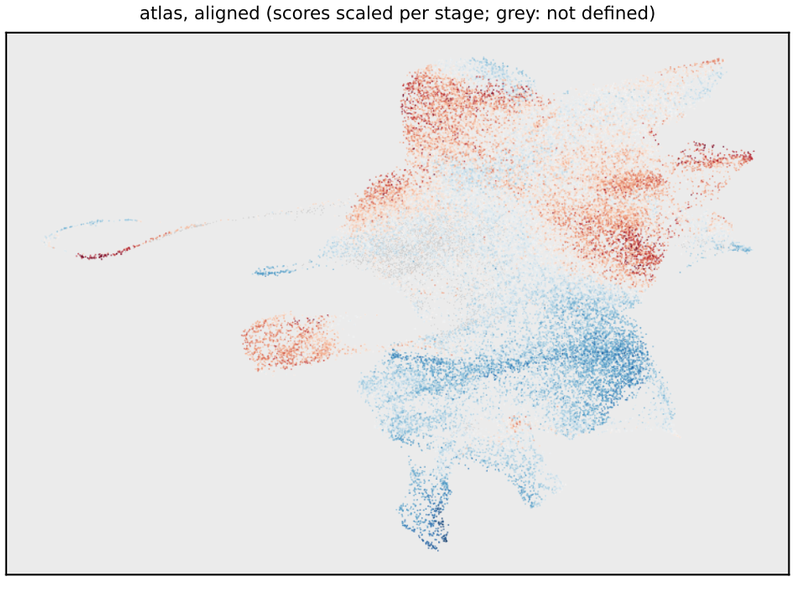
50% epiboly: 25 genes, not supported
active in 6% of cells, library level 0.14



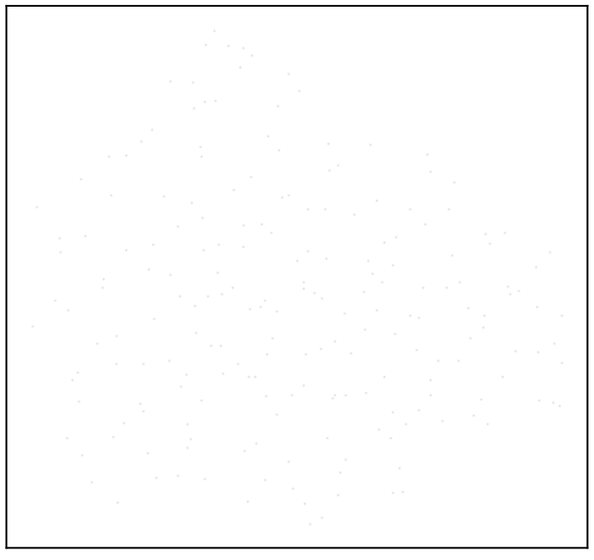
90% epiboly: 30 genes, not supported
active in 11% of cells, library level 0.49



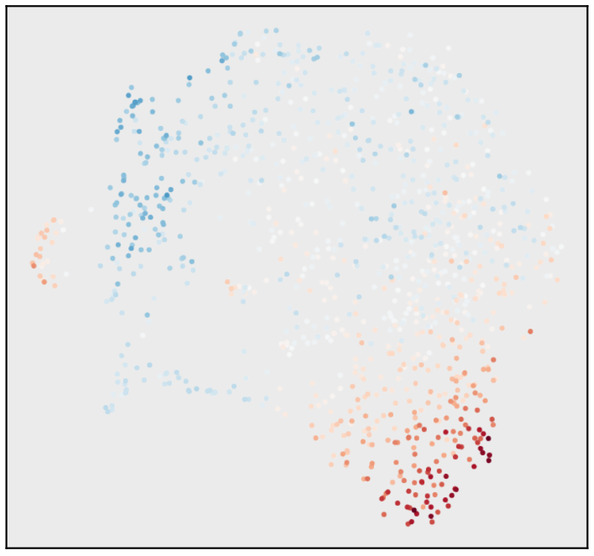
atlas, aligned (scores scaled per stage; grey: not defined)



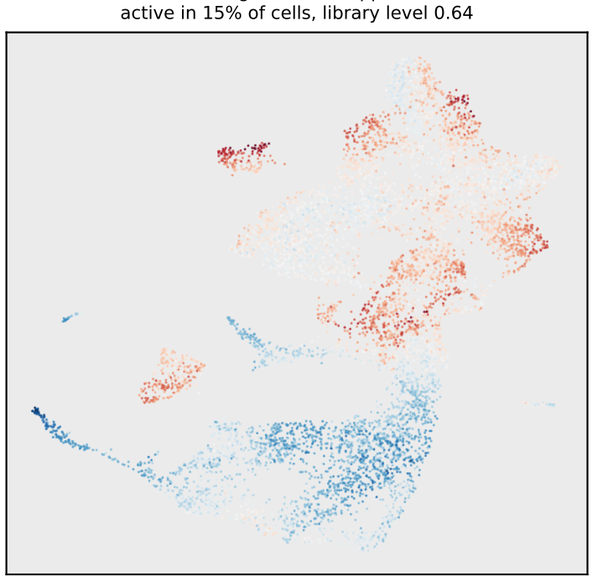
oblong: not defined



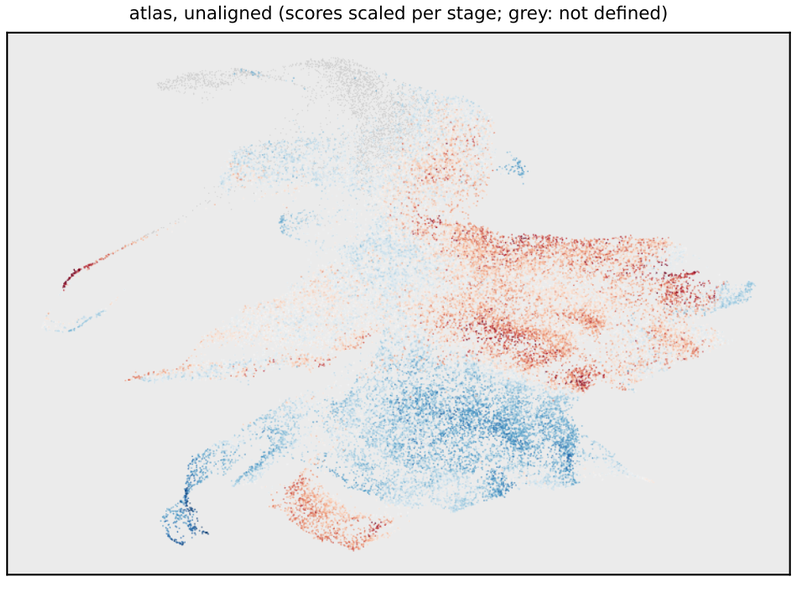
shield: 21 genes, not supported
active in 8% of cells, library level 0.36



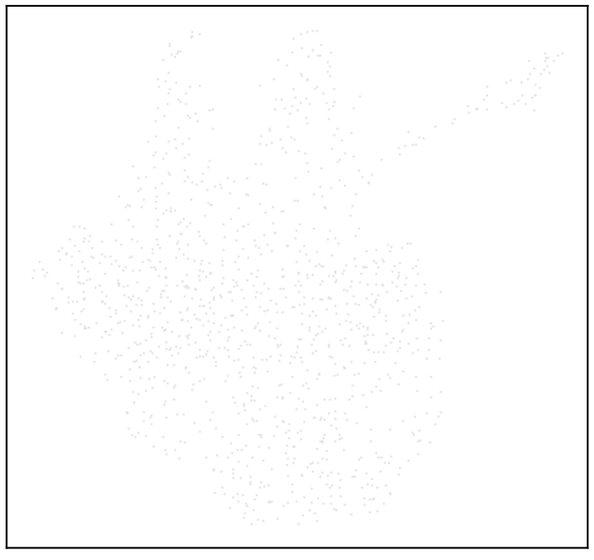
bud: 30 genes, not supported
active in 15% of cells, library level 0.64



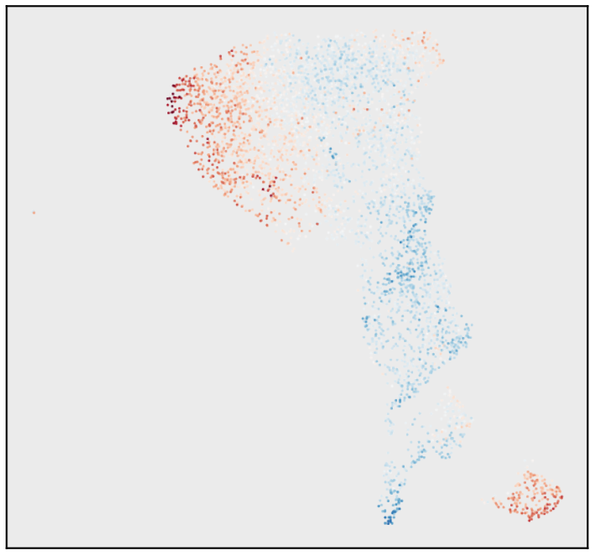
atlas, unaligned (scores scaled per stage; grey: not defined)



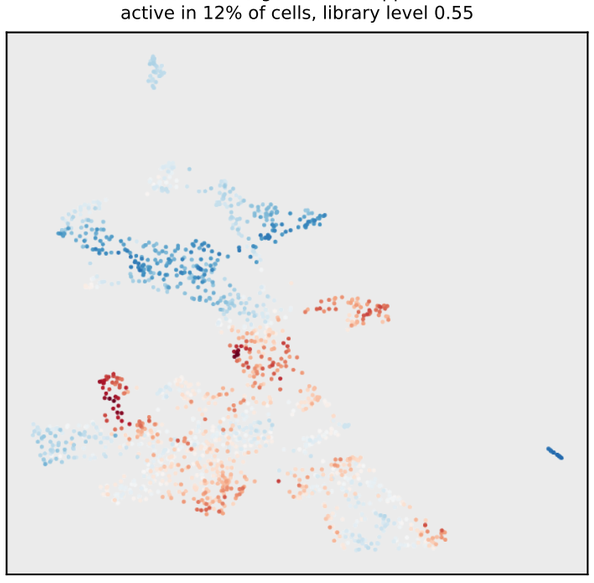
dome: not defined



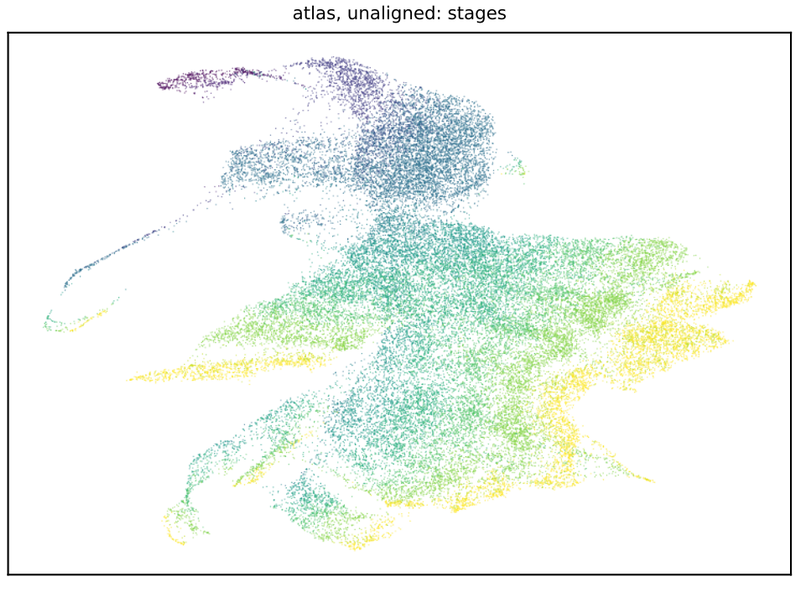
60% epiboly: 28 genes, not supported
active in 10% of cells, library level 0.21



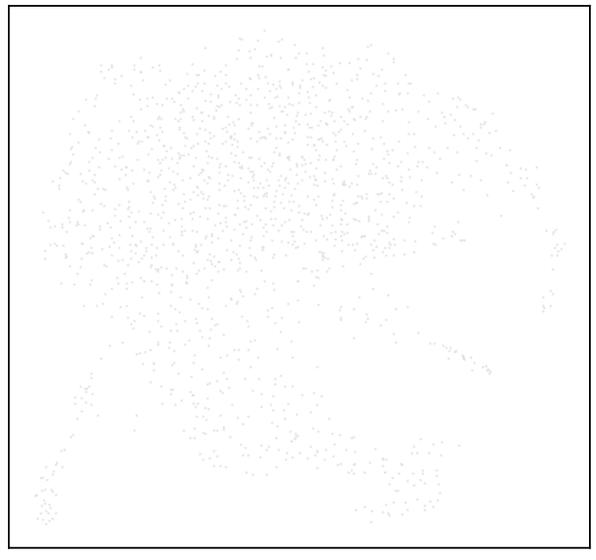
3-somite: 29 genes, not supported
active in 12% of cells, library level 0.55



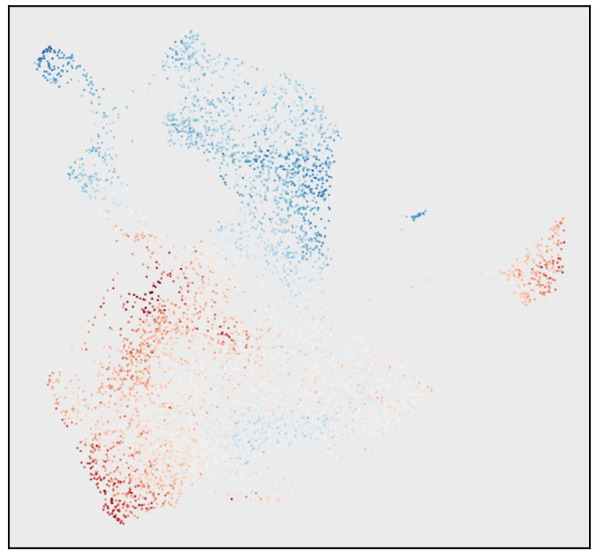
atlas, unaligned: stages



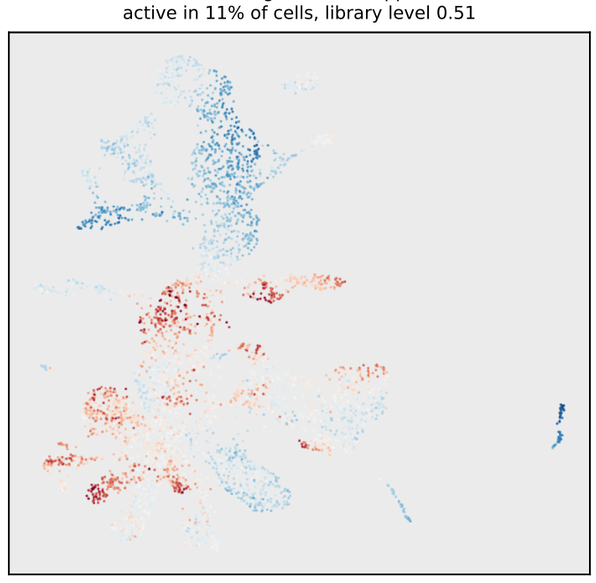
30% epiboly: not defined

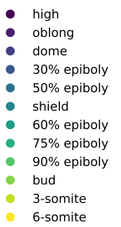


75% epiboly: 30 genes, not supported
active in 9% of cells, library level 0.30



6-somite: 30 genes, not supported
active in 11% of cells, library level 0.51

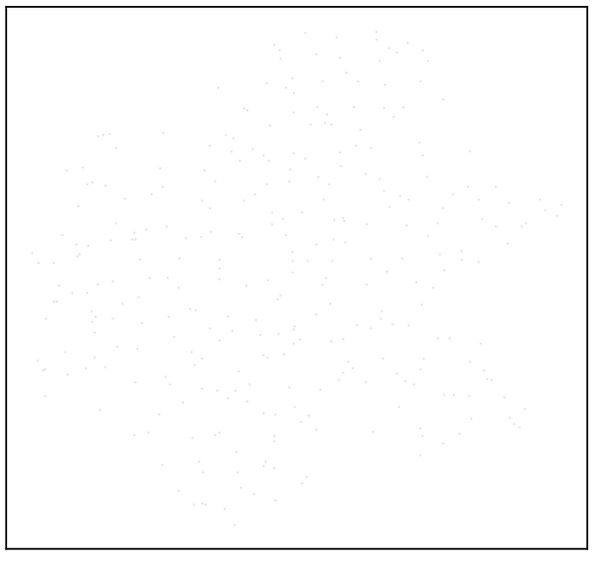




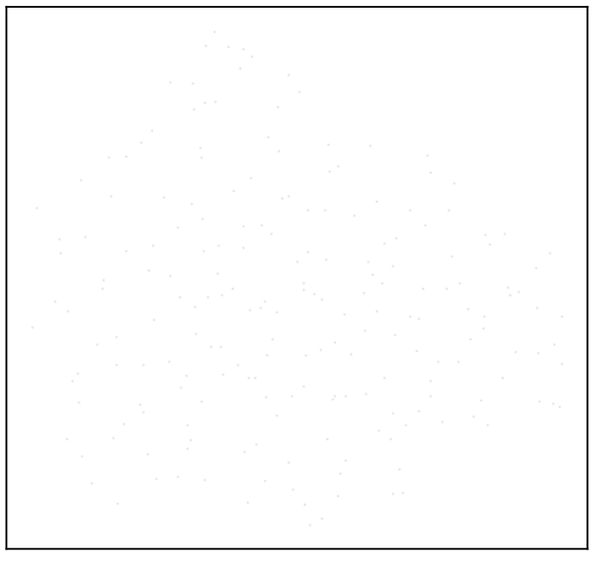
Genes (number of stages): her3 (8), hesx1 (8), msx1b (8), hspb1 (8), dlx3b (8), cdh6 (8), akap12b (8), mdka (8), mdkb (8), si:ch211-152c2.3 (8), sox19a (8), klf6a (8), rxraa (8), her9 (8), olig4 (8), olig3 (8), sp5a (8), asb11 (8), atp1b1a (8), alcamb (8), nkx1.2la (7), olig2 (7), alcama (7), slc2a12 (7), fezf2 (6), nkx2.9 (6), fstl1a (6), nkx6.2 (6), pax3a (6), nkx2.4b (5)

presumptive_neural

high: not defined



oblong: not defined

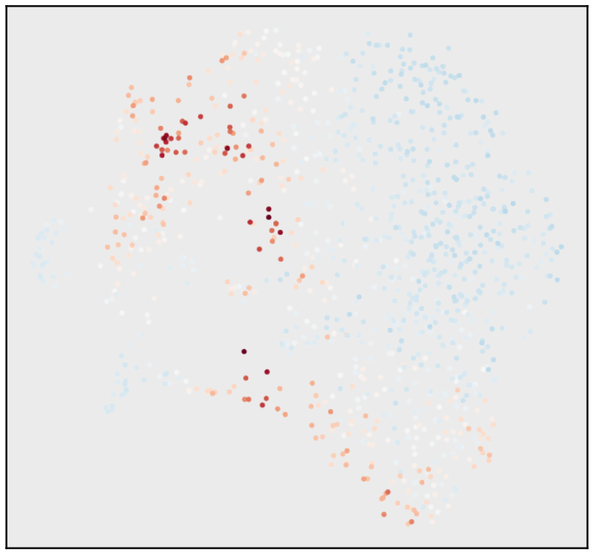


dome: not defined

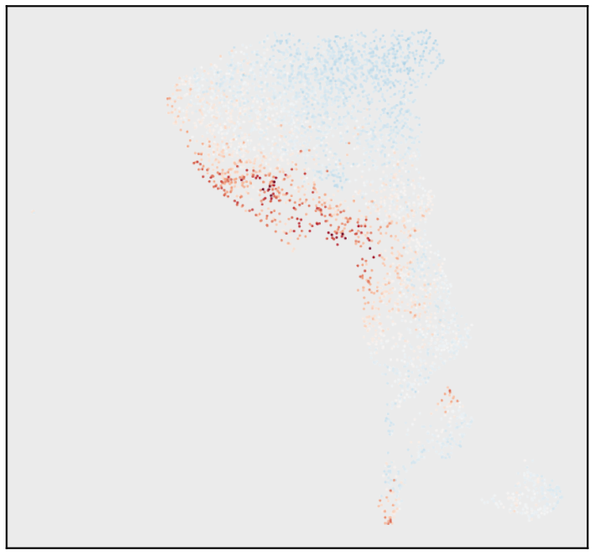
30% epiboly: not defined

50% epiboly: not defined

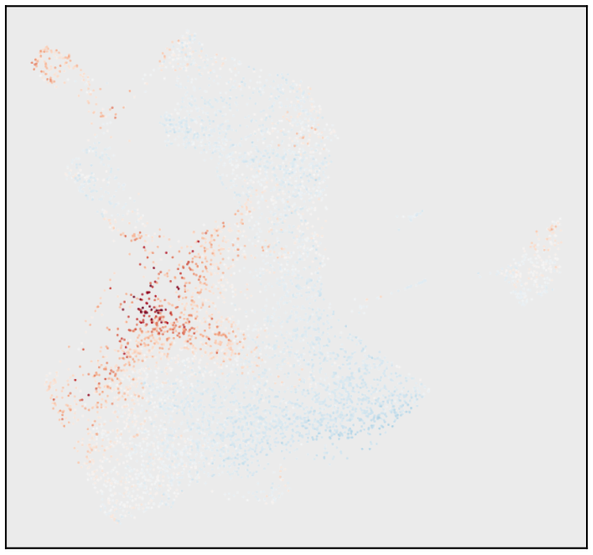
shield: 3 genes, not supported
active in 4% of cells, library level 0.72



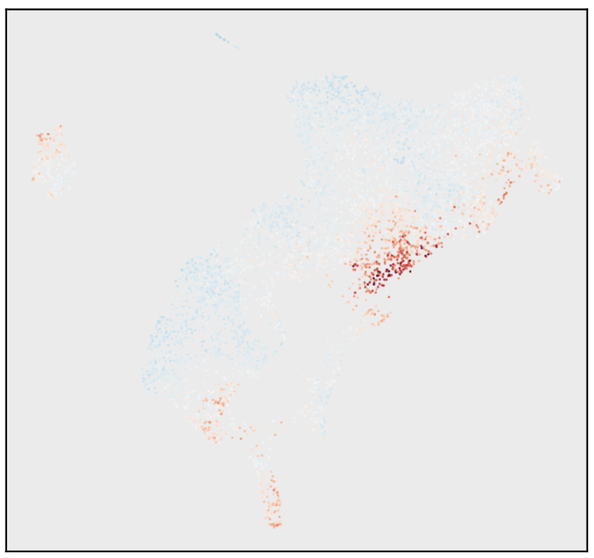
60% epiboly: 7 genes, not supported
active in 6% of cells, library level 0.20



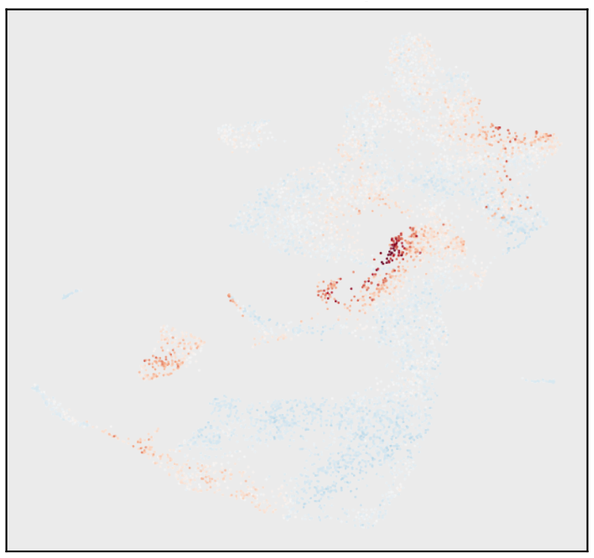
75% epiboly: 9 genes, not supported
active in 8% of cells, library level 0.29



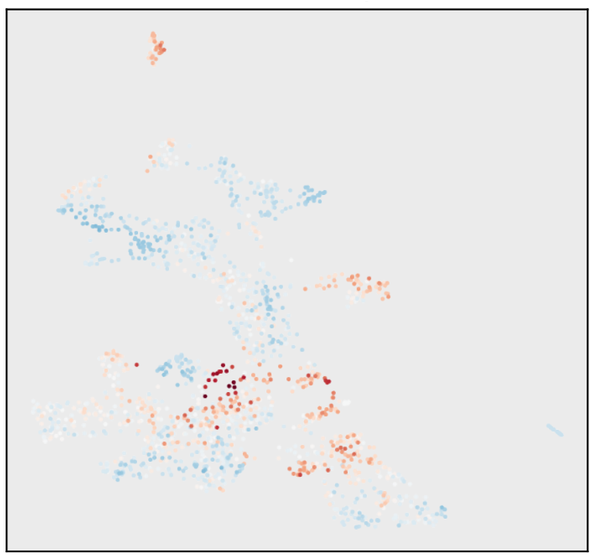
90% epiboly: 9 genes, not supported
active in 5% of cells, library level 0.46



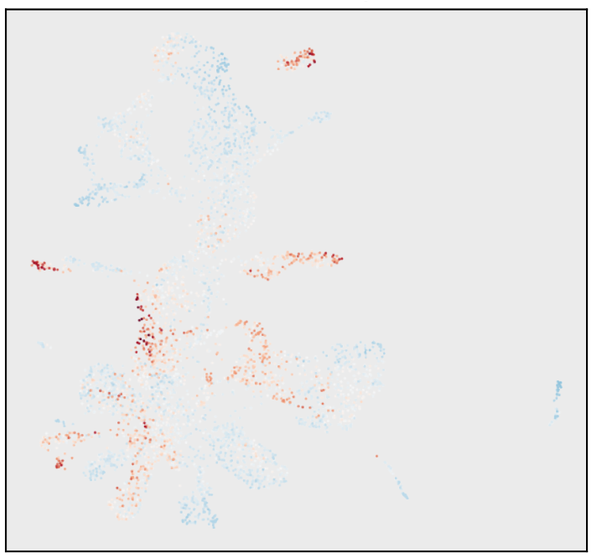
bud: 9 genes, not supported
active in 5% of cells, library level 0.54



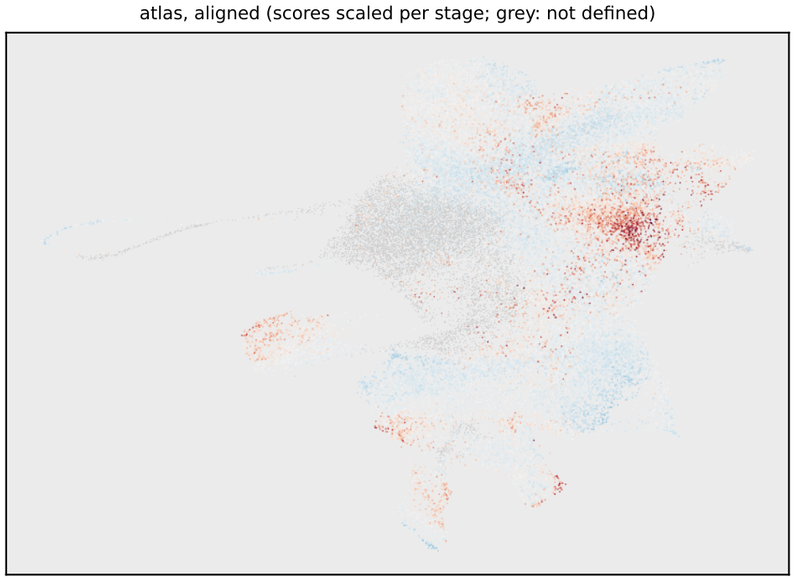
3-somite: 9 genes, not supported
active in 2% of cells, library level 0.56



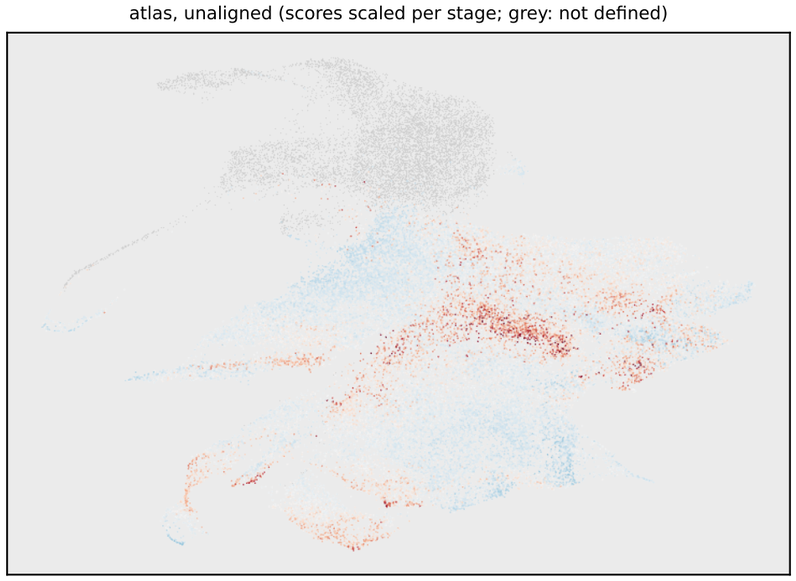
6-somite: 9 genes, not supported
active in 3% of cells, library level 0.64



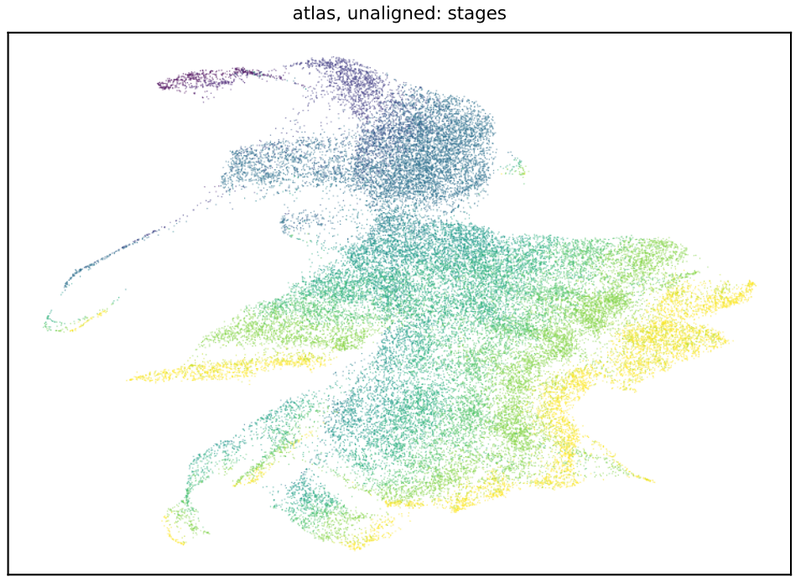
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



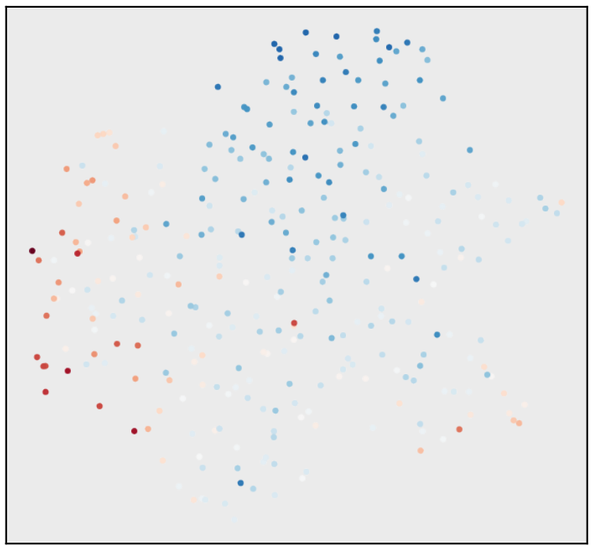
atlas, unaligned: stages



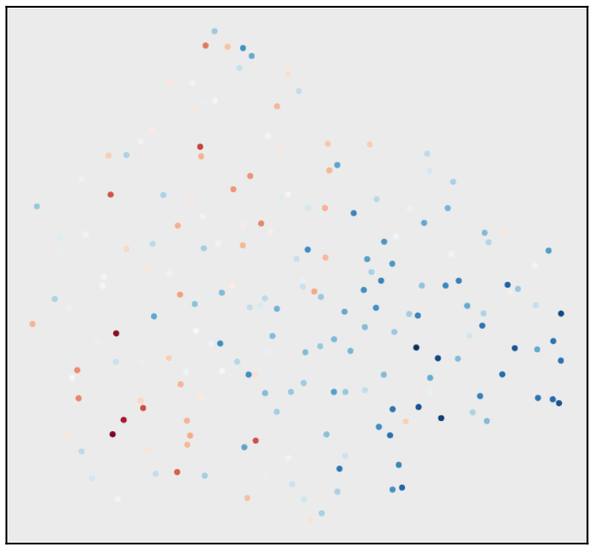
Genes (number of stages): otx2 (7), foxb1a (7), wnt11 (7), hoxb1b (6), nkx6.2 (6), nkx6.1 (6), foxp1b (6), six1b (5), hoxc10a (5)

signal:BMP

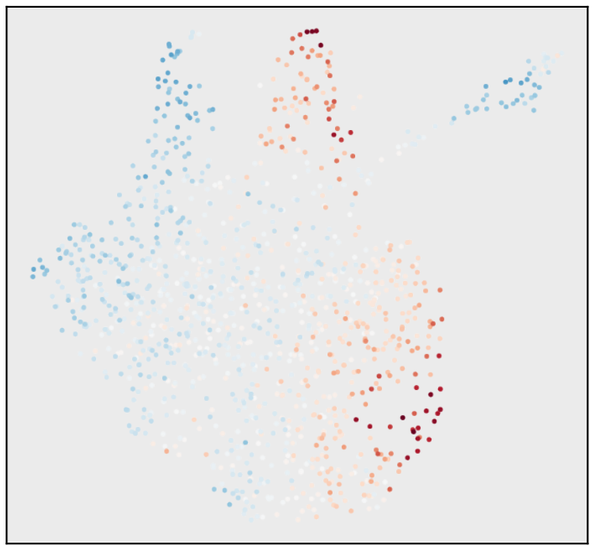
high: 7 genes, supported
active in 1% of cells, library level 0.07



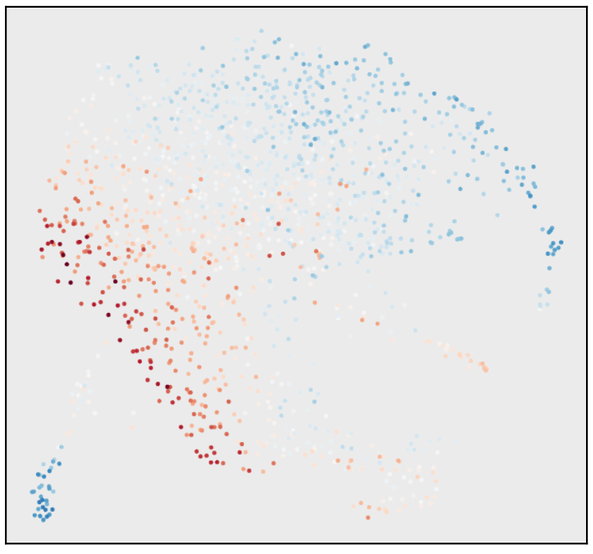
oblong: 5 genes, not supported
active in 1% of cells, library level 0.16



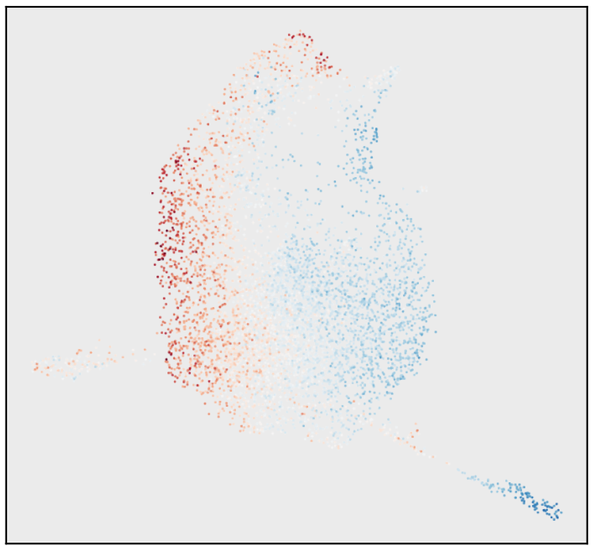
dome: 8 genes, not supported
active in 8% of cells, library level 0.41



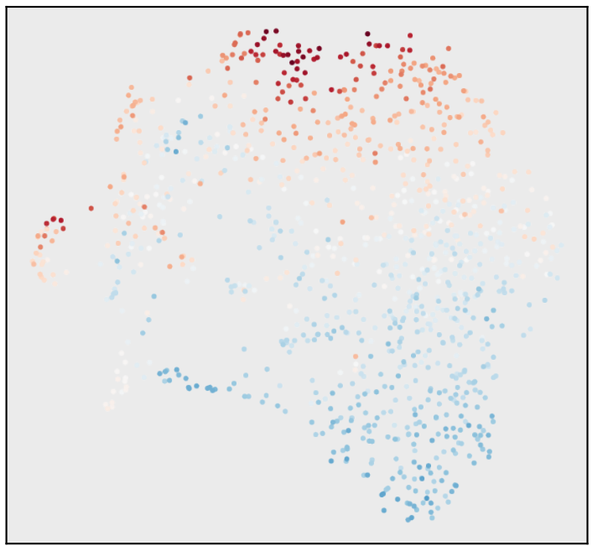
30% epiboly: 8 genes, not supported
active in 14% of cells, library level 0.84



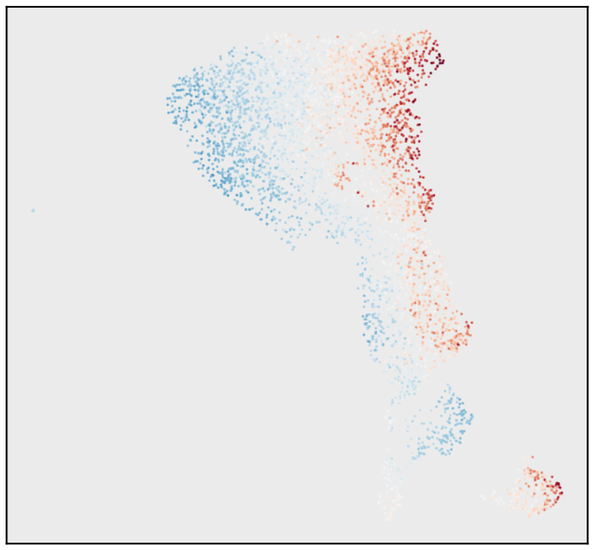
50% epiboly: 8 genes, not supported
active in 15% of cells, library level 0.56



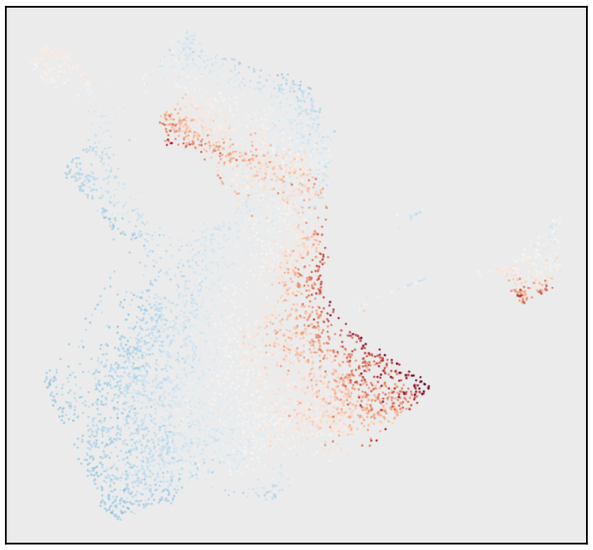
shield: 8 genes, supported
active in 15% of cells, library level 0.75



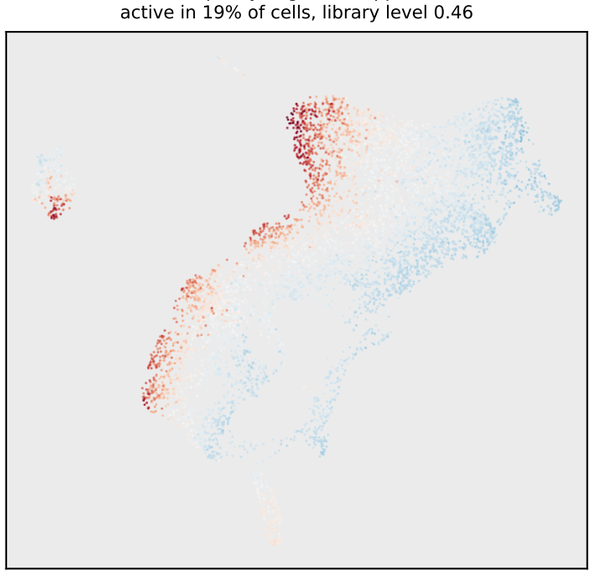
60% epiboly: 8 genes, supported
active in 18% of cells, library level 0.54



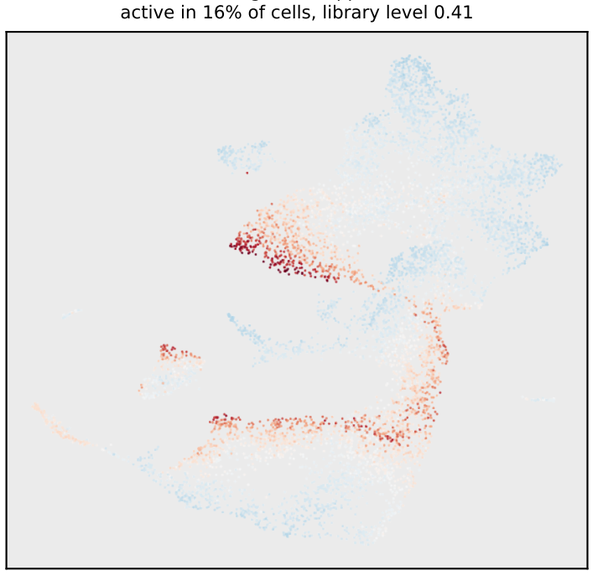
75% epiboly: 8 genes, supported
active in 19% of cells, library level 0.53



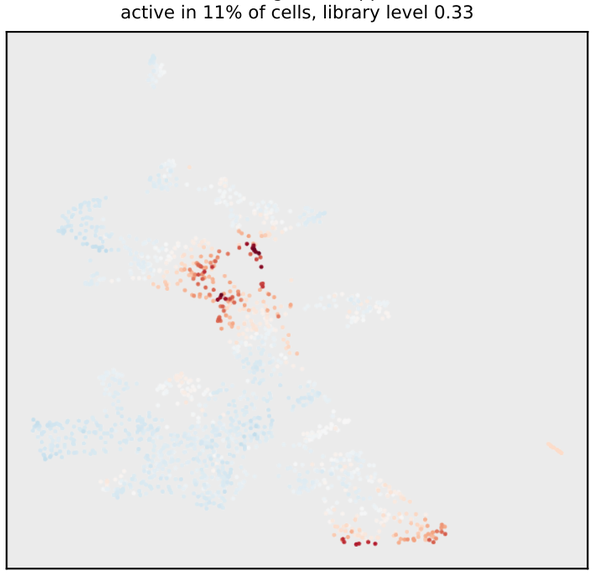
90% epiboly: 8 genes, supported
active in 19% of cells, library level 0.46



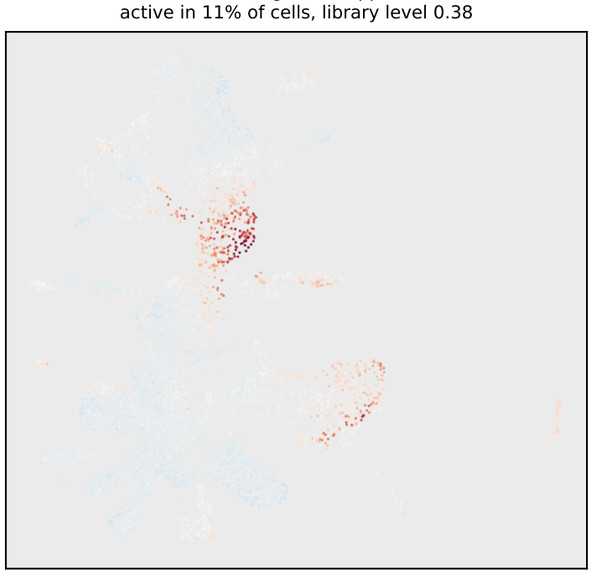
bud: 8 genes, supported
active in 16% of cells, library level 0.41



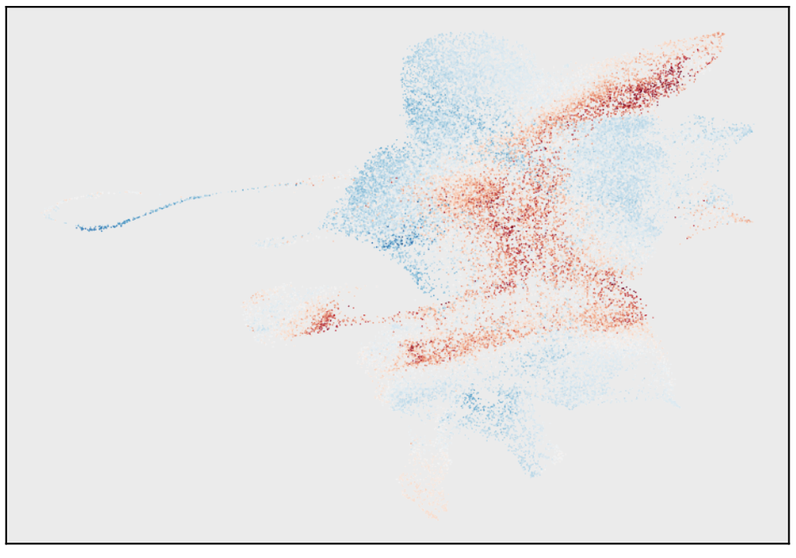
3-somite: 8 genes, supported
active in 11% of cells, library level 0.33



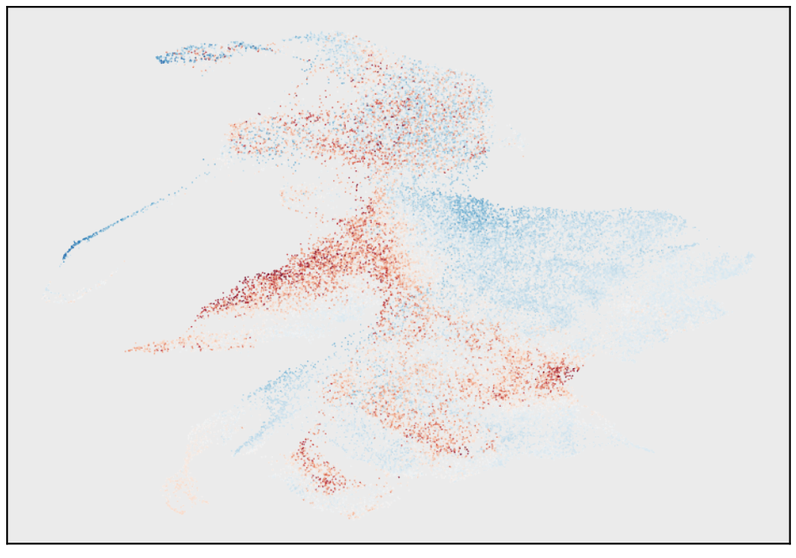
6-somite: 8 genes, supported
active in 11% of cells, library level 0.38



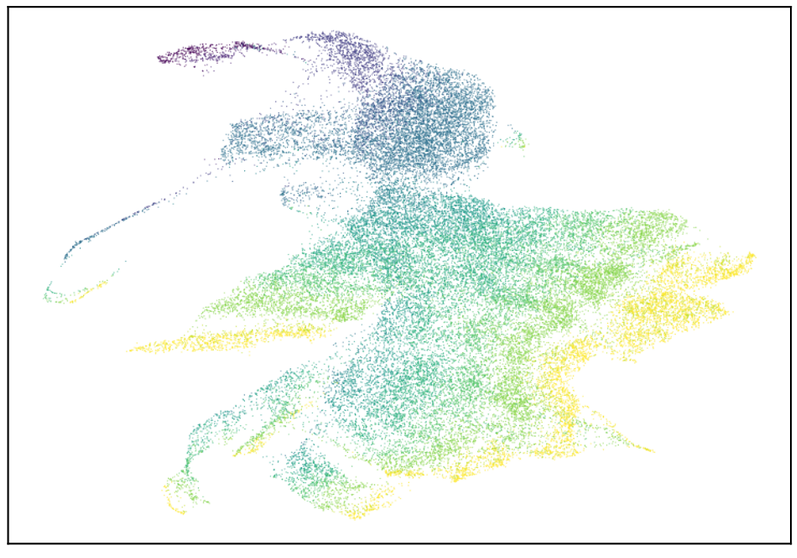
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



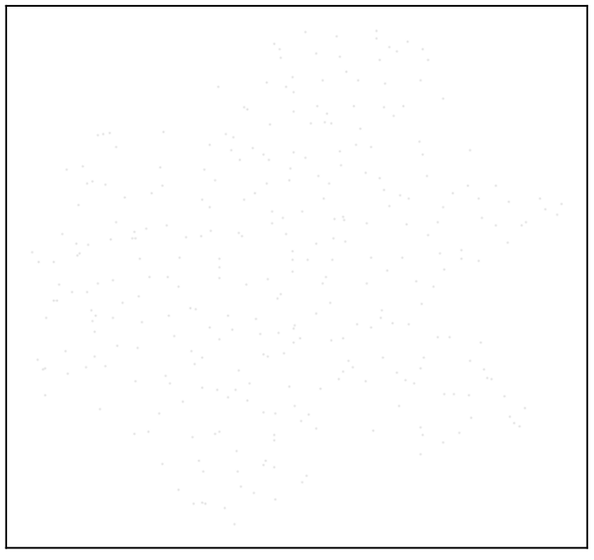
atlas, unaligned: stages



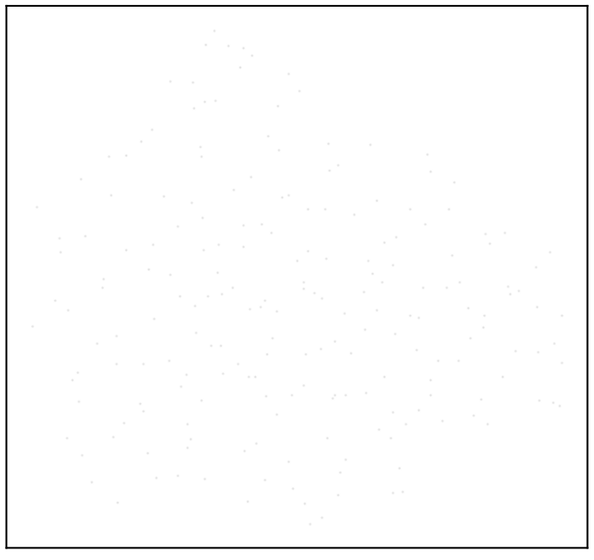
Genes (number of stages): id1 (12), id3 (12), bambia (12), ved (12), vox (12), msx1b (11), vent (11), szl (10)

signal:FGF

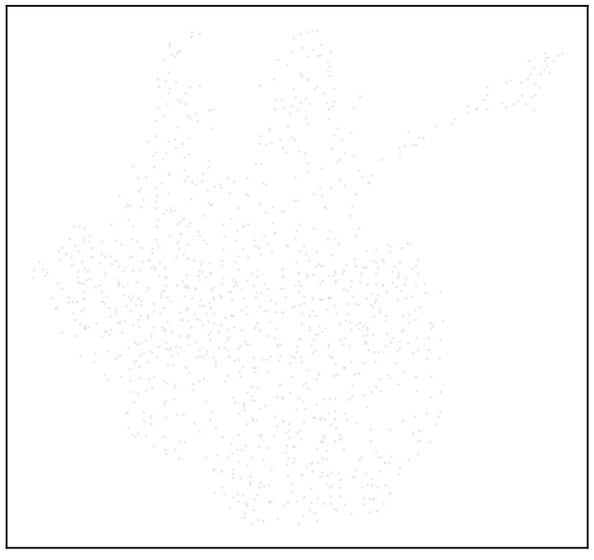
high: not defined



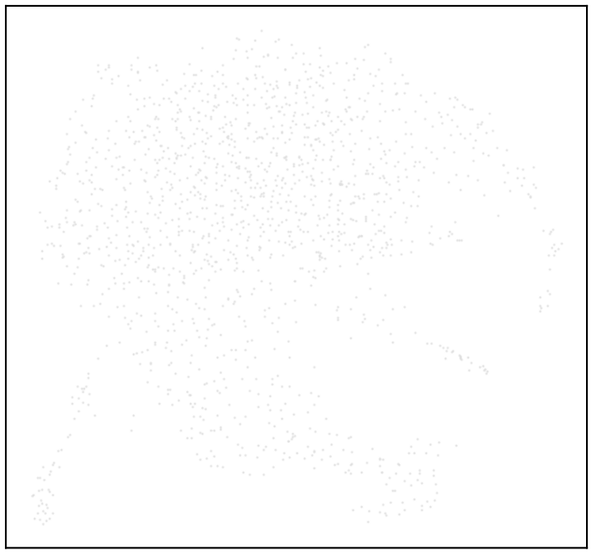
oblong: not defined



dome: not defined

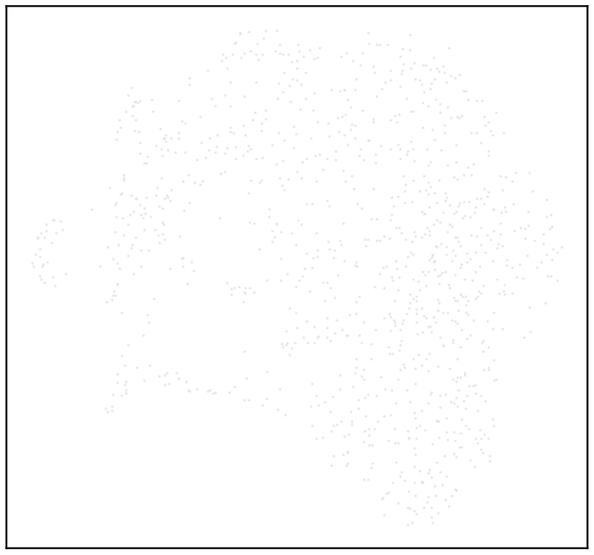


30% epiboly: not defined

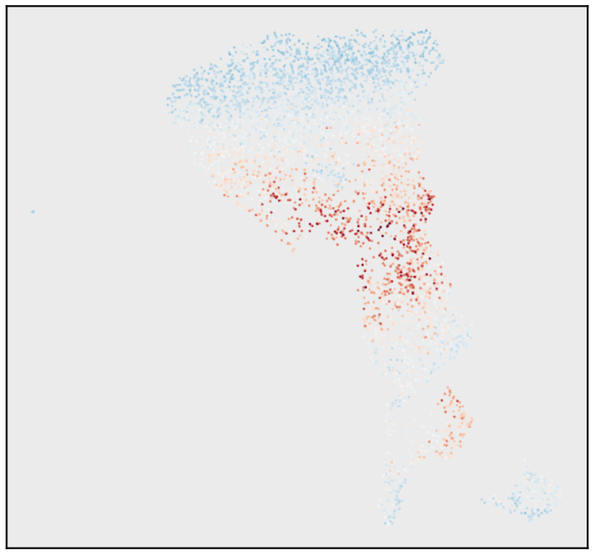


50% epiboly: not defined

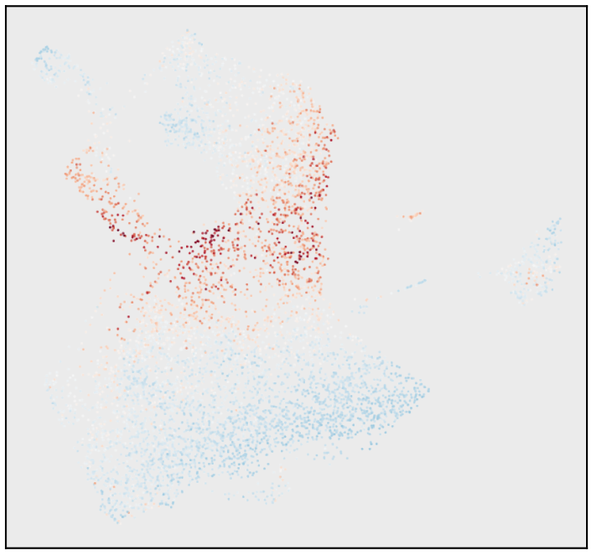
shield: not defined



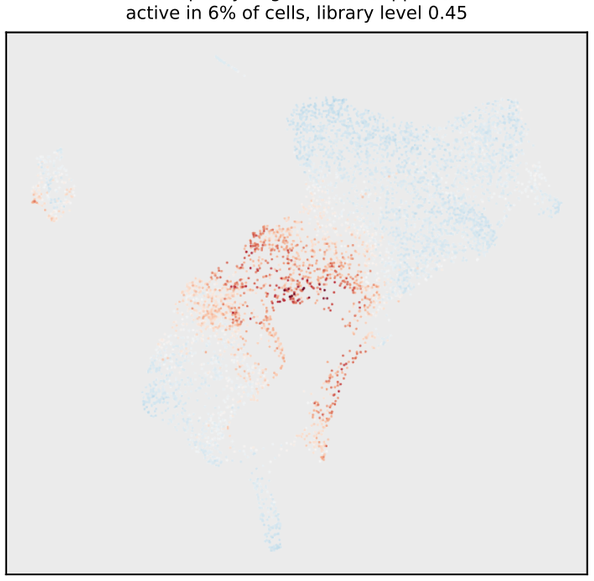
60% epiboly: 2 genes, not supported
active in 7% of cells, library level 0.65



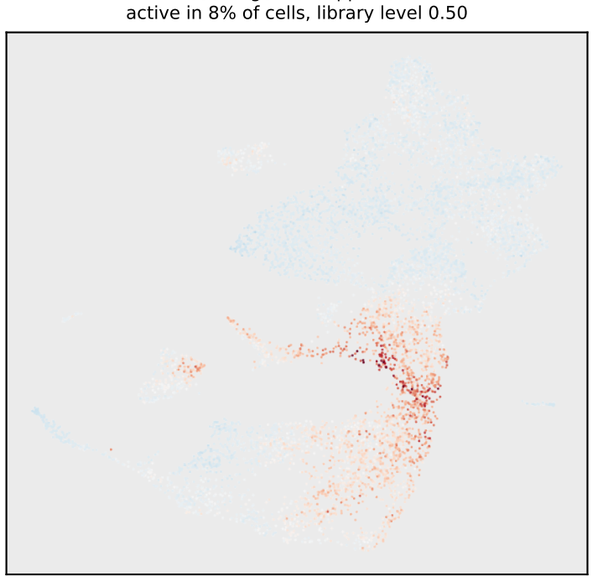
75% epiboly: 2 genes, not supported
active in 6% of cells, library level 0.56



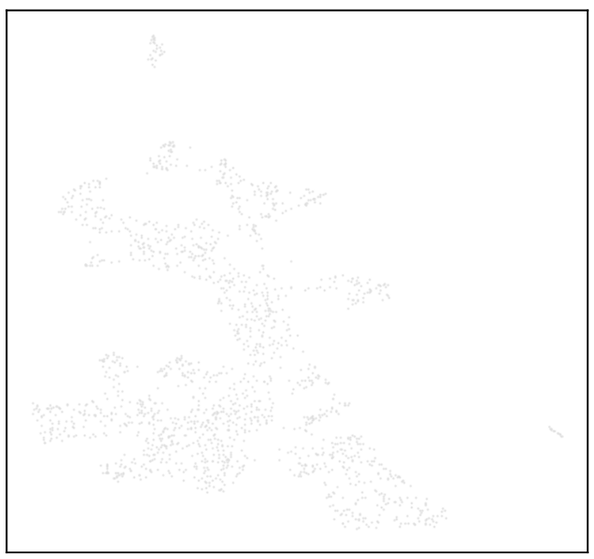
90% epiboly: 2 genes, not supported
active in 6% of cells, library level 0.45



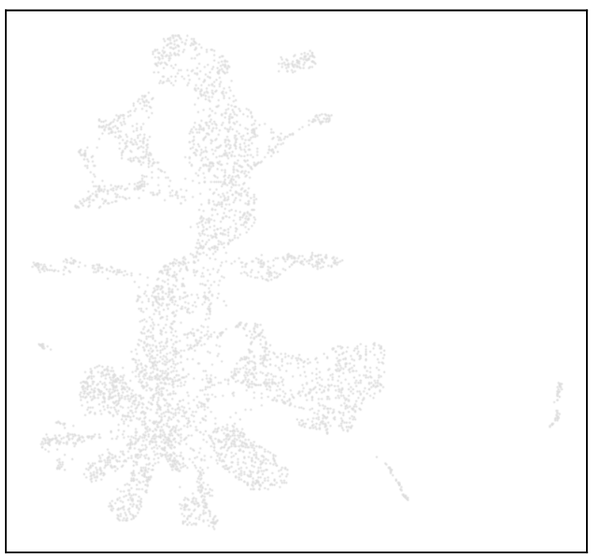
bud: 2 genes, supported
active in 8% of cells, library level 0.50



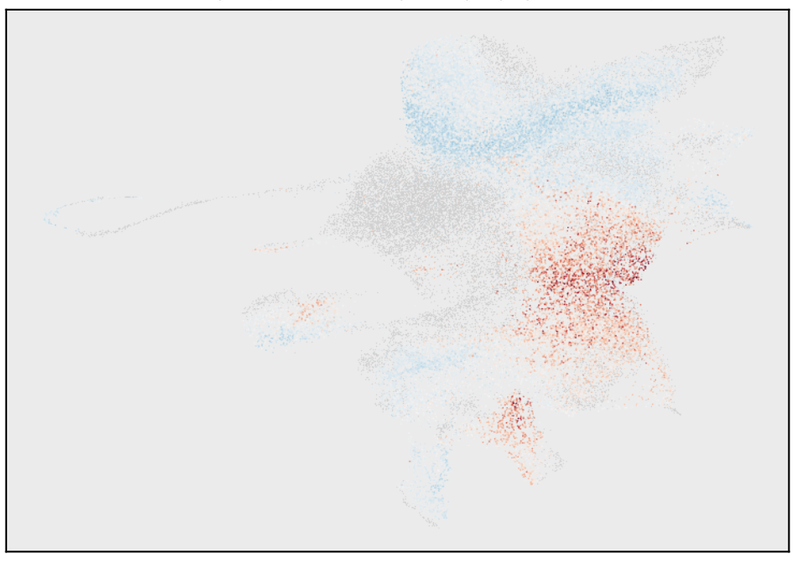
3-somite: not defined



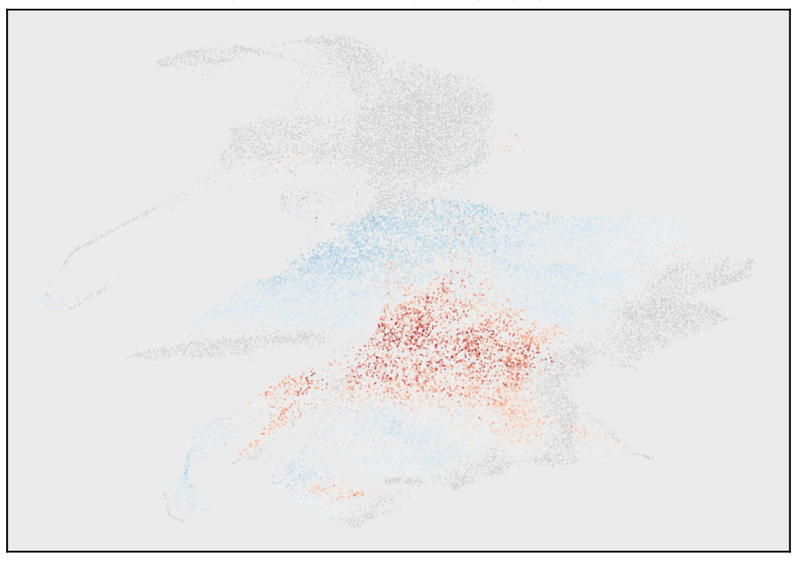
6-somite: not defined



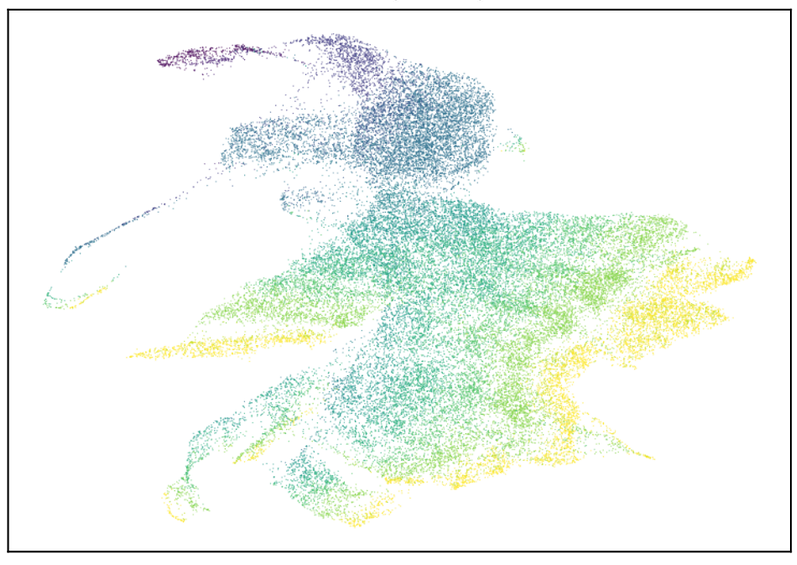
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



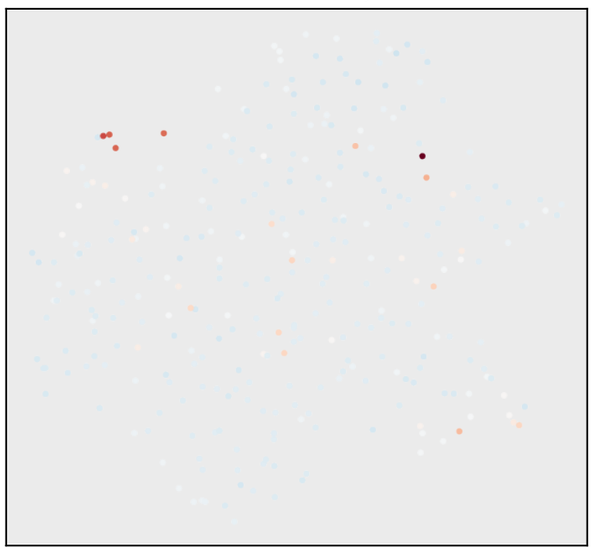
atlas, unaligned: stages



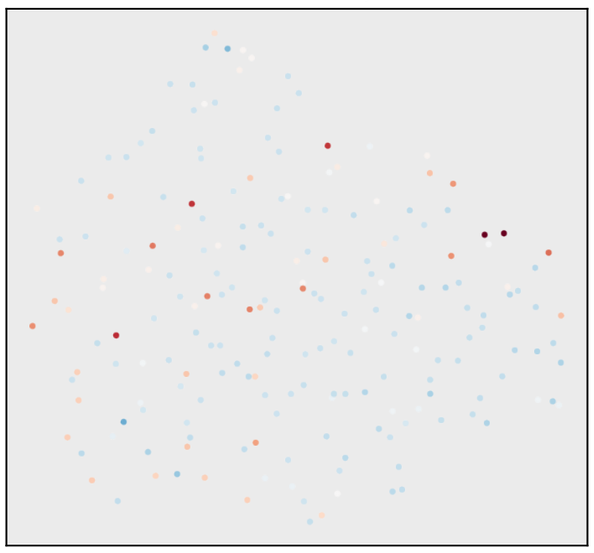
Genes (number of stages): etv4 (4), il17rd (4)

signal:Nodal

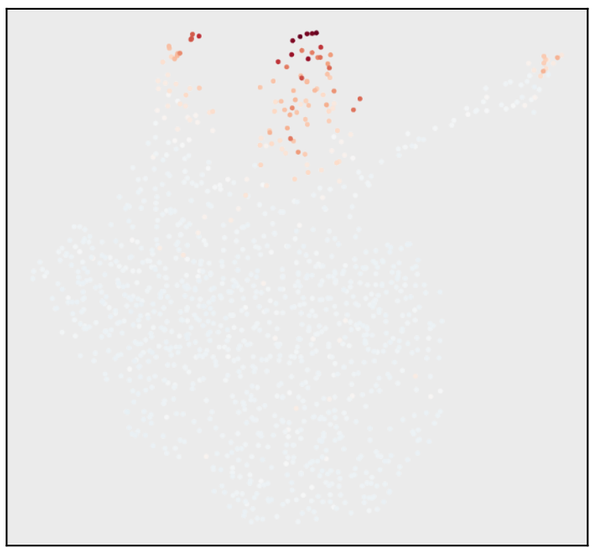
high: 3 genes, not supported
active in 1% of cells, library level 0.14



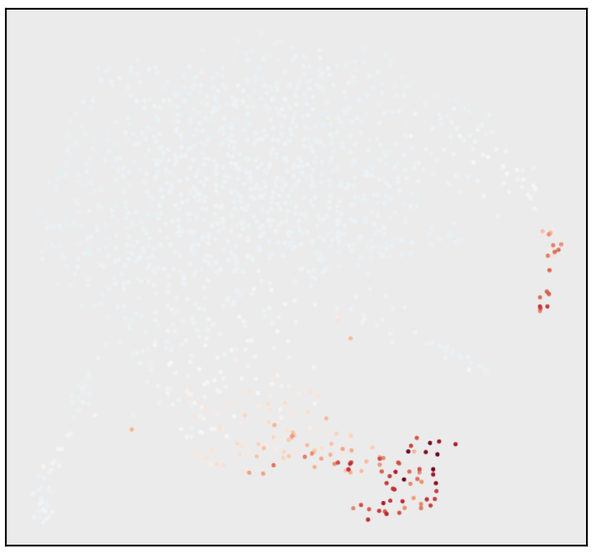
oblong: 3 genes, not supported
active in 1% of cells, library level 0.19



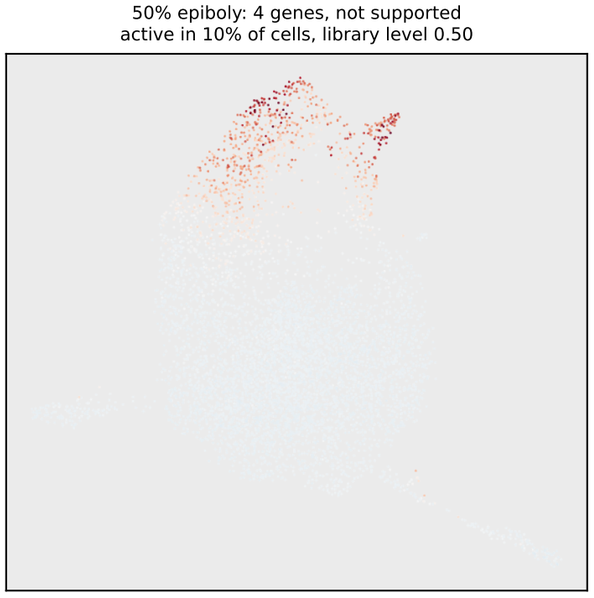
dome: 3 genes, supported
active in 7% of cells, library level 0.56



30% epiboly: 4 genes, supported
active in 7% of cells, library level 0.70

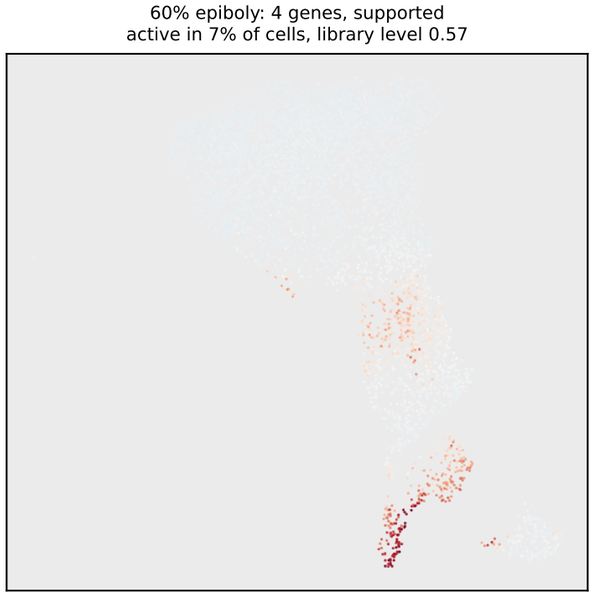


50% epiboly: 4 genes, not supported
active in 10% of cells, library level 0.50



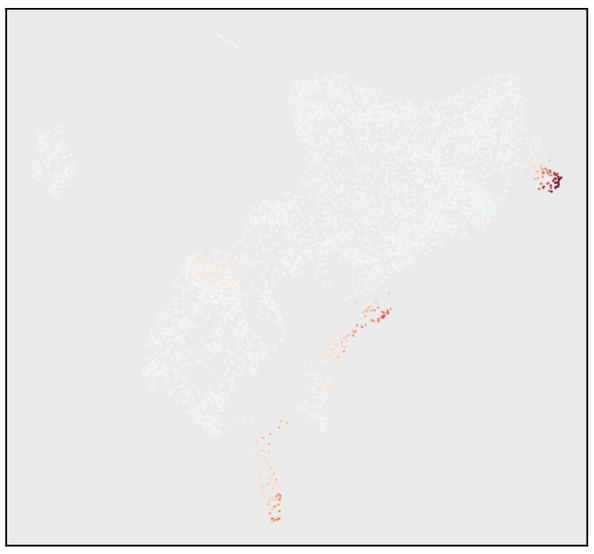
shield: 4 genes, not supported
active in 4% of cells, library level 0.38

60% epiboly: 4 genes, supported
active in 7% of cells, library level 0.57

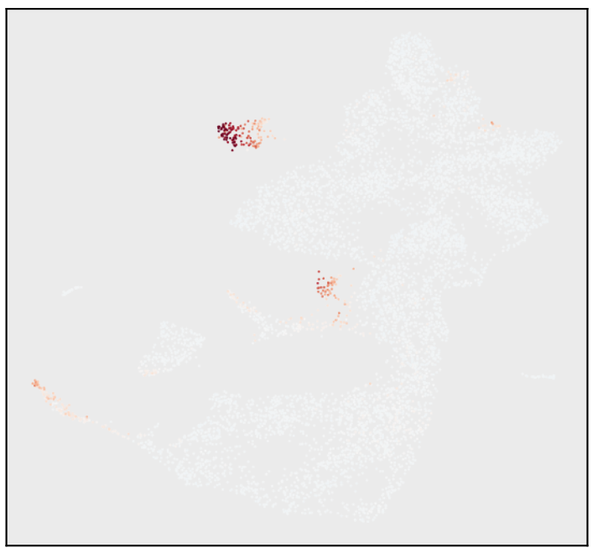


75% epiboly: 4 genes, supported
active in 5% of cells, library level 0.47

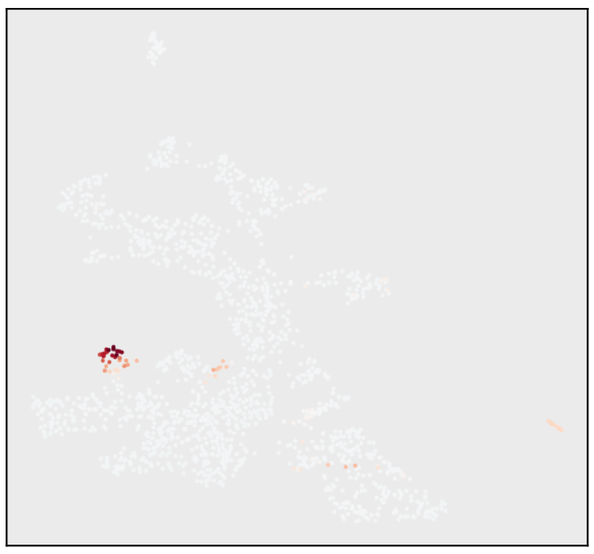
90% epiboly: 3 genes, supported
active in 4% of cells, library level 0.42



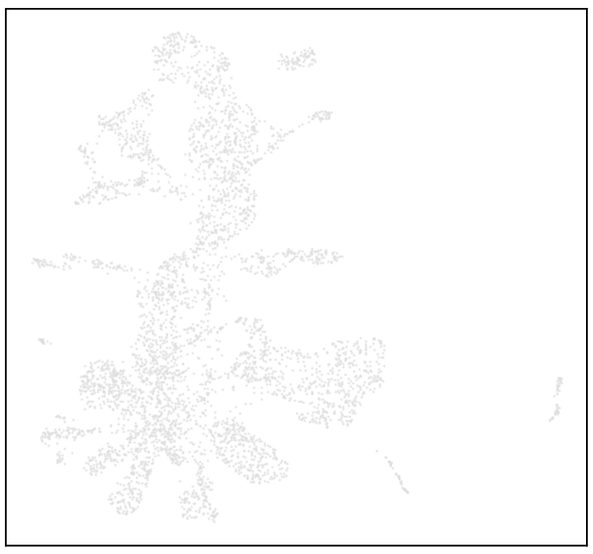
bud: 3 genes, supported
active in 3% of cells, library level 0.45



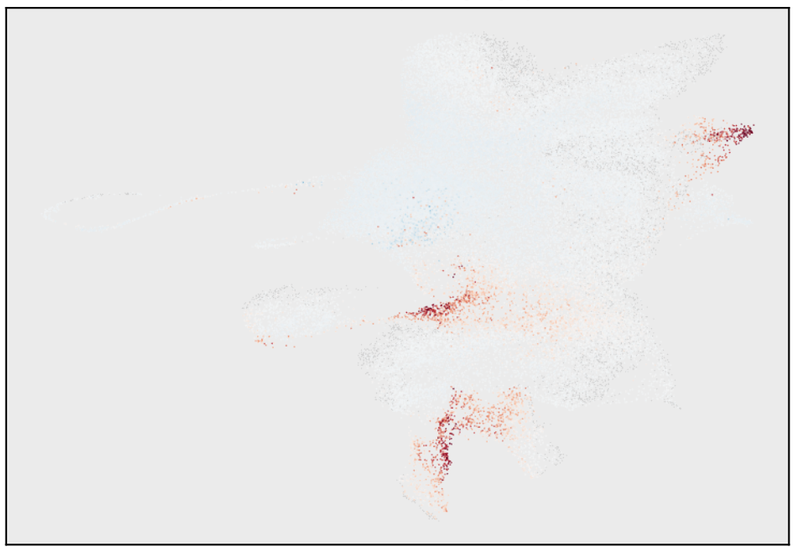
3-somite: 2 genes, supported
active in 3% of cells, library level 0.13



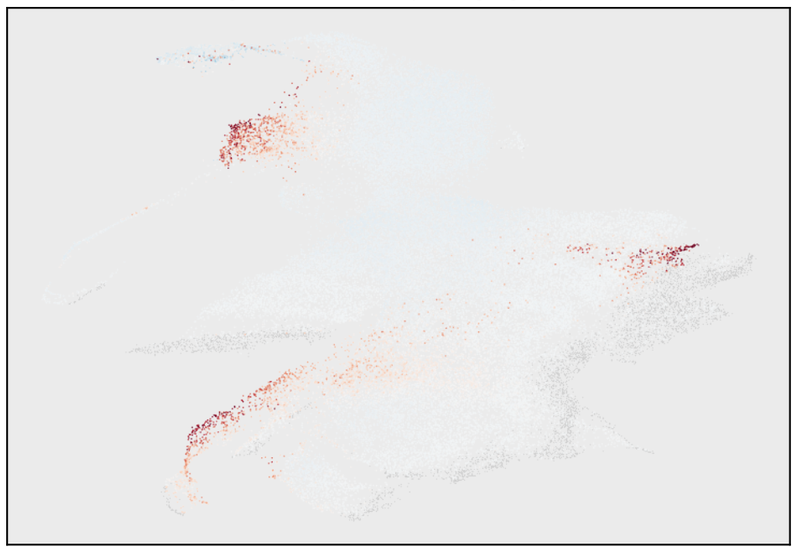
6-somite: not defined



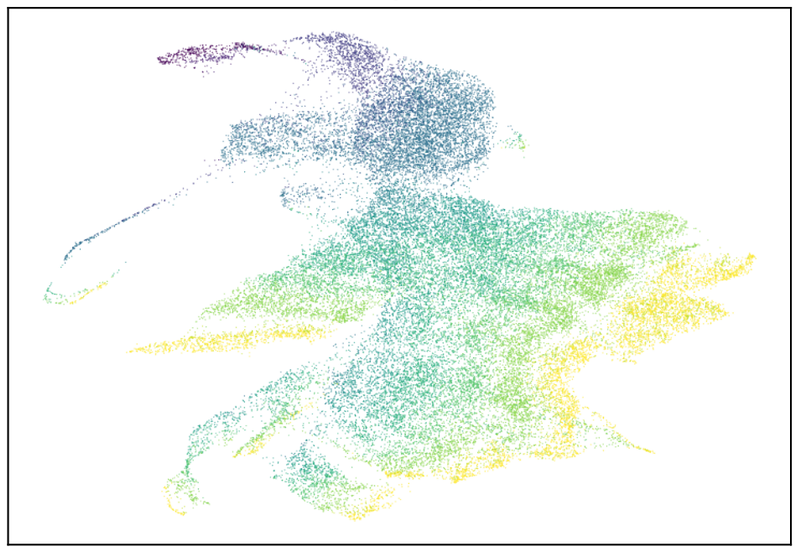
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



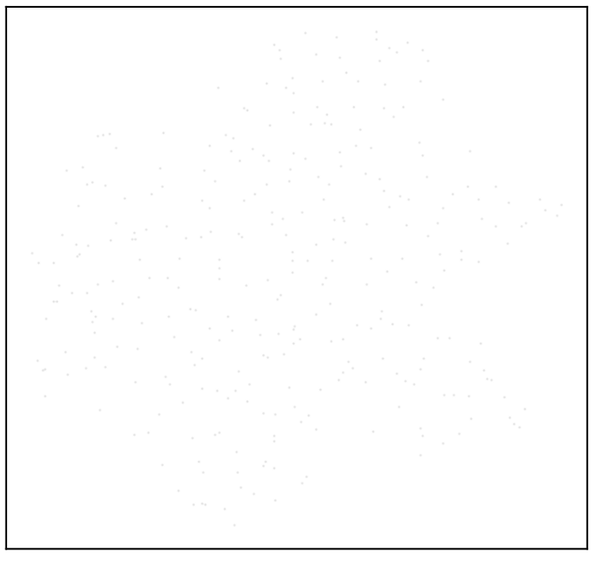
atlas, unaligned: stages



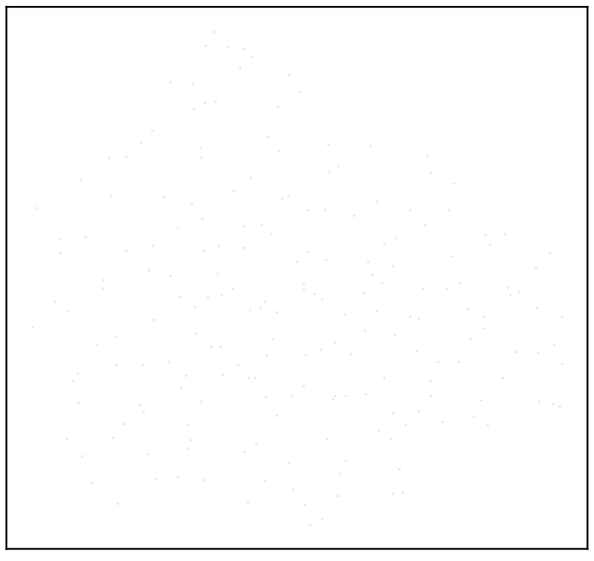
Genes (number of stages): lft1 (11), lft2 (11), ndr1 (8), ndr2 (7)

signal:Notch

high: not defined



oblong: not defined

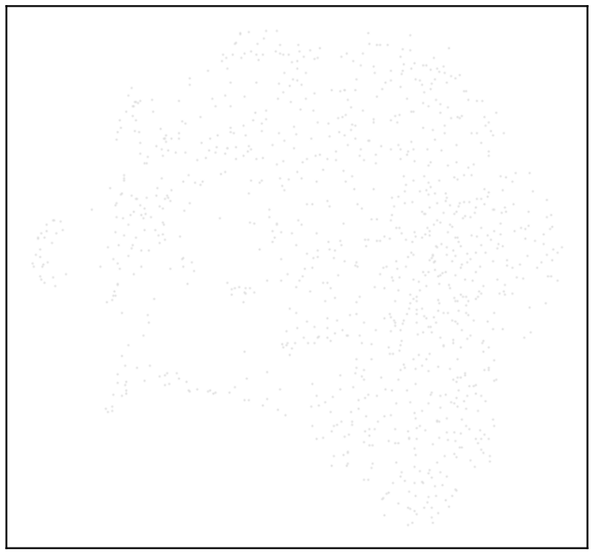


dome: not defined

30% epiboly: not defined

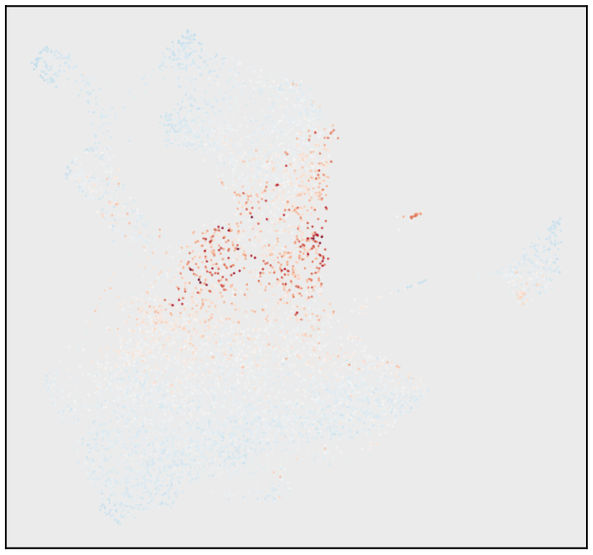
50% epiboly: not defined

shield: not defined

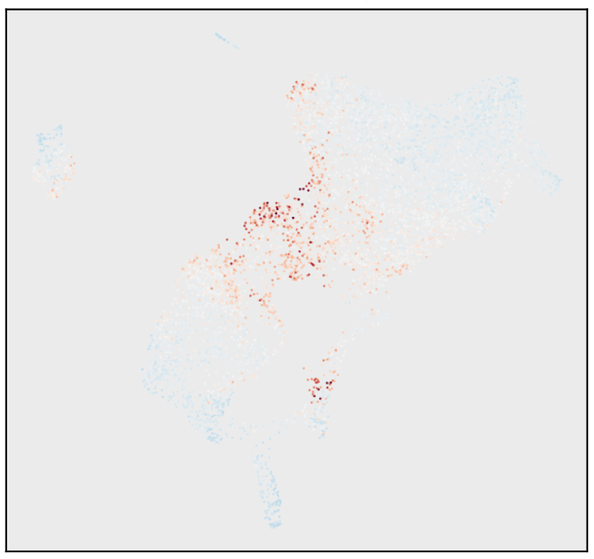


60% epiboly: not defined

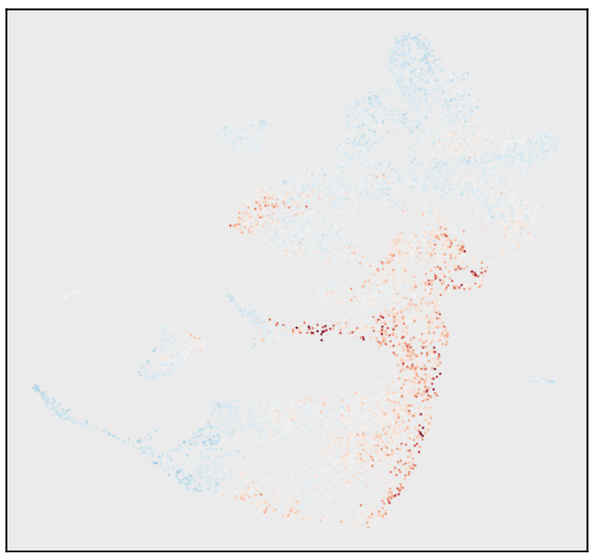
75% epiboly: 9 genes, not supported
active in 4% of cells, library level 0.07



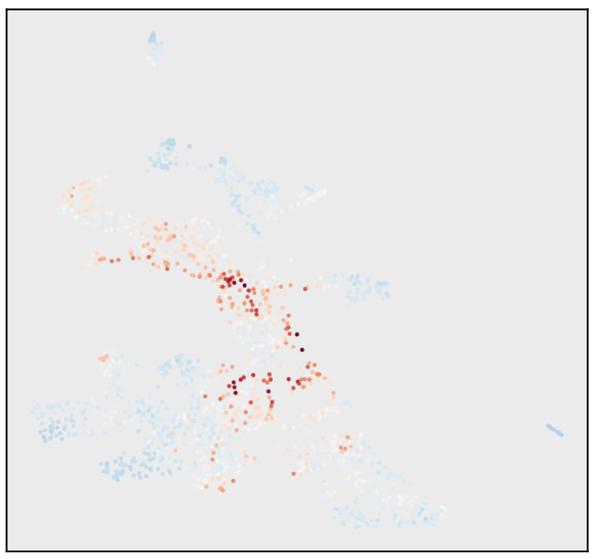
90% epiboly: 9 genes, not supported
active in 3% of cells, library level 0.13



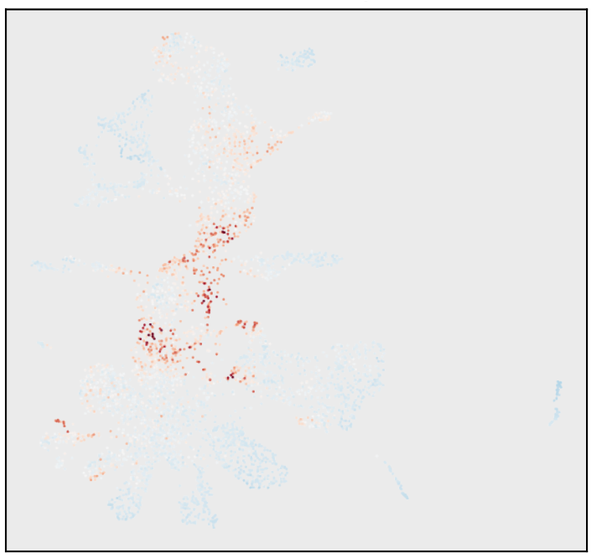
bud: 9 genes, supported
active in 5% of cells, library level 0.36



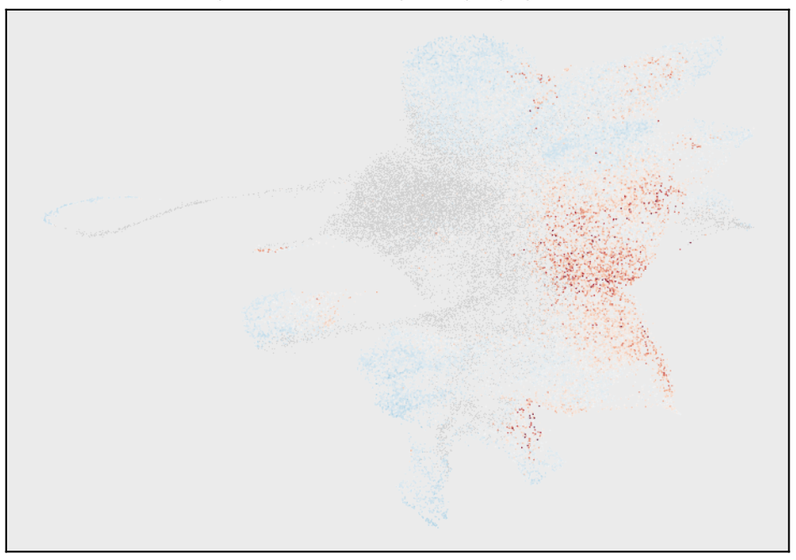
3-somite: 9 genes, supported
active in 7% of cells, library level 0.65



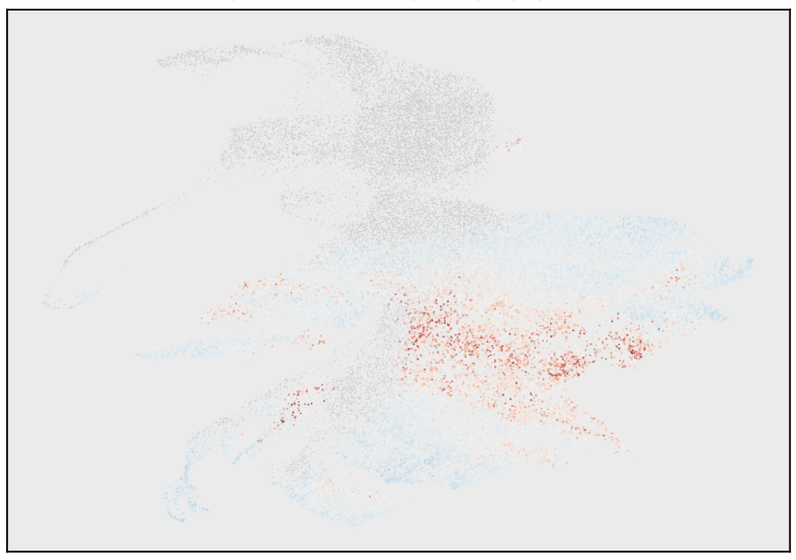
6-somite: 9 genes, supported
active in 7% of cells, library level 0.92



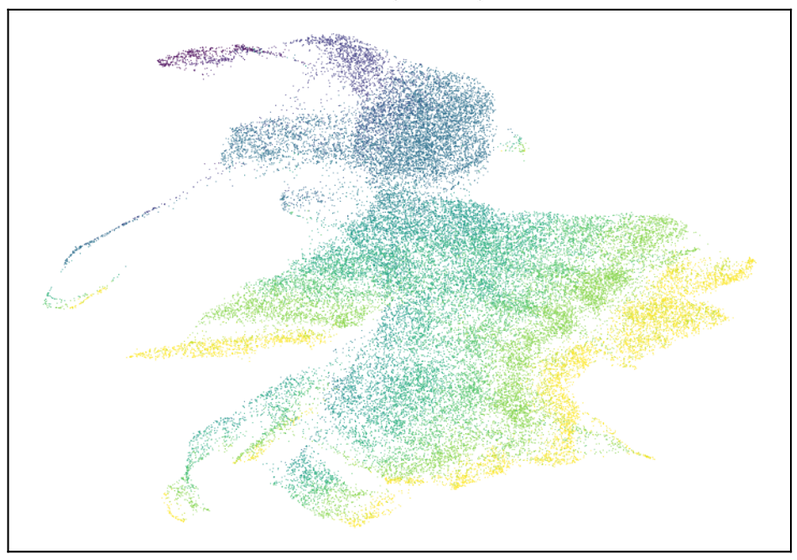
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



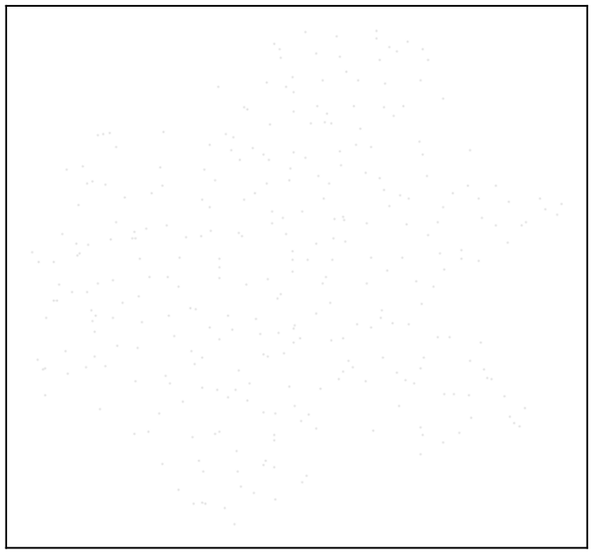
atlas, unaligned: stages



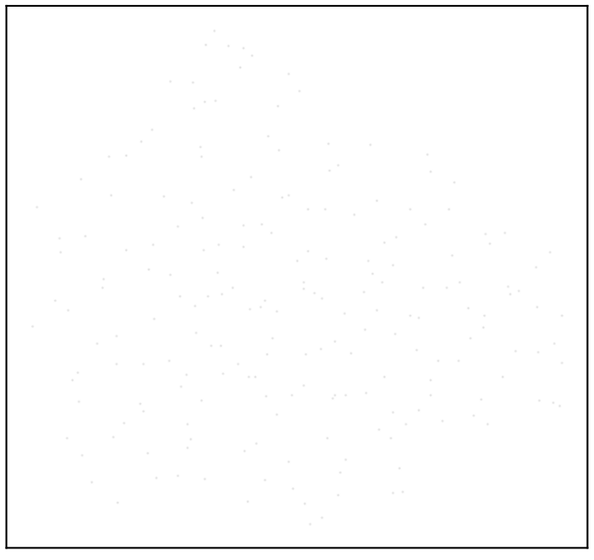
Genes (number of stages): her4.1 (5), her4.2 (5), her4.3 (5), her15.1 (5), her15.2 (5), her12 (5), nrarpa (5), nrarpb (5), hey1 (5)

signal:RA

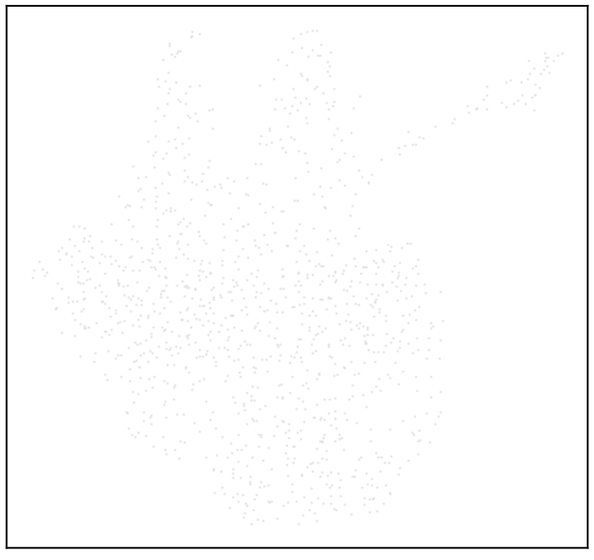
high: not defined



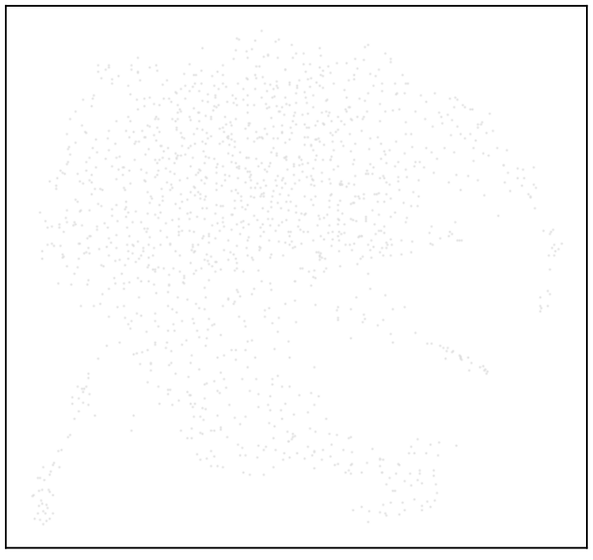
oblong: not defined



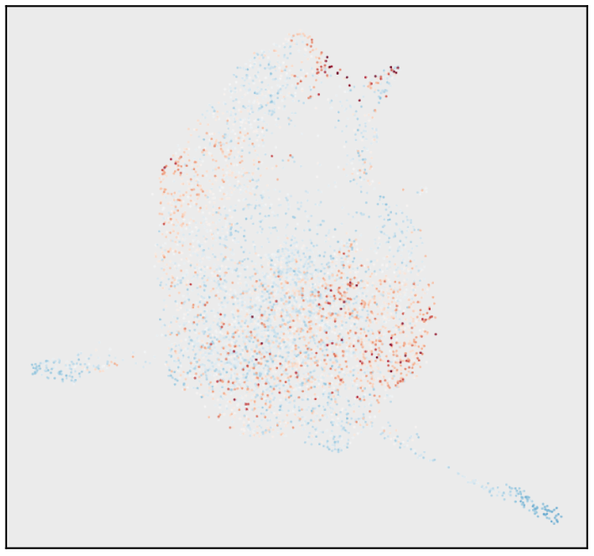
dome: not defined



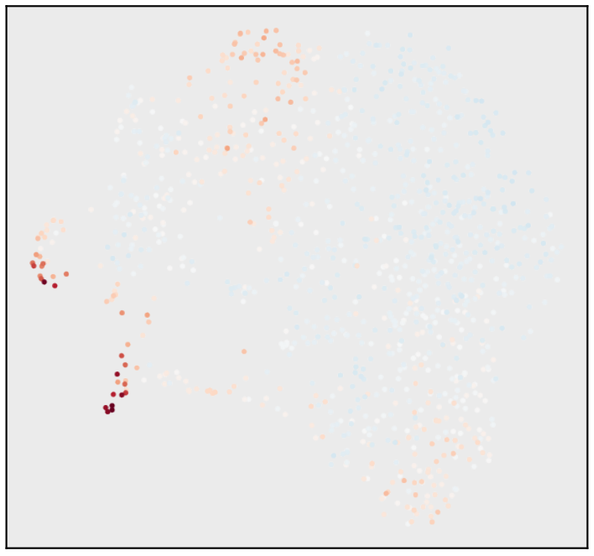
30% epiboly: not defined



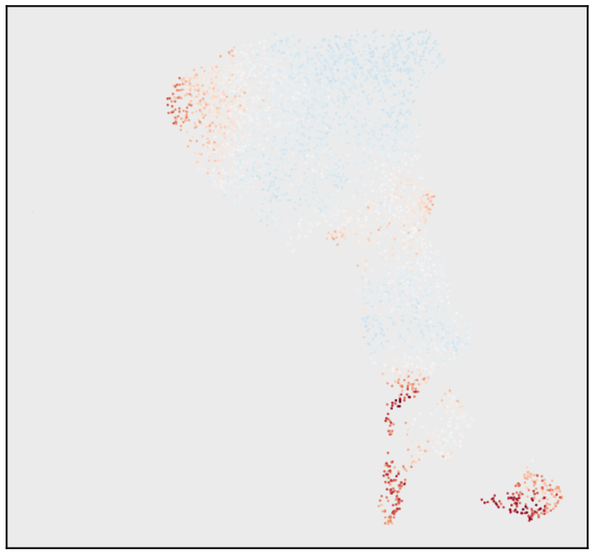
50% epiboly: 2 genes, not supported
active in 1% of cells, library level 0.16



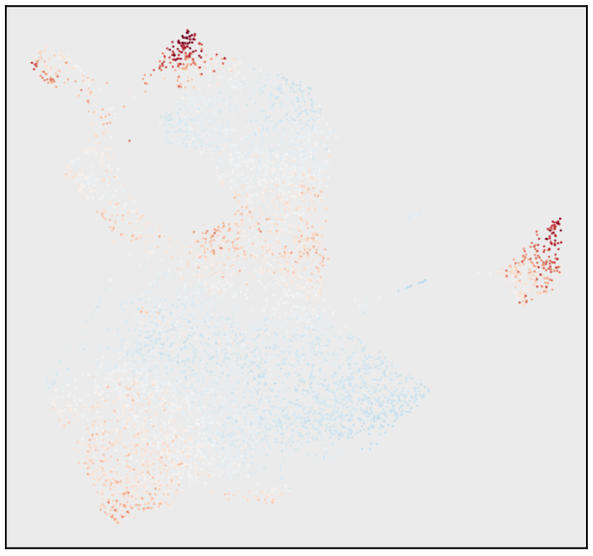
shield: 2 genes, not supported
active in 2% of cells, library level 0.22



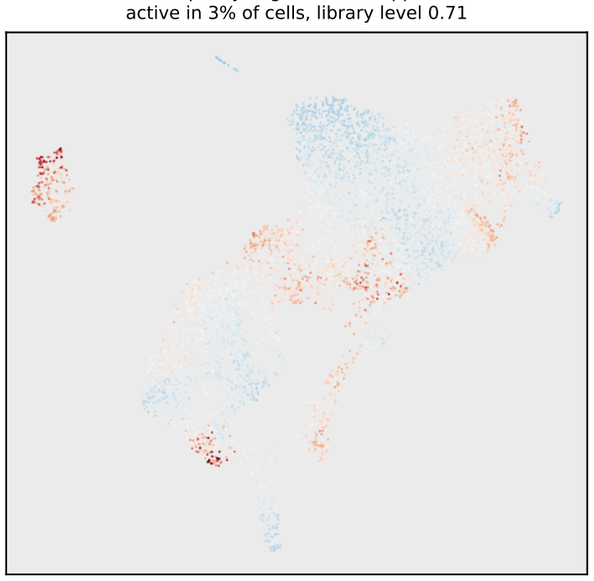
60% epiboly: 2 genes, not supported
active in 8% of cells, library level 0.48



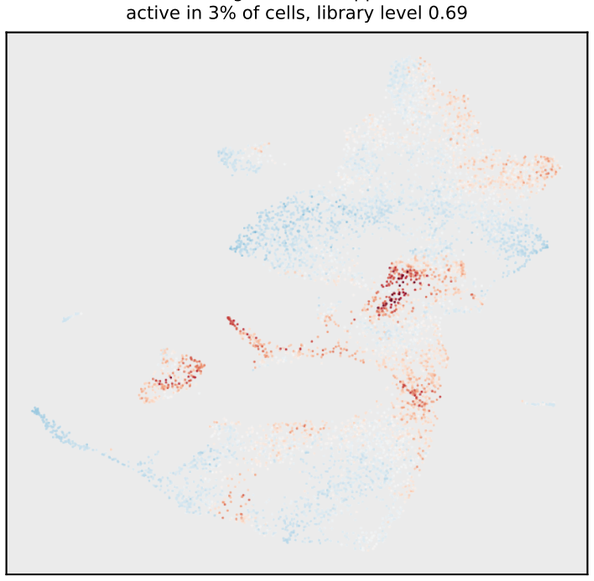
75% epiboly: 3 genes, not supported
active in 6% of cells, library level 0.64



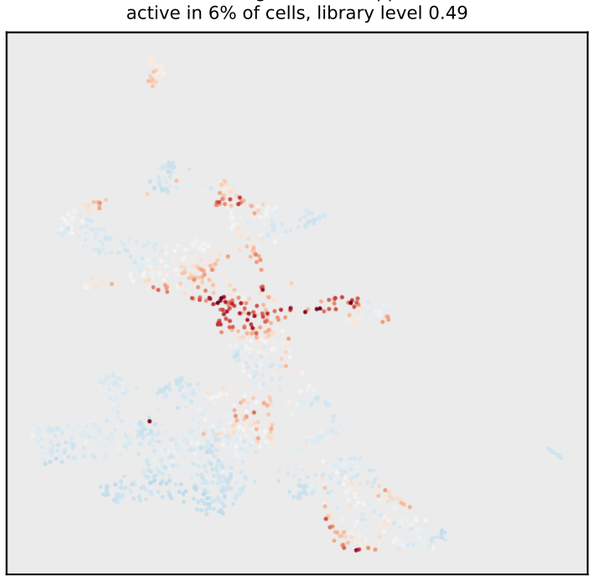
90% epiboly: 3 genes, not supported
active in 3% of cells, library level 0.71



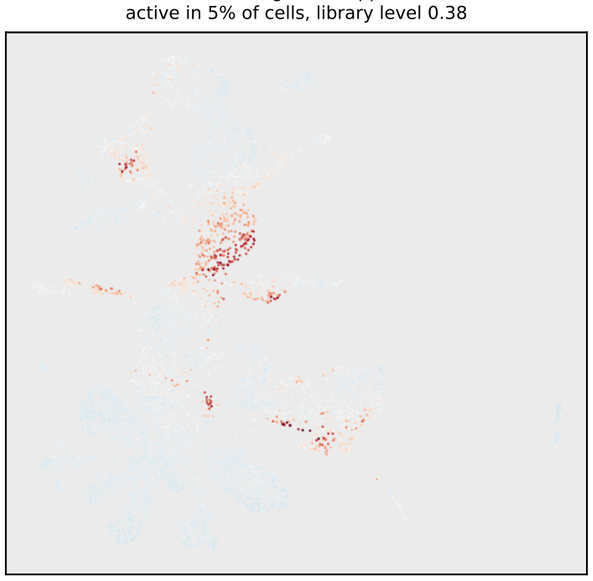
bud: 3 genes, not supported
active in 3% of cells, library level 0.69



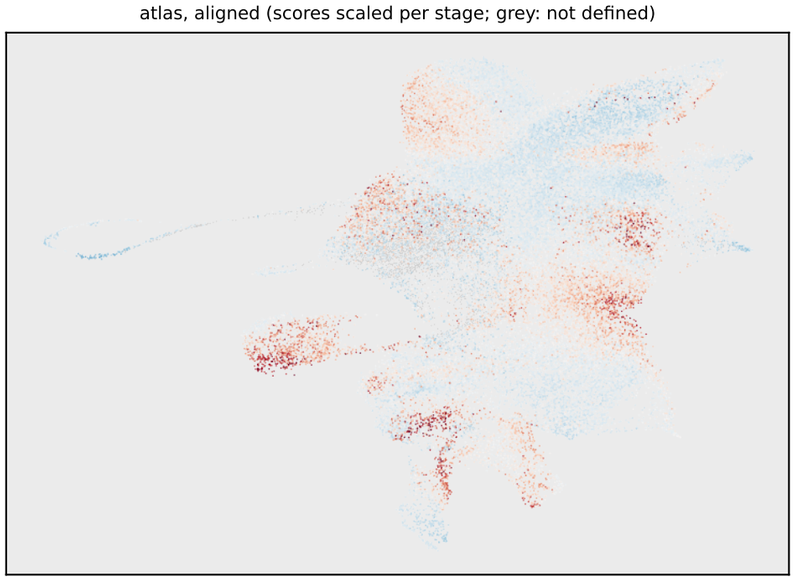
3-somite: 3 genes, not supported
active in 6% of cells, library level 0.49



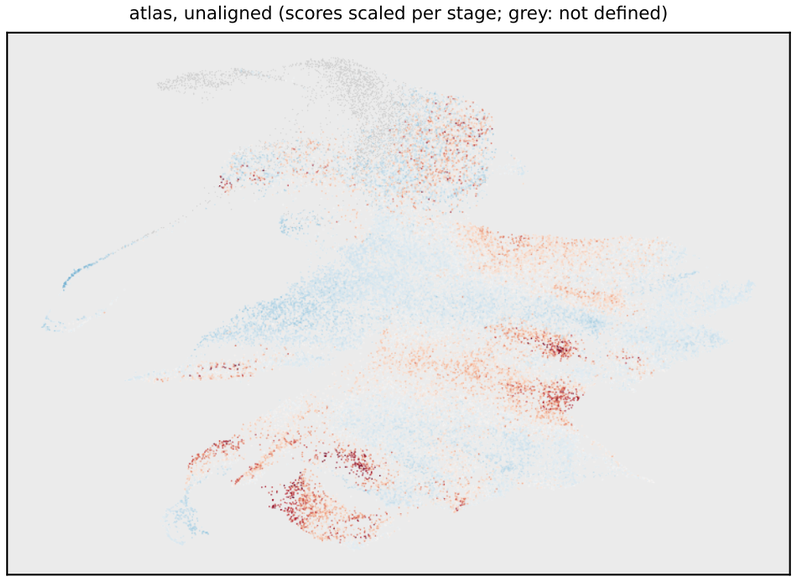
6-somite: 3 genes, supported
active in 5% of cells, library level 0.38



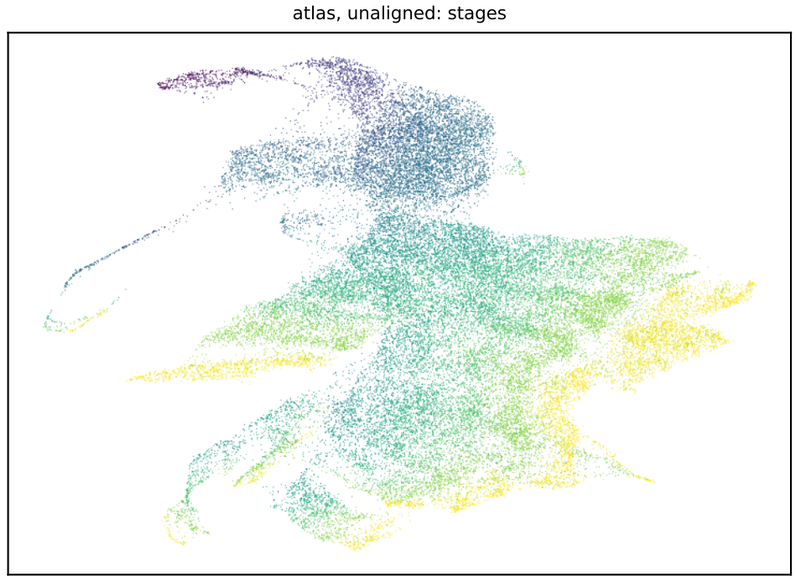
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



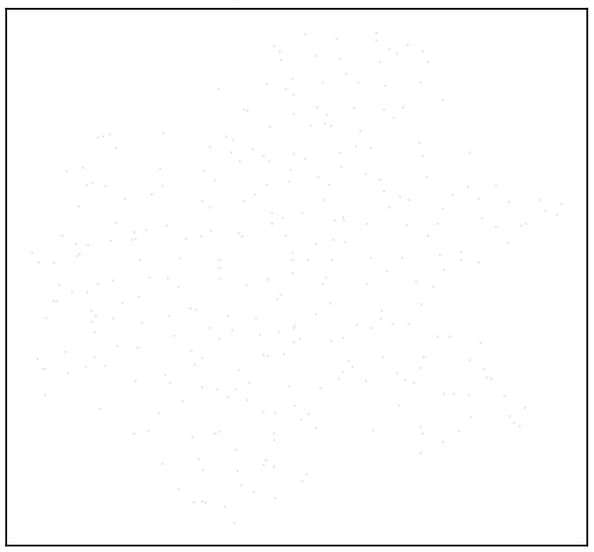
atlas, unaligned: stages



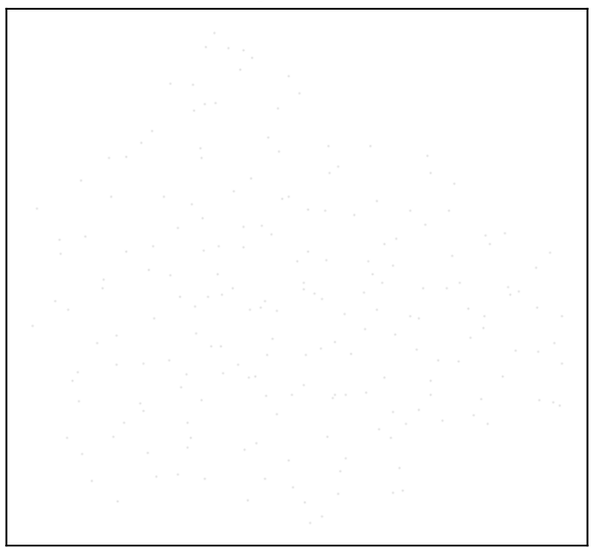
Genes (number of stages): cyp26a1 (8), dhrs3b (8), dhrs3a (5)

signal:Wnt

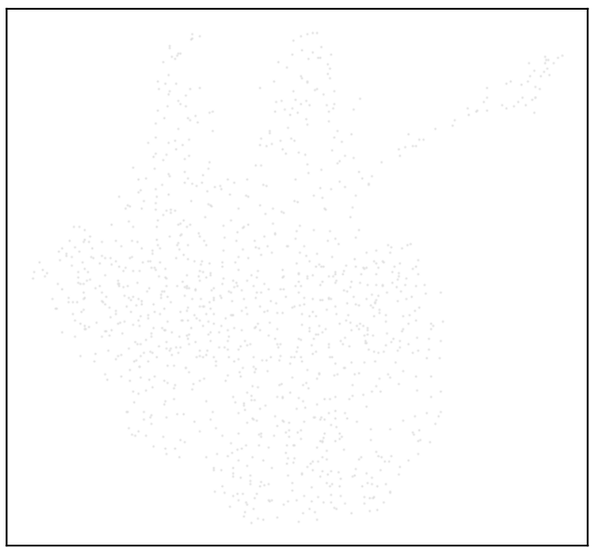
high: not defined



oblong: not defined

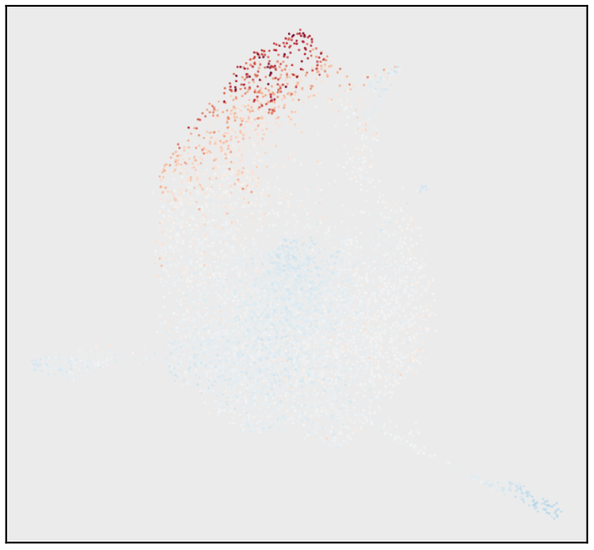


dome: not defined

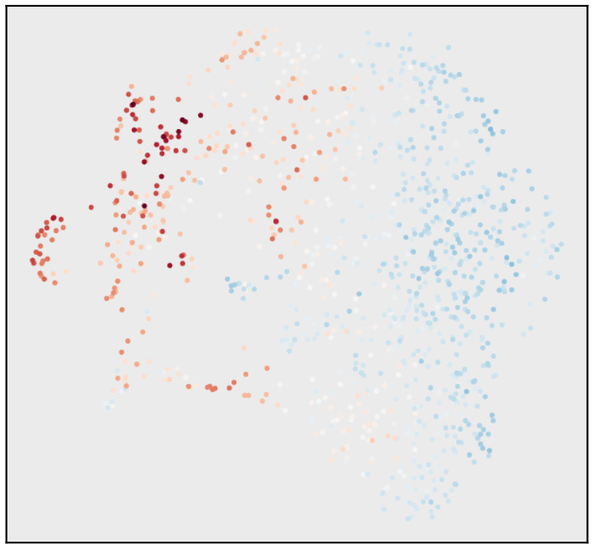


30% epiboly: 4 genes, not supported
active in 2% of cells, library level 0.14

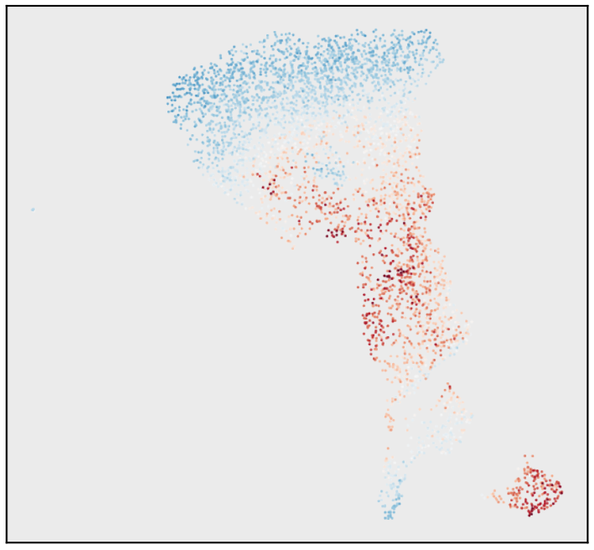
50% epiboly: 4 genes, not supported
active in 6% of cells, library level 0.22



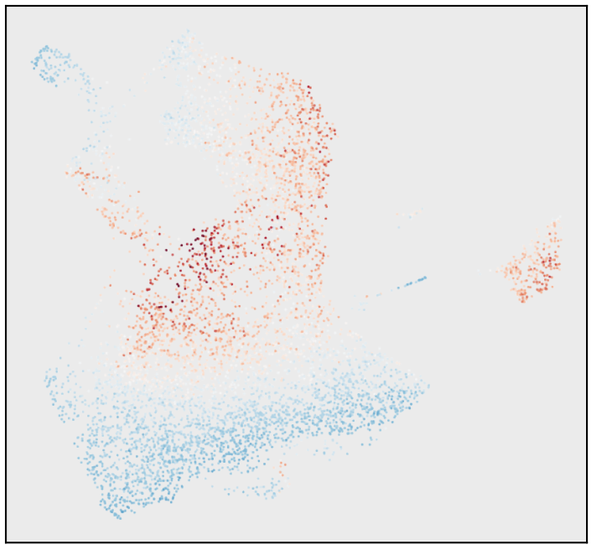
shield: 4 genes, not supported
active in 4% of cells, library level 0.48



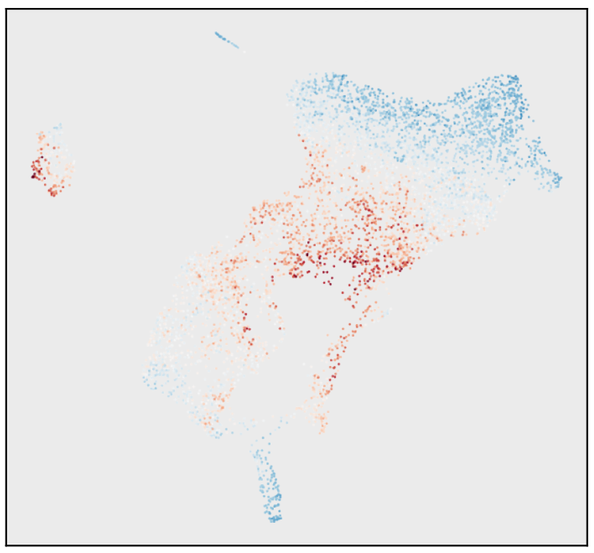
60% epiboly: 4 genes, supported
active in 11% of cells, library level 0.61



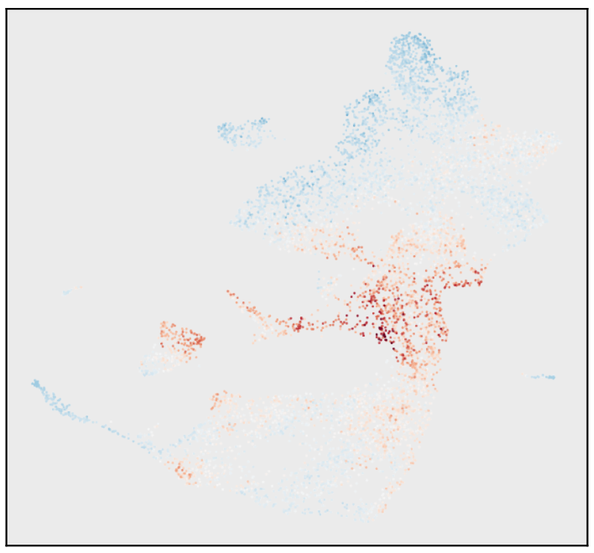
75% epiboly: 4 genes, not supported
active in 9% of cells, library level 0.67



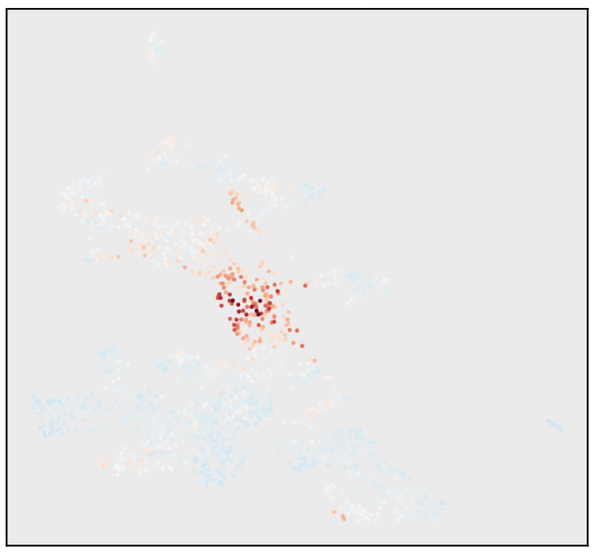
90% epiboly: 4 genes, not supported
active in 6% of cells, library level 0.81



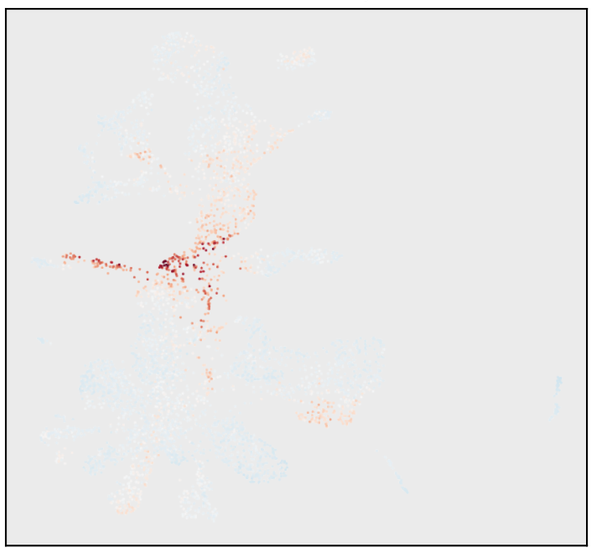
bud: 4 genes, supported
active in 7% of cells, library level 0.67



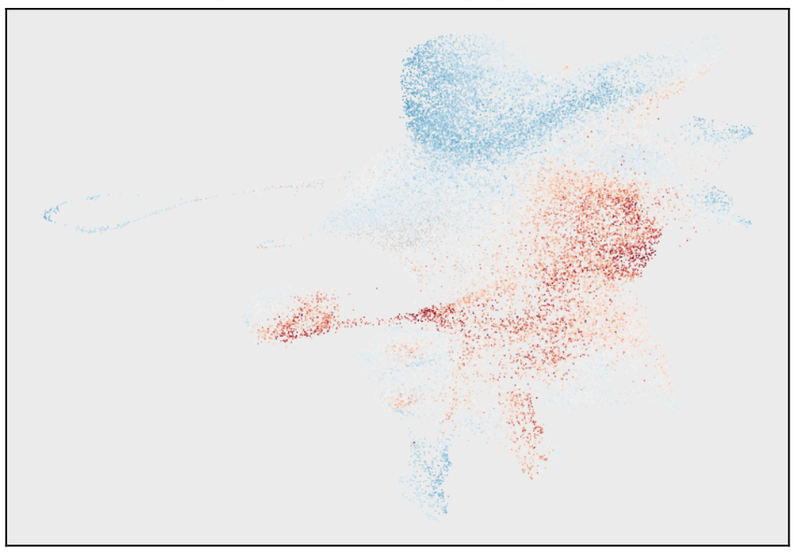
3-somite: 4 genes, not supported
active in 6% of cells, library level 0.25



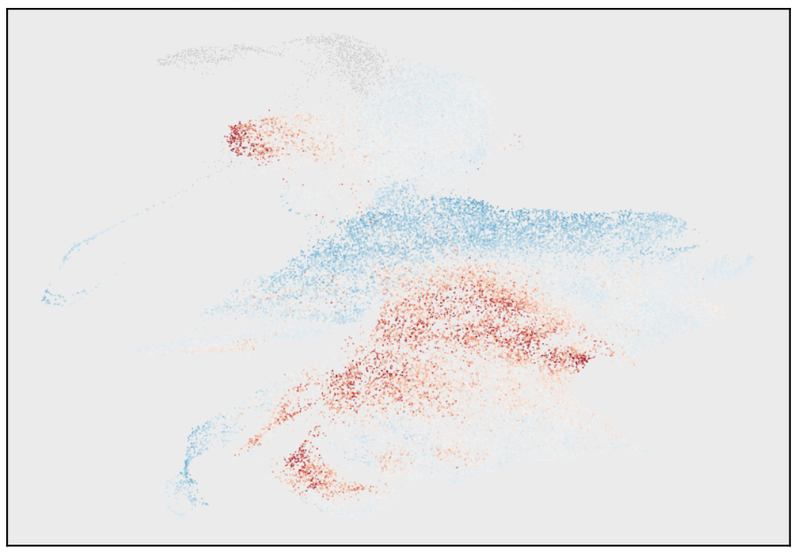
6-somite: 4 genes, supported
active in 5% of cells, library level 0.24



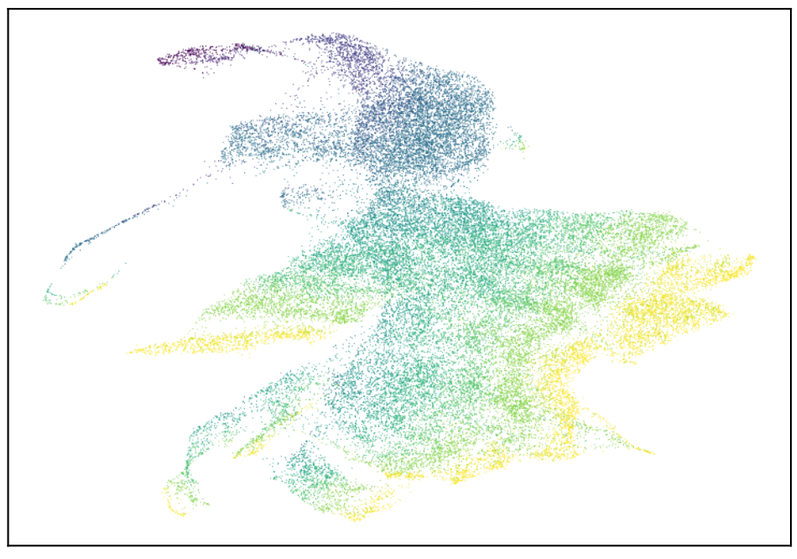
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



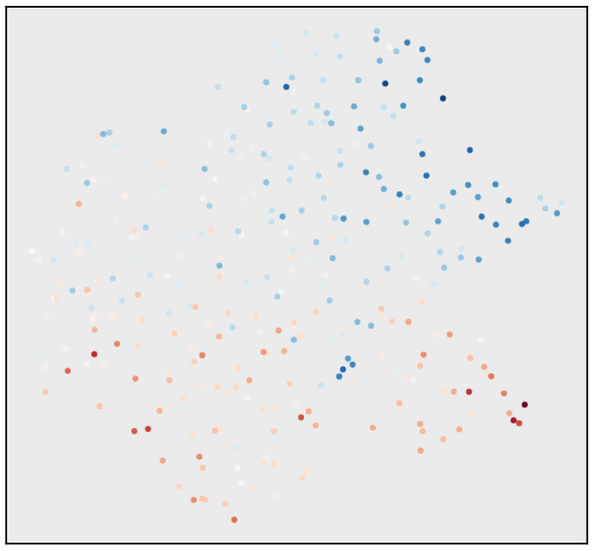
atlas, unaligned: stages



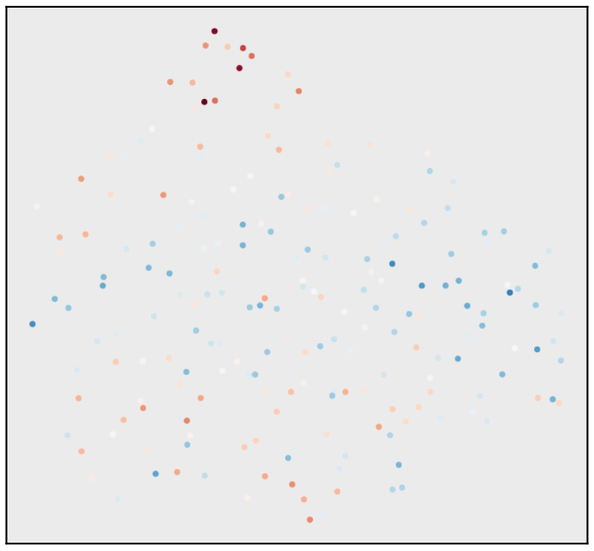
Genes (number of stages): sp5l (9), sp5a (9), notum1a (9), lef1 (9)

state:blastomere

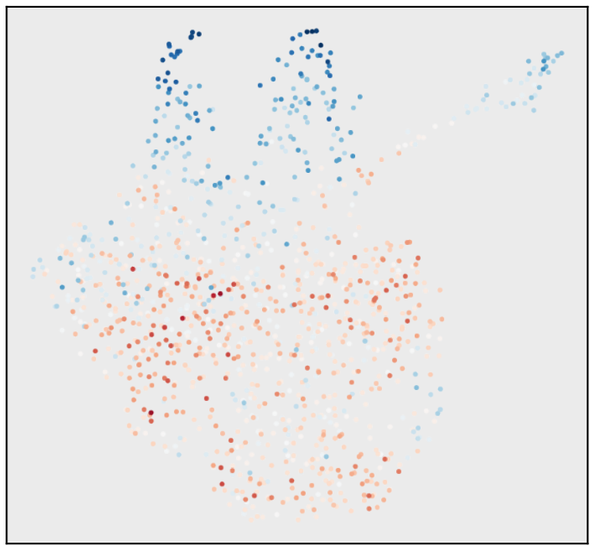
high: 6 genes, not supported
active in 3% of cells, library level 0.28



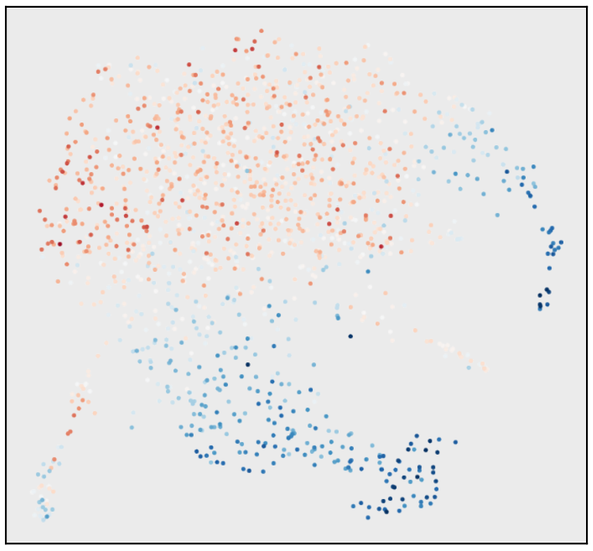
oblong: 6 genes, not supported
active in 2% of cells, library level 0.43



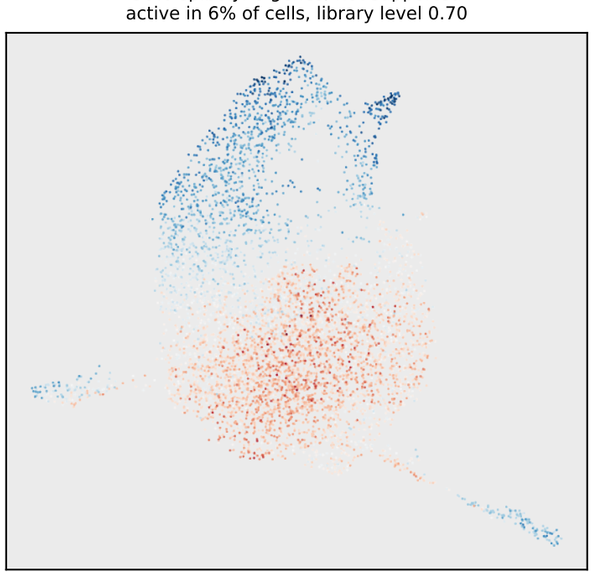
dome: 6 genes, not supported
active in 1% of cells, library level 0.51



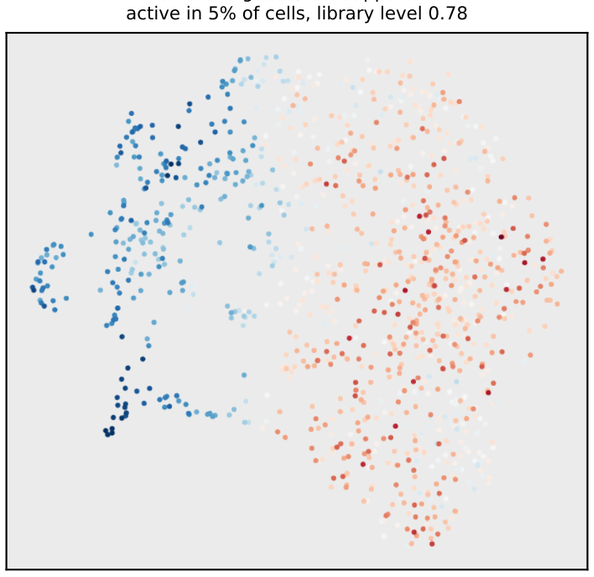
30% epiboly: 6 genes, not supported
active in 1% of cells, library level 0.76



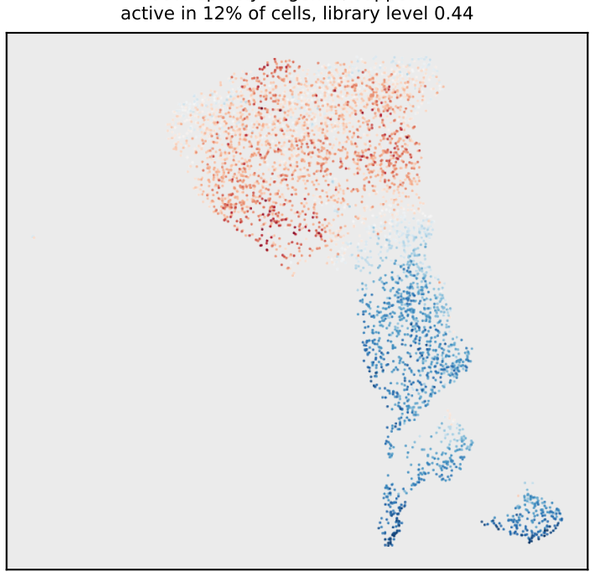
50% epiboly: 6 genes, not supported
active in 6% of cells, library level 0.70



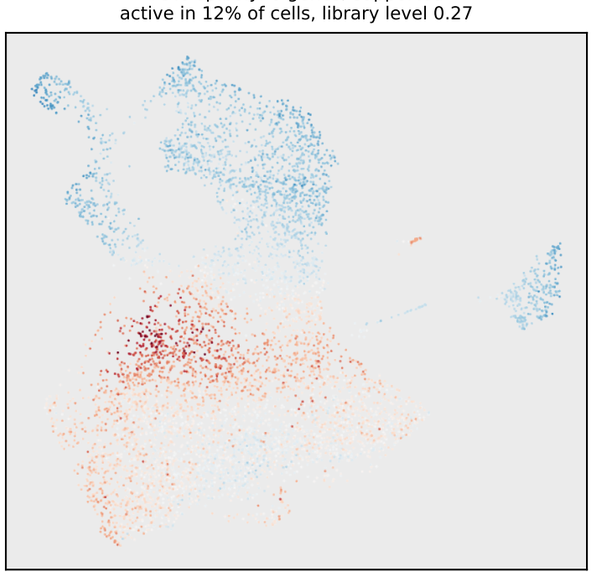
shield: 6 genes, not supported
active in 5% of cells, library level 0.78



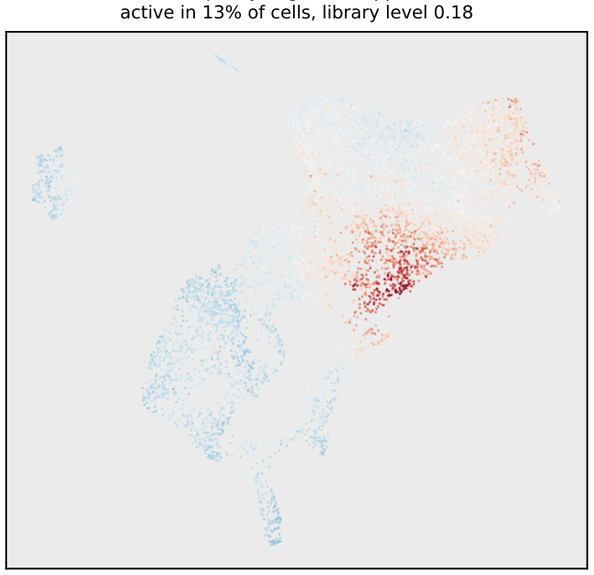
60% epiboly: 6 genes, supported
active in 12% of cells, library level 0.44



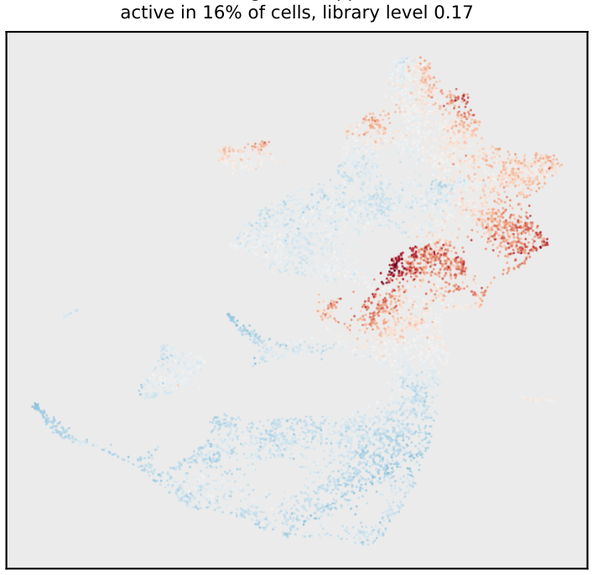
75% epiboly: 6 genes, supported
active in 12% of cells, library level 0.27



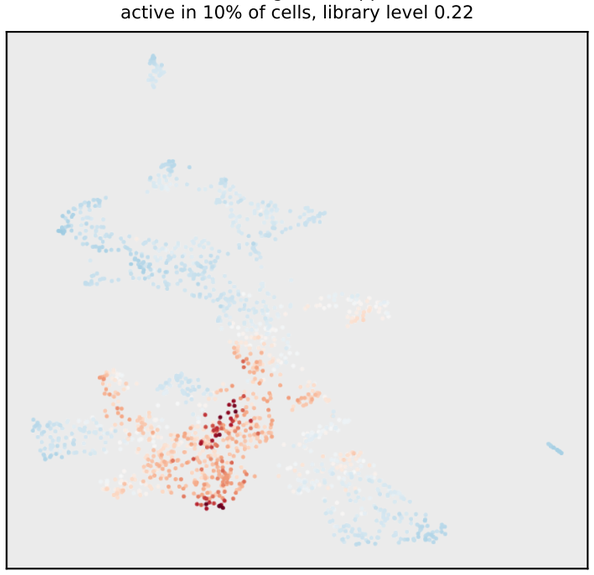
90% epiboly: 6 genes, supported
active in 13% of cells, library level 0.18



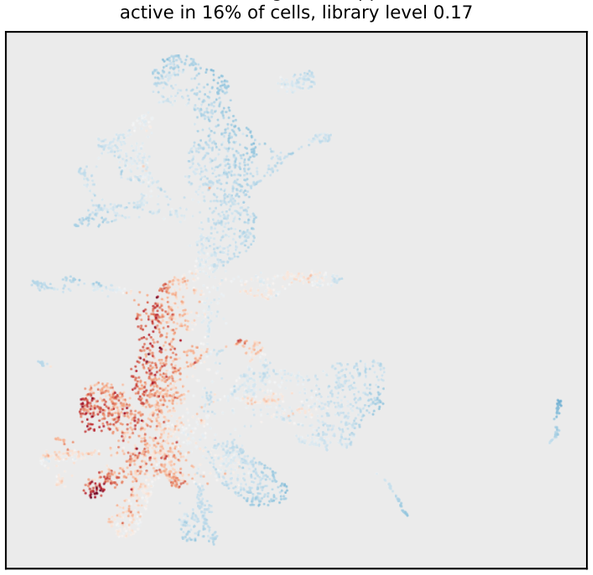
bud: 6 genes, supported
active in 16% of cells, library level 0.17



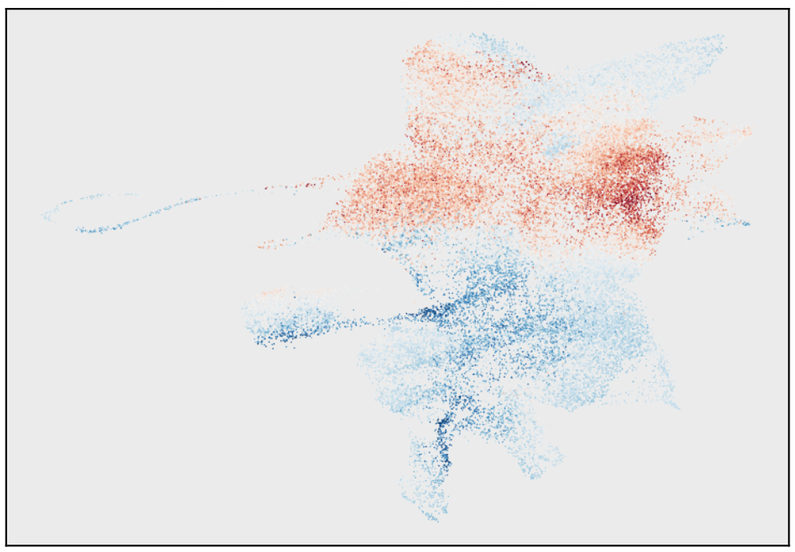
3-somite: 5 genes, supported
active in 10% of cells, library level 0.22



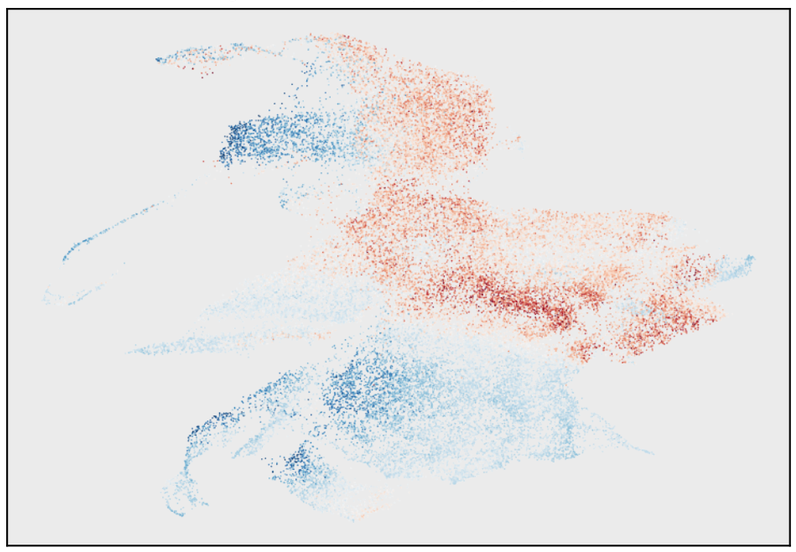
6-somite: 6 genes, supported
active in 16% of cells, library level 0.17



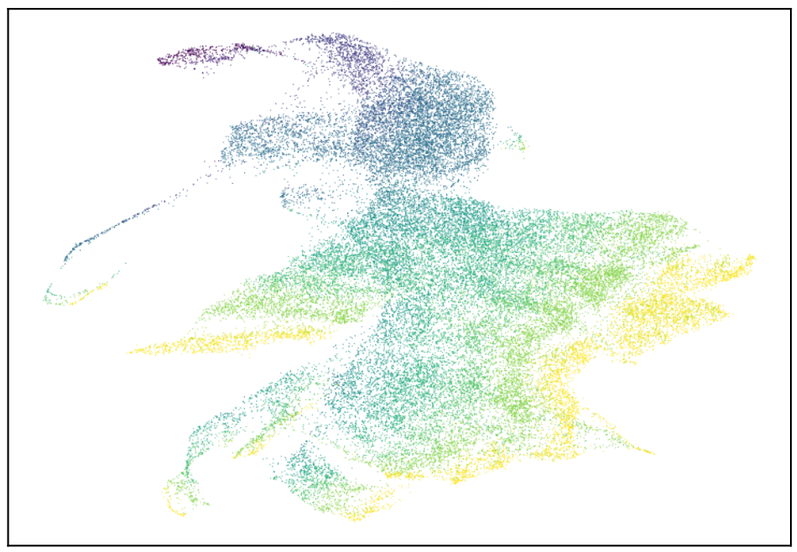
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



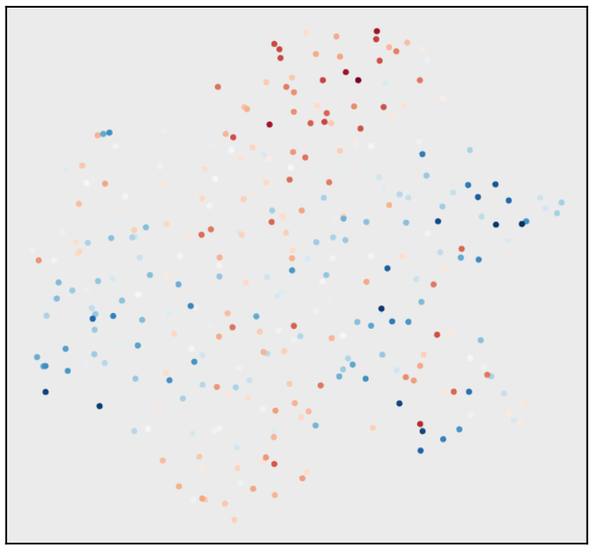
atlas, unaligned: stages



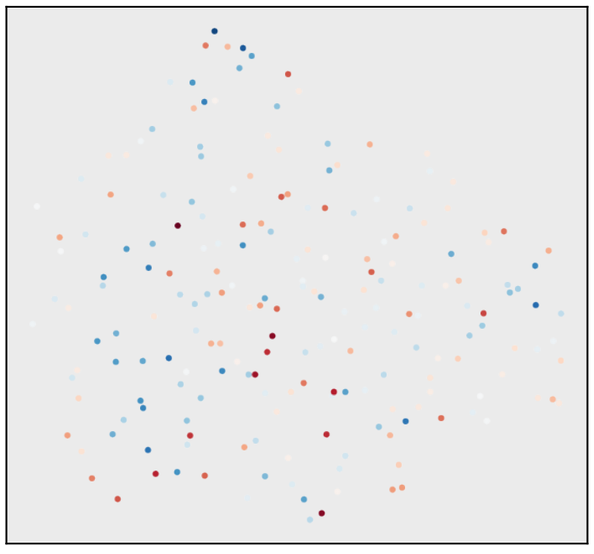
Genes (number of stages): pou5f3 (12), sox19b (12), lin28a (12), sox3 (12), sox19a (12), nanog (11)

state:cell_cycle_g2m

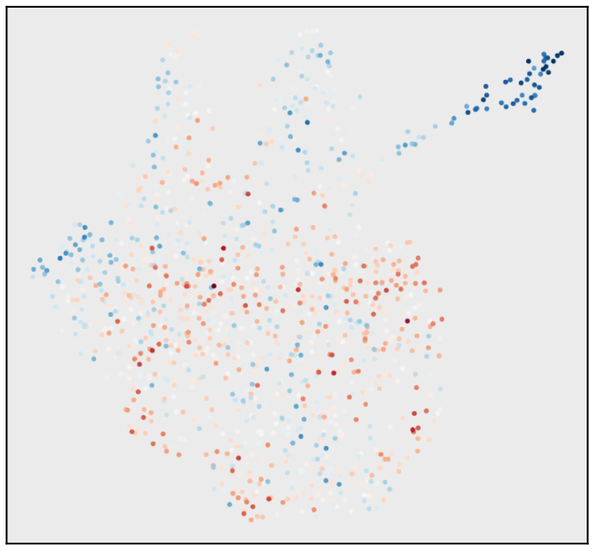
high: 10 genes, not supported
active in 0% of cells, library level 0.45



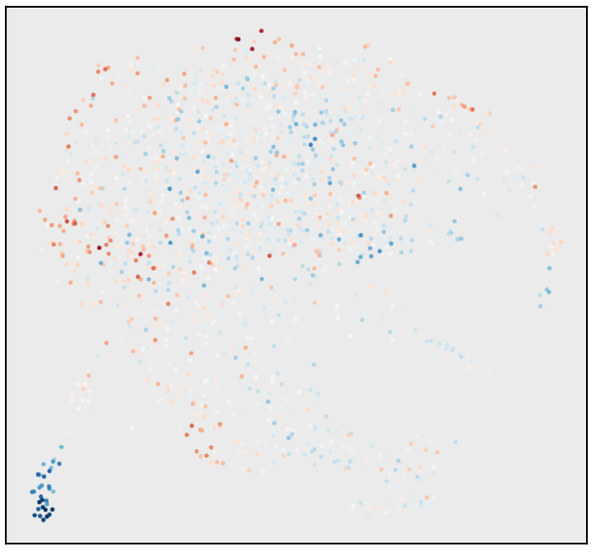
oblong: 10 genes, not supported
active in 0% of cells, library level 0.43



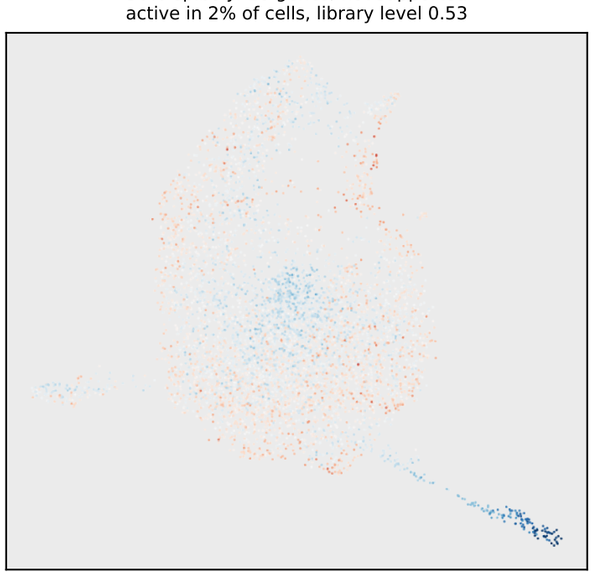
dome: 10 genes, not supported
active in 0% of cells, library level 0.34



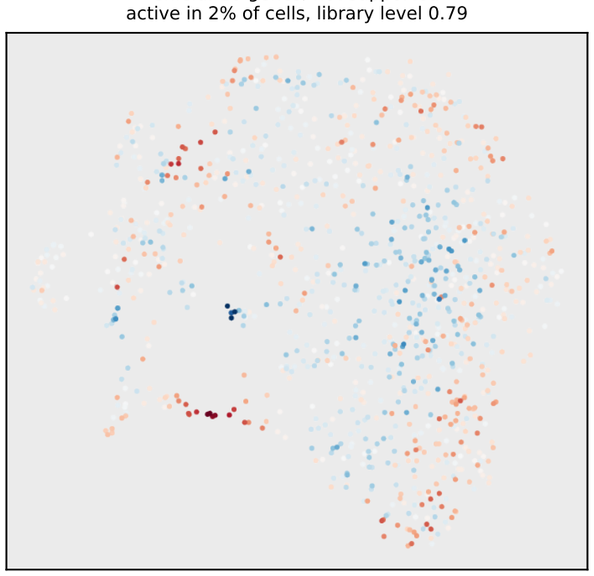
30% epiboly: 10 genes, not supported
active in 1% of cells, library level 0.48



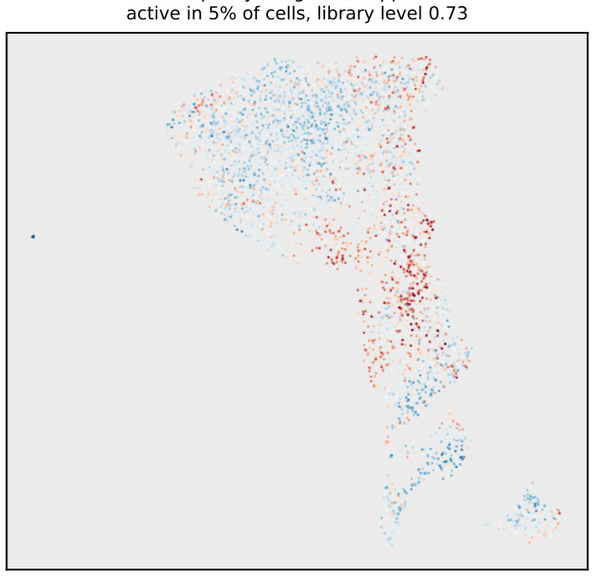
50% epiboly: 10 genes, not supported
active in 2% of cells, library level 0.53



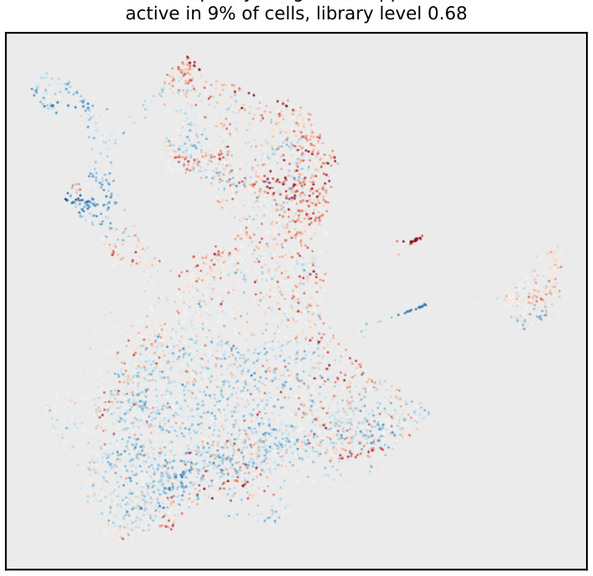
shield: 10 genes, not supported
active in 2% of cells, library level 0.79



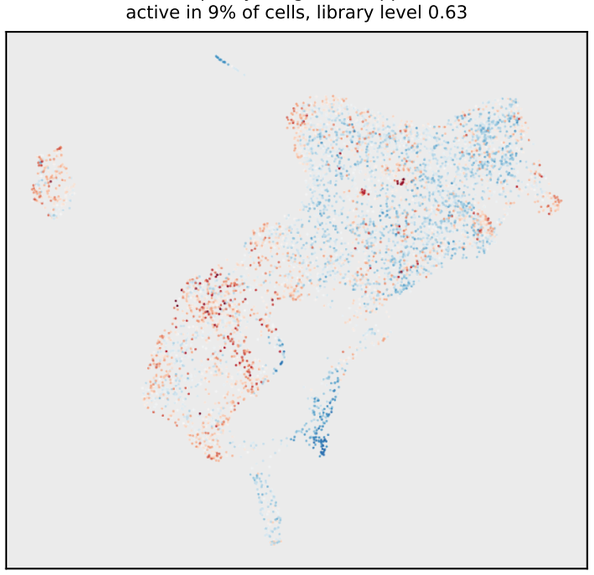
60% epiboly: 10 genes, supported
active in 5% of cells, library level 0.73



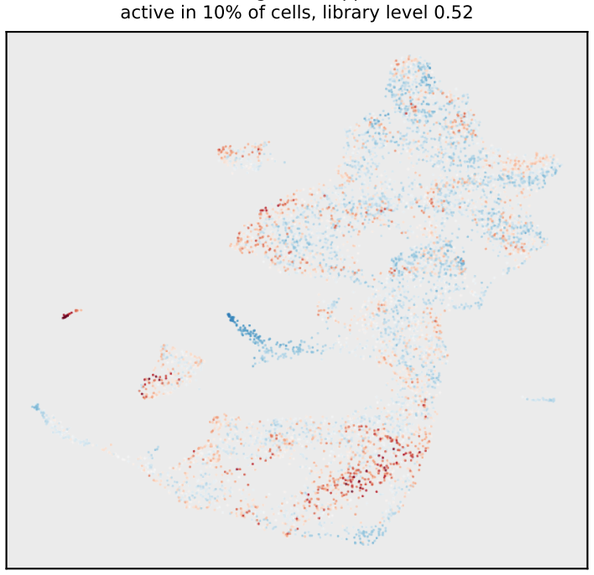
75% epiboly: 10 genes, supported
active in 9% of cells, library level 0.68



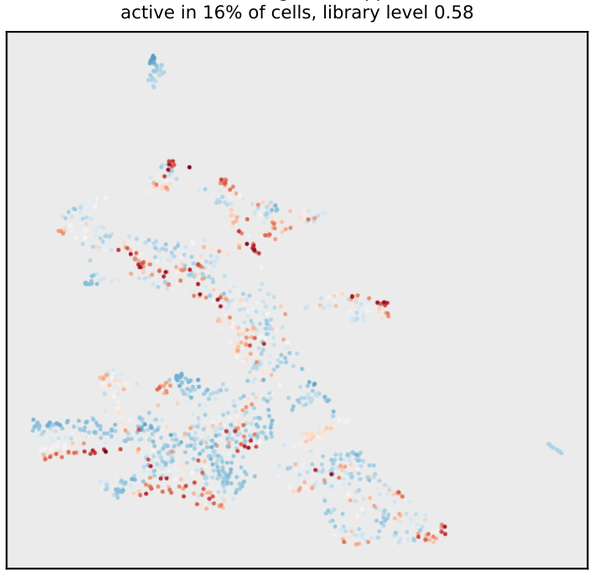
90% epiboly: 10 genes, supported
active in 9% of cells, library level 0.63



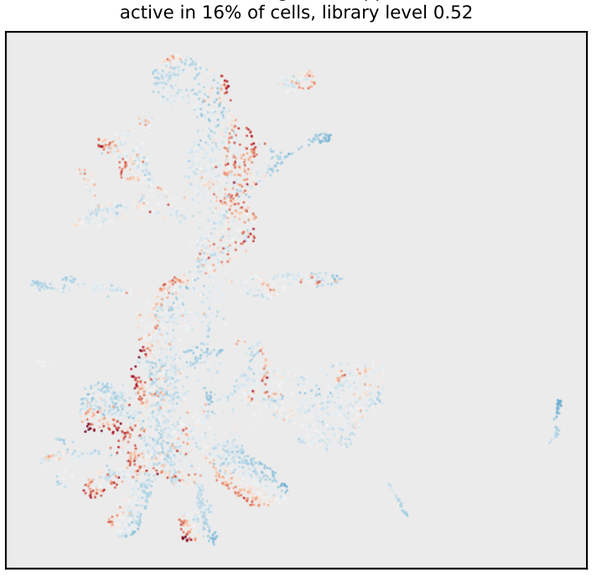
bud: 10 genes, supported
active in 10% of cells, library level 0.52



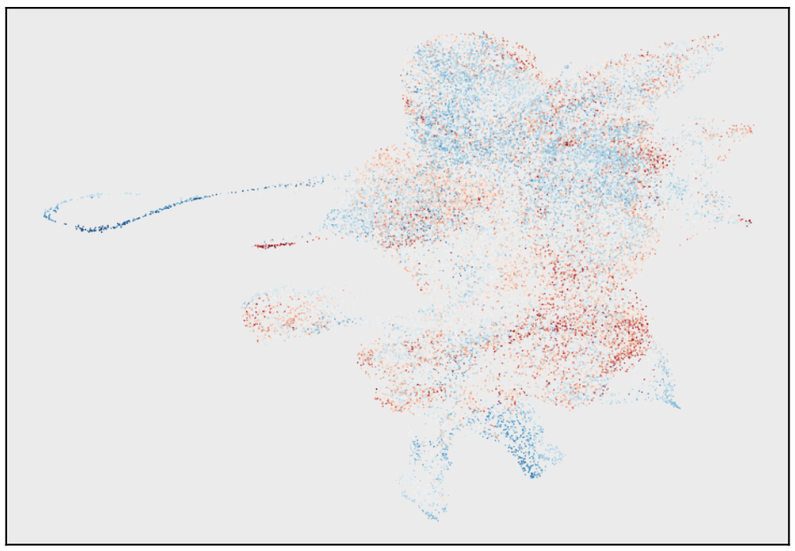
3-somite: 10 genes, supported
active in 16% of cells, library level 0.58



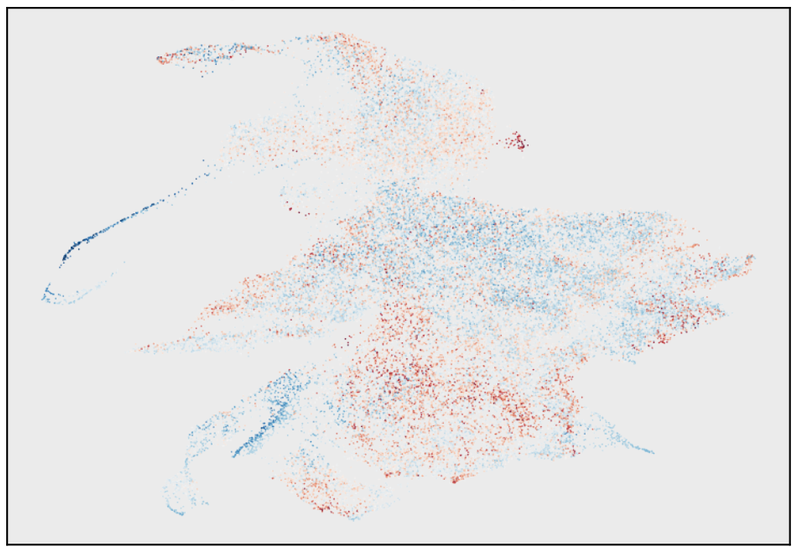
6-somite: 10 genes, supported
active in 16% of cells, library level 0.52



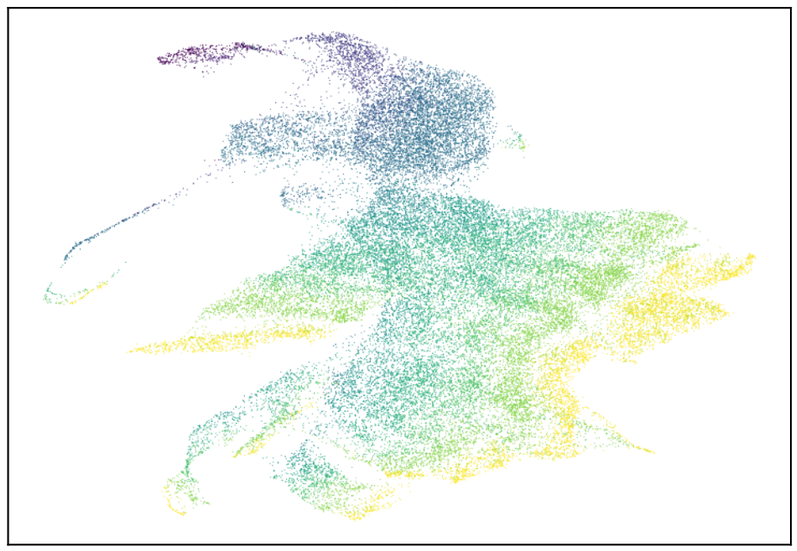
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



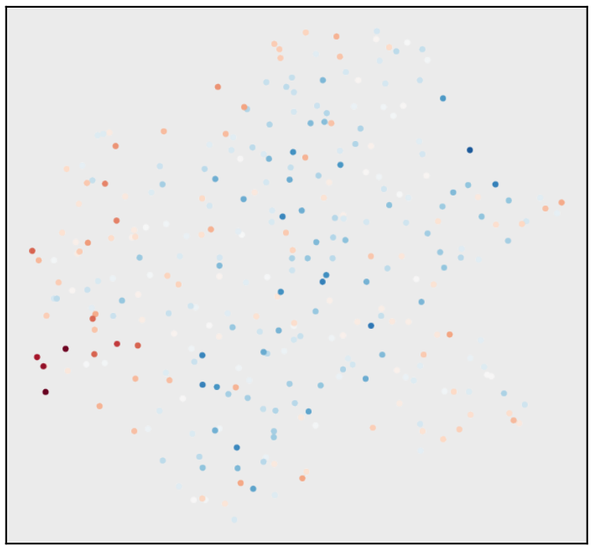
atlas, unaligned: stages



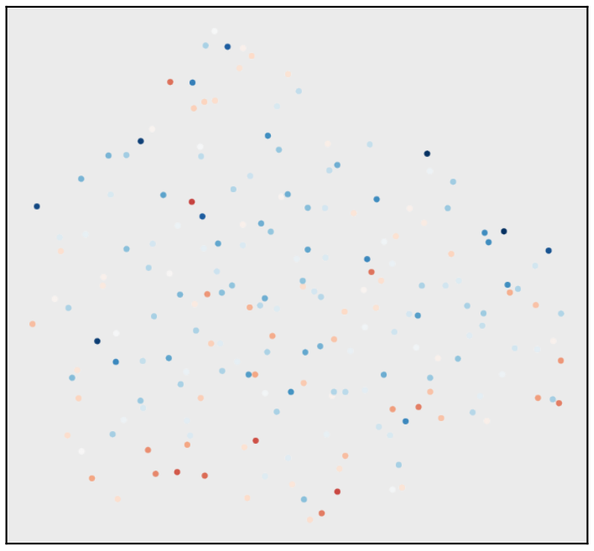
Genes (number of stages): cdk1 (12), ube2c (12), top2a (12), nusap1 (12), cenpf (12), tpx2 (12), aurkb (12), cdc20 (12), birc5a (12), kif11 (12)

state:cell_cycle_s

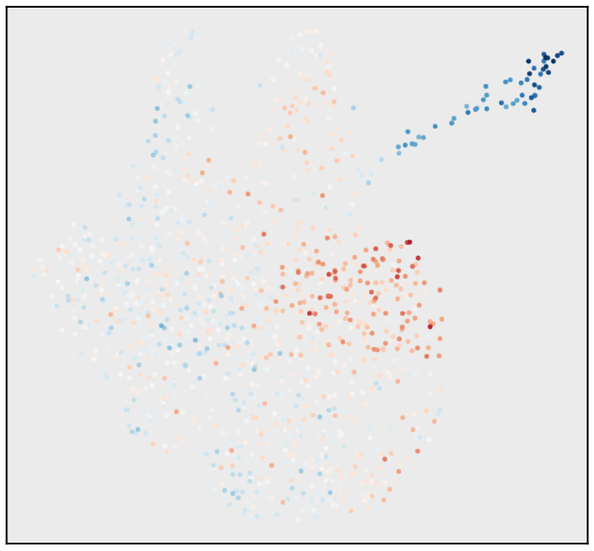
high: 10 genes, not supported
active in 1% of cells, library level 0.56



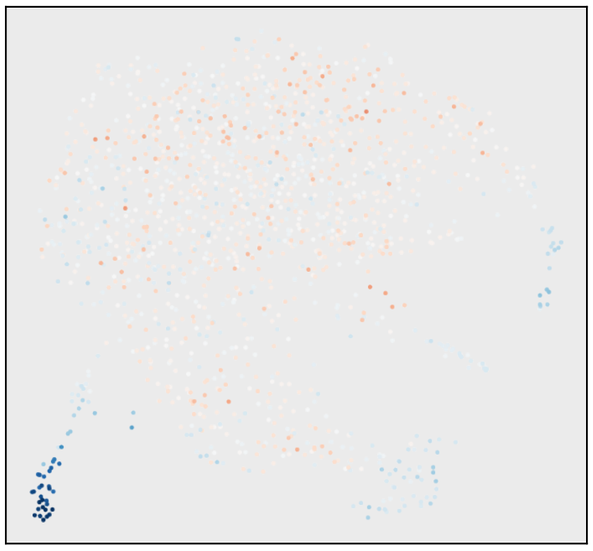
oblong: 10 genes, not supported
active in 0% of cells, library level 0.43



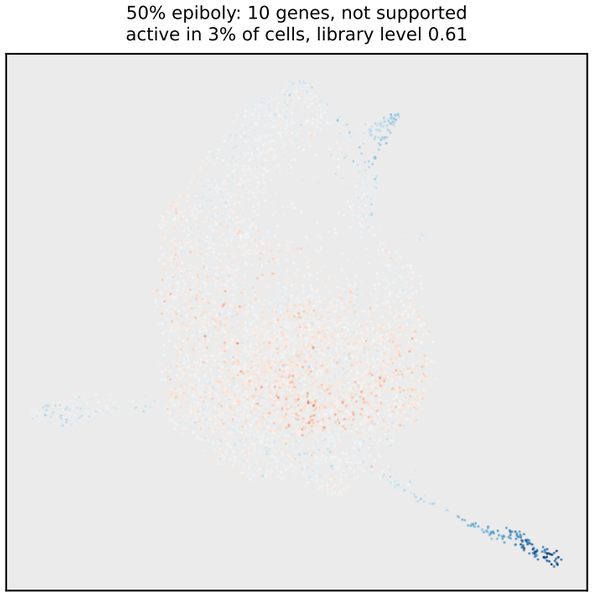
dome: 10 genes, not supported
active in 2% of cells, library level 0.57



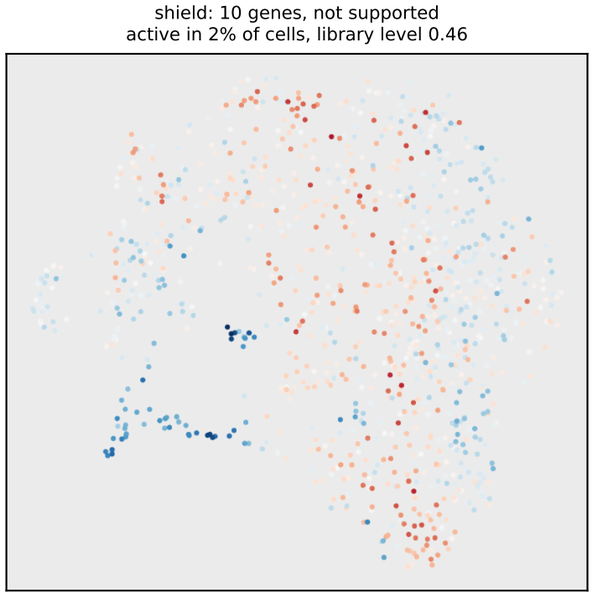
30% epiboly: 10 genes, not supported
active in 0% of cells, library level 0.57



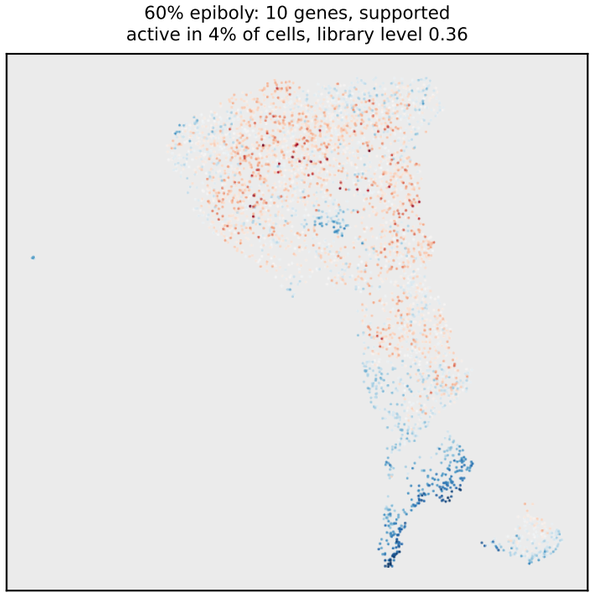
50% epiboly: 10 genes, not supported
active in 3% of cells, library level 0.61



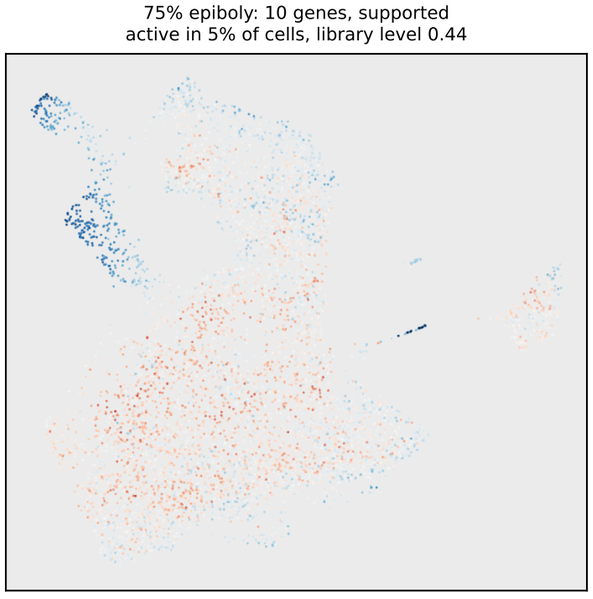
shield: 10 genes, not supported
active in 2% of cells, library level 0.46



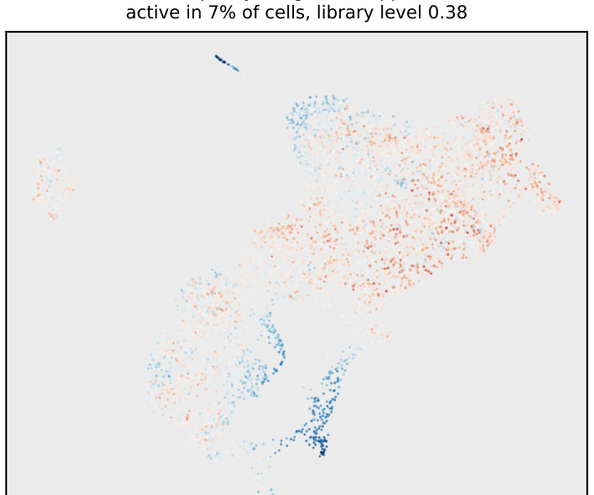
60% epiboly: 10 genes, supported
active in 4% of cells, library level 0.36



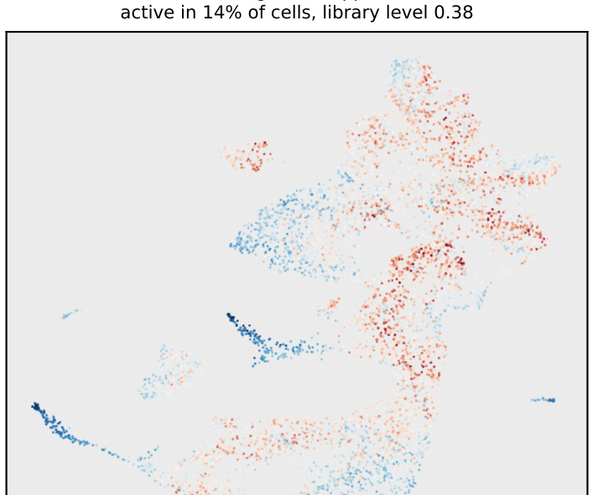
75% epiboly: 10 genes, supported
active in 5% of cells, library level 0.44



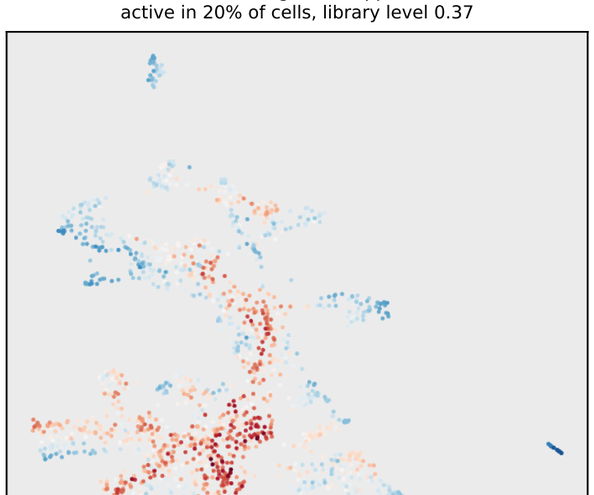
90% epiboly: 10 genes, supported
active in 7% of cells, library level 0.38



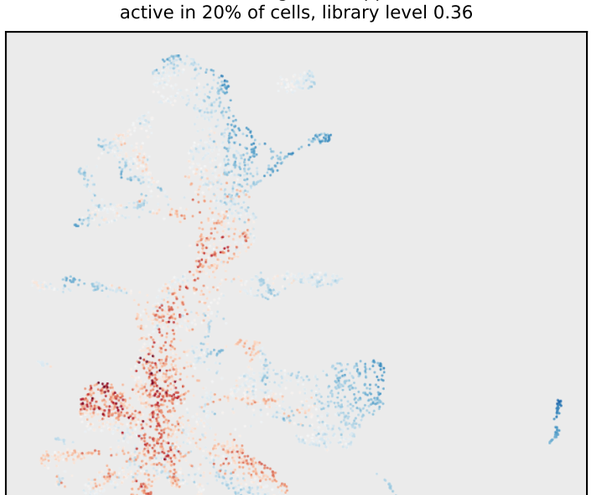
bud: 10 genes, supported
active in 14% of cells, library level 0.38



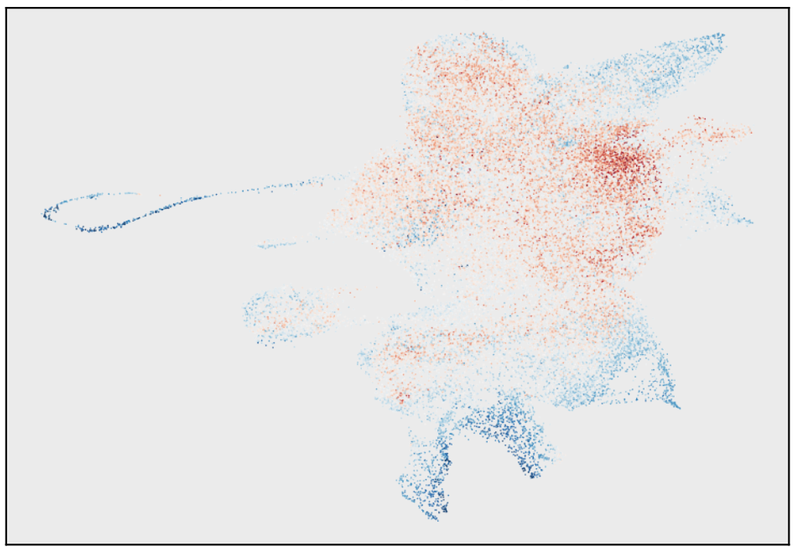
3-somite: 10 genes, supported
active in 20% of cells, library level 0.37



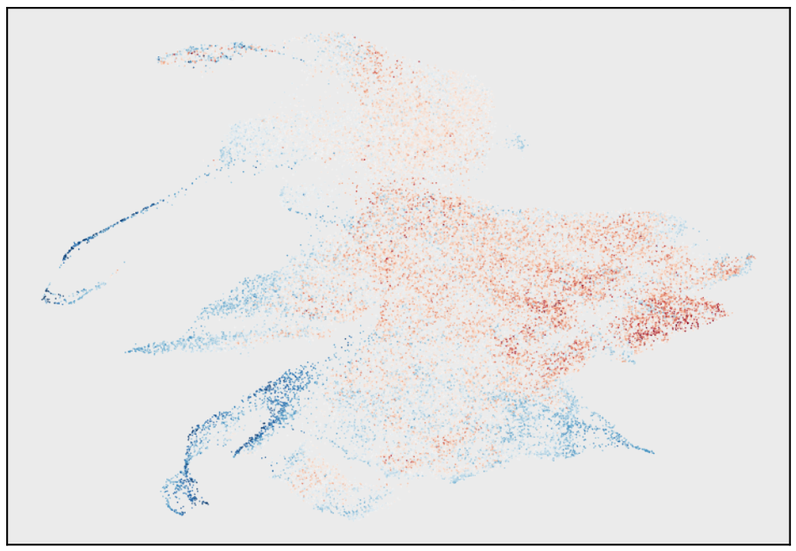
6-somite: 10 genes, supported
active in 20% of cells, library level 0.36



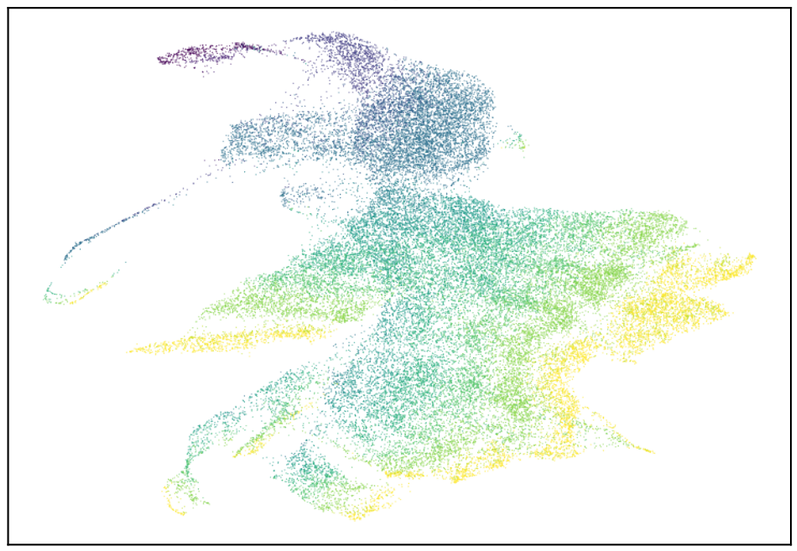
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



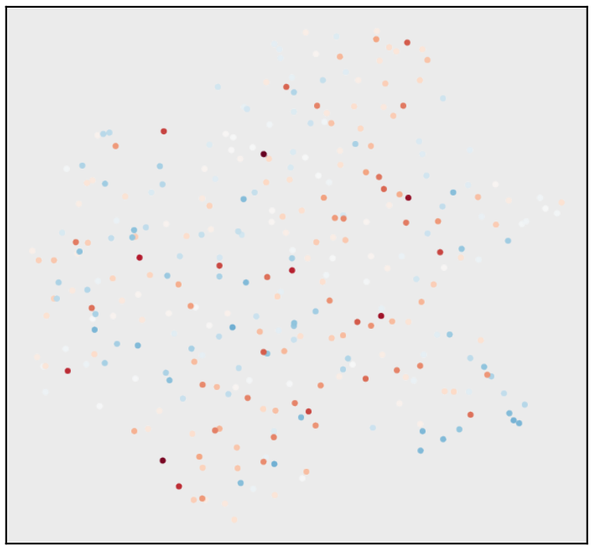
atlas, unaligned: stages



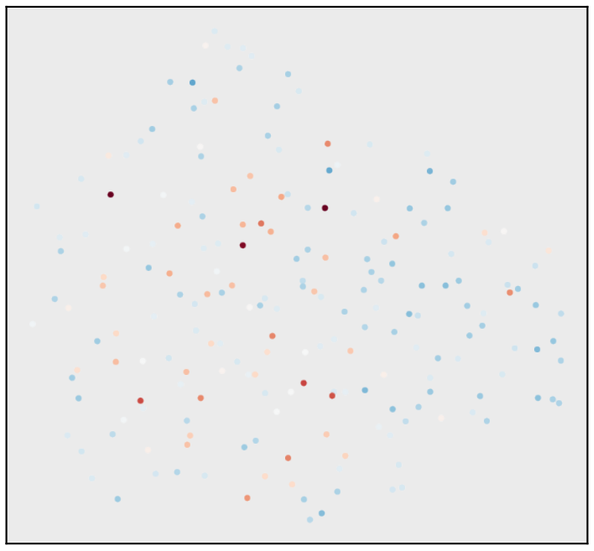
Genes (number of stages): pcna (12), mcm2 (12), mcm5 (12), mcm6 (12), cdc6 (12), slbp (12), tyms (12), mcm4 (12), uhrf1 (12), rrm2 (12)

state:heat_shock

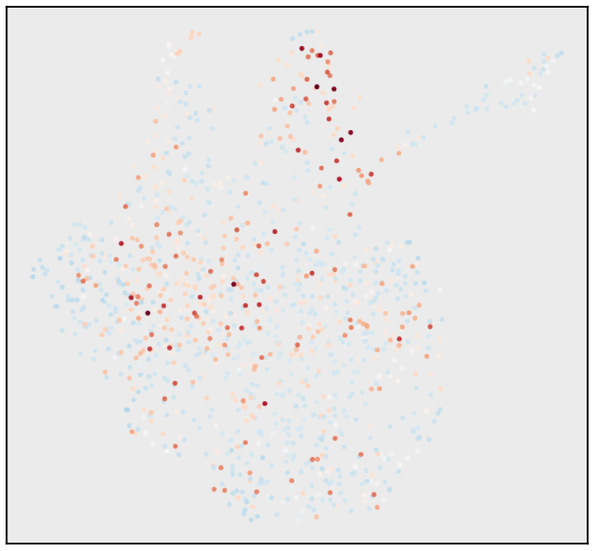
high: 3 genes, not supported
active in 0% of cells, library level 0.05



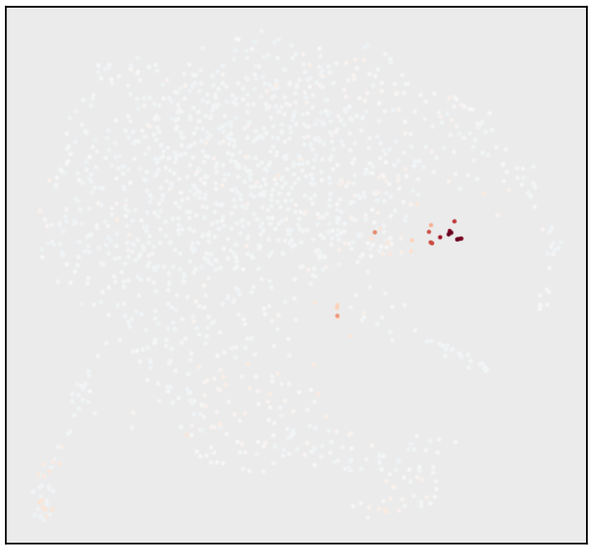
oblong: 2 genes, not supported
active in 0% of cells, library level 0.22



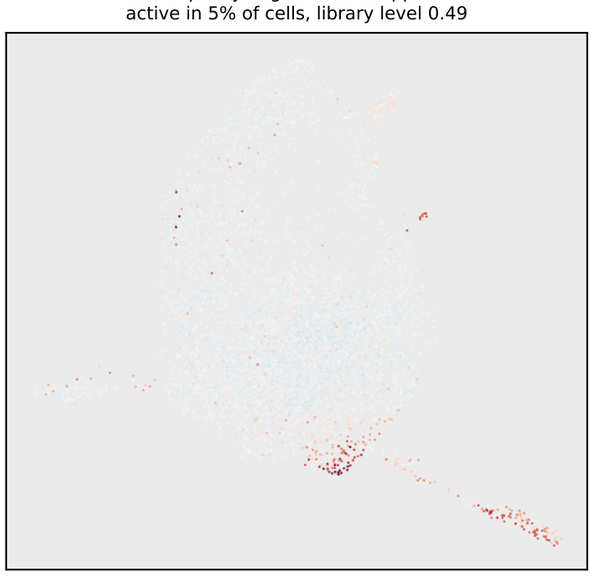
dome: 3 genes, not supported
active in 0% of cells, library level 0.04



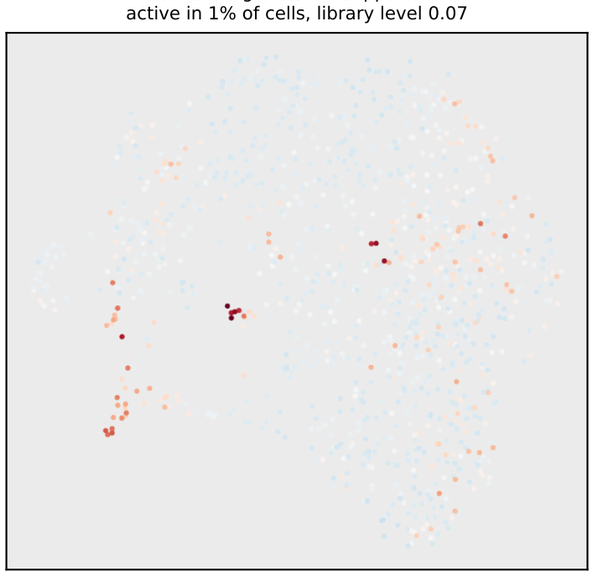
30% epiboly: 4 genes, not supported
active in 1% of cells, library level 0.07



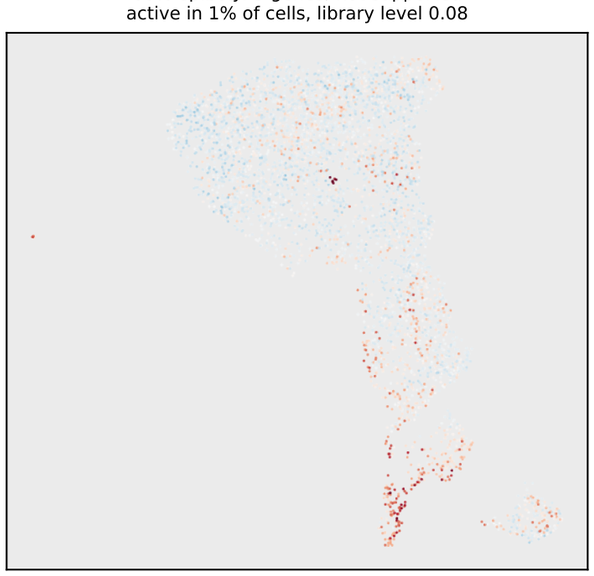
50% epiboly: 5 genes, not supported
active in 5% of cells, library level 0.49



shield: 3 genes, not supported
active in 1% of cells, library level 0.07

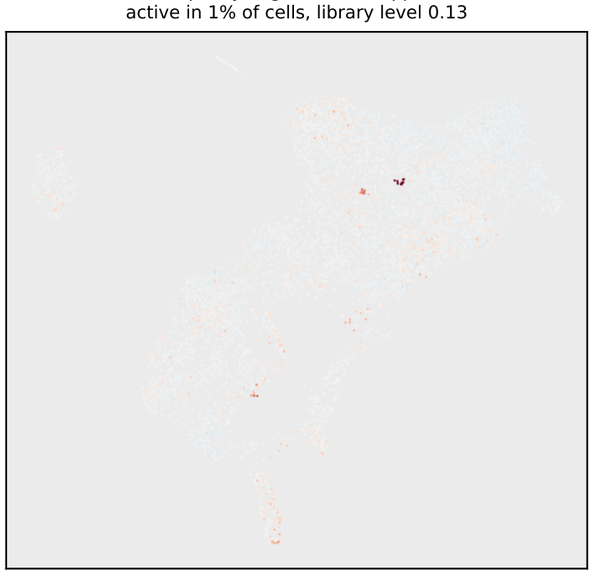


60% epiboly: 5 genes, not supported
active in 1% of cells, library level 0.08

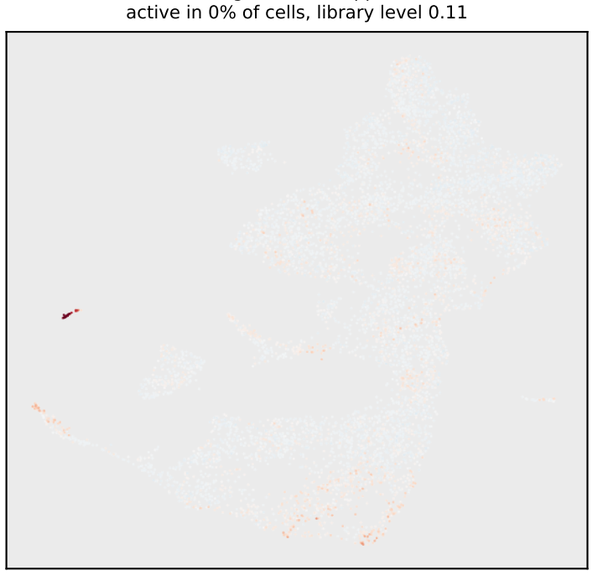


75% epiboly: 5 genes, not supported
active in 1% of cells, library level 0.07

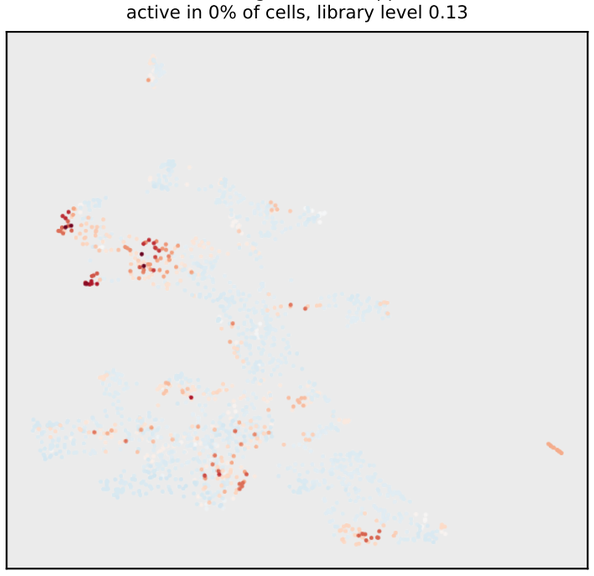
90% epiboly: 5 genes, not supported
active in 1% of cells, library level 0.13



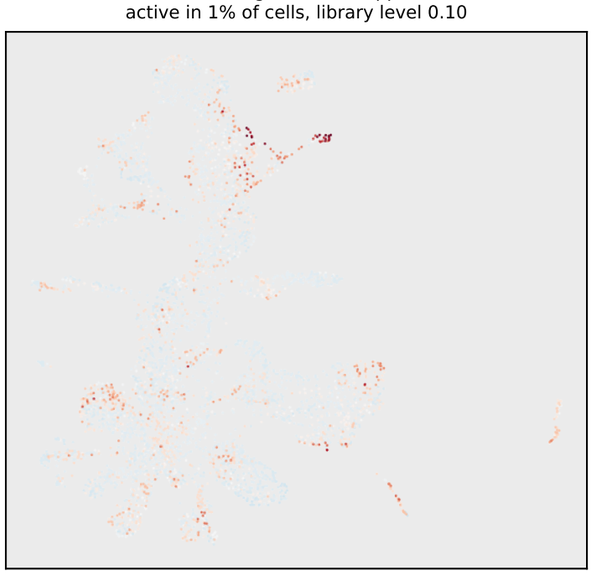
bud: 5 genes, not supported
active in 0% of cells, library level 0.11



3-somite: 3 genes, not supported
active in 0% of cells, library level 0.13

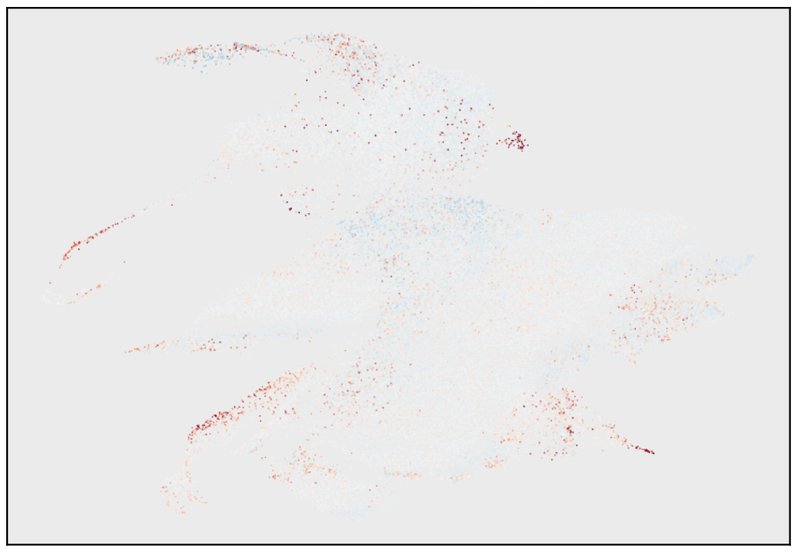


6-somite: 5 genes, not supported
active in 1% of cells, library level 0.10

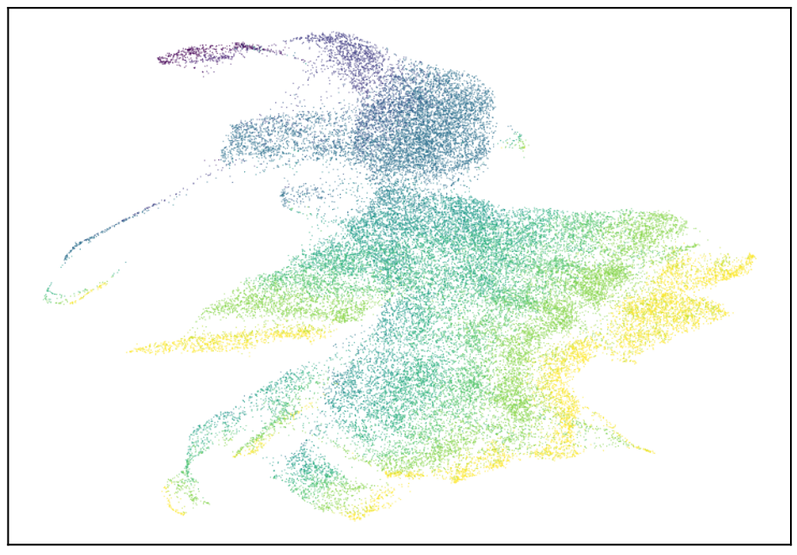


atlas, aligned (scores scaled per stage; grey: not defined)

atlas, unaligned (scores scaled per stage; grey: not defined)



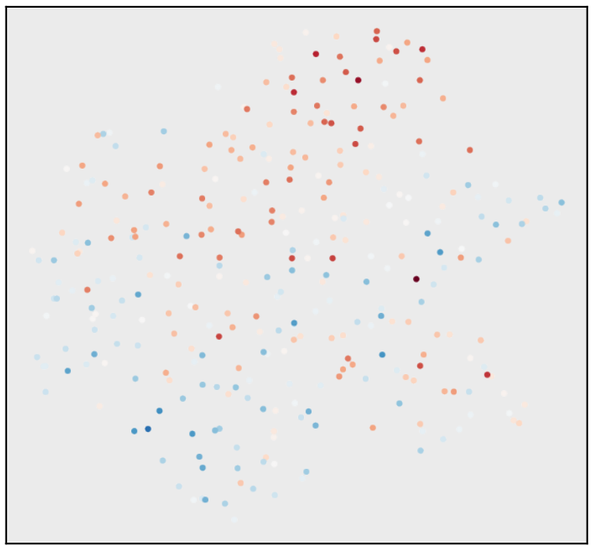
atlas, unaligned: stages



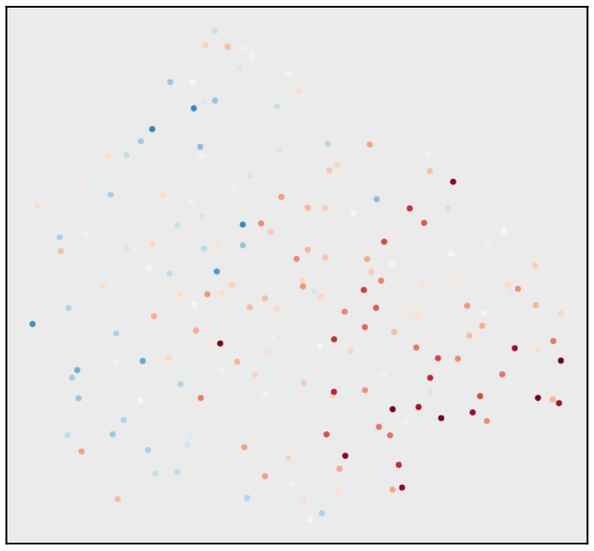
Genes (number of stages): hsp90aa1.2 (12), dnajb1a (11), hsp70.3 (10), bag3 (9), hsph1 (6)

state:maternal

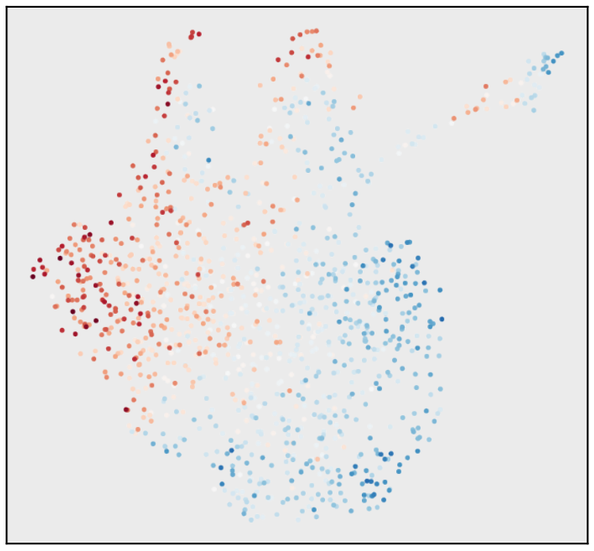
high: 10 genes, not supported
active in 8% of cells, library level 0.87



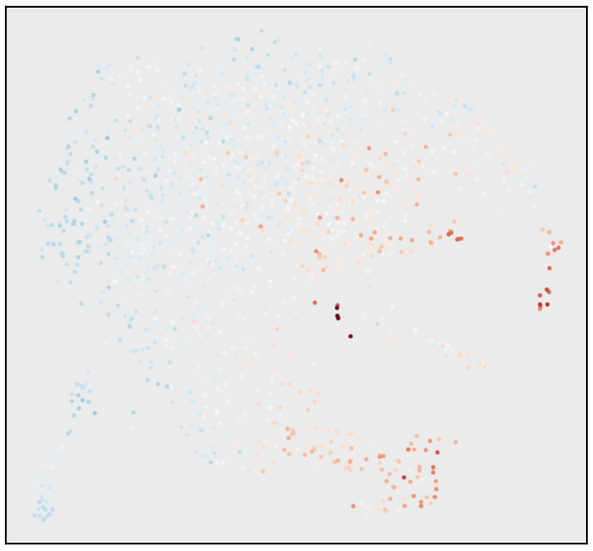
oblong: 10 genes, not supported
active in 8% of cells, library level 0.74



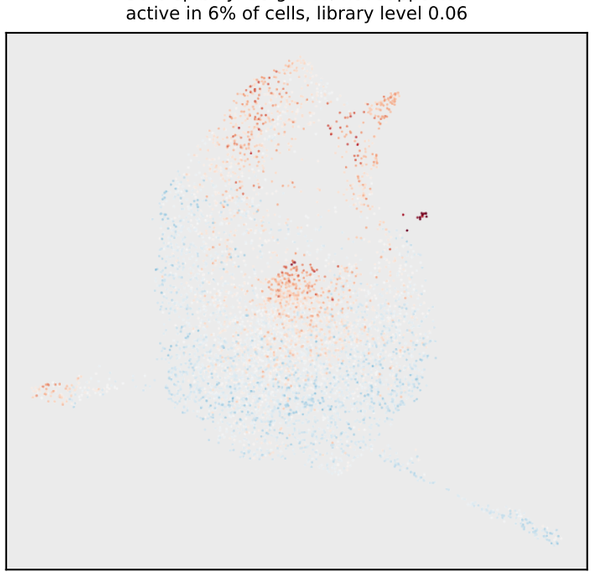
dome: 10 genes, not supported
active in 8% of cells, library level 0.23



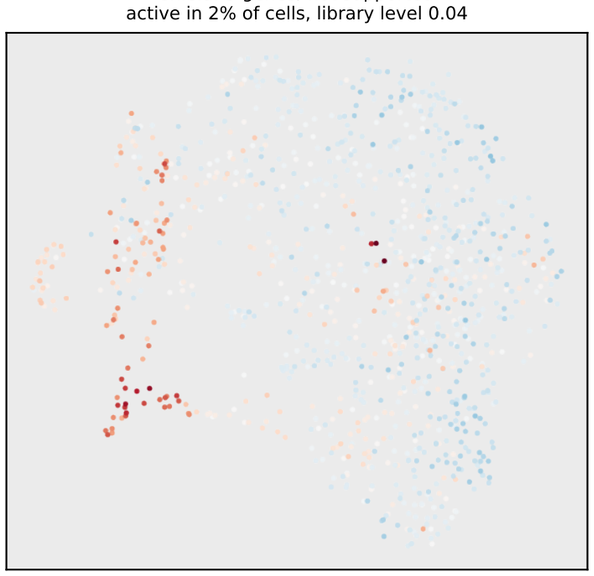
30% epiboly: 10 genes, not supported
active in 2% of cells, library level 0.08



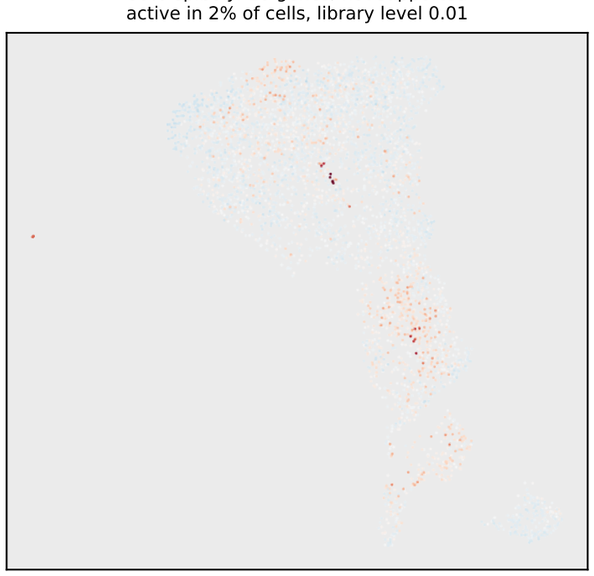
50% epiboly: 10 genes, not supported
active in 6% of cells, library level 0.06



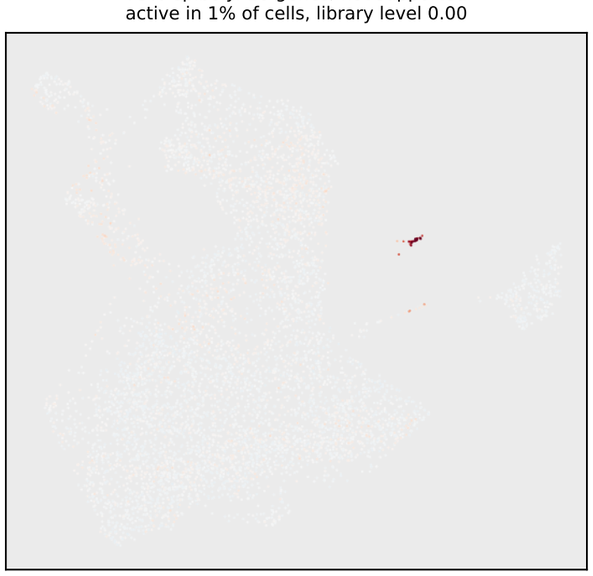
shield: 5 genes, not supported
active in 2% of cells, library level 0.04



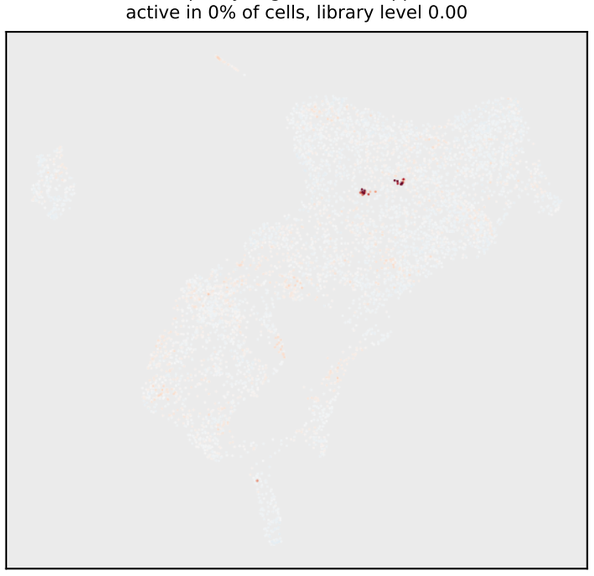
60% epiboly: 10 genes, not supported
active in 2% of cells, library level 0.01



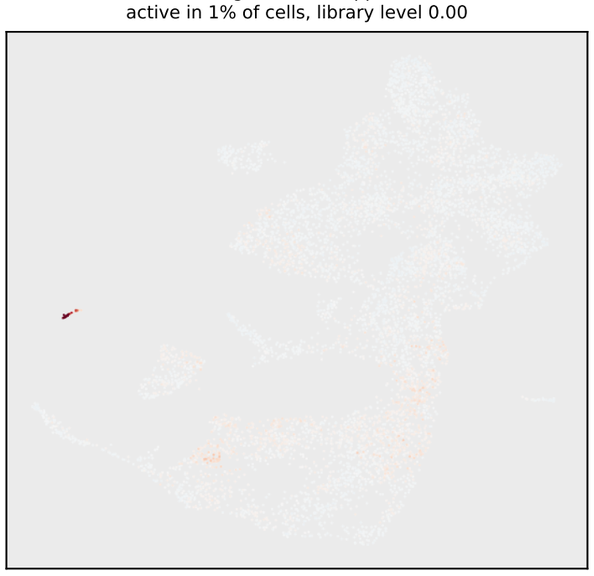
75% epiboly: 10 genes, not supported
active in 1% of cells, library level 0.00



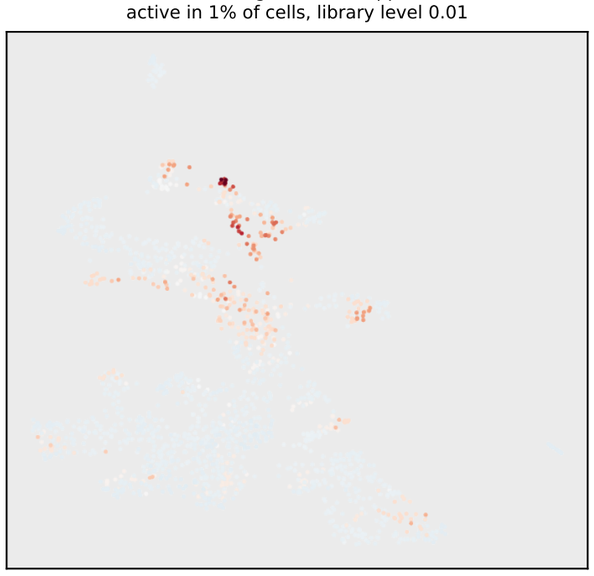
90% epiboly: 7 genes, not supported
active in 0% of cells, library level 0.00



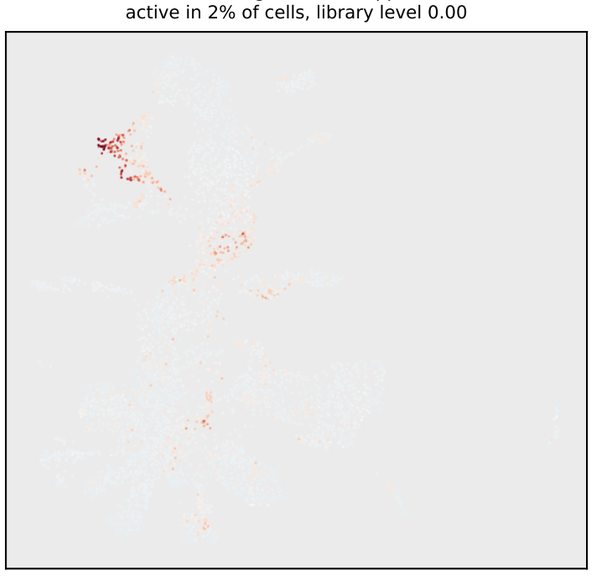
bud: 7 genes, not supported
active in 1% of cells, library level 0.00



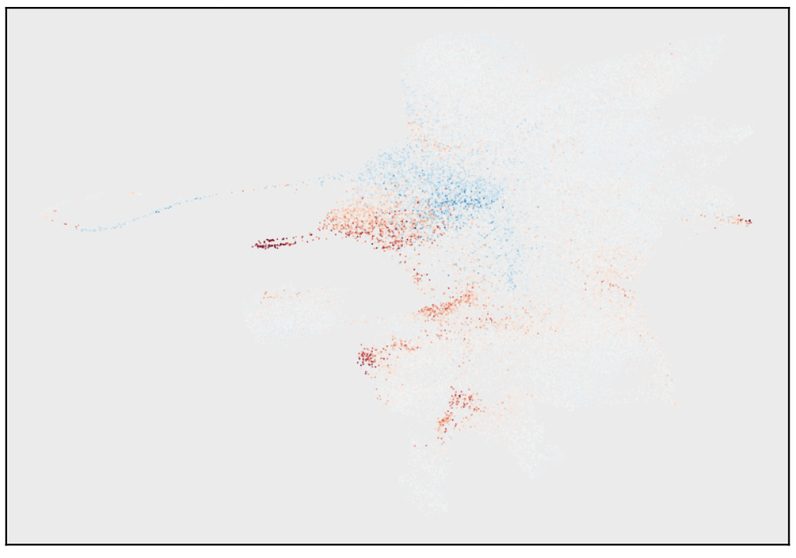
3-somite: 2 genes, not supported
active in 1% of cells, library level 0.01



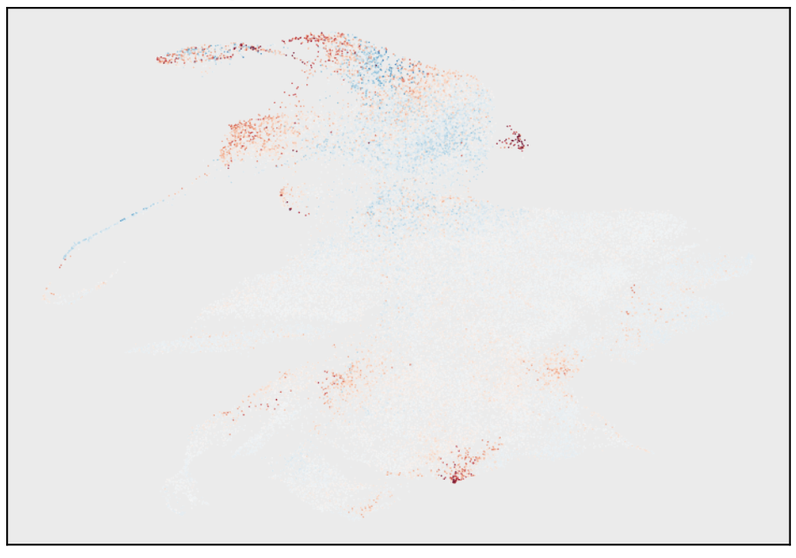
6-somite: 4 genes, not supported
active in 2% of cells, library level 0.00



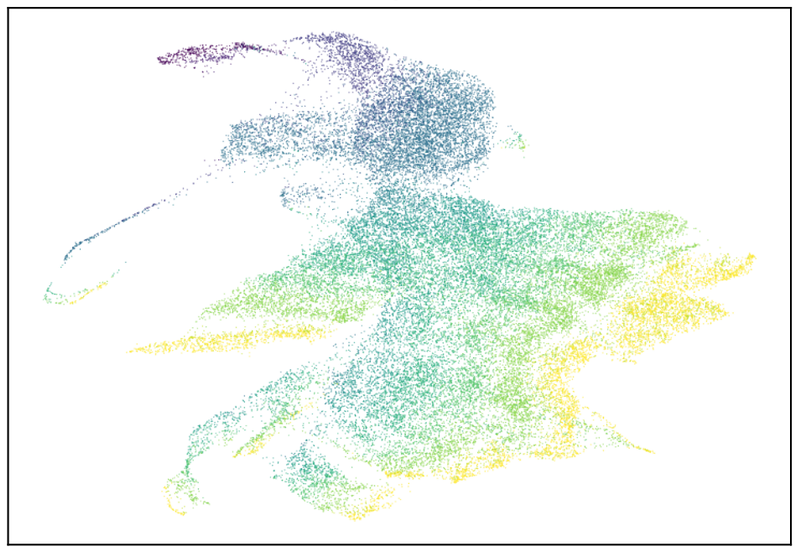
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



atlas, unaligned: stages

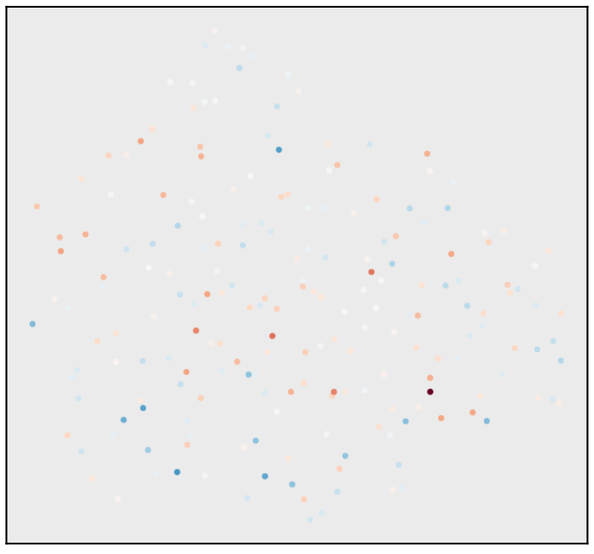


Genes (number of stages): h1m (12), gdf3 (12), cldnd (11), cth1 (10), mos (10), ccna1 (10), kpna7 (9), buc (7), btg4 (7), slbp2 (7)

state:p53_response

high: 10 genes, supported
active in 1% of cells, library level 0.58

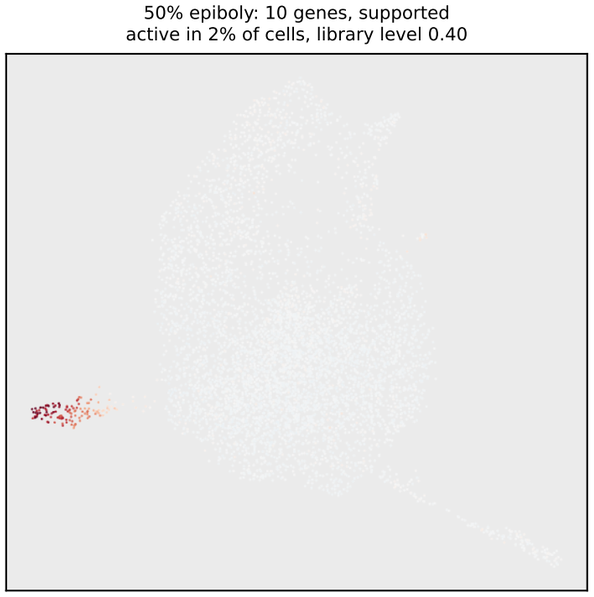
oblong: 9 genes, supported
active in 1% of cells, library level 0.69



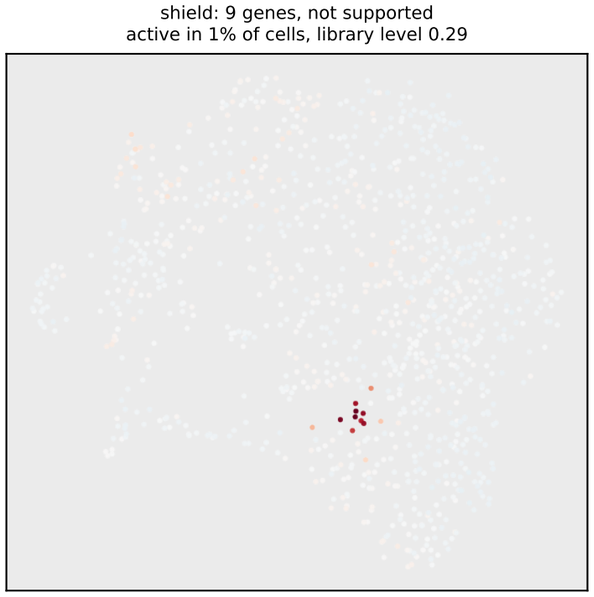
dome: 10 genes, not supported
active in 1% of cells, library level 0.27

30% epiboly: 10 genes, not supported
active in 2% of cells, library level 0.42

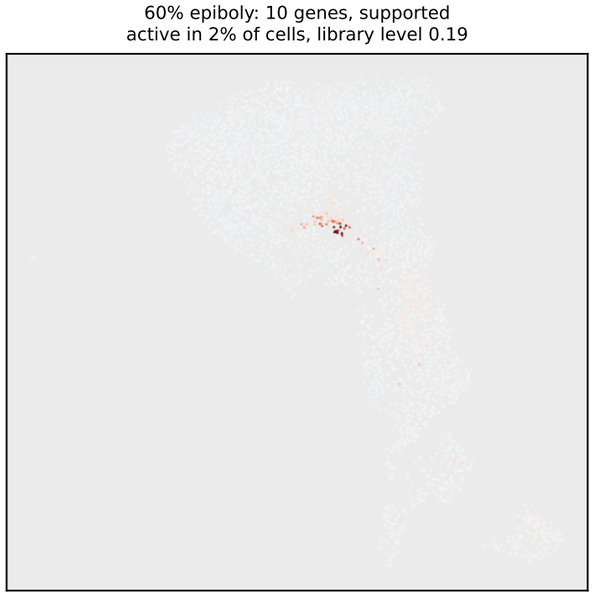
50% epiboly: 10 genes, supported
active in 2% of cells, library level 0.40



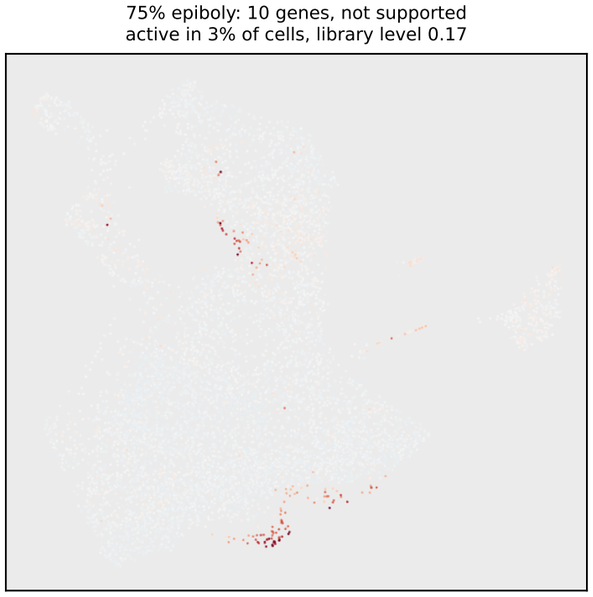
shield: 9 genes, not supported
active in 1% of cells, library level 0.29



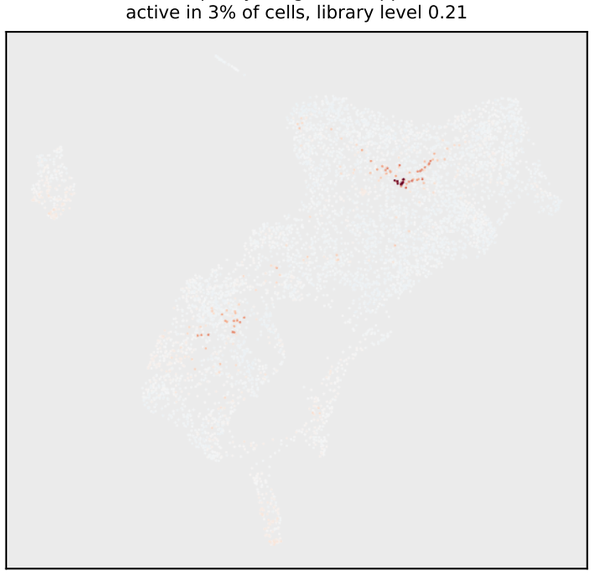
60% epiboly: 10 genes, supported
active in 2% of cells, library level 0.19



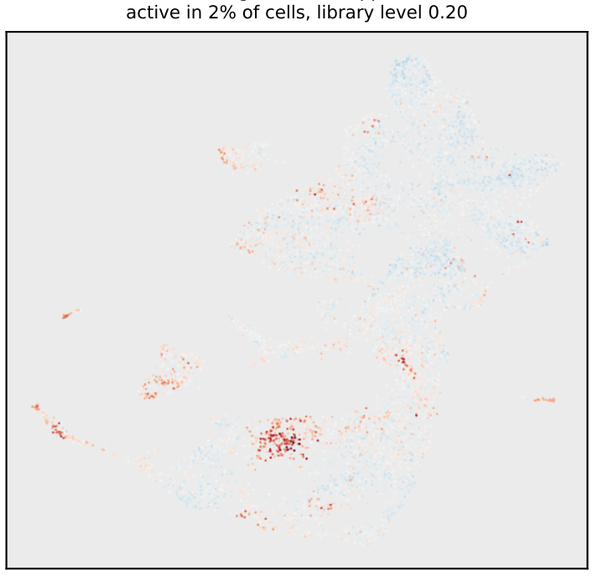
75% epiboly: 10 genes, not supported
active in 3% of cells, library level 0.17



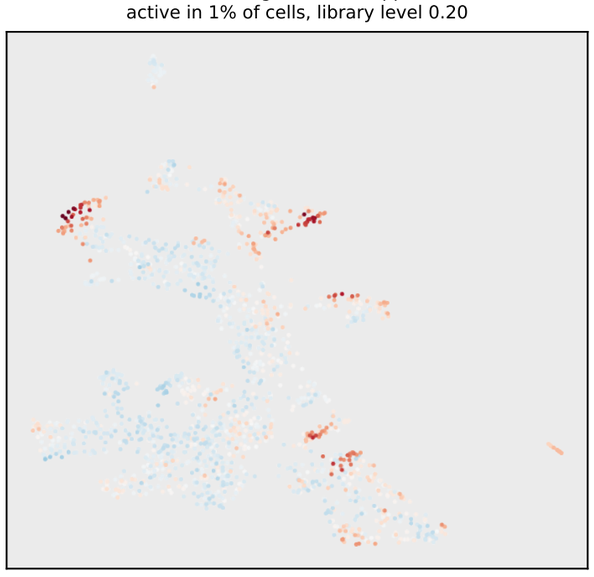
90% epiboly: 10 genes, supported
active in 3% of cells, library level 0.21



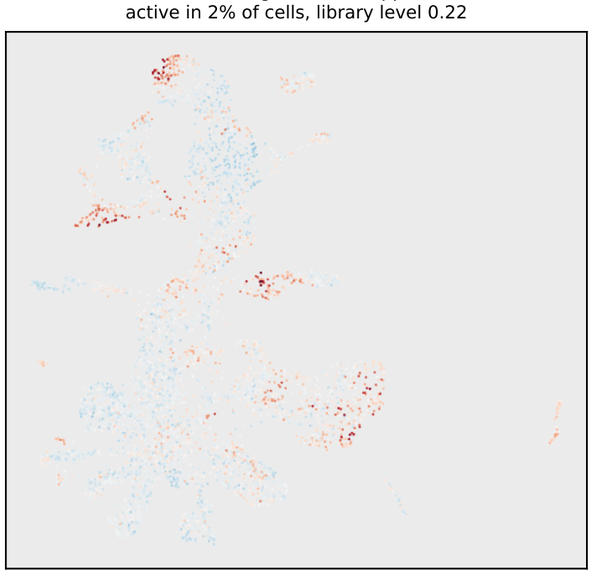
bud: 10 genes, not supported
active in 2% of cells, library level 0.20



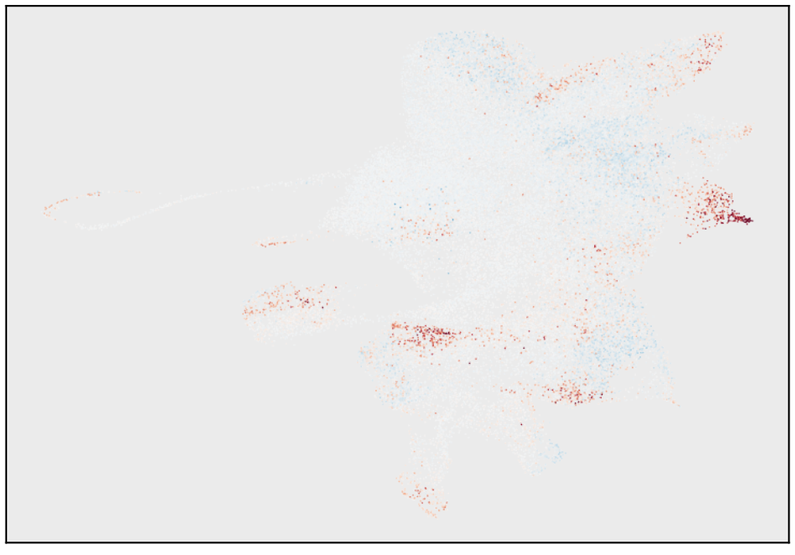
3-somite: 10 genes, not supported
active in 1% of cells, library level 0.20



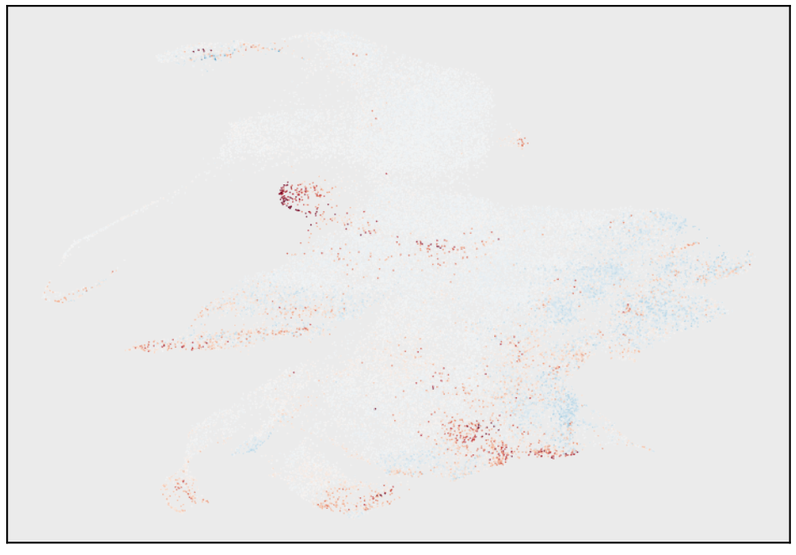
6-somite: 10 genes, not supported
active in 2% of cells, library level 0.22



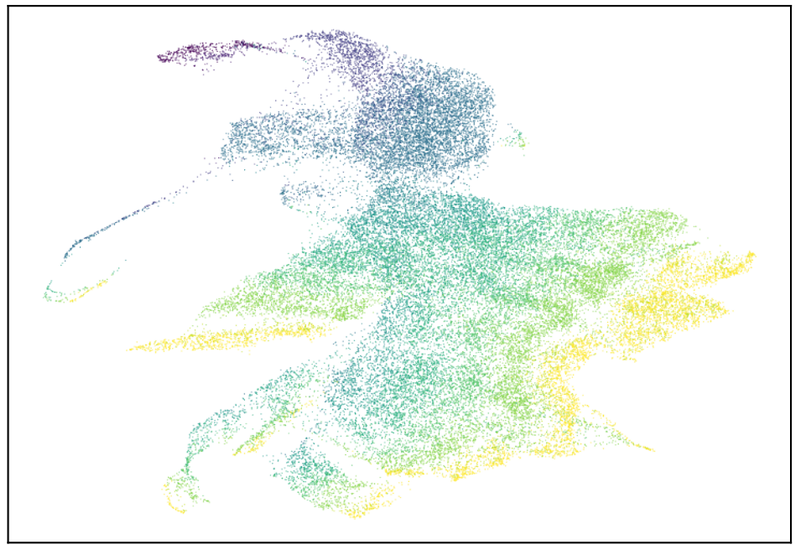
atlas, aligned (scores scaled per stage; grey: not defined)



atlas, unaligned (scores scaled per stage; grey: not defined)



atlas, unaligned: stages



Genes (number of stages): mdm2 (12), cdkn1a (12), phlda3 (12), tp53inp1 (12), rrm2b (12), sesn1 (12), ddb2 (12), plk3 (12), gadd45aa (11), fdxr (11)